

Preface and Acknowledgments

This book contains solutions to the problems presented in *Modern Physics for Scientists and Engineers, Fourth Edition* by Thornton and Rex. It is intended for instructors only and is specifically not intended for general distribution to students. Instructors may wish to share solutions to specific problems with students, but the copying of whole sections or chapters of this book for distribution to students is strongly discouraged.

A *Student Solutions Manual*, which contains solutions to approximately one-fourth of the problems, is available. If you want your students to have that manual, either as a required or an optional text, you can order it through your bookstore (through Cengage Learning).

We would like to thank Stephen T. Thornton who offered suggestions for solutions to many of the problems. We also especially thank Paul Weber of the University of Puget Sound and Thushara Parera of Illinois Wesleyan University who checked these solutions and offered many corrections and suggestions for clarification.

For most physics problems, alternate solutions are possible. We welcome suggestions of alternate solutions and especially identification of any errors in this text, which are ours alone.

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Chapter 2

1. For a particle Newton's second law says $\vec{F} = m\vec{a} = m \left[\frac{d^2x}{dt^2} \hat{i} + \frac{d^2y}{dt^2} \hat{j} + \frac{d^2z}{dt^2} \hat{k} \right]$.

Take the second derivative of each of the expressions in Equation (2.1):

$$\frac{d^2x'}{dt^2} = \frac{d^2x}{dt^2} \quad \frac{d^2y'}{dt^2} = \frac{d^2y}{dt^2} \quad \frac{d^2z'}{dt^2} = \frac{d^2z}{dt^2}. \text{ Substitution into the previous equation gives}$$

$$\vec{F} = m\vec{a} = m \left[\frac{d^2x'}{dt^2} \hat{i} + \frac{d^2y'}{dt^2} \hat{j} + \frac{d^2z'}{dt^2} \hat{k} \right] = \vec{F}'.$$

2. From Equation (2.1) $\vec{p} = m \left[\frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k} \right]$.

$$\text{In a Galilean transformation } \frac{dx'}{dt} = \frac{dx}{dt} - v \quad \frac{dy'}{dt} = \frac{dy}{dt} \quad \frac{dz'}{dt} = \frac{dz}{dt}.$$

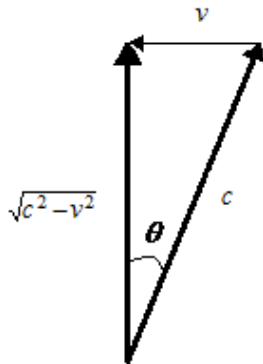
$$\text{Substitution into Equation (2.1) gives } \vec{p} = m \left[\frac{dx'}{dt} + v \right] \hat{i} + \frac{dy'}{dt} \hat{j} + \frac{dz'}{dt} \hat{k} \neq \vec{p}'.$$

However, because $\vec{p}' = m \left[\frac{dx'}{dt} \hat{i} + \frac{dy'}{dt} \hat{j} + \frac{dz'}{dt} \hat{k} \right]$ the same form is clearly retained, given

$$\text{the velocity transformation } \frac{dx'}{dt} = \frac{dx}{dt} - v.$$

3. Using the vector triangle shown, the speed of light coming toward the mirror is $\sqrt{c^2 - v^2}$ and the same on the return trip. Therefore the total time is $t_2 = \frac{\text{distance}}{\text{speed}} = \frac{2\ell_2}{\sqrt{c^2 - v^2}}.$

Notice that $\sin \theta = \frac{v}{c}$, so $\theta = \sin^{-1} \left(\frac{v}{c} \right).$



4. As in Problem 3, $\sin \theta = v_1 / v_2$, so $\theta = \sin^{-1}(v_1 / v_2) = \sin^{-1}\left[\frac{0.350 \text{ m/s}}{1.25 \text{ m/s}}\right] = 16.3^\circ$ and

$$v = \sqrt{v_2^2 - v_1^2} = \sqrt{(1.25 \text{ m/s})^2 - (0.35 \text{ m/s})^2} = 1.20 \text{ m/s}.$$

5. When the apparatus is rotated by 90° , the situation is equivalent, except that we have effectively interchanged ℓ_1 and ℓ_2 . Interchanging ℓ_1 and ℓ_2 in Equation (2.3) leads to Equation (2.4).
6. Let n = the number of fringes shifted; then $n = \frac{\Delta d}{\lambda}$. Because $\Delta d = c(\Delta t' - \Delta t)$, we have

$$n = \frac{c(\Delta t' - \Delta t)}{\lambda} = \frac{v^2(\ell_1 + \ell_2)}{c^2 \lambda}. \text{ Solving for } v \text{ and noting that } \ell_1 + \ell_2 = 22 \text{ m,}$$

$$v = c \sqrt{\frac{n\lambda}{\ell_1 + \ell_2}} = (3.00 \times 10^8 \text{ m/s}) \sqrt{\frac{(0.005)(589 \times 10^{-9} \text{ m})}{22 \text{ m}}} = 3.47 \text{ km/s}.$$

7. Letting $\ell_1 \rightarrow \ell_1 \sqrt{1 - \beta^2}$ (where $\beta = v/c$) the text equation (not currently numbered) for t_1 becomes

$$t_1 = \frac{2\ell_1 \sqrt{1 - \beta^2}}{c(1 - \beta)} = \frac{2\ell_1}{c\sqrt{1 - \beta^2}}$$

which is identical to t_2 when $\ell_1 = \ell_2$ so $\Delta t = 0$ as required.

8. Since the Lorentz transformations depend on c (and the fact that c is the same constant for all inertial frames), different values of c would necessarily lead two observers to different conclusions about the order or positions of two spacetime events, in violation of postulate 1.

9. Let an observer in K send a light signal along the $+x$ -axis with speed c . According to the Galilean transformations, an observer in K' measures the speed of the signal to be

$$\frac{dx'}{dt} = \frac{dx}{dt} - v = c - v. \text{ Therefore the speed of light cannot be constant under the Galilean transformations.}$$

10. From the Principle of Relativity, we know the correct transformation must be of the form (assuming $y = y'$ and $z = z'$):

$$x = ax' + bt'; \quad x' = ax - bt.$$

The spherical wave front equations (2.9a) and (2.9b) give us:

$$ct = (ac + b)t'; \quad ct' = (ac - b)t.$$

Solve the second wave front equation for t' and substitute into the first:

$$ct = \left[\frac{(ac+b)(ac-b)t}{c} \right] \text{ or } c^2 = (ac+b)(ac-b) = a^2c^2 - b^2.$$

Now v is the speed of the origin of the x' -axis. We can find that speed by setting $x' = 0$ which gives $0 = ax - bt$, or $v = x/t = b/a$, or equivalently $b = av$. Substituting this into the equation above for c^2 yields $c^2 = a^2c^2 - a^2v^2 = a^2(c^2 - v^2)$. Solving for a :

$$a = \frac{1}{\sqrt{1 - v^2/c^2}} = \gamma.$$

This expression, along with $b = av$, can be substituted into the original expressions for x and x' to obtain:

$$x = \gamma(x' + vt'); \quad x' = \gamma(x - vt)$$

which in turn can be solved for t and t' to complete the transformation.

11. When $v \ll c$ we find $1 - \beta^2 \rightarrow 1$, so:

$$x' = \frac{x - \beta ct}{\sqrt{1 - \beta^2}} \rightarrow x - \beta ct = x - vt;$$

$$t' = \frac{t - \beta x/c}{\sqrt{1 - \beta^2}} \rightarrow t - \beta x/c \approx t;$$

$$x = \frac{x' + \beta ct'}{\sqrt{1 - \beta^2}} \rightarrow x' + \beta ct' = x' + vt';$$

$$t = \frac{t' + \beta x'/c}{\sqrt{1 - \beta^2}} \rightarrow t' + \beta x'/c \approx t'.$$

12. (a) First we convert to SI units: $95 \text{ km/h} = 26.39 \text{ m/s}$, so

$$\beta = v/c = (26.39 \text{ m/s}) / (3.00 \times 10^8 \text{ m/s}) = 8.8 \times 10^{-8}$$

$$(b) \beta = v/c = (240 \text{ m/s}) / (3.00 \times 10^8 \text{ m/s}) = 8.0 \times 10^{-7}$$

$$(c) v = 2.3v_{\text{sound}} = (2.3 \times 330 \text{ m/s}) \text{ so}$$

$$\beta = v/c = (2.3 \times 330 \text{ m/s}) / (3.00 \times 10^8 \text{ m/s}) = 2.5 \times 10^{-6}$$

(d) Converting to SI units, $27,000 \text{ km/h} = 7500 \text{ m/s}$, so

$$\beta = v/c = (7500 \text{ m/s}) / (3.00 \times 10^8 \text{ m/s}) = 2.5 \times 10^{-5}$$

$$(e) (25 \text{ cm}) / (2 \text{ ns}) = 1.25 \times 10^8 \text{ m/s so } \beta = v/c = (1.25 \times 10^8 \text{ m/s}) / (3.00 \times 10^8 \text{ m/s}) = 0.42$$

$$(f) (1 \times 10^{-14} \text{ m}) / (0.35 \times 10^{-22} \text{ s}) = 2.857 \times 10^8 \text{ m/s , so}$$

$$\beta = v/c = (2.857 \times 10^8 \text{ m/s}) / (3.00 \times 10^8 \text{ m/s}) = 0.95$$

13. From the Lorentz transformations $\Delta t' = \gamma [\Delta t - v\Delta x / c^2]$. But $\Delta t' = 0$ in this case, so solving for v we find $v = c^2 \Delta t / \Delta x$. Inserting the values $\Delta t = t_2 - t_1 = -a / 2c$ and $\Delta x = x_2 - x_1 = a$, we find $v = \frac{c^2 (-a / 2c)}{a} = -c / 2$. We conclude that the frame K' travels at a speed $c/2$ in the $-x$ -direction. Note that there is no motion in the transverse direction.

14. Try setting $\Delta x' = 0 = \gamma (\Delta x - v\Delta t)$. Thus $0 = \Delta x - v\Delta t = a + va / 2c$. Solving for v we find $v = -2c$, which is impossible. There is no such frame K' .

15. For the smaller values of β we use the binomial expansion $\gamma = (1 - \beta^2)^{-1/2} \approx 1 + \beta^2 / 2$.

(a) $\gamma \approx 1 + \beta^2 / 2 = 1 + 3.87 \times 10^{-15}$

(b) $\gamma \approx 1 + \beta^2 / 2 = 1 + 3.2 \times 10^{-13}$

(c) $\gamma \approx 1 + \beta^2 / 2 = 1 + 3.1 \times 10^{-12}$

(d) $\gamma \approx 1 + \beta^2 / 2 = 1 + 3.1 \times 10^{-10}$

(e) $\gamma = (1 - \beta^2)^{-1/2} = (1 - 0.42^2)^{-1/2} = 1.10$

(f) $\gamma = (1 - \beta^2)^{-1/2} = (1 - 0.95^2)^{-1/2} = 3.20$

16. There is no motion in the transverse direction, so $y = z = 3.5$ m.

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}} = \frac{1}{\sqrt{1 - 0.8^2}} = 5/3$$

$$x = \gamma (x' + vt') = \frac{5}{3} (2\text{ m} + 0.8c(0)) = 10/3 \text{ m}$$

$$t = \gamma (t' + vx' / c^2) = \frac{5}{3} (0 + (0.8c)(2 \text{ m}) / c^2) = 8.9 \times 10^{-9} \text{ s}$$

17. (a) $t = \frac{\sqrt{x^2 + y^2 + z^2}}{c} = \frac{\sqrt{(3 \text{ m})^2 + (5 \text{ m})^2 + (10 \text{ m})^2}}{3.00 \times 10^8 \text{ m/s}} = 3.86 \times 10^{-8} \text{ s}$

(b) With $\beta = 0.8$ we find $\gamma = 5/3$. Then $y' = y = 5$ m, $z' = z = 10$ m,

$$x' = \gamma (x - vt) = \frac{5}{3} [3\text{ m} - (2.40 \times 10^8 \text{ m/s})(3.86 \times 10^{-8} \text{ s})] = -10.4 \text{ m}$$

$$t' = \gamma (t - vx / c^2) = \frac{5}{3} [(3.86 \times 10^{-8} \text{ s}) - (2.40 \times 10^8 \text{ m/s})(3 \text{ m}) / (3.00 \times 10^8 \text{ m/s})^2] = 51.0 \text{ ns}$$

(c) $\frac{\sqrt{x'^2 + y'^2 + z'^2}}{t'} = \frac{\sqrt{(-10.4 \text{ m})^2 + (5 \text{ m})^2 + (10 \text{ m})^2}}{51.0 \times 10^{-9} \text{ s}} = 2.994 \times 10^8 \text{ m/s}$ which equals c to within rounding errors.

18. At the point of reflection the light has traveled a distance $L + v\Delta t_1 = c\Delta t_1$. On the return trip it travels $L - v\Delta t_2 = c\Delta t_2$. Then the total time is $\Delta t = \Delta t_1 + \Delta t_2 = \frac{2Lc}{c^2 - v^2} = \frac{2L/c}{1 - v^2/c^2}$.

But from time dilation we know (with $\Delta t' = \text{proper time} = 2L_0/c$) that

$$\Delta t = \gamma \Delta t' = \frac{2L_0/c}{\sqrt{1 - v^2/c^2}}. \text{ Comparing these two results for } \Delta t \text{ we get } \frac{2L/c}{1 - v^2/c^2} = \frac{2L_0/c}{\sqrt{1 - v^2/c^2}}$$

which reduces to $L = L_0 \sqrt{1 - v^2/c^2} = \frac{L_0}{\gamma}$. This is Equation (2.21).

19. (a) With a contraction of 1%, $L/L_0 = 0.99 = \sqrt{1 - v^2/c^2}$. Thus $1 - \beta^2 = (0.99)^2 = 0.9801$.

Solving for β , we find $\beta = 0.14$ or $v = 0.14c$.

(b) The time for the trip in the Earth-based frame is

$$\Delta t = \frac{d}{v} = \frac{5.00 \times 10^6 \text{ m}}{0.14 \times 3.00 \times 10^8 \text{ m/s}} = 1.19 \times 10^{-1} \text{ s}. \text{ With the relativistic factor } \gamma = 1.01$$

(corresponding to a 1% shortening of the ship's length), the elapsed time on the rocket ship is 1% less than the Earth-based time, or a difference of

$$(0.01)1.2 \times 10^{-1} \text{ s} = 1.2 \times 10^{-3} \text{ s}.$$

20. The round-trip distance is $d = 40 \text{ ly}$. Assume the same constant speed $v = \beta c$ for the entire round trip. In the rocket's reference frame the distance is only $d' = d\sqrt{1 - \beta^2}$. Then

$$\text{in the rocket's frame of reference } v = \frac{\text{distance}}{\text{time}} = \frac{d'}{40 \text{ y}} = \frac{40 \text{ ly } \sqrt{1 - \beta^2}}{40 \text{ y}} = c\sqrt{1 - \beta^2}.$$

$$\text{Rearranging } \beta = \frac{v}{c} = \sqrt{1 - \beta^2}. \text{ Solving for } \beta \text{ we find } \beta = \sqrt{0.5}, \text{ or } v = \sqrt{0.5}c \approx 0.71c.$$

$$\text{To find the elapsed time } t \text{ on Earth, we know } t' = 40 \text{ y, so } t = \gamma t' = \frac{1}{\sqrt{1 - \beta^2}} 40 \text{ y} = 56.6 \text{ y}.$$

21. In the muon's frame $T_0 = 2.2 \text{ } \mu\text{s}$. In the lab frame the time is longer; see Equation (2.19):

$$T' = \gamma T_0. \text{ In the lab the distance traveled is } 9.5 \text{ cm} = vT' = v\gamma T_0 = \beta c\gamma T_0, \text{ since } v = \beta c.$$

Therefore $\beta = \frac{9.5 \text{ cm}(\sqrt{1-\beta^2})}{cT_0}$, so $\beta = \frac{v}{c} = \frac{9.5 \text{ cm}(\sqrt{1-\beta^2})}{c(2.2\mu\text{s})}$. Now all quantities are known except β . Solving for β we find $\beta = 1.4 \times 10^{-4}$ or $v = 1.4 \times 10^{-4} c$.

22. Converting the speed to m/s we find $25,000 \text{ mi/h} = 11,176 \text{ m/s}$. From tables the distance is $3.84 \times 10^8 \text{ m}$. In the earth's frame of reference the time is the distance divided by

speed, or $t = \frac{d}{v} = \frac{3.84 \times 10^8 \text{ m}}{11,176 \text{ m/s}} = 34,359 \text{ s}$. In the astronauts' frame the time elapsed is

$$t' = t / \gamma = t \sqrt{1-\beta^2}. \text{ The time difference is } \Delta t = t - t' = t - t \sqrt{1-\beta^2} = t \left[1 - \sqrt{1-\beta^2} \right].$$

$$\text{Evaluating numerically } \Delta t = 34,359 \text{ s} \left[1 - \sqrt{1 - \left(\frac{11,176 \text{ m/s}}{3.00 \times 10^8 \text{ m/s}} \right)^2} \right] = 2.4 \times 10^{-5} \text{ s}.$$

23. $T' = \gamma T_0$, so we know that $\gamma = 5/3 = \frac{1}{\sqrt{1-v^2/c^2}}$. Solving for v we find $v = 4c/5$.

24. $L = L_0 / \gamma$ so clearly $\gamma = 2$ in this case. Thus $2 = \frac{1}{\sqrt{1-v^2/c^2}}$ and solving for v we find

$$v = \frac{\sqrt{3}c}{2}.$$

25. The clocks' rates differ by a factor of $\gamma = 1/\sqrt{1-v^2/c^2}$. Because β is very small we will use the binomial theorem approximation $\gamma \approx 1 + \beta^2/2$. Then the time difference is $\Delta t = t - t' = t - \gamma t = t(\gamma - 1)$. Using $\gamma - 1 \approx \beta^2/2$ and the fact that the time for the trip equals distance divided by speed,

$$\Delta t = t(\beta^2/2) = \frac{8 \times 10^6 \text{ m} \left(\frac{375 \text{ m/s}}{3.00 \times 10^8 \text{ m/s}} \right)^2}{375 \text{ m/s} \cdot 2}$$

$$\Delta t = 1.67 \times 10^{-8} \text{ s} = 16.7 \text{ ns}.$$

26. (a) $L' = L / \gamma = L \sqrt{1-v^2/c^2} = (3.58 \times 10^4 \text{ km}) \sqrt{1-0.94^2} = 1.22 \times 10^4 \text{ km}$

(b) Earth's frame: $t = L / v = \frac{3.58 \times 10^7 \text{ m}}{(0.94)(3.00 \times 10^8 \text{ m/s})} = 0.127 \text{ s}$

Golf ball's frame: $t' = t / \gamma = 0.127 \text{ s} \sqrt{1-0.94^2} = 0.0433 \text{ s}$

27. Spacetime invariant (see Section 2.9): $c^2\Delta t^2 - \Delta x^2 = c^2\Delta t'^2 - \Delta x'^2$. We know $\Delta x = 4$ km,

$$\Delta t = 0, \text{ and } \Delta x' = 5 \text{ km. Thus } \Delta t'^2 = \frac{\Delta x'^2 - \Delta x^2}{c^2} = \frac{(5000 \text{ m})^2 - (4000 \text{ m})^2}{(3.00 \times 10^8 \text{ m/s})^2} = 1.0 \times 10^{-10} \text{ s}^2$$

and $\Delta t' = 1.0 \times 10^{-5} \text{ s}$.

28. (a) Converting $v = 120 \text{ km/h} = 33.3 \text{ m/s}$. Now with $c = 100 \text{ m/s}$, we have

$$\beta = v/c = 0.333 \text{ and } \gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-0.333^2}} = 1.061. \text{ We conclude that the moving}$$

person ages 6.1% slower.

$$(b) L' = L/\gamma = (1 \text{ m})/(1.061) = 0.942 \text{ m.}$$

29. Converting $v = 300 \text{ km/h} = 83.3 \text{ m/s}$. Now with $c = 100 \text{ m/s}$, we have $\beta = v/c = 0.833$

$$\text{and } \gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-0.833^2}} = 1.81. \text{ So the length is } L = L_0/\gamma = 40/1.81 = 22.1 \text{ m.}$$

30. Let subscript 1 refer to firing and subscript 2 to striking the target. Therefore we can see that $x_1 = 1 \text{ m}$, $x_2 = 121 \text{ m}$, and $t_1 = 3 \text{ ns}$.

$$t_2 = t_1 + \frac{\text{distance}}{\text{speed}} = 3 \text{ ns} + \frac{120 \text{ m}}{0.98c} = 3 \text{ ns} + 408 \text{ ns} = 411 \text{ ns.}$$

To find the four primed quantities we can use the Lorentz transformations with the known values of x_1 , x_2 , t_1 , and t_2 . Note that with $v = 0.8c$, $\gamma = \sqrt{1-v^2/c^2} = 5/3$.

$$t'_1 = \gamma(t_1 - vx_1/c^2) = 0.56 \text{ ns}$$

$$t'_2 = \gamma(t_2 - vx_2/c^2) = 147 \text{ ns}$$

$$x'_1 = \gamma(x_1 - vt_1) = 0.47 \text{ m}$$

$$x'_2 = \gamma(x_2 - vt_2) = 37.3 \text{ m}$$

31. Start from the formula for velocity addition, Equation (2.23a): $u_x = \frac{u'_x + v}{1 + vu'_x/c^2}$.

$$(a) u_x = \frac{0.62c + 0.84c}{1 + (0.62c)(0.84c)/c^2} = \frac{1.46c}{1.52} = 0.96c$$

$$(b) u_x = \frac{-0.62c + 0.84c}{1 + (-0.62c)(0.84c)/c^2} = \frac{0.22c}{0.48} = 0.46c$$

32. Velocity addition, Equation (2.24): $u'_x = \frac{u_x - v}{1 - vu_x/c^2}$ with $v = -0.8c$ and $u_x = 0.8c$.

$$u'_x = \frac{0.8c - (-0.8c)}{1 - (-0.8c)(0.8c)/c^2} = \frac{1.6c}{1.64} = 0.976c$$

33. Conversion: $110 \text{ km/h} = 30.556 \text{ m/s}$ and $140 \text{ km/h} = 38.889 \text{ m/s}$. Let $u_x = 30.556 \text{ m/s}$ and $v = -38.889 \text{ m/s}$. Our premise is that $c = 100 \text{ m/s}$. Then by velocity addition,

$$u'_x = \frac{u_x - v}{1 - vu_x / c^2} = \frac{30.556 \text{ m/s} - (-38.889 \text{ m/s})}{1 - (-38.889 \text{ m/s})(30.556 \text{ m/s}) / (100 \text{ m/s})^2} = 62.1 \text{ m/s.}$$

By symmetry each observer sees the other one traveling at the same speed.

34. From Example 2.5 we have $u = \frac{c}{n} \left[\frac{1 + nv/c}{1 + v/nc} \right]$. For light traveling in opposite directions

$$\Delta u = \frac{c}{n} \left[\frac{1 + nv/c}{1 + v/nc} - \frac{1 - nv/c}{1 - v/nc} \right].$$

Because v/c is very small, use the binomial expansion: $\frac{1 + nv/c}{1 + v/nc} = (1 + nv/c)(1 + v/nc)^{-1} \approx (1 + nv/c)(1 - v/nc) \approx 1 + nv/c - v/nc$, where we

have dropped terms of order v^2/c^2 . Similarly $\frac{1 - nv/c}{1 - v/nc} \approx 1 - nv/c + v/nc$. Thus

$$\Delta u \approx \frac{c}{n} \left[(1 + nv/c - v/nc) - (1 - nv/c + v/nc) \right] = \frac{2v}{n} (1 - 1/n^2) = 2v(1 - 1/n^2).$$

Evaluating numerically we find $\Delta u \approx 2(5 \text{ m/s}) \left(1 - \frac{1}{1.33^2} \right) = 4.35 \text{ m/s}$.

35. Clearly the speed of B is just $0.60c$. To find the speed of C use $u_x = 0.60c$ and

$$v = -0.60c: \quad u'_x = \frac{u_x - v}{1 - vu_x / c^2} = \frac{0.60c - (-0.60c)}{1 - (-0.60c)(0.60c) / c^2} = 0.88c.$$

36. We can ignore the 400 km, which is small compared with the Earth-to-moon distance $3.84 \times 10^8 \text{ m}$. The rotation rate is $\omega = 2\pi \text{ rad} \times 100 \text{ s}^{-1} = 2\pi \times 10^2 \text{ rad/s}$. Then the speed across the moon's surface is $v = \omega R = (2\pi \times 10^2 \text{ rad/s})(3.84 \times 10^8 \text{ m}) = 2.41 \times 10^{11} \text{ m/s}$.

37. Classical: $t = \frac{4205 \text{ m}}{0.98c} = 1.43 \times 10^{-5} \text{ s}$. Then $N = N_0 \exp \left[\frac{-(\ln 2)t}{t_{1/2}} \right] = 14.6$

or about 15 muons.

$$\text{Relativistic: } t' = t / \gamma = \frac{1.43 \times 10^{-5} \text{ s}}{5} = 2.86 \times 10^{-6} \text{ s so}$$

$$N = N_0 \exp \left[\frac{-(\ln 2)t}{t_{1/2}} \right] = 2710 \text{ muons.}$$

Because of the exponential nature of the decay curve, a factor of five (shorter) in time results in many more muons surviving.

38. The circumference of the fixed point's rotational path is $2\pi R_E \cos(39^\circ)$, where $R_E = \text{Earth's radius} = 6378 \text{ km}$. Thus the circumference of the path is 31,143 km. The

rotational speed of that point is $v = (31,143 \text{ km}) / 24 \text{ h} = 1298 \text{ km/h} = 360.5 \text{ m/s}$. The

observatory clock runs slow by a factor of $\gamma = \frac{1}{\sqrt{1-\beta^2}} \approx 1 + \beta^2 / 2 = 1 + 7.22 \times 10^{-13}$. In

41.2 h the observatory clock is slow by $(41.2 \text{ h})(7.22 \times 10^{-13}) = 2.9746 \times 10^{-11} \text{ h} = 107 \text{ ns}$.

In 48.6 h it is slow by $(48.6 \text{ h})(7.22 \times 10^{-13}) = 3.5089 \times 10^{-11} \text{ h} = 126 \text{ ns}$. The Eastward-moving clock has a ground speed of $31,143 \text{ km} / 41.2 \text{ h} = 755.9 \text{ km/h} = 210.0 \text{ m/s}$ and thus has a net speed of $210.0 \text{ m/s} + 360.5 \text{ m/s} = 570.5 \text{ m/s}$. For this clock

$\gamma = \frac{1}{\sqrt{1-\beta^2}} \approx 1 + \beta^2 / 2 = 1 + 1.81 \times 10^{-12}$ and in 41.2 hours it runs slow by

$(41.2 \text{ h})(1.81 \times 10^{-12}) = 7.4572 \times 10^{-11} \text{ h} = 268 \text{ ns}$. The Westward-moving clock has a ground speed of $31,143 \text{ km} / 48.6 \text{ h} = 640.8 \text{ km/h} = 178.0 \text{ m/s}$ and thus has a net speed of $360.5 \text{ m/s} - 178.0 \text{ m/s} = 182.5 \text{ m/s}$. For this clock

$\gamma = \frac{1}{\sqrt{1-\beta^2}} \approx 1 + \beta^2 / 2 = 1 + 1.85 \times 10^{-13}$ and in 48.6 hours it runs slow by

$(48.6 \text{ h})(1.85 \times 10^{-13}) = 8.991 \times 10^{-12} \text{ h} = 32 \text{ ns}$. So our prediction is that the Eastward-moving clock is off by $107 \text{ ns} - 269 \text{ ns} = -162 \text{ ns}$, while the Westward-moving clock is off by $126 \text{ ns} - 32 \text{ ns} = 94 \text{ ns}$. These results are correct for special relativity but do not reconcile with those in the table in the text, because general relativistic effects are of the same order of magnitude.

39. The derivations of Equations (2.31) and (2.32) in the beginning of Section 2.10 will suffice. Mary receives signals at a rate f' for t'_1 and a rate f'' for t'_2 . Frank receives signals at a rate f' for t_1 and a rate f'' for t_2 .

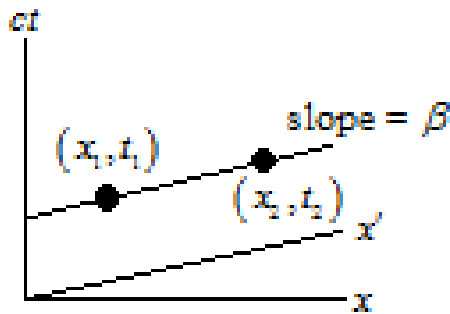
40. $T = t_1 + t_2 = \frac{L}{v} + \frac{L}{c} + \frac{L}{v} - \frac{L}{c} = \frac{2L}{v}$; Frank sends signals at rate f , so Mary receives $f T = 2 f L / v$ signals.

$T' = t'_1 + t'_2 = \frac{2L}{\gamma v}$; Mary sends signals at rate f , so Frank receives $f T' = 2 f L / \gamma v$ signals.

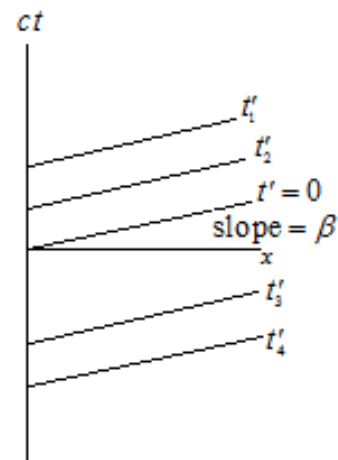
41. $s^2 = x^2 + y^2 + z^2 - c^2 t^2$; Using the Lorentz transformation

$$\begin{aligned} s^2 &= \gamma^2 (x' + vt')^2 + y'^2 + z'^2 - c^2 \gamma^2 (t' + vx' / c^2)^2 \\ &= x'^2 \gamma^2 (1 - v^2 / c^2) + y'^2 + z'^2 - c^2 t'^2 \gamma^2 (1 - v^2 / c^2) \\ &= x'^2 + y'^2 + z'^2 - c^2 t'^2 = s'^2 \end{aligned}$$

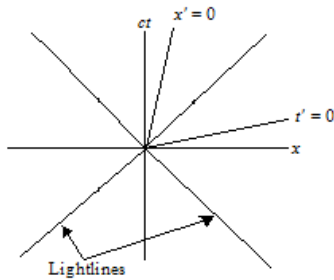
42. For a timelike interval $\Delta s^2 < 0$ so $\Delta x^2 < c^2 \Delta t^2$. We will prove by contradiction. Suppose that there is a frame K' in which the two events were simultaneous, so that $\Delta t' = 0$. Then by the spacetime invariant $\Delta x^2 - c^2 \Delta t^2 = \Delta x'^2 - c^2 \Delta t'^2 = \Delta x'^2$. But because $\Delta x^2 < c^2 \Delta t^2$, this implies $\Delta x'^2 < 0$ which is impossible because $\Delta x'$ is real.
43. As in Problem 42, we know that for a spacelike interval $\Delta s^2 > 0$ so $\Delta x^2 > c^2 \Delta t^2$. Then in a frame K' in which the two events occur in the same place, $\Delta x' = 0$ and $\Delta x^2 - c^2 \Delta t^2 = \Delta x'^2 - c^2 \Delta t'^2 = -c^2 \Delta t'^2$. But because $\Delta x^2 > c^2 \Delta t^2$ we have $c^2 \Delta t'^2 < 0$, which is impossible because $\Delta t'$ is real.
44. In order for two events to be simultaneous in K' , the two events must lie along the x' axis, or along a line parallel to the x' axis. The slope of the x' axis is $\beta = v/c$, so $v/c = \text{slope} = \frac{c\Delta t}{\Delta x}$. Solving for v , we find $v = c^2 \Delta t / \Delta x$. Since the slope of the x' axis must be less than one, we see that $\Delta x > c\Delta t$ so $s^2 = \Delta x^2 - c^2 \Delta t^2 > 0$ is required.



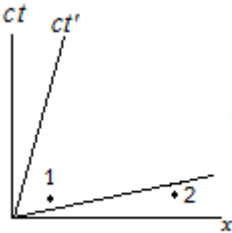
45. parts (a) and (b) To find the equation of the line use the Lorentz transformation. With $t' = 0$ we have $t' = 0 = \gamma(t - vx/c^2)$ or, rearranging, $ct = vx/c = \beta x$. Thus the graph of ct vs. x is a straight line with a slope β .
- (c) Now with t' constant, the Lorentz transformation gives $t' = \gamma(t - vx/c^2)$. Again we solve for ct : $ct = \beta x + ct'/\gamma = \beta x + \text{constant}$. This line is parallel to the $t' = 0$ line we found earlier but shifted by the constant.
- (d) Here both the x' and ct' axes are shifted from their normal (x, ct) orientation and they are not perpendicular.



46. The diagram is shown here. Note that there is only one worldline for light, and it bisects both the x, ct axes and the x', ct' axes. The x' and ct' axes are not perpendicular. This can be seen as a result of the Lorentz transformations, since $x' = 0$ defines the ct' axis and $t' = 0$ defines the x' axis.



47. The diagram shows that the events A and B that occur at the same time in K occur at different times in K'.



48. The Doppler shift gives $\lambda = \lambda_0 \sqrt{\frac{1-\beta}{1+\beta}}$. With numerical values $\lambda_0 = 650 \text{ nm}$ and $\lambda = 540 \text{ nm}$, solving this equation for β gives $\beta = 0.183$. The astronaut's speed is $v = \beta c = 5.50 \times 10^7 \text{ m/s}$. In addition to a red light violation, the astronaut gets a speeding ticket.

49. According to the fixed source (K) the signal and receiver move at speeds c and v , respectively, in opposite directions, so their relative speed is $c + v$. The time interval between receipt of signals is $\Delta t = \lambda / (c + v) = 1 / f_0$. By time dilation $\Delta t' = \frac{\Delta t}{\gamma} = \frac{\lambda}{\gamma(c + v)}$.

Using $\lambda = c / \nu_0$ and $\gamma = \frac{1}{\sqrt{1-v^2/c^2}}$ we find $\Delta t' = \frac{c\sqrt{1-v^2/c^2}}{f_0(c+v)} = \frac{\sqrt{1-\beta^2}}{f_0(1+\beta)}$ and

$$f' = \frac{1}{\Delta t'} = \frac{f_0(1+\beta)}{\sqrt{1-\beta^2}} = f_0 \sqrt{\frac{1+\beta}{1-\beta}}.$$

50. For a fixed source and moving receiver, the length of the wave train is $cT + vT$. Since n waves are emitted during time T , $\lambda = \frac{cT + vT}{n}$ and the frequency $f = c / \lambda$ is $f = \frac{cn}{cT + vT}$.

As in the text $n = f_0 T'_0$ and $T'_0 = T / \gamma$. Therefore $f = \frac{cf_0 T / \gamma}{cT + vT} = \frac{f_0 \sqrt{1 - \beta^2}}{1 + \beta} = f_0 \sqrt{\frac{1 - \beta}{1 + \beta}}$.

51. $f = f_0 \sqrt{\frac{1 - \beta}{1 + \beta}} = (1400 \text{ kHz}) \sqrt{\frac{1 - 0.95}{1 + 0.95}} = 224 \text{ kHz}$

52. The Doppler shift function $f' = f_0 \sqrt{\frac{1 - \beta}{1 + \beta}}$

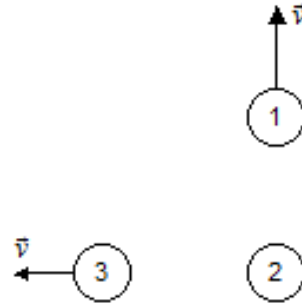


is the rate at which #1 and #2 receive signals from each other and the rate at which #2 and #3 receive signals from each other. But for signals between #1 and #3 the rate is

$$f'' = f' \sqrt{\frac{1 - \beta}{1 + \beta}} = f_0 \frac{1 - \beta}{1 + \beta}.$$

53. The Doppler shift function $f' = f_0 \sqrt{\frac{1 - \beta}{1 + \beta}}$

is the rate at which #1 and #2 receive signals from each other and the rate at which #2 and #3 receive signals from each other. As for #1 and #3 we will assume that these plumbing vans are non-relativistic ($v \ll c$). Otherwise it would be necessary to use the velocity addition law and apply the transverse Doppler shift. From the figure we see



that $f' = \frac{1}{t_0 + (t_2 - t_1)}$. Now $f_0 = 1/t_0$ and

$$t_2 - t_1 = \frac{2x}{c} = \frac{2vt_0 \cos \theta}{c}. \text{ With an angle of } 45^\circ, \cos(45) = 1/\sqrt{2} \text{ and}$$

$$f' = \frac{1}{1/f_0 + (2v \cos \theta)/cf_0} = \frac{f_0}{1 + (2v \cos \theta)/c} = \frac{f_0}{1 + \sqrt{2}v/c}.$$

54. The Doppler shift to higher wavelengths is (with $\lambda_0 = 589 \text{ nm}$) $\lambda = 700 \text{ nm} = \lambda_0 \sqrt{\frac{1 + \beta}{1 - \beta}}$.

Solving for β we find $\beta = 0.171$. Then $t = \frac{v}{a} = \frac{(0.171)(3.00 \times 10^8 \text{ m/s})}{29.4 \text{ m/s}^2} = 1.75 \times 10^6 \text{ s}$

which is 20.25 days. One problem with this analysis is that we have only computed the

time as measured by Earth. We are not prepared to handle the non-inertial frame of the spaceship.

55. Let the instantaneous momentum be in the x-direction and the force be in the y-direction.

Then $d\vec{p} = \vec{F} dt = \gamma m d\vec{v}$ and $d\vec{v}$ is also in the y-direction. So we have $\vec{F} = \gamma m \frac{d\vec{v}}{dt} = \gamma m \vec{a}$.

56. The magnitude of the centripetal force is $\gamma m a = \gamma m \frac{v^2}{r}$ for circular motion. For a charged

particle $F = qvB$, so $qvB = \gamma m \frac{v^2}{r}$ or, rearranging $qBr = \gamma mv = p$. Therefore

$$r = \frac{p}{qB}.$$

When the speed increases the momentum increases, and thus for a given value of B the radius must increase.

57. $\vec{p} = \gamma m \vec{v} = \frac{m \vec{v}}{\sqrt{1 - v^2/c^2}}$ and $\vec{F} = \frac{d\vec{p}}{dt}$. The momentum is the product of two factors that contain the velocity, so we apply the product rule for derivatives:

$$\begin{aligned} \vec{F} &= m \frac{d}{dt} \left[\frac{m \vec{v}}{\sqrt{1 - v^2/c^2}} \right] \\ &= m \left[\frac{d\vec{v}/dt}{\sqrt{1 - v^2/c^2}} + \vec{v} \frac{d}{dt} \left(\frac{1}{\sqrt{1 - v^2/c^2}} \right) \right] \\ &= \gamma m \vec{a} + m \vec{v} \left(-\frac{1}{2} \right) \left(-\frac{2v}{c^2} \right) \gamma^3 \frac{dv}{dt} \\ &= \gamma m \vec{a} + \gamma^3 m \vec{a} \left(\frac{v^2}{c^2} \right) \\ &= \gamma^3 m \vec{a} \left[1 - \frac{v^2}{c^2} + \frac{v^2}{c^2} \right] \\ &= \gamma^3 m \vec{a} \end{aligned}$$

58. From the preceding problem $F = \gamma^3 m a$. We have $a = 10^{19} \text{ m/s}^2$ and $m = 1.67 \times 10^{-27} \text{ kg}$.

$$(a) \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} = \frac{1}{\sqrt{1 - 0.01^2}} = 1.00005$$

$$F = (1.00005)^3 (1.67 \times 10^{-27} \text{ kg}) (10^{19} \text{ m/s}^2) = 1.67 \times 10^{-8} \text{ N}$$

(b) As in (a) $\gamma = 1.005$ and $F = 1.70 \times 10^{-8} \text{ N}$

(c) As in (a) $\gamma = 2.294$ and $F = 2.02 \times 10^{-7} \text{ N}$

(d) As in (a) $\gamma = 7.0888$ and $F = 5.95 \times 10^{-6} \text{ N}$

$$59. p = \gamma mv \text{ with } \gamma = \frac{1}{\sqrt{1-v^2/c^2}} = \frac{1}{\sqrt{1-0.92^2}} = 2.5516;$$

$$m = \frac{p}{\gamma v} = \frac{10^{-16} \text{ kg}\cdot\text{m/s}}{(2.5516)(0.92)(3.00 \times 10^8 \text{ m/s})} = 1.42 \times 10^{-25} \text{ kg}$$

$$60. \text{ The initial momentum is } p_0 = \gamma mv = \frac{1}{\sqrt{1-(0.5)^2}} m(0.5c) = 0.57735mc.$$

$$(a) p/p_0 = 1.01$$

$$1.01 = \frac{\gamma mv}{0.57735mc}$$

$$\gamma v = (1.01)(0.57735c) = 0.58312c$$

$$\text{Substituting for } \gamma \text{ and solving for } v, v = \left[\frac{1}{(.58312c)^2} + \frac{1}{c^2} \right]^{-1/2} = 0.504c.$$

$$(b) \text{ Similarly } v = \left[\frac{1}{(.63509c)^2} + \frac{1}{c^2} \right]^{-1/2} = 0.536c$$

$$(c) \text{ Similarly } v = \left[\frac{1}{(1.1547c)^2} + \frac{1}{c^2} \right]^{-1/2} = 0.756c$$

61. 6.3 GeV protons have $K = 6.3 \times 10^3 \text{ MeV}$ and $E = K + E_0 = 7238 \text{ MeV}$. Then

$$p = \frac{\sqrt{E^2 - E_0^2}}{c} = 7177 \text{ MeV}/c. \text{ Converting to SI units}$$

$$p = 7177 \text{ MeV}/c \left(\frac{1.60 \times 10^{-13} \text{ J}}{\text{MeV}} \right) \left(\frac{c}{3.00 \times 10^8 \text{ m/s}} \right) = 3.83 \times 10^{-18} \text{ kg}\cdot\text{m/s}$$

$$\text{From Problem 56 we have } B = \frac{p}{qr} = \frac{3.83 \times 10^{-18} \text{ kg}\cdot\text{m/s}}{(1.60 \times 10^{-19} \text{ C})(15.2 \text{ m})} = 1.57 \text{ T}.$$

62. Initially Mary throws her ball with velocity (primes showing the measurements are in Mary's frame): $u'_{M_x} = 0$ $u'_{M_y} = -u_0$. After the elastic collision, the signs on the above expressions are reversed, so the change in momentum as measured by Mary is

$$\Delta p'_M = \frac{mu_0}{\sqrt{1-u_0^2/c^2}} - \frac{-mu_0}{\sqrt{1-u_0^2/c^2}} = \frac{2mu_0}{\sqrt{1-u_0^2/c^2}}.$$

Now for Frank's ball, we know $u_{F_x} = 0$ and $u_{F_y} = u_0$. The velocity transformations give

$$\text{for Frank's ball as measured by Mary: } u'_{F_x} = -v \quad u'_{F_y} = u_0 \sqrt{1 - v^2 / c^2}.$$

To find γ for Frank's ball, note that $(u'_{F_x})^2 + (u'_{F_y})^2 = v^2 + u_0^2 (1 - v^2 / c^2)$. Then

$$\gamma = \frac{1}{\sqrt{1 - u'^2_F / c^2}} = \frac{1}{\sqrt{1 - v^2 / c^2 - u_0^2 (1 - v^2 / c^2) / c^2}} = \frac{1}{\sqrt{(1 - u_0^2 / c^2)(1 - v^2 / c^2)}}. \text{ Using}$$

$p' = \gamma mu'$ along with the reversal of velocities in an elastic collision, we find

$$\begin{aligned} \Delta p'_F &= \gamma m(-u_0) \sqrt{1 - v^2 / c^2} - \gamma mu_0 \sqrt{1 - v^2 / c^2} = -2\gamma mu_0 \sqrt{1 - v^2 / c^2} \\ &= \frac{-2mu_0 \sqrt{1 - v^2 / c^2}}{\sqrt{(1 - u_0^2 / c^2)(1 - v^2 / c^2)}} = \frac{-2mu_0}{\sqrt{(1 - u_0^2 / c^2)}} \end{aligned}$$

$$\text{Finally } \Delta p' = \Delta p'_F + \Delta p'_M = \frac{-2mu_0 + 2mu_0}{\sqrt{(1 - u_0^2 / c^2)}} = 0 \text{ as required.}$$

63. To prove by contradiction, suppose that $K = \frac{1}{2} \gamma mv^2$. Then

$$K = E - E_0 = \gamma mc^2 - mc^2 = (\gamma - 1)mc^2 = \frac{1}{2} \gamma mv^2. \text{ This implies } \gamma - 1 = v^2 / 2c^2, \text{ or}$$

$$\gamma = 1 + v^2 / 2c^2, \text{ which is clearly false.}$$

64. The source of the energy is the internal energy associated with the change of state, commonly called that latent heat of fusion L_f . Let m be the mass equivalent of 2 grams

$$\text{and } M \text{ be the mass of ice required. } m = \frac{E}{c^2} = \frac{L_f M}{c^2} \text{ Rearranging}$$

$$M = \frac{mc^2}{L_f} = \frac{(0.002 \text{ kg})(3.00 \times 10^8 \text{ m/s})^2}{334 \times 10^3 \text{ J/kg}} = 5.39 \times 10^8 \text{ kg}$$

65. In general $K = (\gamma - 1)mc^2$, so $\gamma = 1 + \frac{K}{mc^2}$. For 9 GeV electrons:

$$\gamma = 1 + \frac{9000 \text{ MeV}}{0.511 \text{ MeV}} = 1.76 \times 10^4 \text{ Then from the definition of } \gamma \text{ we have}$$

$$\beta = \sqrt{1 - \frac{1}{\gamma^2}} = \sqrt{1 - \frac{1}{(1.76 \times 10^4)^2}} = 1 - 1.6 \times 10^{-9}. \text{ Thus}$$

$$v = (1 - 1.6 \times 10^{-9})c = 0.9999999984c.$$

For 3.1 GeV positrons: $\gamma = 1 + \frac{3100 \text{ MeV}}{0.511 \text{ MeV}} = 6068$ and $\beta = \sqrt{1 - \frac{1}{(6068)^2}} = 1 - 1.4 \times 10^{-8}$.

Thus $v = (1 - 1.4 \times 10^{-8})c = 0.999999986c$.

66. Note that the proton's mass is $938 \text{ MeV}/c^2$. In general $K = (\gamma - 1)mc^2$, so $\gamma = 1 + \frac{K}{mc^2}$.

Then from the definition of γ we have $\beta = \sqrt{1 - \frac{1}{\gamma^2}}$. For the first section $K = 0.750$

MeV, and $\gamma = 1 + \frac{0.750 \text{ MeV}}{938 \text{ MeV}} = 1.00080$ with $\beta = \sqrt{1 - \frac{1}{\gamma^2}} = \beta = \sqrt{1 - \frac{1}{(1.00080)^2}} = 0.040$.

Thus $v = 0.04c$ at the end of the first stage. For the other stages the computations are similar, and we tabulate the results:

$K \text{ (GeV)}$	γ	β
0.400	1.43	0.71
8	9.53	0.994
150	160.9	0.99998
1000	1067	0.9999996

$$67. (a) \quad p = \gamma mu = \frac{(511 \text{ keV}/c^2)(0.020c)}{\sqrt{1 - 0.020^2}} = 10.22 \text{ keV}/c;$$

$$E = \gamma mc^2 = \frac{(511 \text{ keV}/c^2)(c^2)}{\sqrt{1 - 0.02^2}} = 511.102 \text{ keV};$$

$$K = E - E_0 = 511.102 \text{ keV} - 511.00 \text{ keV} = 102 \text{ eV}$$

The results for (b) and (c) follow with similar computations and are tabulated:

β	$p \text{ (keV}/c)$	$E \text{ (keV)}$	$K \text{ (keV)}$
0.20	104.3	521.5	10.5
0.90	1055	1172	661

$$68. \quad E = 2E_0 = \gamma E_0 \text{ so } \gamma = 2. \text{ Then } \beta = \sqrt{1 - \frac{1}{\gamma^2}} = \frac{\sqrt{3}}{2} \text{ and } v = \frac{\sqrt{3}c}{2}.$$

69. For a constant force, work = change in kinetic energy = $Fd = mc^2 / 4$, because

$$25\% = 1/4. \quad d = \frac{mc^2}{4F} = \frac{(80 \text{ kg})(3.00 \times 10^8 \text{ m/s})^2}{4(8 \text{ N})} = 2.25 \times 10^{17} \text{ m} = 23.8 \text{ ly}.$$

70. $E = K + E_0 = 2E_0 + E_0 = 3E_0 = \gamma E_0$ so $\gamma = 3$. Then

$$\beta = \sqrt{1 - \frac{1}{\gamma^2}} = \beta = \sqrt{1 - \frac{1}{3^2}} = \frac{2\sqrt{2}}{3} = 0.943. \text{ Thus } v = 0.943c.$$

71. (a) $E = K + E_0 = 0.1E_0 + E_0 = 1.1E_0 = \gamma E_0$, so $\gamma = 1.1$. Then

$$\beta = \sqrt{1 - \frac{1}{\gamma^2}} = \beta = \sqrt{1 - \frac{1}{1.1^2}} = 0.417 \text{ and } v = 0.417c.$$

(b) As in (a) $\gamma = 2$ and $v = \sqrt{3}c / 2$.

(c) As in (a) $\gamma = 11$ and $v = 0.996c$.

72. $K = E - E_0 = \frac{E_0}{\sqrt{1 - \beta^2}} - E_0$ so $K + E_0 = \frac{E_0}{\sqrt{1 - \beta^2}}$. Rearranging $1 - \beta^2 = \left(\frac{E_0}{K + E_0} \right)^2$ which

$$\text{can be written as } \beta^2 = 1 - \left(\frac{E_0}{K + E_0} \right)^2 \text{ or } \beta = \sqrt{1 - \left(\frac{E_0}{K + E_0} \right)^2}.$$

73. Using $E = \gamma mc^2$ along with $p = \gamma mv$ we see that $\gamma = E / mc^2 = p / mv$. Solving for v/c we find $\beta = v/c = pc / E$.

74. The speed is the same for protons, electrons, or any particle.

$$K = (\gamma - 1)mc^2 = 1.01 \left(\frac{1}{2} mv^2 \right) = 0.505 mc^2 \beta^2 \text{ so } \gamma - 1 = \frac{1}{\sqrt{1 - \beta^2}} - 1 = 0.505 \beta^2.$$

Rearranging and solving for β , we find $\beta = 0.114$ or $v = 0.114c$.

75. Converting $0.1 \text{ ounce} = 2.835 \times 10^{-3} \text{ kg}$.

$$E = mc^2 = (2.835 \times 10^{-3} \text{ kg})(3.00 \times 10^8 \text{ m/s})^2 = 2.55 \times 10^{14} \text{ J}.$$

Eating 10 ounces results in a factor of 100 greater mass-energy increase, or $2.55 \times 10^{16} \text{ J}$. This is a small increase compared with your original mass-energy, but it will tend to increase your weight; depending on how they are prepared, peanuts generally contain about 100 kcal of food energy per ounce.

76. The energy needed equals the kinetic energy of the spaceship.

$$K = (\gamma - 1)mc^2 = \left(\frac{1}{\sqrt{1 - \beta^2}} - 1 \right) mc^2$$

$$= \left(\frac{1}{\sqrt{1 - 0.3^2}} - 1 \right) (10^4 \text{ kg}) (3.00 \times 10^8 \text{ m/s})^2 = 4.35 \times 10^{19} \text{ J}$$

or 4.35% of 10^{21} J .

77. Up to Equation (2.57) the derivation in the text is complete. Then using the integration by parts formula, $\int x dy = xy - \int y dx$ and noting that in this case $x = u$ and $y = \gamma u$, we have $\int u d(\gamma u) = \gamma u^2 - \int \gamma u du$. Thus

$$K = m \int_0^{\gamma u} u d(\gamma u) = \gamma m u^2 - m \int \gamma u du$$

$$= \gamma m u^2 - m \int \frac{u}{\sqrt{1 - u^2 / c^2}} du$$

Using integral tables or simple substitution:

$$K = \gamma m u^2 + m c^2 \sqrt{1 - u^2 / c^2} \Big|_0^u$$

$$= \gamma m u^2 + m c^2 \sqrt{1 - u^2 / c^2} - m c^2$$

$$= \frac{m c^2 + m c^2 (1 - u^2 / c^2)}{\sqrt{1 - u^2 / c^2}} - m c^2$$

$$= \gamma m c^2 - m c^2 = m c^2 (\gamma - 1)$$

78. Converting $0.11 \text{ cal} \cdot \text{g}^{-1} \cdot ^\circ \text{C}^{-1} = 460 \text{ J} \cdot \text{kg}^{-1} \cdot ^\circ \text{C}^{-1} = c_v$

(specific heat at constant volume). From thermodynamics the energy ΔE used to change the temperature by ΔT is $m c_v \Delta T$. Thus

$$\Delta E = (1000 \text{ kg}) \left[(460 \text{ J} / (\text{kg} \cdot ^\circ \text{C})) (0.5^\circ \text{C}) \right] = 2.30 \times 10^5 \text{ J} \text{ and}$$

$$\Delta m = \frac{\Delta E}{c^2} = \frac{2.30 \times 10^5 \text{ J}}{(3.00 \times 10^8 \text{ m/s})^2} = 2.56 \times 10^{-12} \text{ kg} . \text{ The source of this energy is the internal energy of the arrangement of atoms and molecules prior to the collision.}$$

79.

$$\begin{aligned}
 E_b &= [2m_p + 2m_n - m(\text{He})]c^2 \\
 &= [2(1.007276 \text{ u}) + 2(1.008665 \text{ u}) - 4.001505 \text{ u}]c^2 \left(\frac{931.494 \text{ MeV}}{c^2 \cdot \text{u}} \right) = 28.3 \text{ MeV}
 \end{aligned}$$

80.

$$\begin{aligned}
 \Delta E &= [m_n - m_p - m_e]c^2 \\
 &= [1.008665 \text{ u} - 1.007276 \text{ u} - 0.000549 \text{ u}]c^2 \left(\frac{931.494 \text{ MeV}}{c^2 \cdot \text{u}} \right) = 0.782 \text{ MeV}
 \end{aligned}$$

81. $E = K + E_0 = 1 \text{ TeV} + 938 \text{ MeV} \approx 1 \text{ TeV}$;

$$p = \frac{\sqrt{E^2 - E_0^2}}{c} = \frac{\sqrt{(1 \text{ TeV} + 938 \text{ MeV})^2 - (938 \text{ MeV})^2}}{c} = 1.000938 \text{ TeV}/c$$

$$\gamma = \frac{E + E_0}{E_0} = \frac{1.000938 \text{ TeV}}{0.000938 \text{ TeV}} = 1067; \quad \beta^2 = 1 - \frac{1}{\gamma^2} = 1 - 8.78 \times 10^{-7}$$

$$\beta = \sqrt{1 - 8.78 \times 10^{-7}} \approx 1 - 4.39 \times 10^{-7}; \text{ and } v = \beta c \approx 0.999999561c$$

82. (a) $E = \sqrt{p^2 c^2 + E_0^2} = \sqrt{(40 \text{ GeV})^2 + (511 \text{ keV})^2} \approx 40.0 \text{ GeV}$

$$K = E - E_0 = 40.0 \text{ GeV}$$

$$(b) \quad E = \sqrt{p^2 c^2 + E_0^2} = \sqrt{(40 \text{ GeV})^2 + (0.938 \text{ GeV})^2} = 40.011 \text{ GeV}$$

$$K = E - E_0 = 40.011 \text{ GeV} - 0.938 \text{ GeV} = 39.07 \text{ GeV}$$

83. $E = K + E_0 = 200 \text{ MeV} + 106 \text{ MeV} = 306 \text{ MeV}$

$$p = \frac{\sqrt{E^2 - E_0^2}}{c} = \frac{\sqrt{(306 \text{ MeV})^2 - (106 \text{ MeV})^2}}{c} = 287.05 \text{ MeV}/c$$

$$\gamma = \frac{E}{E_0} = \frac{306 \text{ MeV}}{106 \text{ MeV}} = 2.887$$

$$\beta = \sqrt{1 - \frac{1}{\gamma^2}} = 0.938$$

$$\text{so } v = 0.938c$$

84. (a) The mass-energy imbalance occurs because the helium-3 (^3He) nucleus is more tightly bound than the two separate deuterium nuclei (^2H). (Masses from Appendix 8.)

$$\begin{aligned}\Delta E &= ([2m(^2\text{H})] - [m_n + m(^3\text{He})])c^2 \\ &= [2(2.014102 \text{ u}) - (1.008665 \text{ u} + 3.016029 \text{ u})]c^2 \left(\frac{931.494 \text{ MeV}}{c^2 \cdot \text{u}} \right) = 3.27 \text{ MeV}\end{aligned}$$

(b) The initial rest energy is $2m(^2\text{H}) = 2(2.014102 \text{ u}) \left(\frac{931.494 \text{ MeV}}{c^2 \cdot \text{u}} \right) = 3752 \text{ MeV}$.

Thus the answer in (a) is about 0.09% of the initial rest energy.

85. (a) The mass-energy imbalance occurs because the helium-4 (^4He) is more tightly bound than the deuterium (^2H) and tritium nuclei (^3H).

$$\begin{aligned}\Delta E &= ([m(^2\text{H}) + m(^3\text{H})] - [m_n + m(^4\text{He})])c^2 \\ &= [(2.014102 \text{ u} + 3.016029 \text{ u}) - (1.008665 \text{ u} + 4.002603 \text{ u})]c^2 \left(\frac{931.494 \text{ MeV}}{c^2 \cdot \text{u}} \right) \\ &= 17.6 \text{ MeV}\end{aligned}$$

(b) The initial rest energy is $[m(^2\text{H}) + m(^3\text{H})] \cdot c^2 = [(5.030131 \text{ u})] \cdot c^2 \left(\frac{931.494 \text{ MeV}}{c^2 \cdot \text{u}} \right) = 4686 \text{ MeV}$. Thus the answer in (a) is about 0.37% of the initial rest energy.

86. (a) In the inertial frame moving with the negative charges in wire 1, the negative charges in wire 2 are stationary, but the positive charges are moving. The density of the positive charges in wire 2 is thus greater than the density of negative charges, and there is a net attraction between the wires.

(b) By the same reasoning as in (a), note that the positive charges in wire 2 will be stationary and have a normal density, but the negative charges are moving and have an increased density, causing a net attraction between the wires.

(c) There are two facts to be considered. First, (a) and (b) are consistent with the physical result being independent of inertial frame. Second, we know from classical physics that two parallel wires carrying current in the same direction attract each other. That is, the same result is achieved in the “lab” frame.

87. As in the solution to Problem 21 we have $\beta = \frac{v}{c} = \frac{d\sqrt{1-\beta^2}}{ct'}$ where d is the length of the particle track and t' the particle's lifetime in its rest frame. In this problem $t' = 8.2 \times 10^{-11} \text{ s}$

and $d = 24$ mm. Solving the above equation we find $\beta = 0.698$. Then

$$E = \frac{E_0}{\sqrt{1-\beta^2}} = \frac{1672 \text{ MeV}}{\sqrt{1-0.698^2}} = 2330 \text{ MeV}$$

88.

$$\begin{aligned} F &= \frac{dp}{dt} = \frac{d}{dt}(\gamma mv) = \frac{d}{dt} \left[mv \left(1 - \frac{v^2}{c^2} \right)^{-1/2} \right] \\ &= m\gamma \frac{dv}{dt} - \frac{mv}{2} \gamma^3 \left(-\frac{2v}{c^2} \right) \frac{dv}{dt} \\ &= m\gamma \frac{dv}{dt} \left[1 + \frac{v^2 \gamma^2}{c^2} \right] = m\gamma \frac{dv}{dt} \left[1 + \frac{v^2 / c^2}{1 - v^2 / c^2} \right] \\ &= m\gamma \frac{dv}{dt} \left[\frac{1}{1 - v^2 / c^2} \right] = m\gamma^3 \frac{dv}{dt} = m \frac{dv}{dt} \frac{1}{(1 - v^2 / c^2)^{3/2}} \end{aligned}$$

89. (a) The number n received by Frank at f' is half the number sent by Mary at that rate, or $fL / \gamma v$. The detected time of turnaround is

$$t = \frac{n}{f'} = \frac{fL / \gamma v}{v \sqrt{(1-\beta)/(1+\beta)}} = \frac{L \sqrt{1+\beta}}{\gamma v \sqrt{1-\beta}} = \frac{L(1+\beta)}{v} = \frac{L}{v} + \frac{L}{c}.$$

(b) Similarly, the number n' received by Mary at f' is

$$n' = f' \frac{T'}{2} = f \sqrt{\frac{1-\beta}{1+\beta}} \frac{L}{\gamma v} = \frac{f L (1-\beta)}{v}. \quad \text{Her turnaround time is } T' / 2 = L / \gamma v.$$

(c) For Frank, the time t_2 for the remainder of the trip is $t_2 = T - t_1 = L/v - L/c$.

$$\text{Number of signals} = f'' t_2 = f \sqrt{\frac{1+\beta}{1-\beta}} (L/v - L/c) = \frac{fL}{\gamma v}.$$

$$\text{Total number received} = \frac{fL}{\gamma v} + \frac{fL}{\gamma v} = \frac{2fL}{\gamma v}.$$

$$\text{Mary's age} = \frac{\text{Total number received}}{f} = \frac{2L}{\gamma v}.$$

(d) For Mary, $t'_2 = T' - t'_1 = L / \gamma v$.

$$\text{Number of signals} = f'' t'_2 = f \sqrt{\frac{1+\beta}{1-\beta}} \frac{L}{\gamma v} = \frac{fL}{v} (1+\beta).$$

$$\text{Total number received} = \frac{fL}{v} (1-\beta + 1+\beta) = \frac{2fL}{v}.$$

$$\text{Frank's age} = \frac{\text{Total number received}}{f} = \frac{2L}{v}.$$

90. In the fixed frame, the distance is $\Delta x = 8 \text{ ly}$ and the elapsed time is $\Delta t = 10 \text{ y}$, so the interval is $s^2 = \Delta x^2 - c^2 \Delta t^2 = 64 \text{ ly}^2 - 100 \text{ ly}^2 = -36 \text{ ly}^2$. In the moving frame, Mary's clock is at rest, so $\Delta x' = 0$, and the time interval is $\Delta t' = 6 \text{ y}$. Thus the interval is $s'^2 = \Delta x'^2 - c^2 \Delta t'^2 = 0 \text{ ly}^2 - 36 \text{ ly}^2 = -36 \text{ ly}^2$. The results are the same, as they should be, because the spacetime interval is the same for all inertial frames.

91. (a) From Table 2.1 number $= \frac{fL}{v}(1-\beta) = \frac{(52\text{y}^{-1})(4.3 \text{ ly})}{0.8c}(1-0.8) = 55.9 \approx 56$

(b) $t_1 = \frac{L}{v} + \frac{L}{c} = \frac{4.3 \text{ ly}}{0.8c} + \frac{4.3 \text{ ly}}{c} = 9.68 \text{ y}$; number $= f' t_1 = \frac{fL}{v} \sqrt{1-\beta^2} = 167.7 \approx 168$

(c) Frank: $f'' t'_2 = \frac{fL}{v} \sqrt{1-\beta^2} \approx 168$ so the total is $168 + 168 = 336$.

Mary: number $= 2 f L / v = 559$

(d) Frank: $T = 2L / v = 10.75 \text{ y}$ Mary: $T' = 2L / \gamma v = 6.45 \text{ y}$

(e) From part (c), 559 weeks = 10.75 years and 336 weeks = 6.46 years, which agrees with part (d).

92. Notice that the radar is shifted twice, once upon receipt by the speeding car and again upon reemission (reflection) of the beam. For this double shift, the received frequency f is

$$\frac{f}{f_0} = \sqrt{\frac{1-\beta}{1+\beta}} \sqrt{\frac{1-\beta}{1+\beta}} = \frac{1-\beta}{1+\beta}$$

For speeds much less than c , a Taylor series approximation gives an excellent result:

$$\frac{f}{f_0} = \frac{1-\beta}{1+\beta} = (1-\beta)(1+\beta)^{-1} \approx 1-2\beta.$$

Converting 80 mph to 35.8 m/s gives $\beta = 1.19 \times 10^{-7}$, so the received frequency is approximately

$$f = f_0(1-2\beta) = 10.4999975 \text{ GHz}, \text{ which is } 2.5 \text{ kHz less than the original frequency.}$$

93. For this transformation $v = 0.8c$ (so $\gamma = 5/3$), $u'_x = 0$ and $u'_y = 0.8c$. Applying the transformations,

$$u_x = \frac{u'_x + v}{1 + \frac{vu'_x}{c^2}} = \frac{0 + 0.8c}{1 + 0} = 0.8c ; \quad u_y = \frac{u'_y}{\gamma \left(1 + \frac{vu'_x}{c^2}\right)} = \frac{0.8c}{\frac{5}{3}(1 + 0)} = 0.48c$$

The speed is $u = \sqrt{u_x^2 + u_y^2} = 0.93c$, safely under the speed of light.

94. (a) $K = E + E_0 = 250E_0$. Thus $K = 249E_0 = 249(511\text{keV}) = 127.2\text{ MeV}$

(b) $\gamma = 250$

$$\beta = \sqrt{1 - \frac{1}{\gamma^2}} = 0.999992$$

$$v = 0.999992c$$

(c) $p = \frac{\sqrt{E^2 - E_0^2}}{c} = \frac{\sqrt{(250 \times 511\text{ keV})^2 - (511\text{ keV})^2}}{c} = 128\text{ MeV}/c$

95. (a) For the proton: $p = \gamma mu = \frac{1}{\sqrt{1 - 0.9^2}} (938\text{ MeV}/c^2)(0.9c) = 1940\text{ MeV}/c$.

For the electron: $E = \sqrt{p^2 c^2 + E_0^2} = \sqrt{(1940\text{ MeV})^2 + (0.511\text{ MeV})^2} = 1940\text{ MeV}$.

$$\gamma = \frac{E}{E_0} = \frac{1940\text{ MeV}}{0.511\text{ MeV}} = 3797$$

$$\beta = \sqrt{1 - \frac{1}{\gamma^2}} = \sqrt{1 - \frac{1}{3797^2}} = \sqrt{1 - 6.94 \times 10^{-8}} \approx 1 - 3.97 \times 10^{-8}$$

$$v = (1 - 3.97 \times 10^{-8})c$$

(b) For the proton: $K = (\gamma - 1)E_0 = \left(\frac{1}{\sqrt{1 - 0.9^2}} - 1\right)(938\text{ MeV}) = 1214\text{ MeV}$.

For the electron: $\gamma = \frac{K + E_0}{E_0} = \frac{1214\text{ MeV} + 0.511\text{ MeV}}{0.511\text{ MeV}} = 2377$.

$$\beta = \sqrt{1 - \frac{1}{\gamma^2}} = \sqrt{1 - \frac{1}{2377^2}} = \sqrt{1 - 1.77 \times 10^{-7}} \approx 1 - 8.85 \times 10^{-8}$$

$$v = (1 - 8.85 \times 10^{-8})c$$

96. In the frame of the decaying K^0 meson, the pi mesons must recoil with equal speeds in opposite directions in order to conserve momentum. In that reference frame the available kinetic energy is $498\text{ MeV} - 2(140\text{ MeV}) = 218\text{ MeV}$. The pi mesons share this equally,

so each one has a kinetic energy of 109 MeV in that frame. The speed of each pi meson can be found:

$$\gamma = \frac{K + E_0}{E_0} = \frac{109 \text{ MeV} + 140 \text{ MeV}}{140 \text{ MeV}} = 1.779 \text{ so } u = \sqrt{1 - \frac{1}{\gamma^2}} c = 0.827c.$$

The greatest and least speeds in the lab frame are obtained when the pi mesons are released in the forward and backward directions. Then by the velocity addition laws:

$$v_{\max} = \frac{0.9c + 0.827c}{1 + (0.9)(0.827)} = 0.990c$$

$$v_{\min} = \frac{0.9c - 0.827c}{1 - (0.9)(0.827)} = 0.285c$$

97. (a) The round-trip distance is $L_0 = 8.60 \text{ ly}$. Assume the same constant speed $v = \beta c$ for the entire trip. In the rocket's frame, the distance is only $L = L_0 \gamma^{-1} = L_0 \sqrt{1 - \beta^2}$.

Mary will age in the rocket's reference frame a total of 22 y, and in that frame

$$v = \frac{\text{distance}}{\text{time}} = \frac{L}{22\text{y}} = \frac{8.60 \text{ ly} \sqrt{1 - \beta^2}}{22\text{y}} = 0.39c \sqrt{1 - \beta^2}. \text{ Therefore, } \beta = \frac{v}{c} = 0.39 \sqrt{1 - \beta^2}.$$

Solving for β we find $\beta = \sqrt{\frac{0.39^2}{1.00 + 0.39^2}}$, or $\beta = 0.36$ so Mary's speed is $v = 0.36c$.

- (b) To find the elapsed time on Earth, we know that $T_0 = 22\text{y}$, so

$$T = \gamma T_0 = \frac{1}{\sqrt{1 - \beta^2}} 22\text{y} = 23.6 \text{ y}. \text{ Frank will be 53.6 years old when Mary returns at the age of 52 y.}$$

98. With $v = 0.995c$ then $\beta = 0.995$ and $\gamma = 10.01$. The distance out to the star is

$L_0 = 5.98 \text{ ly}$. In the rocket's reference frame, the distance is only

$$L = L_0 \gamma^{-1} = \frac{5.98 \text{ ly}}{10.01} = 0.597 \text{ ly}.$$

- (a) The time out to the star in Mary's frame is $\text{time} = \frac{\text{distance}}{\text{speed}} = \frac{0.597 \text{ ly}}{0.995c} = 0.6 \text{ y}$. The

time for the return journey will be the same. When you include the 3 years she spends at the star, her total journey will take 4.2 y.

- (b) To find the elapsed time on Earth for the outbound journey, we know that $T_0 = 0.6\text{y}$ so $T = \gamma T_0 = (10.01)0.6 \text{ y} = 6.006 \text{ y}$. The return journey will take an equal time. The 3 years the spaceship orbits the star will be equivalent for both observers. Therefore Frank will measure a total elapsed time of 15.012 y, which makes him 10.8 years older than her.

99. (a) The Earth to moon distance is 3.84×10^8 m. The rotation rate is

$\omega = 2\pi \text{ rad} \times 0.030 \text{ s}^{-1} = 1.885 \times 10^{-1} \text{ rad/s}$. Then the speed across the moon's surface is

$$v = \omega R = (1.885 \times 10^{-1} \text{ rad/s})(3.82 \times 10^8 \text{ m}) = 7.24 \times 10^7 \text{ m/s}.$$

- (b) The required speed can be found using $c = \omega R = 2\pi f R$ which requires the frequency

$$\text{to be } f = \frac{c}{2\pi R} = \frac{3.00 \times 10^8 \text{ m/s}}{2\pi(3.84 \times 10^8 \text{ m})} = 0.124 \text{ Hz}.$$

100. With the data given, $\beta = 3.28 \times 10^{-6}$, which is very small. We will use the binomial approximation theorem. From Equation (2.21), we know that

$$L = L_0 \gamma^{-1} \cdot \gamma^{-1} = \sqrt{1 - \beta^2} \approx 1 - \beta^2 / 2.$$

- (a) The percentage of length contraction would be:

$$\begin{aligned} \% \text{ change} &= \frac{[L_0 - L]}{L_0} \times 100\% = \frac{[L_0 - L_0 \gamma^{-1}]}{L_0} \times 100\% \\ &= [1 - \gamma^{-1}] \times 100\% = [1 - (1 - \beta^2 / 2)] \times 100\% \\ &= \frac{\beta^2}{2} \times 100\% = 5.37 \times 10^{-10} \end{aligned}$$

- (b) The clocks' rates differ by a factor $\gamma = 1 / \sqrt{1 - \beta^2}$. The clock on the SR-71 measures the proper time and Equation 2.19 tells us that $T' = \gamma T_0$ so the time difference is $\Delta t = T' - T_0 = \gamma T_0 - T_0 = T_0(\gamma - 1)$. Using $\gamma - 1 \approx \beta^2 / 2$ and the fact that the time for the trip in the SR-71 equals the distance divided by the speed

$$\begin{aligned} \Delta t &= t(\beta^2 / 2) = \frac{3.2 \times 10^6 \text{ m}}{983 \text{ m/s}} \frac{\left(\frac{983 \text{ m/s}}{3.00 \times 10^8 \text{ m/s}} \right)^2}{2} \\ &= 1.75 \times 10^{-8} \text{ s} = 17.5 \text{ ns} \end{aligned}$$

101. As the spaceship is approaching the observer, we will make use of Equation (2.32),

$$f' = \frac{\sqrt{1 + \beta}}{\sqrt{1 - \beta}} f_0 \text{ with the prime indicating the Doppler shifted frequency (or wavelength).}$$

This equation indicates that the Doppler shifted frequency will be larger than the frequency measured in a frame where the observer is at rest with respect to the source. Since $c = \lambda f$, this means the Doppler shifted wavelengths will be lower. As given in the problem, we see that the difference in wavelengths for an observer at rest with respect to the source is $\Delta \lambda_0 = 0.5974 \text{ nm}$. We want to find a speed so that the Doppler shifted difference is reduced to $\Delta \lambda' = 0.55 \text{ nm}$. We have

$$\begin{aligned}
\Delta\lambda' &= \lambda'_2 - \lambda'_1 = \frac{c}{f'_2} - \frac{c}{f'_1} = c \left(\frac{f'_1 - f'_2}{f'_1 f'_2} \right) \\
&= c \left[\frac{\left(\frac{\sqrt{1+\beta}}{\sqrt{1-\beta}} \right) f_{0,1} - \left(\frac{\sqrt{1+\beta}}{\sqrt{1-\beta}} \right) f_{0,2}}{\frac{1+\beta}{1-\beta} f_{0,1} f_{0,2}} \right] = c \left[\left(\frac{\sqrt{1-\beta}}{\sqrt{1+\beta}} \right) \left(\frac{f_{0,1} - f_{0,2}}{f_{0,1} f_{0,2}} \right) \right] \\
&= \left[\left(\frac{\sqrt{1-\beta}}{\sqrt{1+\beta}} \right) \left(\frac{c}{f_{0,2}} - \frac{c}{f_{0,1}} \right) \right] = \left[\left(\frac{\sqrt{1-\beta}}{\sqrt{1+\beta}} \right) (\lambda_{0,2} - \lambda_{0,1}) \right] \\
&= \left[\left(\frac{\sqrt{1-\beta}}{\sqrt{1+\beta}} \right) (\Delta\lambda_0) \right]
\end{aligned}$$

We can complete the algebra to show that $\beta = \frac{\Delta\lambda_0^2 - (\Delta\lambda')^2}{\Delta\lambda_0^2 + (\Delta\lambda')^2} = 0.0825$ so that

$$v = 2.47 \times 10^7 \text{ m/s}.$$

102. As we know that the quasars are moving away at high speeds, we make use of Equation (2.33) and the equation $c = \lambda f$. Using a prime to indicate the Doppler shifted frequency

(or wavelength), Equation (2.33) indicates that frequency is given by $f' = \frac{\sqrt{1-\beta}}{\sqrt{1+\beta}} f_0$, or

$$\begin{aligned}
\frac{f_0}{f'} &= \frac{\sqrt{1+\beta}}{\sqrt{1-\beta}} \text{ so} \\
z &= \frac{(\lambda' - \lambda_0)}{\lambda_0} = \left(\frac{\lambda'}{\lambda_0} - 1 \right) = \left(\frac{c/f'}{c/f_0} - 1 \right) \\
&= \left(\frac{f_0}{f'} - 1 \right) = \left(\frac{\sqrt{1+\beta}}{\sqrt{1-\beta}} - 1 \right)
\end{aligned}$$

Therefore $(z+1)^2 = \frac{1+\beta}{1-\beta}$. We can complete the algebra to show that

$$\beta = \frac{(z+1)^2 - 1}{(z+1)^2 + 1} \text{ and thus } v = \left[\frac{(z+1)^2 - 1}{(z+1)^2 + 1} \right] c. \text{ For the values of } z \text{ given, } v = 0.787c \text{ for}$$

$$z = 1.9 \text{ and } v = 0.944c \text{ for } z = 4.9.$$

103. (a) In the frame of the decaying K^0 meson, the pi mesons must recoil with equal momenta in opposite directions in order to conserve momentum. In that reference frame the available kinetic energy is $498 \text{ MeV} - 2(135 \text{ MeV}) = 228 \text{ MeV}$.

(b) The pi mesons share this equally, so each one has a kinetic energy of 114 MeV in that frame. The energy of each pi meson is $E = K + E_0 = 114 \text{ MeV} + 135 \text{ MeV} = 249 \text{ MeV}$.

The momentum of each pi meson can be found:

$$p = \frac{\sqrt{E^2 - E_0^2}}{c} = \frac{\sqrt{(249 \text{ MeV})^2 - (135 \text{ MeV})^2}}{c} = 209.2 \text{ MeV}/c$$

104. (a) For each second, the energy used is $3.9 \times 10^{26} \text{ J}$. The mass used in each second is

$$m = \frac{E}{c^2} = \frac{3.9 \times 10^{26} \text{ J}}{(3.0 \times 10^8 \text{ m/s})^2} = 4.3 \times 10^9 \text{ kg}$$

(b) The time to use this mass is

$$\Delta t = \frac{m_{\text{sun}}}{\Delta m / \Delta t} \times \text{efficiency} = \frac{2.0 \times 10^{30} \text{ kg}}{4.3 \times 10^9 \text{ kg/s}} \times 0.007 = 3.3 \times 10^{18} \text{ s} \approx 1.0 \times 10^{11} \text{ y.}$$

That is 20 times longer than the expected lifetime of the sun.

105. a) The planet's orbital speed is $v_p = \frac{2\pi r}{T} = 1.30 \times 10^4 \text{ m/s}$. If the system's center of mass is at rest, conservation of momentum gives the star's speed:

$$m_p v_p = m_s v_s, \text{ so } v_s = \frac{m_p v_p}{m_s} = \frac{1.90 \times 10^{27}}{1.99 \times 10^{30}} 1.30 \times 10^4 \text{ m/s} = 12.4 \text{ m/s}$$

(b) The redshifted wavelength is $\lambda = \lambda_0 \sqrt{\frac{1+\beta}{1-\beta}} \approx 550 \text{ nm} + 2.3 \times 10^{-5} \text{ nm}$

Similarly, the blueshifted wavelength is $\lambda = \lambda_0 \sqrt{\frac{1-\beta}{1+\beta}} \approx 550 \text{ nm} - 2.3 \times 10^{-5} \text{ nm}$

These small differences can be detected using modern spectroscopic tools.

106. To a good approximation, the shift is the same in each direction, so the redshift for the approaching light is half the difference, or 0.0045 nm. Using this in the equation

$\lambda = \lambda_0 \sqrt{\frac{1-\beta}{1+\beta}}$ leads to a speed parameter $\beta = 6.9 \times 10^{-6}$, or $v = \beta c = 2070 \text{ m/s}$. The speed

is equal to the circumference $2\pi R$ divided by the period, so the period is

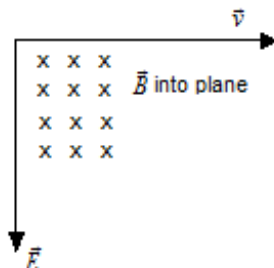
$$T = \frac{2\pi R}{v} = \frac{2\pi (6.96 \times 10^8 \text{ m})}{2070 \text{ m/s}} = 2.11 \times 10^6 \text{ s} = 24.5 \text{ days.}$$

Chapter 3

1. The required field is homogeneous within the desired region and decreases in magnitude to zero as rapidly as possible outside that region. The magnitude of the field is $B = E / v_0$. The best design is an electromagnet with flat, parallel pole faces that are large compared with the distance between them. But no matter what the design, it is impossible to eliminate edge effects.

2. $eE = evB$

$$B = \frac{E}{v} = \frac{(2.5 \times 10^5 \text{ V/m})}{(2.2 \times 10^6 \text{ m/s})} = 0.11 \text{ T}$$



3. Assume the speed is exact. Non-relativistically use the energy obtained by accelerating through a potential difference causes an increase in kinetic energy:

$$eV = \Delta\left(\frac{1}{2}mv^2\right)$$

$$V = \frac{mv^2}{2e} = \frac{(9.1094 \times 10^{-31} \text{ kg})(1.80 \times 10^7 \text{ m/s})^2}{2(1.6022 \times 10^{-19} \text{ C})} = 921.06 \text{ V}$$

Relativistically $eV = K = (\gamma - 1)mc^2$:

$$V = \left[\frac{1}{\sqrt{1 - \left(\frac{1.80 \times 10^7 \text{ m/s}}{2.9979 \times 10^8 \text{ m/s}} \right)^2}} - 1 \right] \left[\frac{511 \text{ keV}/c^2}{e} \right] \cdot c^2 = 923.59 \text{ V}.$$

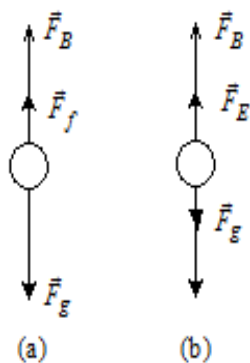
The results differ by about 2.5 volts, or about 0.27%. Relativity is required only if that level of precision is needed.

4. $eE = evB$ so $E = vB = (4.0 \times 10^6 \text{ m/s})(1.2 \times 10^{-2} \text{ T}) = 4.8 \times 10^4 \text{ V/m}$

$$y = \frac{1}{2}at^2 = \frac{1}{2}\left(\frac{F}{m}\right)\left(\frac{\ell}{v_0}\right)^2 = \frac{1}{2}\left(\frac{eE}{m}\right)\left(\frac{\ell}{v_0}\right)^2 = \frac{eE\ell^2}{2mv_0^2}$$

$$= \frac{(1.602 \times 10^{-19} \text{ C})(4.8 \times 10^4 \text{ V/m})(0.02 \text{ m})^2}{2(9.109 \times 10^{-31} \text{ kg})(4.0 \times 10^6 \text{ m/s})^2} = 1.0552 \times 10^{-1} \text{ m} = 10.6 \text{ cm}$$

5. The forces are shown below.



6. At terminal velocity the net force is zero, so $F_f = fv_t = mg$ and $v_t = mg / f$.

$$7. \quad v_t = \frac{mg}{f} = \frac{mg}{6\pi\eta r} \quad m = \rho(\text{volume}) = \frac{4}{3}\pi\rho r^3$$

$$v_t = \left(\frac{4}{3}\pi\rho r^3\right)\left(\frac{g}{6\pi\eta r}\right) = \frac{2g\rho r^2}{9\eta} \quad \text{Solving for } r: \quad r = 3\sqrt{\frac{\eta v_t}{2g\rho}}$$

$$8. \quad (a) \quad r = 3\sqrt{\frac{\eta v_t}{2g\rho}} = 3\sqrt{\frac{(1.82 \times 10^{-5} \text{ kg}\cdot\text{m}^{-1}\cdot\text{s}^{-1})(1.3 \times 10^{-3} \text{ m/s})}{2(9.80 \text{ m/s}^2)(900 \text{ kg/m}^3)}} = 3.47 \mu\text{m}$$

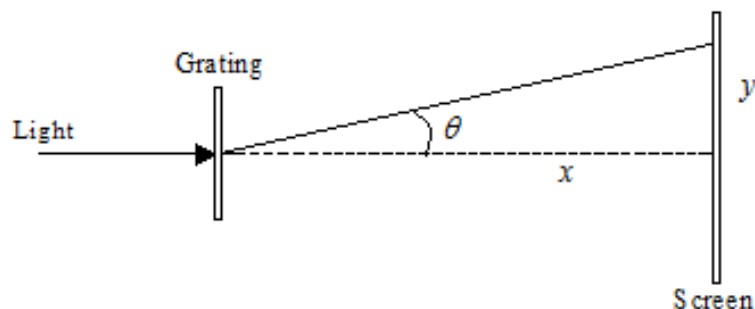
$$(b) \quad m = \rho V = \frac{4}{3}\pi\rho r^3 = \frac{4}{3}\pi(900 \text{ kg/m}^3)(3.47 \times 10^{-6} \text{ m})^3 = 1.58 \times 10^{-13} \text{ kg}$$

$$(c) \quad f = \frac{mg}{v_t} = \frac{(1.58 \times 10^{-13} \text{ kg})(9.80 \text{ m/s}^2)}{1.3 \times 10^{-3} \text{ m/s}} = 1.19 \times 10^{-9} \text{ kg/s}$$

$$9. \quad \text{Lyman: } \lambda = \left[R_H \left(1 - \frac{1}{\infty^2} \right) \right]^{-1} = R_H^{-1} = (1.096776 \times 10^7 \text{ m}^{-1})^{-1} = 91.2 \text{ nm}$$

$$\text{Balmer: } \lambda = \left[R_H \left(\frac{1}{2^2} - \frac{1}{\infty^2} \right) \right]^{-1} = 4R_H^{-1} = 4(1.096776 \times 10^7 \text{ m}^{-1})^{-1} = 364.7 \text{ nm}$$

10. $d = (400 \text{ mm}^{-1})^{-1} = 2.5 \mu\text{m} = 2500 \text{ nm}$ and $\lambda = d \sin \theta$ in first order. Also $\tan \theta = y / x$ so $y = x \tan \theta = (3.0 \text{ m}) \tan [\sin^{-1}(\lambda / d)]$.



$$\text{Red: } y = (3.0 \text{ m}) \tan \left[\sin^{-1} \left(\frac{656.5 \text{ nm}}{2500 \text{ nm}} \right) \right] = 81.6 \text{ cm}$$

$$\text{Blue-green: } y = (3.0 \text{ m}) \tan \left[\sin^{-1} \left(\frac{486.3 \text{ nm}}{2500 \text{ nm}} \right) \right] = 59.5 \text{ cm}$$

$$\text{Violet: } y = (3.0 \text{ m}) \tan \left[\sin^{-1} \left(\frac{434.2 \text{ nm}}{2500 \text{ nm}} \right) \right] = 52.9 \text{ cm}$$

11. $d = (420 \text{ mm}^{-1})^{-1} = 2.381 \mu\text{m}$; $\lambda = 656.5 \text{ nm}$ for red. We know $n\lambda = d \sin \theta$ so $\theta = \sin^{-1} \left(\frac{n\lambda}{d} \right)$. For $n = 1$ (first order) we find

$$y = x \tan \theta = (2.8 \text{ m}) \tan \left[\sin^{-1} \left(\frac{656.5 \text{ nm}}{2381 \text{ nm}} \right) \right] = 80.3 \text{ cm}.$$

Similarly for $n = 2$ we find $y = 185.1 \text{ cm}$, and for $n = 3$ we find $y = 412.2 \text{ cm}$.

Therefore the separations are:

between $n = 1$ and $n = 2$, $\Delta y = 185.1 \text{ cm} - 80.3 \text{ cm} = 104.8 \text{ cm}$;

between $n = 2$ and $n = 3$, $\Delta y = 412.2 \text{ cm} - 185.1 \text{ cm} = 227.1 \text{ cm}$.

12. Use Equation (3.13) with $n = 4$ for the Brackett series and $n = 5$ for the Pfund series. The largest wavelengths occur for the smallest values of k .

$$\text{Brackett: } \lambda = \left[R_{\text{H}} \left(\frac{1}{4^2} - \frac{1}{k^2} \right) \right]^{-1} = \left[(1.096776 \times 10^7 \text{ m}^{-1}) \left(\frac{1}{4^2} - \frac{1}{k^2} \right) \right]^{-1}.$$

For $k = 5$, $\lambda = 4.052 \mu\text{m}$; for $k = 6$, $\lambda = 2.626 \mu\text{m}$; for $k = 7$, $\lambda = 2.166 \mu\text{m}$;
for $k = 8$, $\lambda = 1.945 \mu\text{m}$.

$$\text{Pfund: } \lambda = \left[R_{\text{H}} \left(\frac{1}{5^2} - \frac{1}{k^2} \right) \right]^{-1} = \left[(1.096776 \times 10^7 \text{ m}^{-1}) \left(\frac{1}{5^2} - \frac{1}{k^2} \right) \right]^{-1}.$$

For $k = 6$, $\lambda = 7.460 \mu\text{m}$; for $k = 7$, $\lambda = 4.654 \mu\text{m}$; for $k = 8$, $\lambda = 3.741 \mu\text{m}$;
for $k = 9$, $\lambda = 3.297 \mu\text{m}$.

13. Beginning with Equation (3.10) with $n = 1$, we have $\lambda = d \sin \theta$ with $d = 0.20 \text{ mm}$.

Therefore $\frac{d\lambda}{d\theta} = d \cos \theta$. Assuming the spectrum was viewed in the forward direction,

then $\theta \approx 0$ and $\cos \theta \approx 1$ so $\frac{d\lambda}{d\theta} \approx \frac{\Delta\lambda}{\Delta\theta} = d$. An angle of 0.50 minutes of arc corresponds to $1.45 \times 10^{-4} \text{ rad}$ so

$$\Delta\lambda = d\Delta\theta = (0.20 \times 10^{-3} \text{ m}) 1.45 \times 10^{-4} \text{ rad} = 29 \text{ nm}.$$

14. (a) Use Equation (3.13) $\frac{1}{\lambda} = R_H \left(\frac{1}{n^2} - \frac{1}{k^2} \right)$ with $R_H = 1.096776 \times 10^7 \text{ m}^{-1}$ and $n = 2$ for

the Balmer series. With $k = 3$ we find $\frac{1}{\lambda} = 1.096776 \times 10^7 \text{ m}^{-1} \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = 656.47 \text{ nm}$.

This wavelength, found with the smallest value of k , is referred to as the hydrogen alpha or H_α line. Using the equation above with $k = 4$ we find the hydrogen beta or H_β wavelength equals 486.27 nm. Similarly with $k = 5$ the hydrogen gamma or H_γ wavelength equals 434.17 nm, and finally with $k = 6$ the hydrogen delta or H_δ wavelength equals 410.29 nm.

(b) Because only three wavelengths are observed, the source must be moving relative to the detector. The wavelengths have increased, so the source must be moving away from the detector. [See Equation (2.34) for example.] The H_α line at 656.47 nm is redshifted out of the visible.

(c) Because the wavelengths are known and $c = \lambda f$, we will use the reciprocal of Equation (2.33). We select this equation since the wavelengths are larger and thus the

source is receding from the observer. $\frac{\lambda_{\text{observed}}}{\lambda_{\text{source}}} = \sqrt{\frac{1+\beta}{1-\beta}}$. Using algebra to solve the

equation for β and substituting the smallest wavelength gives

$$\beta = \frac{\left(\frac{\lambda_{\text{observed}}}{\lambda_{\text{source}}} \right)^2 - 1}{\left(\frac{\lambda_{\text{observed}}}{\lambda_{\text{source}}} \right)^2 + 1} = \frac{\left(\frac{453.4 \text{ nm}}{410.29 \text{ nm}} \right)^2 - 1}{\left(\frac{453.4 \text{ nm}}{410.29 \text{ nm}} \right)^2 + 1} = 0.0995.$$

Therefore the speed is $v = 0.10c$ or $v = 3.0 \times 10^7 \text{ m/s}$. Using other pairs of wavelengths gives a similar result. If the object is rotating, then one side would be moving toward and another side away from the detector so a range of wavelengths would be observed. This effect can be used to determine the rotation speeds.

15. (a) Use Equation (3.13) $\frac{1}{\lambda} = R_H \left(\frac{1}{n^2} - \frac{1}{k^2} \right)$ with $R_H = 1.096776 \times 10^7 \text{ m}^{-1}$ and $n = 3$ for the

Paschen series. With $k = 4$ we find $\frac{1}{\lambda} = 1.096776 \times 10^7 \text{ m}^{-1} \left(\frac{1}{3^2} - \frac{1}{4^2} \right)$; $\lambda = 1875.63 \text{ nm}$.

With $k = 5$, $\lambda = 1282.17 \text{ nm}$; with $k = 6$, $\lambda = 1094.12 \text{ nm}$;

with $k = 7$, $\lambda = 1005.22 \text{ nm}$; with $k = 8$, $\lambda = 954.86 \text{ nm}$.

(b) The observed spectral lines have been Doppler shifted. It might appear as if the wavelengths have been blueshifted since the largest observed wavelength is *smaller* than the largest expected wavelength using the Paschen series. However it is more likely that the wavelengths have been redshifted and the calculated wavelength just below 1000 nm in part (a) corresponds to the observed wavelength of 1046.1 nm.

(c) Using the formula from Problem 14 and noting from part (b) that the star is receding

$$\text{from the detector } \beta = \frac{\left(\frac{\lambda_{\text{observed}}}{\lambda_{\text{source}}} \right)^2 - 1}{\left(\frac{\lambda_{\text{observed}}}{\lambda_{\text{source}}} \right)^2 + 1} = \frac{\left(\frac{1334.5 \text{ nm}}{1282.17 \text{ nm}} \right)^2 - 1}{\left(\frac{1334.5 \text{ nm}}{1282.17 \text{ nm}} \right)^2 + 1} = 0.040 \text{ or } v = 1.20 \times 10^7 \text{ m/s}.$$

16. (a) To obtain a charge of +1 with three quarks requires two charges of $+2e/3$ and one of charge $-e/3$. Three quarks with charge $+e/3$ would violate the Pauli Exclusion Principle for spin 1/2 particles.

(b) To obtain a charge of zero we could have either two $+e/3$ and one $-2e/3$ or one $+2e/3$ and two $-e/3$. At this point in the text there is no reason to prefer either choice. (The latter turns out to be correct.)

17. (a) $\lambda_{\text{max}} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{4.2 \text{ K}} = 0.69 \text{ mm}$

(b) $\lambda_{\text{max}} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{293 \text{ K}} = 9.89 \text{ } \mu\text{m}$

(c) $\lambda_{\text{max}} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{2500 \text{ K}} = 1.16 \text{ } \mu\text{m}$

(d) $\lambda_{\text{max}} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{9000 \text{ K}} = 0.322 \text{ } \mu\text{m}$

18. (a) $T = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{1.50 \times 10^{-14} \text{ m}} = 1.932 \times 10^{11} \text{ K}$

(b) $T = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{1.50 \times 10^{-9} \text{ m}} = 1.932 \times 10^6 \text{ K}$

(c) $T = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{640 \times 10^{-9} \text{ m}} = 4528 \text{ K}$

$$(d) T = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{1 \text{ m}} = 2.898 \times 10^{-3} \text{ K}$$

$$(e) T = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{204 \text{ m}} = 1.42 \times 10^{-5} \text{ K}$$

19. (a) $\frac{P_1}{P_0} = \frac{\sigma T_1^4}{\sigma T_0^4}$; so $P_1 = P_0 \frac{T_1^4}{T_0^4} = \left(\frac{2300 \text{ K}}{900 \text{ K}} \right)^4 P_0 = 42.7 P_0$; The power increases by a factor of 42.7.

(b) To double the power output, the ratio of temperatures to the fourth power must equal

2. $\left(\frac{T_1}{900 \text{ K}} \right)^4 = 2$. Solving we find $T_1 = 1070 \text{ K}$.

20. (a) $\lambda_{\text{max}} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{310 \text{ K}} = 9.35 \text{ } \mu\text{m}$

(b) At this temperature the power per unit area is

$$R = \sigma T^4 = (5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^4)(310 \text{ K})^4 = 524 \text{ W/m}^2. \text{ The total surface area of a cylinder is } 2\pi r(r+h) = 2\pi(0.13 \text{ m})(1.75 \text{ m} + 0.13 \text{ m}) = 1.54 \text{ m}^2 \text{ so the total power is } P = (524 \text{ W/m}^2)(1.54 \text{ m}^2) = 807 \text{ W}.$$

(c) The total energy radiated in one day is the power multiplied by the time;

$$E = P \cdot t = (807 \text{ W}) \cdot (86400 \text{ s}) = 6.97 \times 10^7 \text{ J}.$$

$$2000 \text{ kcal} = (2 \times 10^6 \text{ cal}) \cdot (4.186 \text{ J/cal}) = 8.37 \times 10^6 \text{ J}.$$

There are several assumptions. First, a cylinder may overestimate the total surface area; second, radiation is minimized by hair covering and clothing.

21. $\lambda_{\text{max}} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{200,273 \text{ K}} = 1.447 \times 10^{-8} \text{ m}$

22. We know from Example 3.8 that in the long-wavelength limit, the two expressions are the same. By comparing the expressions, we can also note that the Rayleigh-Jeans spectral distribution will be larger than the Planck expression for a given λ and T .

[Compare graphs or calculations using Equations (3.22) and (3.23) one can obtain using Excel, Mathcad, or similar program.] So we want

$$(0.95) \left(\frac{2\pi ckT}{\lambda^4} \right) = \left(\frac{2\pi c^2 h}{\lambda^5} \right) \left(\frac{1}{[e^{hc/\lambda kT} - 1]} \right). \text{ Simplifying we find}$$

$$(0.95) \left(\frac{\lambda kT}{hc} \right) = \frac{1}{[e^{hc/\lambda kT} - 1]}. \text{ Let } \frac{\lambda kT}{hc} = x \text{ and this simplifies to } (0.95)x = \frac{1}{[e^{1/x} - 1]}. \text{ We}$$

can solve this transcendental equation to find $x = 9.83$ so $9.83 = \frac{\lambda kT}{hc}$. Substituting numerical values for the constants we find

$$\lambda = \frac{9.83hc}{kT} = \frac{9.83(1.986 \times 10^{-25} \text{ J} \cdot \text{m})}{1.38 \times 10^{-23} \text{ J} \cdot \text{K}^{-1} (5800 \text{ K})} = 2.44 \times 10^{-5} \text{ m} = 24.4 \text{ } \mu\text{m}.$$

23. $\lambda_{\text{max}} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{3000 \text{ K}} = 966 \text{ nm}$ which is in the near infrared.

24. The graph is a characteristic Planck law curve with a maximum at $\lambda = 966 \text{ nm}$ (see Problem 23).

(a) Numerical integration of the $\mathcal{I}(\lambda, T)$ function shows that approximately 8.1% of the radiated power is between 400 nm and 700 nm. Details of the calculation are:

$$2\pi c^2 h \int_{4 \times 10^{-7}}^{7 \times 10^{-7}} \exp\left(-\frac{hc}{\lambda kT}\right) \lambda^{-5} d\lambda = (3.74 \times 10^{-16}) \int_{4 \times 10^{-7}}^{7 \times 10^{-7}} \exp\left(-\frac{4.798 \times 10^{-6}}{\lambda}\right) \lambda^{-5} d\lambda \quad \text{That is}$$

$$= 3.71 \times 10^5 \text{ W/m}^2$$

the power per unit area emitted over visible wavelengths. Over all wavelengths we know the power per unit area is $R = \sigma T^4 = 4.59 \times 10^6 \text{ W/m}^2$. Therefore the fraction emitted in the visible is $\frac{3.7128 \times 10^5 \text{ W/m}^2}{4.593 \times 10^6 \text{ W/m}^2} = 0.081$.

(b) Using computed numerical intensity values $\frac{\mathcal{I}(400 \text{ nm}, T)}{\mathcal{I}(966 \text{ nm}, T)} \approx 0.073$;

$$\frac{\mathcal{I}(700 \text{ nm}, T)}{\mathcal{I}(966 \text{ nm}, T)} \approx 0.754.$$

25. In this limit $\exp(hc / \lambda kT) \gg 1$, so $\mathcal{I}(\lambda, T) \approx \frac{2\pi c^2 h}{\lambda^5} \exp\left(-\frac{hc}{\lambda kT}\right)$. The exponential goes to zero faster than λ^5 , so the intensity approaches zero in this limit.

26. (a) At this temperature the power per unit area is

$$R = \sigma T^4 = (5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}) (293 \text{ K})^4 = 419 \text{ W/m}^2. \text{ For the basketball, a sphere of radius } r = 12.5 \text{ cm, we obtain } P = R(4\pi r^2) = (419 \text{ W/m}^2)(4\pi)(0.125 \text{ m})^2 = 82.3 \text{ W}.$$

(b) At this temperature the power per unit area is

$$R = \sigma T^4 = (5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}) (310 \text{ K})^4 = 524 \text{ W/m}^2. \text{ Assume the body is roughly cylindrical with a radius of 13 cm and a height of 1.65 m. The total surface area of a cylinder is } 2\pi r(r + h) = 2\pi(0.13 \text{ m})(1.78 \text{ m}) = 1.45 \text{ m}^2 \text{ so the total power is}$$

$$P = (524 \text{ W/m}^2)(1.45 \text{ m}^2) = 760 \text{ W. Numerical values will vary depending on estimates of the human body size.}$$

$$27. \lambda_{\max} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{310 \text{ K}} = 9.35 \mu\text{m}$$

28. Taking derivatives $\frac{\partial^2 \psi}{\partial t^2} = \frac{\partial^2 a}{\partial t^2} \sin\left(\frac{n\pi x}{L}\right)$; $\frac{\partial^2 \psi}{\partial x^2} = -a \frac{n^2 \pi^2}{L^2} \sin\left(\frac{n\pi x}{L}\right)$. Substituting these values into the wave equation produces $\frac{1}{c^2} \frac{\partial^2 a}{\partial t^2} \sin\left(\frac{n\pi x}{L}\right) - \left(-a \frac{n^2 \pi^2}{L^2} \sin\left(\frac{n\pi x}{L}\right)\right) = 0$ which simplifies to $\frac{\partial^2 a}{\partial t^2} = -a \frac{n^2 \pi^2 c^2}{L^2} = -\omega^2 a$ where $\omega = \frac{n\pi c}{L}$. Since $\lambda = \frac{2L}{n}$ for this system and $c = \lambda f$, then $\omega = 2\pi f$.

29. Let $r = \sqrt{n_x^2 + n_y^2 + n_z^2}$ be the radius of a three-dimensional number space with the n_i the three components of that space. Then let dN be the number of allowed states between r and $r + dr$. This corresponds to the number of points in a spherical shell of number space, which is $dN = \frac{1}{8}(4\pi r^2)dr$ where we have used the fact that $4\pi r^2 dr$ is the “volume” of the shell (area $4\pi r^2$ by thickness dr), and the $1/8$ is due to the fact that only positive numbers n_i are allowed, so only $1/8$ of the space is available. Also

$$r^2 = n_x^2 + n_y^2 + n_z^2 = \frac{\Omega^2 L^2}{\pi^2 c^2} = \frac{4L^2 f^2}{c^2} \text{ or } r = \frac{2Lf}{c}. \text{ Then from this } dr = \left(\frac{2L}{c}\right)df. \text{ Putting}$$

$$\text{everything together: } dN = \frac{1}{8}(4\pi r^2)dr = \frac{\pi}{2} r^2 dr = \frac{\pi}{2} \left(\frac{2Lf}{c}\right)^2 \frac{2L}{c} df = \frac{4\pi L^3}{c^3} f^2 df$$

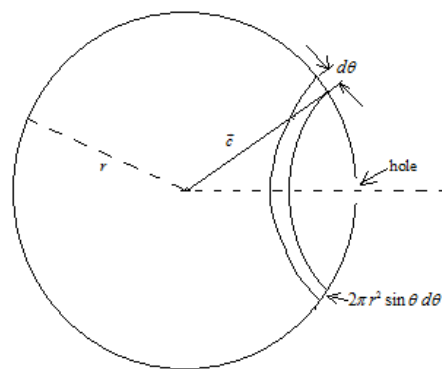
30. From the diagram at right we see that the average x-component of the velocity ($\vec{c}$) of electromagnetic radiation within the cavity is

$$\langle c_x \rangle = \frac{\int_0^{\pi/2} (c \cos \theta) 2\pi r^2 \sin \theta d\theta}{\int_0^{\pi/2} 2\pi r^2 \sin \theta d\theta}.$$

$$\text{Letting } u = \cos \theta \text{ we have } \langle c_x \rangle = \frac{c \int_0^1 u du}{\int_0^1 du} = \frac{c}{2}.$$

On average only one-half of the photons are traveling to the right. Thus the mean velocity of photons traveling to the right is $c/4$.

$$\text{Therefore power} = (\text{intensity})(\text{area}) = \frac{c}{4} dU(\Delta A).$$



31. For classical oscillators the Maxwell-Boltzmann distribution gives

$n(E) = A \exp(-E/kT) = A \exp(-\beta E)$ where $E = nhf$ and $\beta = 1/kT$. The mean energy

is $\bar{E} = \frac{\sum_{n=0}^{\infty} E n(E)}{\sum_{n=0}^{\infty} n(E)} = \frac{\sum_{n=0}^{\infty} nhf \exp(-\beta nhf)}{\sum_{n=0}^{\infty} \exp(-\beta nhf)}$. Notice that $\bar{E} = \frac{\partial}{\partial \beta} \ln \sum_{n=0}^{\infty} \exp(-\beta nhf)$. Now

letting $x = \exp(-\beta hf)$ we see that by Taylor series

$$\sum_{n=0}^{\infty} \exp(-\beta nhf) = 1 + x + x^2 + \dots = (1-x)^{-1} \text{ for } x^2 < 1.$$

$$\bar{E} = \frac{\partial}{\partial \beta} \ln(1-x)^{-1} = -\frac{\partial}{\partial \beta} \ln(1-x) = \frac{hf \exp(-\beta hf)}{1 - \exp(-\beta hf)}$$

$$\bar{E} = \frac{hf}{\exp(hf/kT) - 1}. \text{ Using the result of Problem 29 (along with a factor of 2 for two}$$

photon polarizations) we can see that

$$U(f, T) = 2 \frac{4\pi}{c^3} f^2 \frac{hf}{\exp(hf/kT) - 1} = \frac{8\pi hf^3/c^3}{\exp(hf/kT) - 1}.$$

To change from U to $\mathcal{I}$ requires the factor $c/4$ (Problem 30), and changing from a frequency distribution requires a factor c/λ^2 , because with $f = c/\lambda$ we have

$|df| = (c/\lambda^2)d\lambda$. Putting these together

$$\mathcal{I}(\lambda, T) = \frac{8\pi h/\lambda^3}{\exp(hf/kT) - 1} \left(\frac{c}{\lambda^2}\right) \left(\frac{c}{4}\right) = \frac{2\pi c^2 h}{\lambda^5} \frac{1}{\exp(hc/\lambda kT) - 1}.$$

32. Energy per photon $= hf = (6.626 \times 10^{-34} \text{ J}\cdot\text{s})(98.1 \times 10^6 \text{ s}^{-1}) = 6.50 \times 10^{-26} \text{ J}$

$$(5.0 \times 10^4 \text{ J/s}) \frac{1 \text{ photon}}{6.50 \times 10^{-26} \text{ J}} = 7.69 \times 10^{29} \text{ photons/s}$$

33. (a) Energy per photon $= hf = (6.626 \times 10^{-34} \text{ J}\cdot\text{s})(1100 \times 10^3 \text{ s}^{-1}) = 7.29 \times 10^{-28} \text{ J}$

$$(180 \text{ J/s}) \frac{1 \text{ photon}}{7.29 \times 10^{-28} \text{ J}} = 2.47 \times 10^{29} \text{ photons/s}$$

(b) energy per photon $= h \frac{c}{\lambda} = (6.626 \times 10^{-34} \text{ J}\cdot\text{s}) \left(\frac{3.00 \times 10^8 \text{ m/s}}{8 \times 10^{-9} \text{ m}} \right) = 2.48 \times 10^{-17} \text{ J}$

$$(180 \text{ J/s}) \frac{1 \text{ photon}}{2.48 \times 10^{-17} \text{ J}} = 7.26 \times 10^{18} \text{ photons/s}$$

(c) $(180 \text{ J/s}) \frac{1 \text{ photon}}{4 \text{ MeV}} \frac{1 \text{ MeV}}{1.60 \times 10^{-13} \text{ J}} = 2.81 \times 10^{14} \text{ photons/s}$

$$34. f_t = \frac{\phi}{h} = \frac{2.93 \text{ eV}}{4.136 \times 10^{-15} \text{ eV} \cdot \text{s}} = 7.08 \times 10^{14} \text{ Hz}; \quad eV_0 = \frac{hc}{\lambda} - \phi \text{ so } V_0 = \frac{1}{e} \left[\frac{hc}{\lambda} - \phi \right];$$

$$V_0 = \frac{1}{e} \left[\frac{1240 \text{ eV} \cdot \text{nm}}{380 \text{ nm}} - 2.93 \text{ eV} \right] = 0.333 \text{ V}$$

$$35. \lambda_t = \frac{hc}{\phi} = \frac{1240 \text{ eV} \cdot \text{nm}}{4.64 \text{ eV}} = 267.2 \text{ nm}.$$

If the wavelength is halved (to $\lambda = 133.6 \text{ nm}$), then

$$K = \frac{hc}{\lambda} - \phi = \frac{1240 \text{ eV} \cdot \text{nm}}{133.6 \text{ nm}} - 4.64 \text{ eV} = 4.64 \text{ eV}$$

$$36. \text{ Notice that } \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{532 \text{ nm}} = 2.33 \text{ eV} > \phi, \text{ so photoelectrons will be produced.}$$

$$\left(2 \times 10^{-3} \text{ J/s} \right) \left(\frac{10^{-5} \text{ electrons}}{1 \text{ photon}} \right) \left(\frac{1 \text{ photon}}{2.33 \text{ eV}} \right) \left(\frac{1 \text{ eV}}{1.60 \times 10^{-19} \text{ J}} \right) \left(\frac{1.60 \times 10^{-19} \text{ C}}{\text{electron}} \right) = 8.58 \text{ nA}$$

$$37. \phi = \frac{hc}{\lambda_t} = \frac{1240 \text{ eV} \cdot \text{nm}}{270 \text{ nm}} = 4.59 \text{ eV}; \quad K = 2.0 \text{ eV} = hf - \phi;$$

$$f = \frac{K + \phi}{h} = \frac{2.0 \text{ eV} + 4.59 \text{ eV}}{4.136 \times 10^{-15} \text{ eV} \cdot \text{s}} = 1.59 \times 10^{15} \text{ Hz}$$

$$38. E = 100 \left(\frac{hc}{\lambda} \right) = 100 \frac{1240 \text{ eV} \cdot \text{nm}}{610 \text{ nm}} = 203 \text{ eV}$$

$$39. eV_{01} = hc / \lambda_1 - \phi \text{ and } eV_{02} = hc / \lambda_2 - \phi. \text{ Subtracting these equations and rearranging we find}$$

$$h = \frac{e(V_{02} - V_{01})}{c \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right)} = \frac{e(2.3 \text{ V} - 1.0 \text{ V})}{(3.00 \times 10^8 \text{ m/s}) \left(\frac{1}{207 \text{ nm}} - \frac{1}{260 \text{ nm}} \right)} = 4.40 \times 10^{-15} \text{ eV} \cdot \text{s}.$$

This is about 6% from the accepted value. For the work function we use the first set of data (the second set should give the same result):

$$\phi = \frac{hc}{\lambda_1} - eV_{01} = \frac{(4.40 \times 10^{-15} \text{ eV} \cdot \text{s})(3.00 \times 10^8 \text{ m/s})}{260 \times 10^{-9} \text{ m}} - 1.0 \text{ eV} = 4.1 \text{ eV}$$

40. $hf = \phi + K$ or $\frac{hc}{\lambda} = \phi + K = \phi + \frac{1}{2}mv_{\max}^2$; Using 4.63 eV as the work function for tungsten (see table 3.3) and 9.11×10^{-31} kg as the mass of the electron, one can solve for λ .

$$\lambda = \frac{hc}{\left(\phi + \frac{1}{2}mv_{\max}^2\right)} = \left(\frac{1240 \text{ eV} \cdot \text{nm}}{4.63 \text{ eV} + \frac{1}{2}(9.11 \times 10^{-31} \text{ kg})(1.4 \times 10^6 \text{ m/s})^2 \frac{1 \text{ eV}}{1.6 \times 10^{-19} \text{ J}}} \right)$$

$$= 1.21 \times 10^2 \text{ nm} = 121 \text{ nm}$$

41. $\lambda_{\min} = \frac{1.24 \times 10^{-6} \text{ V} \cdot \text{m}}{V} = \frac{1.24 \times 10^{-6} \text{ V} \cdot \text{m}}{30 \text{ kV}} = 0.0413 \text{ nm}$

42. $\lambda = \frac{hc}{K} = \frac{1240 \text{ eV} \cdot \text{nm}}{5 \times 10^{10} \text{ eV}} = 2.48 \times 10^{-17} \text{ m}$. A photon produced by bremsstrahlung is still an x ray, even though this falls outside the normal range for x rays.

43. $\lambda = \frac{1240 \text{ eV} \cdot \text{nm}}{2.5 \times 10^4 \text{ eV}} = 0.0496 \text{ nm}$

44. Because the electron is accelerated through a potential difference, it has kinetic energy equal to $K = eV_0$. As described in the text, Equation (3.37) gives the minimum wavelength (assuming the work function is small). So

$$\lambda_{\min} = \left(\frac{hc}{e} \right) \left(\frac{1}{V_0} \right) = \frac{1.240 \times 10^{-6} \text{ V} \cdot \text{m}}{3.5 \times 10^4 \text{ V}} = 3.54 \times 10^{-11} \text{ m} = 3.54 \times 10^{-2} \text{ nm}.$$

The work function for tungsten from Table 3.3 is 4.63 eV. If we include the work function, the energy available for the photon and the wavelength of the photon will change by 4.63 parts out of 35000 or about 0.013% which is very small.

45. From Figure (3.19) we observe that the two characteristic spectral lines occur at wavelengths of $6.4 \times 10^{-11} \text{ m}$ and $7.2 \times 10^{-11} \text{ m}$. Rearrange Equation (3.37) to solve for the potential V_0 ; $V_0 = \frac{hc}{e \lambda_{\min}} = \frac{1.24 \times 10^{-6} \text{ V} \cdot \text{m}}{7.2 \times 10^{-11} \text{ m}} = 17.2 \text{ kV}$ and we have used the larger of the two wavelengths in order to determine the minimum potential difference.

46. $\Delta\lambda = \frac{h}{mc}(1 - \cos\theta)$ so at maximum $\cos\theta = -1$ and

$$\frac{\Delta\lambda}{\lambda} = \frac{2h}{\lambda mc} = \frac{2hc}{mc^2 \lambda} = \frac{2(1240 \text{ eV} \cdot \text{nm})}{(511.0 \text{ keV})(480 \text{ nm})} = 1.01 \times 10^{-5}.$$

This corresponds to $\Delta\lambda = 4.85 \times 10^{-12} \text{ m}$ and therefore is not easily observed.

47. The maximum change in the photon's energy is obtained in backscattering ($\theta = 180^\circ$), so

$1 - \cos \theta = 2$ and $\Delta\lambda = \frac{2h}{mc} = 4.853 \times 10^{-12} \text{ m}$. The photon's original wavelength was

$$\lambda = \frac{hc}{E} = \frac{1240 \text{ eV} \cdot \text{nm}}{40000 \text{ eV}} = 0.0310 \text{ nm} = 3.10 \times 10^{-11} \text{ m} \text{ and the new wavelength is}$$

$\lambda' = \lambda + \Delta\lambda = 3.585 \times 10^{-11} \text{ m}$. The electron's recoil energy equals the change in the photon's energy, or

$$K = \frac{hc}{\lambda} - \frac{hc}{\lambda'} = \frac{1240 \text{ eV} \cdot \text{nm}}{3.10 \times 10^{-2} \text{ nm}} - \frac{1240 \text{ eV} \cdot \text{nm}}{3.585 \times 10^{-2} \text{ nm}} = 5411 \text{ eV} = 5.41 \text{ keV}$$

48. Use the Compton scattering formula but with the proton's mass and $\theta = 90^\circ$:

$$\Delta\lambda = \frac{h}{mc}(1 - \cos \theta) = \frac{h}{mc} = \frac{hc}{mc^2} = \frac{1240 \text{ eV} \cdot \text{nm}}{938.27 \text{ MeV}} = 1.32 \text{ fm}.$$

49. We find the Compton wavelength using $\lambda_c = \frac{h}{mc} = \frac{hc}{mc^2} = \frac{1240 \text{ eV} \cdot \text{nm}}{938.27 \text{ MeV}} = 1.32 \text{ fm}$. The

photon energy is $E = \frac{hc}{\lambda_c} = 938 \text{ MeV}$. In principle this could be observed, but the energy requirements are high.

$$50. \frac{\Delta\lambda}{\lambda} = 0.004 = \frac{\lambda_c}{\lambda}(1 - \cos \theta) \quad \text{so} \quad \lambda = 250\lambda_c(1 - \cos \theta);$$

$$(a) \lambda = 250(2.43 \times 10^{-12} \text{ m})(1 - \cos 30^\circ) = 8.14 \times 10^{-11} \text{ m}$$

$$(b) \lambda = 250(2.43 \times 10^{-12} \text{ m})(1 - \cos 90^\circ) = 6.08 \times 10^{-10} \text{ m}$$

$$(c) \lambda = 250(2.43 \times 10^{-12} \text{ m})(1 - \cos 170^\circ) = 1.21 \times 10^{-9} \text{ m}$$

51. By conservation of energy we know the electron's recoil energy equals the energy lost by

$$\text{the photon: } K = \frac{hc}{\lambda} - \frac{hc}{\lambda'} = \frac{hc(\lambda' - \lambda)}{\lambda'\lambda} = \frac{hc\Delta\lambda}{\lambda'\lambda}.$$

$$\text{Using } \lambda' = \lambda + \Delta\lambda \text{ we have } K = \frac{hc\Delta\lambda}{\lambda(\lambda + \Delta\lambda)} = \frac{hf\Delta\lambda}{\lambda + \Delta\lambda} = \frac{hf\Delta\lambda}{\lambda(1 + \Delta\lambda/\lambda)} = \frac{(\Delta\lambda/\lambda)hf}{1 + \frac{\Delta\lambda}{\lambda}}.$$

$$\text{Conservation of } p_x: p_e \cos \phi + \frac{h}{\lambda'} \cos \theta = \frac{h}{\lambda};$$

$$p_e \cos \phi = \frac{h}{\lambda} - \frac{h}{\lambda'} \cos \theta. \quad (1)$$

$$\text{Conservation of } p_y: p_e \sin \phi - \frac{h}{\lambda'} \sin \theta = 0;$$

$$p_e \sin \phi = \frac{h}{\lambda'} \sin \theta \quad (2)$$

Dividing equation (2) by equation (1), $\tan \phi = \frac{\frac{h}{\lambda'} \sin \theta}{\frac{h}{\lambda} - \frac{h}{\lambda'} \cos \theta}$.

Using $\lambda' = \lambda + \frac{h}{mc}(1 - \cos \theta)$ we have: $\tan \phi = \frac{\frac{h \sin \theta}{\lambda + \frac{h}{mc}(1 - \cos \theta)}}{\frac{h}{\lambda} - \frac{h \cos \theta}{\lambda + \frac{h}{mc}(1 - \cos \theta)}}$.

Multiplying numerator and denominator by $\lambda \left[\lambda + \frac{h}{mc}(1 - \cos \theta) \right]$, we have:

$$\tan \phi = \frac{\lambda h \sin \theta}{\lambda h + \frac{h^2}{mc}(1 - \cos \theta) - \lambda h \cos \theta} = \frac{\lambda \sin \theta}{\left(\lambda + \frac{h}{mc} \right)(1 - \cos \theta)}.$$

Using the trig identity: $\frac{\sin \theta}{(1 - \cos \theta)} = \cot \left(\frac{\theta}{2} \right)$, we find:

$$\tan \phi = \frac{\lambda}{\lambda + \frac{h}{mc}} \cot \left(\frac{\theta}{2} \right) = \frac{1}{1 + \frac{h}{mc\lambda}} \cot \left(\frac{\theta}{2} \right) = \frac{1}{1 + \frac{hf}{mc^2}} \cot \left(\frac{\theta}{2} \right).$$

Inverting the equation gives $\cot \phi = \left[1 + \frac{hf}{mc^2} \right] \tan \left(\frac{\theta}{2} \right)$.

52. $\lambda' = \lambda + \lambda_c (1 - \cos \theta) = \frac{hc}{E} + \lambda_c (1 - \cos \theta);$

$$\lambda' = \frac{1240 \text{ eV} \cdot \text{nm}}{650 \times 10^3 \text{ eV}} + (2.43 \times 10^{-3} \text{ nm})(1 - \cos 110^\circ) = 5.17 \text{ pm}$$

$$E' = \frac{hc}{\lambda'} = \frac{1240 \text{ eV} \cdot \text{nm}}{5.17 \times 10^{-3} \text{ nm}} = 2.40 \times 10^5 \text{ eV} = 240 \text{ keV}$$

By conservation of energy we find: $K_e = E - E' = 650 \text{ keV} - 240 \text{ keV} = 410 \text{ keV}$ which agrees with the K formula in the previous problem. Also from the previous

problem we have: $\cot \phi = \left[1 + \frac{hf}{mc^2} \right] \tan \left(\frac{\theta}{2} \right) = \left[1 + \frac{650 \text{ keV}}{511 \text{ keV}} \right] \tan \left(\frac{110^\circ}{2} \right) = 3.245$ so

$$\phi = 17.1^\circ.$$

53. For $\theta = 90^\circ$ we know $\lambda' = \lambda + \lambda_c = 2.00243 \text{ nm}; \quad \frac{\Delta \lambda}{\lambda} = \frac{\lambda_c}{\lambda} = \frac{2.43 \times 10^{-3} \text{ nm}}{2 \text{ nm}} = 1.22 \times 10^{-3}$

54. $E = 2mc^2 = 2(938.3 \text{ MeV}) = 1877 \text{ MeV}$. This energy could come from a particle accelerator.

55. a) To find the minimum energy consider the zero-momentum frame. Let E_e be the energy of the electron in that frame, and E_0 is the rest energy of the electron. From conservation of energy and momentum:

$$\text{momentum: } \frac{hf}{c} = p_e = \frac{\sqrt{E_e^2 - E_0^2}}{c} \text{ or } hf = \sqrt{E_e^2 - E_0^2}.$$

$$\text{energy: } hf + E_e = 3E_0 \text{ or } hf = 3E_0 - E_e.$$

Squaring and subtracting these two equations gives: $0 = -10E_0^2 + 6E_eE_0$ which simplifies

to $E_e = \frac{5}{3}E_0$. This tells us that for the transformation from the lab frame to the zero-momentum frame, $\gamma = 5/3$ and $v = 0.8c$. Then from the momentum equation we have in

$$\text{the zero-momentum frame: } hf = \sqrt{\frac{25E_0^2}{9} - E_0^2} = \frac{4}{3}E_0.$$

In the lab the photon's energy is obtained using a Doppler shift:

$$hf_{\text{lab}} = hf \sqrt{\frac{1+\beta}{1-\beta}} = \frac{4}{3}E_0 \sqrt{\frac{1+0.8}{1-0.8}} = 4E_0 = 2.04 \text{ MeV}$$

b) The proton's rest energy is Mc^2 . Now as in (a) we let the proton's energy in the lab frame be E_p and conservation of momentum and energy give

$$\text{momentum: } hf = \sqrt{E_p^2 - (Mc^2)^2}; \quad \text{energy: } hf + E_p = 2E_0 + Mc^2.$$

Squaring and subtracting, we find $E_p = \frac{(Mc^2)^2 + 2E_0^2 + 2E_0Mc^2}{2E_0 + Mc^2}$. This is very close to

$E_p = Mc^2$. Therefore the zero-momentum and lab frames are equivalent, and we conclude $hf_{\text{lab}} \approx 2E_0 = 1.02 \text{ MeV}$.

56. The maximum energy transfer occurs when $\theta = 180^\circ$ so that

$$\Delta\lambda = (h/mc)(1 - \cos\theta) = 2h/mc. \text{ By conservation of energy the kinetic energy of the}$$

$$\text{electron is } K = E - E' = \frac{hc}{\lambda} - \frac{hc}{\lambda'} = \frac{hc}{\lambda} - \frac{hc}{\lambda + \Delta\lambda}. \text{ Multiplying through by } \lambda(\lambda + \Delta\lambda) \text{ we}$$

find $\lambda(\lambda + \Delta\lambda)K = hc(\lambda + \Delta\lambda) - hc\lambda = hc\Delta\lambda$ or $\lambda^2K + \lambda\Delta\lambda K - hc\Delta\lambda = 0$. This is a quadratic equation that with numerical values can be solved for λ to find

$$\lambda = 1.20 \times 10^{-11} \text{ m}. \text{ Then } E = \frac{hc}{\lambda} = \frac{1.24 \text{ keV} \cdot \text{nm}}{1.20 \times 10^{-2} \text{ nm}} = 104 \text{ keV}.$$

57. The power of the light received is intensity times area, or

$$P = IA = (4.0 \times 10^{-11} \text{ W/m}^2) \left[\pi (4.5 \times 10^{-3} \text{ m})^2 \right] = 2.5 \times 10^{-15} \text{ W} . \text{ The energy of each}$$

$$\text{photon is } E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(3.00 \times 10^8 \text{ m/s})}{5.50 \times 10^{-7} \text{ m}} = 3.6 \times 10^{-19} \text{ J} . \text{ Thus, the number}$$

$$\text{of photons per second is } 2.5 \times 10^{-15} \text{ J/s} \times \frac{1 \text{ photon}}{3.6 \times 10^{-19} \text{ J}} = 6900 \text{ photons} .$$

$$58. (a) \lambda_{\text{max}} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{1358 \text{ K}} = 2.13 \times 10^{-6} \text{ m}$$

(b) The peak is well into the infrared part of the spectrum. However, the shape of the “tail” of the distribution implies that more red photons are emitted than from any other part of the visible spectrum, and so we see red.

59. To find the asteroid mass m , note that Earth (matter) would supply an equal mass m to the

process, so $2mc^2 = \frac{GM_E^2}{2R_E}$. Solving for m we find:

$$m = \frac{GM_E^2}{4R_E c^2} = \frac{(6.67 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(5.98 \times 10^{24} \text{ kg})^2}{4(6378 \times 10^3 \text{ m})(3.00 \times 10^8 \text{ m/s})^2} = 1.04 \times 10^{15} \text{ kg} .$$

$$\text{Then } r = \left[\frac{3m}{4\pi\rho} \right]^{1/3} = \left[\frac{3(1.04 \times 10^{15} \text{ kg})}{4\pi(5000 \text{ kg/m}^3)} \right]^{1/3} = 3.68 \text{ km which is relatively small.}$$

Evaluating the energy:

$$E = \frac{GM_E^2}{2R_e} = \frac{(6.67 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(5.98 \times 10^{24} \text{ kg})^2}{2(6378 \times 10^3 \text{ m})} = 1.87 \times 10^{32} \text{ J} ;$$

$$\frac{E}{\text{nuclear arsenals}} = \frac{1.87 \times 10^{32} \text{ J}}{5000(4.2 \times 10^{15} \text{ J})} = 8.9 \times 10^{12} . \text{ There is a lot of energy in the annihilation process!}$$

60. For maximum recoil energy the scattering angle is $\theta = 180^\circ$ and $\phi = 0$. Then as usual

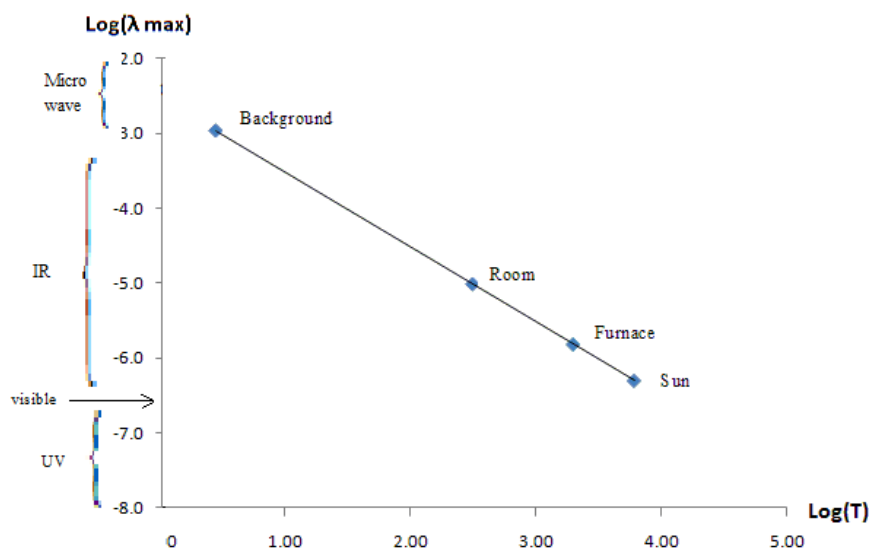
$$\Delta\lambda = \frac{2h}{mc} . \text{ Using the result of Problem 56,}$$

$$K = \frac{\Delta\lambda / \lambda}{1 + \Delta\lambda / \lambda} hf = \frac{2h / mc\lambda}{1 + 2h / mc\lambda} hf = \frac{2hf / mc^2}{1 + 2hf / mc^2} hf .$$

We can rearrange this equation: $K\left(1 + \frac{2hf}{mc^2}\right) = \frac{2(hf)^2}{mc^2}$ or

$\left(\frac{2}{mc^2}\right)(hf)^2 - \left(\frac{2K}{mc^2}\right)(hf) - K = 0$. This constitutes a quadratic equation in hf which can be solved with the given value of $K = 100 \text{ keV}$ numerically to yield $hf = 217 \text{ keV}$.

61. See graph below.



62. (a) For $\theta = 180^\circ$ we have $\lambda' - \lambda = 2h/mc$. Therefore $E' = \frac{hc}{\lambda'} = \frac{hc}{\lambda + \frac{2h}{mc}}$.

(b) With $\lambda = \frac{hc}{E}$ we find

$$E' = \frac{hc}{\frac{hc}{E} + \frac{2h}{mc}} = \frac{1}{\frac{1}{1 \times 10^5 \text{ eV}} + \frac{2}{511000 \text{ eV}}} = 71.9 \text{ keV}.$$

63. (a) The energy per second detection capability of the detector in the Hubble Space Telescope would be $(2.0 \times 10^{-20} \text{ W/m}^2) 0.30 \text{ m}^2 = 6.0 \times 10^{-21} \text{ J/s}$. Each photon at 486 nm

$$\text{would have energy } E = hf = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34} \text{ J}\cdot\text{s})(3.0 \times 10^8 \text{ m/s})}{486 \times 10^{-9} \text{ m}} = 4.09 \times 10^{-19} \text{ J}.$$

Therefore the average number of photons per second would be

$$n = \frac{6.0 \times 10^{-21} \text{ J/s}}{4.09 \times 10^{-19} \text{ J/photon}} = 0.015 \text{ photons/s}$$

and long exposure times would be required.

(b) If 30th magnitude corresponds to $2 \times 10^{-20} \text{ W/m}^2$ and each change in magnitude by one represents a change in brightness of $100^{1/5} = 2.5119$ then we must increase by a factor of $(100^{1/5})^{24} = (2.5119)^{24} = 3.98 \times 10^9$. So 6th magnitude corresponds to $7.96 \times 10^{-11} \text{ W/m}^2$.

If the diameter of the pupil is 6.5 mm, then the power reaching the retina is

$$P = 7.96 \times 10^{-11} \text{ W/m}^2 \left[\pi (3.25 \times 10^{-3} \text{ m})^2 \right] = 2.64 \times 10^{-15} \text{ J/s.}$$

With each photon carrying energy $4.09 \times 10^{-19} \text{ J}$, then $2.64 \times 10^{-15} \text{ J/s} = 4.09 \times 10^{-19} \text{ J} \left(n \frac{\text{photons}}{\text{s}} \right)$ and

$$n = 6.45 \times 10^3 \text{ photons each second.}$$

64. The laser's power is $25 \text{ kW} = 2.5 \times 10^4 \text{ J/s}$. The energy of a single photon is

$$E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(3.00 \times 10^8 \text{ m/s})}{1.06 \times 10^{-6} \text{ m}} = 1.88 \times 10^{-19} \text{ J. Thus,}$$

$$2.5 \times 10^4 \text{ J/s} \frac{1 \text{ photon}}{1.88 \times 10^{-19} \text{ J}} = 1.3 \times 10^{23} \text{ photons/s.}$$

65. The laser's power is $25 \text{ kW} = 2.5 \times 10^4 \text{ J/s}$. The energy of a single photon is

$$E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(3.00 \times 10^8 \text{ m/s})}{1.06 \times 10^{-6} \text{ m}} = 1.88 \times 10^{-19} \text{ J. Thus,}$$

$$2.5 \times 10^4 \text{ J/s} \frac{1 \text{ photon}}{1.88 \times 10^{-19} \text{ J}} = 1.3 \times 10^{23} \text{ photons/s}$$

66. From Example 3.5, the sun uses $3.91 \times 10^{26} \text{ J/s}$. At that rate, the sun's lifetime is

$$2.0 \times 10^{30} \text{ kg} \frac{5 \times 10^6 \text{ J/kg}}{3.91 \times 10^{26} \text{ J/s}} = 2.6 \times 10^{10} \text{ s or about 820 years! This estimate is off, because the sun uses nuclear energy, not chemical.}$$

$$67. (a) \lambda_{\text{max}} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{9600 \text{ K}} = 3.01 \times 10^{-7} \text{ m} = 301 \text{ nm}$$

This is actually in the ultraviolet part of the spectrum, but the shape of the distribution function guarantees that most of the visible photons will be in the violet-blue region, which explains the star's blue appearance.

(b) Assuming a perfect blackbody,

$$P = \sigma T^4 A = (5.67 \times 10^{-8} \text{ W/(m}^2 \cdot \text{K}^4))(9600 \text{ K})^4 \left[4\pi (1.6 \times 6.96 \times 10^8 \text{ m})^2 \right] \\ = 7.51 \times 10^{27} \text{ W}$$

This is about 19 times the value for the sun, reported in Example 3.5.

68. Kinetic energy is related to momentum (nonrelativistically) by $K = p^2 / 2m$, so an electron of mass m and kinetic energy K has momentum $p = \sqrt{2mK}$. Thus the magnitude of the electron's change in momentum is $\Delta p = \sqrt{2mK_f} - \sqrt{2mK_i} = 7.5 \times 10^{-24}$ (SI units). By conservation, this is the momentum of the recoiling nucleus. The nucleus then has kinetic energy $K = \frac{(\Delta p)^2}{2m} = 9.2 \times 10^{-23}$ J = 5.8×10^{-4} eV. This is negligible, compared with the energy (5 keV) of the x ray produced in the process.

69. For the 10-keV gamma ray, the electron absorbs half this amount, so its kinetic energy is 5 keV. That is less than 1 percent of its rest energy, so we'll use a non-relativistic

calculation $K = \frac{1}{2}mv^2$ so

$$v = \sqrt{2K/m} = \sqrt{2(5000 \text{ eV})(1.60 \times 10^{-19} \text{ J/eV})/9.11 \times 10^{-31} \text{ kg}} = 4.2 \times 10^7 \text{ m/s}.$$

Half of the 300-MeV gamma ray's energy is 150 MeV, which is highly relativistic. In

this case $K = (\gamma - 1)E_0$ so $\gamma - 1 = \frac{K}{E_0} = \frac{150 \text{ MeV}}{0.511 \text{ MeV}} = 294$ and $\gamma = 295$. Solving for

speed v , $\gamma = 295 = \frac{1}{\sqrt{1 - \beta^2}}$, which yields $\beta = 0.999994$. Thus $v = 0.999994 c$.

70. If n gamma rays strike the target, the energy deposited is $Q = nE_0$, where $E_0 = 100$ MeV is the energy of each gamma ray. Converting to SI units,

$$E_0 = 100 \text{ MeV} \times \frac{1.60 \times 10^{-13} \text{ J}}{1 \text{ MeV}} = 1.60 \times 10^{-11} \text{ J}. \text{ The relationship between thermal energy}$$

and temperature is $Q = mc_V \Delta T$, where c_V is the specific heat. Solving for n :

$$Q = nE_0 = mc_V \Delta T; \quad n = \frac{mc_V \Delta T}{E_0} = \frac{(2.5 \text{ kg})(430 \text{ J/(kg} \cdot \text{K)})(0.010 \text{ K})}{1.60 \times 10^{-11} \text{ J}} = 6.7 \times 10^{11}$$

Chapter 4

1. With more than one electron we are almost forced into some kind of Bohr-like orbits. This was the dilemma faced by physicists in the early 20th Century.

2. Non-relativistically $K = \frac{1}{2}mv^2$ and $v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2(7.7 \text{ MeV})}{3727 \text{ MeV}/c^2}} = 6.4281 \times 10^{-2} c$.

Relativistically $K = (\gamma - 1)mc^2$ so $\gamma = 1 + K / mc^2$ and $\gamma = 1 + \frac{7.7 \text{ MeV}}{3727 \text{ MeV}} = 1.002066$

which gives $\beta = \sqrt{1 - \frac{1}{\gamma^2}} = 6.4181 \times 10^{-2} c$. The difference is about $10^{-4} c$ or about 0.16% of the velocity.

3. Conserving momentum and energy:

$$M_\alpha v_\alpha = M_\alpha v'_\alpha + m_e v'_e \quad (1)$$

$$M_\alpha v_\alpha^2 = M_\alpha v'^2_\alpha + m_e v'^2_e \quad (2)$$

From (1) we see $v'_\alpha = v_\alpha - \frac{m_e}{M_\alpha} v'_e$ which, when substituted into (2) gives

$$M_\alpha v_\alpha^2 = M_\alpha \left[v_\alpha - \frac{m_e}{M_\alpha} v'_e \right]^2 + m_e v'^2_e.$$

This can be solved to find $v'_e \left[1 + \frac{m_e}{M_\alpha} \right] = 2v_\alpha$.

But with $m_e \ll M_\alpha$ we have $v'_e \approx 2v_\alpha$.

4. $P(\theta) = \exp\left(-\frac{80^2}{1^2}\right) = 3 \times 10^{-2780}$

Therefore multiple scattering does not provide an adequate explanation.

5. (a) With $Z_1 = 2$, $Z_2 = 79$, and $\theta = 1^\circ$ we have

$$b = \frac{Z_1 Z_2 e^2}{8\pi\epsilon_0 K} \cot\left(\frac{\theta}{2}\right) = \frac{(2)(79)(1.44 \times 10^{-9} \text{ eV} \cdot \text{m})}{2(7.7 \times 10^6 \text{ eV})} \cot(0.5^\circ) = 1.69 \times 10^{-12} \text{ m}$$

$$(b) \text{ For } \theta = 90^\circ \quad b = \frac{Z_1 Z_2 e^2}{8\pi\epsilon_0 K} \cot\left(\frac{\theta}{2}\right) = \frac{(2)(79)(1.44 \times 10^{-9} \text{ eV} \cdot \text{m})}{2(7.7 \times 10^6 \text{ eV})} \cot(45^\circ) = 1.48 \times 10^{-14} \text{ m}$$

6. $f = \pi n t \left(\frac{Z_1 Z_2 e^2}{8\pi\epsilon_0 K} \right)^2 \cot^2\left(\frac{\theta}{2}\right)$. For the two different angles everything is the same except

the angles, so the ratio is $\frac{f(1^\circ)}{f(2^\circ)} = \frac{\cot^2(0.5^\circ)}{\cot^2(1.0^\circ)} = 4.00$.

7. The fraction f is proportional to nt and to Z^2 from Equation (4.12). However, the question states that the number of scattering nuclei per unit area (equal to nt) is the same

for either target. Therefore $\frac{N(\text{Au})}{N(\text{Al})} = \frac{n(\text{Au})t(79)^2}{n(\text{Al})t(13)^2} = \frac{79^2}{13^2} = 36.93$.

8. From Example 4.2 we know $n = 5.90 \times 10^{28} \text{ m}^{-3}$. Thus

$$\begin{aligned} f &= \pi n t \left(\frac{Z_1 Z_2 e^2}{8\pi\epsilon_0 K} \right)^2 \cot^2 \left(\frac{\theta}{2} \right) \\ &= \pi (5.90 \times 10^{28} \text{ m}^{-3}) (10^{-8} \text{ m}) \left(\frac{(2)(79)(1.44 \times 10^{-9} \text{ eV} \cdot \text{m})}{2(5 \times 10^6 \text{ eV})} \right)^2 \cot^2(4^\circ) \\ &= 1.96 \times 10^{-4} \end{aligned}$$

9. (a) With all other parameters equal the number depends only on the scattering angles, so

$$\frac{f(90^\circ)}{f(50^\circ)} = \frac{\cot^2(45^\circ)}{\cot^2(25^\circ)} = 0.217 \text{ so the number scattered through angles greater than } 90^\circ \text{ is}$$

$$(10000)(0.217) = 2170.$$

(b) Similarly: $\frac{f(70^\circ)}{f(50^\circ)} = \frac{\cot^2(35^\circ)}{\cot^2(25^\circ)} = 0.4435$ and $\frac{f(80^\circ)}{f(50^\circ)} = \frac{\cot^2(40^\circ)}{\cot^2(25^\circ)} = 0.3088$. The numbers for the two angles are thus 4435 and 3088 and the number scattered between 70° and 80° is $4435 - 3088 = 1347$.

10. From the Rutherford scattering result, the number detected through a small angle is

inversely proportional to $\sin^4 \left(\frac{\theta}{2} \right)$. Thus $\frac{n(50^\circ)}{n(6^\circ)} = \frac{\sin^4(3^\circ)}{\sin^4(25^\circ)} = 2.35 \times 10^{-4}$ and if they

count 2000 at 6° the number counted at 50° will be $(2000)(2.35 \times 10^{-4}) = 0.47$ which is insufficient.

11. In each case all the kinetic energy is changed to potential energy:

$K = -\Delta V = |V| = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 (r_1 + r_2)}$ where r_1 and r_2 are the radii of the particular particles (as the problem indicates the particles just touch).

(a) $Z_1 = 2$ and $Z_2 = 13$ for Al, $Z_2 = 79$ for Au;

$$\text{Al: } K = \frac{(2)(13)(1.44 \times 10^{-9} \text{ eV} \cdot \text{m})}{2.6 \times 10^{-15} \text{ m} + 3.6 \times 10^{-15} \text{ m}} = 6.04 \text{ MeV}$$

$$\text{Au: } K = \frac{(2)(79)(1.44 \times 10^{-9} \text{ eV} \cdot \text{m})}{2.6 \times 10^{-15} \text{ m} + 7.0 \times 10^{-15} \text{ m}} = 23.7 \text{ MeV}$$

(b) Now $Z_1 = 1$ and for the two different values of Z_2 :

$$\text{Al: } K = \frac{(1)(13)(1.44 \times 10^{-9} \text{ eV} \cdot \text{m})}{1.3 \times 10^{-15} \text{ m} + 3.6 \times 10^{-15} \text{ m}} = 3.82 \text{ MeV}$$

$$\text{Au: } K = \frac{(1)(79)(1.44 \times 10^{-9} \text{ eV} \cdot \text{m})}{1.3 \times 10^{-15} \text{ m} + 7.0 \times 10^{-15} \text{ m}} = 13.7 \text{ MeV}$$

12. (a) The maximum Coulomb force is at the surface and equal to $2Z_2e^2 / 4\pi\epsilon_0 R^2$. Then

$$\Delta p = F \Delta t = \frac{2Z_2e^2}{4\pi\epsilon_0 R^2} \frac{2R}{v} = \frac{4Z_2e^2}{4\pi\epsilon_0 Rv}. \text{ For maximum deflection}$$

$$\theta \approx \tan \theta = \frac{\Delta p}{p} = \frac{1}{mv} \frac{4Z_2e^2}{4\pi\epsilon_0 Rv} = \frac{2Z_2e^2}{4\pi\epsilon_0 KR}, \text{ for small angles.}$$

$$(b) \theta = \frac{(2)(79)(1.44 \text{ eV} \cdot \text{nm})}{(8 \text{ MeV})(0.135 \text{ nm})} = 2.11 \times 10^{-4} \text{ rad} = 0.012^\circ$$

$$13. (a) \theta = \frac{(2)(79)(1.44 \text{ eV} \cdot \text{nm})}{(10 \text{ MeV})(0.1 \text{ nm})} = 2.28 \times 10^{-4} \text{ rad} = 0.013^\circ$$

(b) These results are comparable in magnitude with those obtained by electron scattering (Example 4.1).

$$14. (a) v = \frac{e}{\sqrt{4\pi\epsilon_0 mr}} = \frac{ec}{\sqrt{4\pi\epsilon_0 mc^2 r}} = \frac{\sqrt{1.44 \text{ eV} \cdot \text{nm}}}{\sqrt{(511000 \text{ eV})(1.2 \times 10^{-6} \text{ nm})}} c = 1.53c$$

which is not an allowed speed.

$$(b) E = -\frac{e^2}{8\pi\epsilon_0 r} = -\frac{1.44 \text{ eV} \cdot \text{nm}}{2(1.2 \times 10^{-6} \text{ nm})} = -600 \text{ keV}$$

(c) Clearly (a) is not allowed and (b) is too much energy.

$$15. (a) v = \frac{e}{\sqrt{4\pi\epsilon_0 mr}} = \frac{ec}{\sqrt{4\pi\epsilon_0 mc^2 r}} = \frac{\sqrt{1.44 \text{ eV} \cdot \text{nm}}}{\sqrt{(938 \times 10^6 \text{ eV})(0.05 \text{ nm})}} c = 1.75 \times 10^{-4} c$$

or $v = 5.25 \times 10^4 \text{ m/s}$.

$$(b) E = -\frac{e^2}{8\pi\epsilon_0 r} = -\frac{1.44 \text{ eV} \cdot \text{nm}}{2(0.05 \text{ nm})} = -14.4 \text{ eV}$$

(c) The “nucleus” is too light to be fixed, and there is no way to reconcile this model with the results of Rutherford scattering.

16. For hydrogen:

$$v = \frac{e}{\sqrt{4\pi\epsilon_0 m r}} = \frac{ec}{\sqrt{4\pi\epsilon_0 m c^2 r}} = \frac{\sqrt{1.44 \text{ eV} \cdot \text{nm}}}{\sqrt{(511 \times 10^3 \text{ eV})(0.0529 \text{ nm})}} c = 7.30 \times 10^{-3} c$$

$$= 2.19 \times 10^6 \text{ m/s}$$

$$a = \frac{v^2}{r} = \frac{(2.19 \times 10^6 \text{ m/s})^2}{5.29 \times 10^{-11} \text{ m}} = 9.07 \times 10^{22} \text{ m/s}^2$$

For the hydrogen-like Li^{++} we know: $F = \frac{Ze^2}{4\pi\epsilon_0 r^2} = \frac{mv^2}{r}$ or $v^2 = \frac{Ze^2}{4\pi\epsilon_0 r m}$. We also

$$\text{know } r = \frac{4\pi\epsilon_0 \hbar^2}{Zme^2} = \frac{a_0}{Z}.$$

$$v^2 = \frac{Z^2 e^2}{4\pi\epsilon_0 a_0 m} = \frac{3^2 (1.44 \text{ eV} \cdot \text{nm})}{(5.29 \times 10^{-11} \text{ m})(511 \times 10^3 \text{ eV}/c^2)} = 4.79 \times 10^{-4} c^2 \text{ or}$$

$v = 2.19 \times 10^{-2} c = 6.57 \times 10^6 \text{ m/s}$. This is a factor of 3 greater than the speed for

hydrogen. The acceleration is: $a = \frac{v^2}{r} = \frac{(6.57 \times 10^6 \text{ m/s})^2}{(5.29 \times 10^{-11} \text{ m})/3} = 2.45 \times 10^{24} \text{ m/s}^2$

17. Both forces are attractive. The magnitude of the gravitational force:

$$F_g = \frac{Gm_1 m_2}{r^2} = \frac{(6.67 \times 10^{-11} \frac{\text{N} \cdot \text{m}^2}{\text{kg}^2})(9.11 \times 10^{-31} \text{ kg})(1.67 \times 10^{-27} \text{ kg})}{(5.3 \times 10^{-11} \text{ m})^2} = 3.6 \times 10^{-47} \text{ N}$$

The magnitude of the electrical force:

$$F_e = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} = \frac{e^2}{4\pi\epsilon_0} \frac{1}{r^2} = \frac{(2.31 \times 10^{-28} \text{ J} \cdot \text{m})}{(5.3 \times 10^{-11} \text{ m})^2} = 8.2 \times 10^{-8} \text{ N}$$

The electrical force is much larger. The ratio is $F_e / F_g = \frac{8.2 \times 10^{-8} \text{ N}}{3.6 \times 10^{-47} \text{ N}} = 2.28 \times 10^{39}$

18. The total energy of the atom is $-e^2 / (8\pi\epsilon_0 r)$. Differentiating with respect to time:

$$\frac{dE}{dt} = \frac{e^2}{8\pi\epsilon_0 r^2} \frac{dr}{dt}. \text{ Equating this result with the given equation from electromagnetic}$$

theory gives: $\frac{e^2}{8\pi\epsilon_0 r^2} \frac{dr}{dt} = -\frac{1}{4\pi\epsilon_0} \frac{2e^2}{3c^3} \left(\frac{d^2 r}{dt^2} \right)^2$. This simplifies to $\frac{e^2}{2r^2} \frac{dr}{dt} = -\frac{2e^2}{3c^3} \left(\frac{d^2 r}{dt^2} \right)^2$.

In a circular orbit $\frac{d^2 r}{dt^2}$ is just the centripetal acceleration, which is also given by

$$a = \frac{F}{m} = \frac{e^2}{4\pi\epsilon_0 m r^2}.$$

Substituting: $\frac{e^2}{2r^2} \frac{dr}{dt} = -\frac{2e^2}{3c^3} \left(\frac{e^2}{4\pi\epsilon_0 mr^2} \right)^2$ or $\frac{dr}{dt} = -\frac{4e^4}{(4\pi\epsilon_0)^2 3m^2 c^3 r^2}$.

Solving by separation of variables: $dt = -\left(4\pi\epsilon_0\right)^2 \frac{3m^2 c^3}{4e^4} r^2 dr$ which can be integrated:

$$t = -\left(4\pi\epsilon_0\right)^2 \frac{3m^2 c^3}{4e^4} \int_{a_0}^0 r^2 dr = \left(4\pi\epsilon_0\right)^2 \frac{m^2 c^3}{4e^4} a_0^3. \text{ Inserting numerical values we find}$$

$$t = 1.55 \times 10^{-11} \text{ s.}$$

19. $f = \frac{c}{\lambda} = Z^2 R_c \left(\frac{1}{n_l^2} - \frac{1}{n_u^2} \right)$; $f(K_\alpha) = Z^2 R_c \left(1 - \frac{1}{4} \right)$ and $f(K_\beta) = Z^2 R_c \left(1 - \frac{1}{9} \right)$ and

$$f(L_\alpha) = Z^2 R_c \left(\frac{1}{4} - \frac{1}{9} \right).$$

With $1 - 1/4 + (1/4 - 1/9) = 1 - 1/9$ we see

$$f(K_\alpha) + f(L_\alpha) = f(K_\beta).$$

20. (a) As in Problem 16, $v = 2.19 \times 10^6 \text{ m/s}$ and

$$L = mvr = (9.11 \times 10^{-31} \text{ kg})(2.19 \times 10^6 \text{ m/s})(5.29 \times 10^{-11} \text{ m}) = 1.0554 \times 10^{-34} \text{ kg} \cdot \text{m}^2/\text{s}$$

Notice that $L = \hbar$.

(b) Using Equation (4.31) for v and Equation (4.24) for r_n we find

$$L = mv_2 r_2 = m \left(\frac{1}{n} \frac{\hbar}{ma_0} \right) (n^2 a_0) = n\hbar = 2\hbar \text{ as expected.}$$

21. $hc = (4.135669 \times 10^{-15} \text{ eV} \cdot \text{s})(299792458 \text{ m/s}) = 1239.8 \text{ eV} \cdot \text{nm}$;

$$\frac{e^2}{4\pi\epsilon_0} = \frac{(1.6021733 \times 10^{-19} \text{ C})^2}{4\pi(8.8541878 \times 10^{-12} \text{ F/m})} \left(\frac{1 \text{ eV}}{1.6021733 \times 10^{-19} \text{ N} \cdot \text{m}} \right) \frac{10^9 \text{ nm}}{\text{m}} = 1.4400 \text{ eV} \cdot \text{nm}$$

$$mc^2 = (510.99906 \text{ keV}/c^2)c^2 = 511.00 \text{ keV};$$

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{me^2} = \frac{4\pi(8.8541878 \times 10^{-12} \text{ F/m})(1.05457 \times 10^{-34} \text{ J} \cdot \text{s})^2}{(9.1093897 \times 10^{-31} \text{ kg})(1.6021733 \times 10^{-19} \text{ C})^2}$$

$$= 5.2918 \times 10^{-11} \text{ m} = 5.2918 \times 10^{-2} \text{ nm}$$

$$E_0 = \frac{e^2}{8\pi\epsilon_0 a_0} = -\frac{(1.6021733 \times 10^{-19} \text{ C})^2}{8\pi(8.8541878 \times 10^{-12} \text{ F/m})(5.2917725 \times 10^{-11} \text{ m})}$$

$$= 2.179874 \times 10^{-18} \text{ J} = 13.606 \text{ eV}$$

22. For a hydrogen-like atom $E = \frac{-Z^2 E_0}{n^2} = -Z^2 E_0$ for $n = 1$.

$$\text{H: } E = -E_0 = -13.6 \text{ eV}$$

$$\text{He}^+: E = -4E_0 = -54.4 \text{ eV}$$

$$\text{Li}^{++}: E = -9E_0 = -122.5 \text{ eV}$$

The binding energy is larger for atoms with larger Z values due to the greater attractive force between the nucleus and electron.

23. The photon energy is $E = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{410 \text{ nm}} = 3.02 \text{ eV}$. This is the energy difference

between the two states in hydrogen. From Figure 4.16, we see $E_3 = -1.51 \text{ eV}$ so the initial state must be $n = 2$ in order to have a large enough difference. This energy difference exists between $n = 2$ (with $E_2 = -3.40 \text{ eV}$) and $n = 6$ (with $E_6 = -0.38 \text{ eV}$).

24. The photon energy is $E = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{95 \text{ nm}} = 13.05 \text{ eV}$. This can only be a transition to

$n = 1$ ($E_1 = -13.6 \text{ eV}$) and because of the energy difference it comes from $n = 5$ with $E_5 = -0.54 \text{ eV}$.

25. In general the ground state energy is $Z^2 E_0$.

(a) $E = 1^2 E_0 = 13.6 \text{ eV}$. Using the reduced mass does not change this result to three significant digits.

(b) $E = 2^2 E_0 = 54.4 \text{ eV}$

(c) $E = 4^2 E_0 = 218 \text{ eV}$

26. Following the strategy of Example 4.8, we use Equation (4.37) to determine appropriate Rydberg constants for atomic hydrogen, deuterium, and tritium. The Balmer series has n lower equal to 2. The α refers to $n_u = 3$; β refers to $n_u = 4$, etc. Then we use Equation

(4.30) to calculate the “isotope shifted” wavelengths. $\frac{1}{\lambda} = R_\infty \left(\frac{1}{n_\ell^2} - \frac{1}{n_u^2} \right)$

The example with n upper equal to 3 is completed in Example 4.8; the results are repeated here:

$$\lambda(\text{H}_\alpha, \text{hydrogen}) = 656.47 \text{ nm}; \lambda(\text{H}_\alpha, \text{deuterium}) = 656.29 \text{ nm};$$

$$\lambda(\text{H}_\alpha, \text{tritium}) = 656.23 \text{ nm}$$

$$\lambda(\text{H}_\beta, \text{hydrogen}) = 486.27 \text{ nm}; \lambda(\text{H}_\beta, \text{deuterium}) = 486.14 \text{ nm};$$

$$\lambda(\text{H}_\beta, \text{tritium}) = 486.10 \text{ nm}$$

$$\lambda(\text{H}_\gamma, \text{hydrogen}) = 434.17 \text{ nm}; \lambda(\text{H}_\gamma, \text{deuterium}) = 434.05 \text{ nm}$$

$$\lambda(\text{H}_\gamma, \text{tritium}) = 434.02 \text{ nm}$$

$$\lambda(\text{H}_\delta, \text{hydrogen}) = 410.29 \text{ nm}; \lambda(\text{H}_\delta, \text{hydrogen}) = 410.18 \text{ nm}$$

$$\lambda(\text{H}_\delta, \text{hydrogen}) = 410.15 \text{ nm}$$

27. We know from Equations (4.25) and (4.29) that $E_1 = -hcR_\infty$. We must adjust the Rydberg constant to account for the finite mass of the nucleus. Use the calculations of example 4.8 that provide the constants for deuterium $R_D = 0.99973R_\infty$ and tritium

$R_T = 0.99982R_\infty$. Then we see

$$E_{1,D} = hc(0.99973)R_\infty = 1239.8 \text{ eV} \cdot \text{nm}(0.99973)(1.097373 \times 10^{-2} \text{ nm}^{-1}) = 13.602 \text{ eV}$$

$$E_{1,T} = hc(0.99982)R_\infty = 1239.8 \text{ eV} \cdot \text{nm}(0.99982)(1.097373 \times 10^{-2} \text{ nm}^{-1}) = 13.603 \text{ eV}$$

In Problem 21 we found that $E_{1,H} = 13.606 \text{ eV}$.

28. (a) It is only the first four lines of the Balmer series, with wavelengths 656.5 nm, 486.3 nm, 434.2 nm, and 410.2 nm.

(b) To determine the energy levels in ionized helium, perform the same analysis as in the text but with e^2 replaced by $Ze^2 = 2e^2$. This results in an extra factor of $Z^2 = 4$ in the energy, so the revised Rydberg-Ritz equation is

$$\frac{1}{\lambda} = \frac{4E_0}{hc} \left(\frac{1}{n_l^2} - \frac{1}{n_u^2} \right) = (4.377 \times 10^7 \text{ m}^{-1}) \left(\frac{1}{n_l^2} - \frac{1}{n_u^2} \right). \text{ We need the wavelength to be}$$

between 400 and 700 nm. The combinations of n_l and n_u that work are tabulated below:

n_l	n_u	λ (nm)	comments
3	4	470	
4	6	658	
4	7	543	
4	8	487	etc. with $n_l = 4$ to $n_u = 13$
4	13	404	but $n_l = 4$ and $n_u > 13$ gives $\lambda < 400 \text{ nm}$
5	12	691	
5	13	670	etc. with $n_l = 5$ all the way to...
5	∞	571	a series limit

29. From Equation (4.31), $v_n = (1/n)(\hbar / ma_0)$ which gives the speed in the $n = 3$ state as $v = 7.30 \times 10^5 \text{ m/s}$. The radius of the orbit is $n^2 a_0 = 9a_0$. Then from kinematics:

$$\text{number of revolutions} = \frac{vt}{2\pi r} = \frac{(7.30 \times 10^5 \text{ m/s})(10^{-8} \text{ s})}{2\pi(9)(5.29 \times 10^{-11} \text{ m})} = 2.44 \times 10^6.$$

30. The energy of each photon is $hc/\lambda = \left(\frac{1239.8 \text{ eV} \cdot \text{nm}}{397 \text{ nm}} \right) = 3.12 \text{ eV}$. Looking at the

energy difference between levels in hydrogen (with $E_n = -E_0/n^2$) we see that

$E_7 - E_2 = 3.12 \text{ eV}$, which matches the photon energy to three significant figures.

Therefore this laser can promote the atom to the $n = 7$ level.

31. We must use the reduced mass for the muon:

$$\mu = \frac{mM}{m+M} = \frac{(106 \text{ MeV}/c^2)(938 \text{ MeV}/c^2)}{106 \text{ MeV}/c^2 + 938 \text{ MeV}/c^2} = 95.2 \text{ MeV}/c^2$$

$$(a) a_0 = \frac{4\pi\epsilon_0\hbar^2}{\mu e^2} = \frac{(6.58 \times 10^{-16} \text{ eV} \cdot \text{s})^2}{(1.44 \times 10^{-9} \text{ eV} \cdot \text{m})(95.2 \times 10^6 \text{ eV}/c^2)} \frac{(3.00 \times 10^8 \text{ m/s})^2}{c^2} = 2.84 \times 10^{-13} \text{ m}$$

$$(b) E = \frac{e^2}{8\pi\epsilon_0 a_0} = \frac{(1.44 \times 10^{-9} \text{ eV} \cdot \text{m})}{2(2.84 \times 10^{-13} \text{ m})} = 2535 \text{ eV}$$

$$(c) \text{ First series: } \lambda = \frac{hc}{E} = \frac{1240 \text{ eV} \cdot \text{nm}}{2535 \text{ eV}} = 0.49 \text{ nm}$$

$$\text{Second series: } \lambda = \frac{4hc}{E} = \frac{4(1240 \text{ eV} \cdot \text{nm})}{2535 \text{ eV}} = 1.96 \text{ nm}$$

$$\text{Third series: } \lambda = \frac{9hc}{E} = \frac{9(1240 \text{ eV} \cdot \text{nm})}{2535 \text{ eV}} = 4.40 \text{ nm}$$

32. The reduced mass for this system is $\mu = \frac{mm}{m+m} = m/2$ where m is the mass of each

$$\text{particle. Then } r = \frac{4\pi\epsilon_0\hbar^2}{\mu e^2} = 2a_0 \text{ and } E = -\frac{e^2}{8\pi\epsilon_0 r} = -\frac{e^2}{8\pi\epsilon_0(2a_0)} = -\frac{E_0}{2} = -6.8 \text{ eV}.$$

33. (a) As shown in Problem 32, the radius of the orbit is $2a_0$.

(b) With $E_0 = 6.8 \text{ eV}$ (see Problem 32) we have

$$\Delta E = E_2 - E_1 = -\frac{E_0}{2^2} - \left(-\frac{E_0}{1^2}\right) = \frac{3E_0}{4} = 5.1 \text{ eV}. \text{ Then } \lambda = \frac{hc}{\Delta E} = \frac{1240 \text{ eV} \cdot \text{nm}}{5.1 \text{ eV}} = 243 \text{ nm}.$$

34. (a) $r_2 - r_1 = 4a_0 - a_0 = 3a_0 = 1.59 \times 10^{-10} \text{ m}$

(b) $r_6 - r_5 = 36a_0 - 25a_0 = 11a_0 = 5.83 \times 10^{-10} \text{ m}$

(c) $r_{11} - r_{10} = 121a_0 - 100a_0 = 21a_0 = 1.11 \times 10^{-9} \text{ m}$

(d) Successive radii (different by 1) would have a difference given by:

$$r_m - r_n = \left[(n+1)^2 - n^2\right]a_0 = \left[n^2 + 2n + 1 - n^2\right]a_0 = (2n+1)a_0. \text{ For large values of } n \text{ this is approximately } r_m - r_n = (2n+1)a_0 \approx 2na_0.$$

35. hydrogen: $\frac{1}{\lambda} = R_H \left(\frac{1}{4} - \frac{1}{n_u^2}\right)$; helium: $\frac{1}{\lambda} = Z^2 R_{\text{He}} \left(\frac{1}{4^2} - \frac{1}{n_u^2}\right) = R_{\text{He}} \left(\frac{1}{4} - \frac{4}{n_u^2}\right)$

We see that the lines match very well when n_u is even for helium but not when it is odd. For example, when $n_u = 6$, for the ionized helium, then the $n_u = 3$ state for hydrogen will be very similar. In the same fashion, the $n_u = 8$ level for the ionized helium will be very similar to the $n_u = 4$ level for hydrogen. Also, all the “matched” lines differ slightly

because of the different Rydberg constant (which is due to the different reduced masses). They differ by a factor of $R_{\text{He}} / R_{\text{H}} = 0.99986 / 0.99946 = 1.0004$.

36. As mentioned in the text below Equation (4.38), if each atom can be treated as single-electron atoms (and the problem states we can make this assumption), then

$$R = \frac{1}{1 + \frac{m}{M}} R_{\infty} \text{ where } R_{\infty} = 1.0973731534 \times 10^7 \text{ m}^{-1} \text{ and } m = 0.0005485799 \text{ u.}$$

$$^4\text{He} (M = 4.0026 \text{ u}), R = 0.999863 R_{\infty} = 1.097223 \times 10^7 \text{ m}^{-1} \text{ (off by 0.14\%).}$$

$$^{39}\text{K} (M = 38.963708 \text{ u}), R = 0.9999859 R_{\infty} = 1.097358 \times 10^7 \text{ m}^{-1} \text{ (off by 0.0014\%)}$$

$$^{238}\text{U} (M = 238.05078 \text{ u}), R = 0.9999977 R_{\infty} = 1.097371 \times 10^7 \text{ m}^{-1} \text{ (off by 0.00023\%)}$$

37. The derivation of the Rydberg equation follows as in the text. Because He^+ is hydrogen-like it works with $R = Z^2 R_{\text{He}}$ and $R_{\text{He}} = 1.097223 \times 10^7 \text{ m}^{-1}$ as in Problem 36. Then

$$R = 4(1.097223 \times 10^7 \text{ m}^{-1}) = 4.38889 \times 10^7 \text{ m}^{-1}.$$

38. For L_{α} we have $\lambda = \frac{c}{f} = \frac{36}{5R(Z - 7.4)^2}.$

$$Z = 43: \lambda = \frac{36}{5R(43 - 7.4)^2} = 0.52 \text{ nm};$$

$$Z = 61: \lambda = \frac{36}{5R(61 - 7.4)^2} = 0.23 \text{ nm};$$

$$Z = 75: \lambda = \frac{36}{5R(75 - 7.4)^2} = 0.14 \text{ nm}$$

39. For Pb $Z = 82$ and for Bi $Z = 83$. For the K_{α} lines $\lambda = \frac{4}{3R(Z - 1)^2}.$

$$\text{Pb: } \lambda = \frac{4}{3R(81)^2} = 18.52 \text{ pm}; \quad \text{Bi: } \lambda = \frac{4}{3R(82)^2} = 18.07 \text{ pm}$$

Therefore $\Delta\lambda = 0.45 \text{ pm}$, which is less than the specified resolution, so it will not work.

40. $f(K_{\beta}) = \frac{c}{\lambda(K_{\beta})} = cR(Z - 1)^2 \left(\frac{1}{1^2} - \frac{1}{3^2} \right) = \frac{8cR}{9} (Z - 1)^2$ This is higher than the K_{α}

frequency by a factor of $(8/9)(3/4) = 32/27$, which seems to be in agreement with Figure 4.19.

41. Helium: $\lambda(K_\alpha) = \frac{4}{3R(Z-1)^2} = 122 \text{ nm}$; $\lambda(K_\beta) = \frac{9}{8R(Z-1)^2} = 103 \text{ nm}$

Lithium: $\lambda(K_\alpha) = \frac{4}{3R(Z-1)^2} = 30.4 \text{ nm}$; $\lambda(K_\beta) = \frac{9}{8R(Z-1)^2} = 25.6 \text{ nm}$

42. Refer to Figure 4.19 and Equation (4.41). λ is inversely proportional to $(Z-1)^2$ for the K series and to $(Z-7.4)^2$ for the L series.

(a) $\frac{\lambda(\text{U})}{\lambda(\text{C})} = \frac{(6-1)^2}{(92-1)^2} = 0.0030$

(b) $\frac{\lambda(\text{Pt})}{\lambda(\text{Ca})} = \frac{(20-7.4)^2}{(78-7.4)^2} = 0.032$

43. The longest wavelengths are produced for an electron vacancy in the K-shell. We begin with Equation (4.43) and notice that the longest wavelengths will occur when the expression on the right of the equal sign is as small as possible; thus we choose n equal to 2, 3, and 4. The series limit occurs for $n = \infty$. Molybdenum (Mo) has $Z = 42$. Then

$$\frac{1}{\lambda_{K_\alpha}} = R(Z-1)^2 \left(1 - \frac{1}{n^2}\right) = (1.09737 \times 10^7 \text{ m}^{-1})(41)^2 \left(1 - \frac{1}{2^2}\right) = 72.28 \text{ pm}^{-1}$$

In a similar fashion, we find $\lambda(K_\beta) = 60.99 \text{ pm}$; $\lambda(K_\gamma) = 57.82 \text{ pm}$ and the K series limit is $\lambda = 54.21 \text{ pm}$.

44. The longest wavelengths occur for an electron vacancy in K shell. The two longest wavelengths correspond to the K_α and the K_β . Use Equation (4.41) and rearrange it to solve for $(Z-1)^2$. Then

$$(Z-1)^2 = \frac{4}{3} \frac{1}{R\lambda_{K_\alpha}} = \frac{4}{3(1.09737 \times 10^7 \text{ m}^{-1})(0.155 \times 10^{-9} \text{ m})} = 783.89$$

$(Z-1) = 28$ or $Z = 29$. Using the expression for λ_{K_β} from problem 40, we have

$$(Z-1)^2 = \frac{9}{8} \frac{1}{R\lambda_{K_\beta}}$$

Using the second wavelength given, 0.131 nm, we find

$$(Z-1)^2 = \frac{9}{8} \frac{1}{R\lambda_{K_\beta}} = \frac{9}{8(1.09737 \times 10^7 \text{ m}^{-1})(0.131 \times 10^{-9} \text{ m})} = 782.58$$

$(Z-1) = 27.97$ or $Z = 29$. Therefore we can conclude the target must be copper.

45. Non-relativistically $45 \text{ eV} = \frac{1}{2}mv_1^2$ so $v_1 = \sqrt{\frac{2(45 \text{ eV})}{511 \text{ keV}/c^2}} = 0.0133c = 3.98 \times 10^6 \text{ m/s}$.

From elementary physics

$v_2' = \frac{2m_1}{m_1 + m_2} v_1 = \frac{2(0.0005486 \text{ u})}{0.0005486 \text{ u} + 200.59 \text{ u}} (3.98 \times 10^6 \text{ m/s}) = 21.8 \text{ m/s}$ where we have used the molar mass of mercury. Then

$$K_2' = \frac{1}{2} m_2 v_2'^2 = \frac{1}{2} (200.59 \text{ u}) (931.49 \text{ MeV/u} \cdot c^2) \left(\frac{c^2}{9 \times 10^{16} \text{ m}^2/\text{s}^2} \right) (21.7 \text{ m/s})^2$$

$$= 4.89 \times 10^{-4} \text{ eV}$$

46. Without the negative potential an electron with any energy, no matter how small, could drift into the collector plate. As a result the electron could give up its kinetic energy to a Hg atom and still contribute to the plate current. The Franck-Hertz curve would not show the distinguishing periodic drops, but rather would rise monotonically.

47. Using $\Delta E = hc / \lambda$ we find $h = \frac{\lambda \Delta E}{c} = \frac{(254 \text{ nm})(4.88 \text{ eV})}{3.00 \times 10^8 \text{ m/s}} = 4.13 \times 10^{-15} \text{ eV} \cdot \text{s}.$

48. 3.6 eV, 4.6 eV, 2(3.6 eV) = 7.2 eV, 3.6 eV + 4.6 eV = 8.2 eV, etc. with other combinations giving 10.8 eV, 11.8 eV, 12.6 eV, 14.4 eV, 15.4 eV, 16.2 eV, 16.4 eV, 17.2 eV, 18.0 eV.

49. Magnesium: $\lambda(K_\alpha) = \frac{4}{3R(Z-1)^2} = 1.00 \text{ nm}; \lambda(L_\alpha) = \frac{36}{5R(Z-7.4)^2} = 31.0 \text{ nm}$

Iron: $\lambda(K_\alpha) = \frac{4}{3R(Z-1)^2} = 0.194 \text{ nm}; \lambda(L_\alpha) = \frac{36}{5R(Z-7.4)^2} = 1.90 \text{ nm}$

50. K_α is a transition from $n = 2$ to $n = 1$ and K_β is from $n = 3$ to $n = 1$. We know those wavelengths in the Lyman series are 121.6 nm and 102.6 nm, respectively. The redshift factor (λ / λ_0) is (with $\beta = 1/6$) $\sqrt{\frac{1+\beta}{1-\beta}} = \sqrt{\frac{1+1/6}{1-1/6}} = 1.183$. Then the redshifted wavelengths are higher by 18.3% in each case. The observed wavelengths are:
 $K_\alpha: \lambda = (1.183)(121.6 \text{ nm}) = 143.9 \text{ nm}$ and $K_\beta: \lambda = (1.183)(102.6 \text{ nm}) = 121.4 \text{ nm}.$

51. $f = \pi n t \left(\frac{Z_1 Z_2 e^2}{8\pi \epsilon_0 K} \right)^2 \cot^2 \left(\frac{\theta}{2} \right)$

(a) $f(1^\circ) = \pi (5.90 \times 10^{28} \text{ m}^{-3}) (0.32 \times 10^{-6} \text{ m}) \left(\frac{(2)(79)(1.44 \times 10^{-9} \text{ eV} \cdot \text{m})}{2(8 \times 10^6 \text{ eV})} \right)^2 \cot^2(0.5^\circ)$

$$= 0.157$$

$f(2^\circ) = \pi (5.90 \times 10^{28} \text{ m}^{-3}) (0.32 \times 10^{-6} \text{ m}) \left(\frac{(2)(79)(1.44 \times 10^{-9} \text{ eV} \cdot \text{m})}{2(8 \times 10^6 \text{ eV})} \right)^2 \cot^2(1^\circ)$

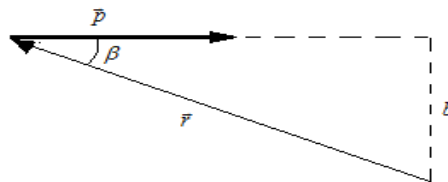
$$= 0.0394$$

The fraction scattered between 1° and 2° is $0.157 - 0.0394 = 0.118$.

$$(b) \frac{f(1^\circ)}{f(10^\circ)} = \frac{\cot^2(0.5^\circ)}{\cot^2(5^\circ)} = 100.5; \quad \frac{f(1^\circ)}{f(90^\circ)} = \frac{\cot^2(0.5^\circ)}{\cot^2(45^\circ)} = 1.31 \times 10^4$$

52. From classical mechanics we know that L is conserved for central forces.

$$L = |L| = |r \times p| = rp_0 \sin \beta. \text{ But } r \sin \beta = b \text{ so } L = p_0 b = mv_0 b.$$



53. If the positions of the electron and proton are respectively (along a line) r_e and r_M then

$$\text{putting the center of mass at } R = 0 \text{ we have } R = 0 = \frac{mr_e + Mr_M}{m + M}$$

$$\text{or } r_M = -\left(\frac{m}{M} r_e\right). \quad r = r_e - r_M = r_e - \left(-\frac{m}{M} r_e\right) = r_e \left(1 + \frac{m}{M}\right) \text{ which gives } r_e = \frac{M}{M + m} r.$$

Similarly we can show that $r_M = \left(\frac{m}{M + m}\right) r$. From these two expressions we can see

that $\frac{r_M}{r_e} = \frac{m}{M}$. Now the centripetal force on the electron and the nucleus must be equal (in

magnitude) by Newton's third law so we have

$$\frac{mv_e^2}{r_e} = \frac{Mv_M^2}{r_M} \text{ or } \frac{v_M^2}{v_e^2} = \frac{mr_M}{Mr_e} = \frac{m^2}{M^2} \text{ so } \frac{v_M}{v_e} = \frac{m}{M}. \text{ The total angular momentum of the}$$

atom is the sum of the electron and nuclear orbital angular momenta, so

$$L = mv_e r_e + Mv_M r_M = mv_e r_e + M \left(\frac{m}{M} v_e\right) \left(\frac{m}{M} r_e\right) = mv_e r_e \left(1 + \frac{m}{M}\right) = mv_e r \text{ where the last}$$

substitution comes from using the expression for r_e from above. Because the total L must

equal $n\hbar$ we find $v_e = \frac{n\hbar}{mr}$. This is the equivalent of Equation (4.22b). Now the

centripetal force depends on the distance of the electron from the center of mass while the Coulomb force depends on r , the distance between the electron and the nucleus. So

$$\text{Equation (4.18) becomes: } \left(\frac{1}{4\pi\epsilon_0}\right) \left(\frac{e^2}{r^2}\right) = \frac{mv_e^2}{r_e} \text{ or } v_e^2 = \left(\frac{e^2}{4\pi\epsilon_0}\right) \left(\frac{1}{r^2}\right) \left(\frac{r_e}{m}\right) = \left(\frac{n\hbar}{mr}\right)^2.$$

This can be solved to give $mr_e = \frac{4\pi\epsilon_0 n^2 \hbar^2}{e^2}$ or $m \left(\frac{M}{M + m}\right) r = \frac{4\pi\epsilon_0 n^2 \hbar^2}{e^2}$. Finally this can

be solved for r and the subscript added to account for allowed orbits. We find

$$r_n = \frac{4\pi\epsilon_0 n^2 \hbar^2}{mM} = \frac{4\pi\epsilon_0 n^2 \hbar^2}{\mu e^2} \text{ with } \mu = \frac{mM}{M+m} = \frac{m}{1+m/M}.$$

This is the modification to Equation (4.24).

$$54. (a) f = \frac{me^4}{4\epsilon_0^2 \hbar^3} = \frac{(9.11 \times 10^{-31} \text{ kg})(1.60 \times 10^{-19} \text{ C})^4}{4(8.85 \times 10^{-12} \text{ F/m})^2 (6.626 \times 10^{-34} \text{ J}\cdot\text{s})^3} = 6.55 \times 10^{15} \text{ Hz}$$

(b) As determined in previous problems $v = 2.19 \times 10^6 \text{ m/s}$ and $r = a_0 = 5.29 \times 10^{-11} \text{ m}$.

$$f = \frac{v}{2\pi r} = \frac{2.19 \times 10^6 \text{ m/s}}{2\pi (5.29 \times 10^{-11} \text{ m})} = 6.59 \times 10^{15} \text{ Hz}$$

(c) We know $E = -\frac{e^2}{8\pi\epsilon_0 a_0}$ and $K = \frac{e^2}{8\pi\epsilon_0 a_0} = |E|$.

(d) Because $K = nhf_{\text{orb}}/2 = hf_{\text{orb}}/2$ for $n = 1$ we have

$$f_{\text{orb}} = \frac{2K}{h} = \frac{2(13.6 \text{ eV})}{4.136 \times 10^{-15} \text{ eV}\cdot\text{s}} = 6.58 \times 10^{15} \text{ Hz} \text{ which agrees with (a) and (b) to within rounding errors.}$$

55. We start with $K = nhf_{\text{orb}}/2$. From classical mechanics we have for a circular orbit

$$f = v/2\pi r, \text{ or } r = v/2\pi f : L = mvr = mv \left(\frac{v}{2\pi f} \right) = \left(\frac{mv^2}{2} \right) \left(\frac{1}{\pi f} \right). \text{ Using } K = \frac{1}{2}mv^2,$$

$$L = \frac{K}{\pi f} = \frac{nhf}{2\pi f} = \frac{nh}{2\pi} = n\hbar$$

56. From problem 32 we know that the reduced mass of the system $\mu = m/2$ where m is the mass of either particle. Then from Equation (4.37), we see that the effective Rydberg constant would be $R_{\text{eff}} = R_{\infty}/2$. We know that

$$E_n = -\frac{(hc)(R_{\text{eff}})}{n^2} = -\frac{(1239.8 \text{ eV}\cdot\text{nm})\left(\left(\frac{1}{2}\right)1.09737 \times 10^{-2} \text{ nm}^{-1}\right)}{n^2} = \frac{6.803 \text{ eV}}{n^2}. \text{ Therefore}$$

$E_1 = -6.803 \text{ eV}$, $E_2 = -1.701 \text{ eV}$, $E_3 = -0.756 \text{ eV}$, and $E_4 = 0.425 \text{ eV}$. As in other problems, we know the K_{α} wavelength occurs for an $n_u = 2$ to an $n_{\ell} = 1$ and the L_{α} wavelength occurs for an $n_u = 3$ to an $n_{\ell} = 2$ transition, etc. Further, we know that

$$\lambda = \frac{hc}{E_u - E_{\ell}}. \text{ For example, we have } \lambda(K_{\alpha}) = \frac{1239.8 \text{ eV}\cdot\text{nm}}{-1.701 \text{ eV} - (-6.803 \text{ eV})} = 243.0 \text{ nm}. \text{ In}$$

a similar fashion we can show that $\lambda(K_{\beta}) = 205.0 \text{ nm}$, $\lambda(L_{\alpha}) = 1312 \text{ nm}$, and

$$\lambda(L_{\beta}) = 971.6 \text{ nm}.$$

57. The longest wavelength in the Lyman series is 121.57 nm, using the value of R_H . Then

from the Doppler-shift formula, $\frac{\lambda}{\lambda_0} = \frac{137.15 \text{ nm}}{121.57 \text{ nm}} = \sqrt{\frac{1+\beta}{1-\beta}}$. We can solve to find

$$\beta = 0.12 \text{ or } v = 3.6 \times 10^7 \text{ m/s.}$$

58. (a) The assumptions identified in the statement of the problem lead to the following

$$\text{expression for the force on each electron: } \left| \left(\frac{1}{4\pi\epsilon_0} \right) \left(\frac{Ze^2}{r^2} \right) - \left(\frac{1}{4\pi\epsilon_0} \right) \left(\frac{e^2}{(2r)^2} \right) \right| = |F_{\text{net}}|,$$

where the terms are of opposite sign since the force of the nucleus is attractive whereas the force of the other electron is repulsive. The above expression simplifies to:

$$\left| \left(\frac{e^2}{4\pi\epsilon_0 r^2} \right) \left(Z - \frac{1}{4} \right) \right| = |F_{\text{net}}|.$$

$$(b) \left| \left(\frac{e^2}{4\pi\epsilon_0 r^2} \right) \left(Z - \frac{1}{4} \right) \right| = |F_{\text{net}}| = \frac{mv^2}{r} \text{ or } \left(\frac{e^2}{4\pi\epsilon_0 mr} \right) \left(Z - \frac{1}{4} \right) = v^2.$$

(c) Bohr's rule for angular momentum: $L = mvr = n\hbar = (1)\hbar$ in the ground state. Rearrange this equation to solve for v and square that expression. $v = \hbar / mr$ which gives

$v^2 = (\hbar / mr)^2$. This is substituted into the expression from part (b). Then solve for r to

$$\text{find } r = \frac{4\pi\epsilon_0 \hbar^2}{me^2 (Z - \frac{1}{4})} = \frac{\epsilon_0 \hbar^2}{\pi me^2 (Z - \frac{1}{4})}.$$

(d) The total energy of the atom is the sum of the kinetic energy term for each of the electrons (easily found using the answer to part (b)) and the sum of the potential energy of each electron, with respect to the nucleus, and the potential energy of the electron pair. Each can be found using the expression for r found in part (c). The general form of the

potential energy is given by $V = -\left(\frac{1}{4\pi\epsilon_0} \right) \frac{Ze^2}{r}$, using the appropriate r and Z and the

appropriate sign.

$$\begin{aligned} E = K + V &= 2\left(\frac{1}{2}mv^2\right) - 2\left(\frac{1}{4\pi\epsilon_0}\right)\frac{Ze^2}{r} + \left(\frac{1}{4\pi\epsilon_0}\right)\frac{e^2}{2r} \\ &= m\left(\frac{e^2}{4\pi\epsilon_0 mr}\right)\left(Z - \frac{1}{4}\right) - \left(\left(\frac{e^2}{4\pi\epsilon_0 r}\right)(2Z - \frac{1}{2})\right) \\ &= \left(\frac{e^2}{4\pi\epsilon_0 r}\right)\left(Z - \frac{1}{4} - 2Z + \frac{1}{2}\right) \\ &= -\left(\frac{e^2}{4\pi\epsilon_0 r}\right)\left(Z - \frac{1}{4}\right) \end{aligned}$$

Now substitute the expression for r and simplify:

$$\begin{aligned} E &= -\left(\frac{e^2}{4\pi\epsilon_0 r}\right)\left(Z - \frac{1}{4}\right) = -\left(\frac{e^2}{4\pi\epsilon_0}\right)\left(Z - \frac{1}{4}\right)\left(\frac{\epsilon_0 h^2}{\pi m e^2 (Z - \frac{1}{4})}\right)^{-1} \\ &= -\left(\frac{m e^4}{8\epsilon_0^2 h^2}\right)\left(2Z - \frac{1}{2}\right)\left(Z - \frac{1}{4}\right) \end{aligned}$$

You may notice the first parenthesis in the answer is E_0 . See Equation (4.26).

(d) The calculation for helium, $Z = 2$, is shown.

$$\begin{aligned} E_{\text{He}} &= \left(\frac{m e^4}{8\epsilon_0^2 h^2}\right)\left(2Z - \frac{1}{2}\right)\left(Z - \frac{1}{4}\right) \\ &= (E_0)\left(4 - \frac{1}{2}\right)\left(2 - \frac{1}{4}\right) \\ &= (13.6 \text{ eV})(3.5)(1.75) \\ &= 83.3 \text{ eV} \end{aligned}$$

The calculation for lithium ($Z = 3$) follows in a similar fashion and gives

$E_{\text{Li}} = 205.7 \text{ eV}$. These are fairly close to the measured values but not precise enough to rank with the Bohr model. Thus, our statement that the Bohr model cannot be applied to multi-electron atoms is fair.

59. The result from dimensional analysis is the same as the exact result [Equation (4.24)] gives for the ground state. In general, dimensional analysis might be expected to give the correct answer to within a multiplicative constant.

60. (a) Performing a binomial expansion on the expression in parentheses for a transition from the $n + 1$ level to the n level gives:

$$\frac{1}{n^2} - \frac{1}{(n+1)^2} = \frac{1}{n^2} - \frac{1}{n^2} \left(1 + \frac{1}{n}\right)^{-2} \approx \frac{1}{n^2} - \frac{1}{n^2} \left(1 - \frac{2}{n} + \dots\right)$$

$$\text{or } \frac{1}{n^2} - \frac{1}{(n+1)^2} \approx \frac{1}{n^2} - \frac{1}{n^2} + \frac{2}{n^3} = \frac{2}{n^3}. \text{ Therefore using Equation (4.30) } \lambda = n^3 / 2R. \text{ (b)}$$

Performing the derivative gives the same result. $E = -\frac{E_0}{n^2}$ so $\frac{dE}{dn} = \frac{2E_0}{n^3}$. With

$$\frac{dE}{dn} = \frac{\Delta E}{\Delta n} = \frac{2E_0}{n^3} \text{ and } \Delta E = hc / \lambda, (E_0 / hc) = R, \text{ and } \Delta n = 1, \text{ we have } \lambda = n^3 / 2R.$$

(c) $\lambda = (100)^3 / 2(1.097373 \times 10^7 \text{ m}^{-1}) = 4.556 \text{ cm}$ using the approximation and

$\frac{1}{\lambda} = 1.097373 \times 10^7 \text{ m}^{-1} \left(\frac{1}{100^2} - \frac{1}{101^2} \right) = 21.6226 \text{ m}^{-1}$ which gives $\lambda = 4.625 \text{ cm}$ using the exact formula.

61. Using Equation (4.40) for the K_α x ray, $f = \frac{3cR}{4}(42-1)^2 = 4.15 \times 10^{18} \text{ Hz}$, so the wavelength is $\lambda = c/f = 7.2 \times 10^{-11} \text{ m}$. Using Equation (4.43) with $n = 3$ for the K_β x ray, the wavelength is $\lambda = 6.1 \times 10^{-11} \text{ m}$.

Both of these results compare favorably with the graph. We can compute the frequency of the L_α x ray using Equation (4.44) and then use that frequency to find the wavelength, which is about $5.5 \times 10^{-10} \text{ m}$. This value does not place the wavelength on the graph. You would want to use a less energetic electron beam to generate the x rays in order to observe the L series.

Chapter 5

1. Starting with Equation (5.1), with $n = 1$ and $\theta_1 = 12.5^\circ$, $\sin \theta_1 = \frac{\lambda}{2d} = 0.216$.

$$\text{Second order: } \sin \theta_2 = \frac{2\lambda}{2d} = 2 \sin \theta_1; \quad \theta_2 = \sin^{-1}(2 \sin \theta_1) = \sin^{-1}(2(0.216)) = 25.6^\circ;$$

$$\theta_3 = \sin^{-1}(3 \sin \theta_1) = 40.4^\circ$$

2. Use $\lambda = 0.207 \text{ nm}$ and we know from the text (Example 5.1) that $d = 0.282 \text{ nm}$ for NaCl.

$$n = 1: \quad \sin \theta = \frac{n\lambda}{2d} = \frac{\lambda}{2d} = 0.367; \quad \theta = 21.3^\circ$$

$$n = 2: \quad \sin \theta = \frac{n\lambda}{2d} = \frac{\lambda}{d} = 0.734; \quad \theta = 47.2^\circ$$

$$\Delta\lambda = 47.2^\circ - 21.3^\circ = 25.9^\circ$$

3. For $n = 1$ we have $\lambda = 2d \sin \theta = 2(0.314 \text{ nm})(\sin 12.8^\circ) = 0.139 \text{ nm}$.

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{0.139 \text{ nm}} = 8.92 \text{ keV. The largest order } n \text{ is the largest integer for which}$$

$$\frac{n\lambda}{2d} < 1 \text{ or } n < \frac{2d}{\lambda} = 4.52 \text{ so we can observe up through } n = 4.$$

4. As in Davisson-Germer scattering $n\lambda = D \sin \phi$, so

$$\phi = \sin^{-1}\left(\frac{\lambda}{D}\right) = \sin^{-1}\left(\frac{hc}{ED}\right) = \sin^{-1}\left(\frac{1240 \text{ eV} \cdot \text{nm}}{(10^5 \text{ eV})(0.24 \text{ nm})}\right) = 3.0^\circ$$

5. $\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{(1.2 \text{ kg})(6.0 \text{ m/s})} = 9.2 \times 10^{-35} \text{ m}$. No, the wavelength of the water waves

depends on the medium; they are strictly mechanical waves.

6. Using the mean speed from kinetic theory (also in Chapter 9)

$$\bar{v} = \frac{4}{\sqrt{2\pi}} \sqrt{\frac{kT}{m}} = \frac{4}{\sqrt{2\pi}} \sqrt{\frac{(1.38 \times 10^{-23} \text{ J/K})(310.15 \text{ K})}{28(1.66 \times 10^{-27} \text{ kg})}} = 484.2 \text{ m/s};$$

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{28(1.66 \times 10^{-27} \text{ kg})(484.2 \text{ m/s})} = 2.94 \times 10^{-11} \text{ m which is roughly 3\% of the size}$$

of the molecule.

7. Using the approach of Example 5.2, and with $K = e\Delta V$ we have, assuming relativistic effects are small, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{hc}{\sqrt{2(mc^2)K}}$. If we do not assume that relativistic

$$\text{effects are small, then } \lambda = \frac{h}{p} = \frac{hc}{\sqrt{K^2 + 2(mc^2)K}}. \text{ See problem 11 for details. When}$$

$K = 40\text{keV}$, if we use the non-relativistic formula then

$$\lambda = \frac{1240\text{eV} \cdot \text{nm}}{\sqrt{2(0.511 \times 10^6 \text{eV})(40 \times 10^3 \text{eV})}} = 6.13 \times 10^{-3} \text{nm} = 6.13 \text{ pm.}$$
 Using the relativistic

formula, we find $\lambda = \frac{1240\text{eV} \cdot \text{nm}}{\sqrt{(40 \times 10^3 \text{eV})^2 + 2(0.511 \times 10^6 \text{eV})(40 \times 10^3 \text{eV})}} = 6.02 \text{ pm.}$ This

represents a 2% difference. Clearly when $K = 100\text{keV}$, we must use the relativistic approach and we find $\lambda = 3.70 \times 10^{-3} \text{nm} = 3.70 \text{ pm.}$

8. The resolution will be comparable to the de Broglie wavelength. The energy of the microscope requires a relativistic treatment, so

$$\begin{aligned} \lambda &= \frac{h}{p} = \frac{hc}{\sqrt{K^2 + 2(mc^2)K}} \\ &= \frac{1240\text{eV} \cdot \text{nm}}{\sqrt{(3.0 \times 10^6 \text{eV})^2 + 2(0.511 \times 10^6 \text{eV})(3.0 \times 10^6 \text{eV})}} \\ &= 3.57 \times 10^{-4} \text{nm} = 0.357 \text{ pm} \end{aligned}$$

9. $(50 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV}) = 8.01 \times 10^{-18} \text{ J};$

$$\lambda = \frac{h}{\sqrt{2mK}} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{\sqrt{2(9.109 \times 10^{-31} \text{ kg})(8.01 \times 10^{-18} \text{ J})}} = 1.73 \times 10^{-10} \text{ m.}$$

This is the same result as in the textbook's example.

10. When $E \gg E_0$ then $E \approx pc$ for the particle; $E = pc$ for a photon. Therefore the electron's energy is approximately equal to the photon energy.

If $E = 2E_0$ then we cannot use $E \approx pc$. The exact expression is $p = \frac{\sqrt{E^2 - E_0^2}}{c} = \frac{\sqrt{3}E_0}{c}$

for the electron's momentum. Then the de Broglie wavelength is $\lambda = \frac{h}{p} = \frac{hc}{\sqrt{3}E_0}$. If the

photon has the same wavelength, its energy is $E = \frac{hc}{\lambda} = \sqrt{3}E_0$.

11. (a) Relativistically $p = \frac{\sqrt{E^2 - E_0^2}}{c} = \frac{\sqrt{(K + mc^2)^2 - (mc^2)^2}}{c} = \frac{\sqrt{K^2 + 2Kmc^2}}{c};$

$$\lambda = \frac{h}{p} = \frac{hc}{\sqrt{K^2 + 2Kmc^2}}$$

- (b) Non-relativistically, as in the text $\lambda = \frac{h}{\sqrt{2mK}} = \frac{hc}{\sqrt{2mc^2K}}.$

12. The rest energy of the electron is very small compared to the total energy.

$$p = \frac{\sqrt{E^2 - E_0^2}}{c} = 50 \text{ GeV}/c; \quad \lambda = \frac{h}{p} = \frac{1240 \text{ eV} \cdot \text{nm}}{50 \text{ GeV}} = 2.48 \times 10^{-17} \text{ m};$$

$$\text{fraction} = \frac{2.48 \times 10^{-17} \text{ m}}{2 \times 10^{-15} \text{ m}} = 0.012.$$

13. (a) For photons, kinetic energy equals total energy and de Broglie wavelength is

$$\text{wavelength: } K = E = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{0.13 \text{ nm}} = 9.54 \text{ keV}.$$

- (b) The energy is low enough that we can use the non-relativistic formula:

$$K = \frac{h^2}{2m\lambda^2} = \frac{(hc)^2}{2mc^2\lambda^2} = \frac{(1240 \text{ eV} \cdot \text{nm})^2}{2(511 \times 10^3 \text{ eV})(0.13 \text{ nm})^2} = 89.0 \text{ eV}$$

$$(c) \quad K = \frac{h^2}{2m\lambda^2} = \frac{(hc)^2}{2mc^2\lambda^2} = \frac{(1240 \text{ eV} \cdot \text{nm})^2}{2(939 \times 10^6 \text{ eV})(0.13 \text{ nm})^2} = 0.048 \text{ eV}$$

$$(d) \quad K = \frac{h^2}{2m\lambda^2} = \frac{(hc)^2}{2mc^2\lambda^2} = \frac{(1240 \text{ eV} \cdot \text{nm})^2}{2(3727 \times 10^6 \text{ eV})(0.13 \text{ nm})^2} = 1.22 \times 10^{-2} \text{ eV}$$

14. (a) As in Problem 6, we use the mean speed formula from kinetic theory:

$$\bar{v} = \frac{4}{\sqrt{2\pi}} \sqrt{\frac{kT}{m}} = \frac{4}{\sqrt{2\pi}} \sqrt{\frac{(1.38 \times 10^{-23} \text{ J/K})(5 \text{ K})}{(1.675 \times 10^{-27} \text{ kg})}} = 324 \text{ m/s};$$

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{(1.675 \times 10^{-27} \text{ kg})(324 \text{ m/s})} = 1.22 \text{ nm}.$$

$$(b) \quad \bar{v} = \frac{4}{\sqrt{2\pi}} \sqrt{\frac{kT}{m}} = \frac{4}{\sqrt{2\pi}} \sqrt{\frac{(1.38 \times 10^{-23} \text{ J/K})(0.01 \text{ K})}{(1.675 \times 10^{-27} \text{ kg})}} = 14.5 \text{ m/s};$$

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{(1.675 \times 10^{-27} \text{ kg})(14.5 \text{ m/s})} = 27.3 \text{ nm}$$

15. From the accelerating potential we know $K = eV = 3.00 \text{ keV}$ so $E = K + E_0 = 514 \text{ keV}$.

$$p = \frac{\sqrt{E^2 - E_0^2}}{c} = \frac{\sqrt{(514 \text{ keV})^2 - (511 \text{ keV})^2}}{c} = 55.4 \text{ keV}/c;$$

$$\lambda = \frac{h}{p} = \frac{hc}{pc} = \frac{1240 \text{ eV} \cdot \text{nm}}{55.4 \times 10^3 \text{ eV}} = 22.4 \text{ pm}$$

16. We use the relativistic formula derived in Problem 11: $\lambda = \frac{h}{p} = \frac{hc}{\sqrt{K^2 + 2Kmc^2}}.$

$$(a) \quad \lambda = 0.194 \text{ nm}$$

$$(b) \quad \lambda = 6.13 \times 10^{-2} \text{ nm}$$

$$(c) \quad \lambda = 1.94 \times 10^{-2} \text{ nm}$$

$$(d) \lambda = 6.02 \times 10^{-3} \text{ nm} \quad (e) \lambda = 1.64 \times 10^{-3} \text{ nm} \quad (f) \lambda = 2.77 \times 10^{-4} \text{ nm}$$

From Example 5.1 in the text, we know the spacing of the NaCl lattice is 0.282 nm. Thus even the lowest energy electrons here could be used to probe the crystal structure.

$$17. (a) \lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{32(1.661 \times 10^{-27} \text{ kg})(480 \text{ m/s})} = 2.60 \times 10^{-11} \text{ m}$$

$$(b) \lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{(6.5 \times 10^{-13} \text{ kg})(10^{-5} \text{ m/s})} = 1.02 \times 10^{-16} \text{ m}$$

18. (a) Using the relativistic formula from Problem 11,

$$\lambda = \frac{h}{p} = \frac{hc}{\sqrt{K^2 + 2Kmc^2}} = \frac{1240 \text{ eV} \cdot \text{nm}}{\sqrt{(10^{12} \text{ eV})^2 + 2(10^{12} \text{ eV})(938 \times 10^6 \text{ eV})}} = 1.24 \times 10^{-18} \text{ m}$$

$$(b) \lambda = \frac{h}{p} = \frac{hc}{\sqrt{K^2 + 2Kmc^2}} = \frac{1240 \text{ eV} \cdot \text{nm}}{\sqrt{(7.0 \times 10^{12} \text{ eV})^2 + 2(7.0 \times 10^{12} \text{ eV})(938 \times 10^6 \text{ eV})}} = 1.77 \times 10^{-19} \text{ m}$$

Note: For both parts (a) and (b), the same answers (to three significant figures) may be obtained using the relativistic approximation $K = pc$.

$$19. d = D \sin(\phi/2) = (0.23 \text{ nm}) \sin 16^\circ = 0.063 \text{ nm};$$

$$\lambda = D \sin \phi = (0.23 \text{ nm}) \sin 32^\circ = 0.122 \text{ nm}; \quad p = \frac{h}{\lambda} = \frac{hc}{\lambda c} = \frac{1240 \text{ eV} \cdot \text{nm}}{(0.122 \text{ nm})c} = 10.2 \text{ keV}/c;$$

$$E = \sqrt{p^2 c^2 + E_0^2} = \sqrt{(10.2 \text{ keV})^2 + (511 \text{ keV})^2} = 511.102 \text{ keV}; \quad K = E - E_0 = 102 \text{ eV}$$

20. Beginning with Equation (5.7) and with $V_0 = 48 \text{ V}$ and $D = 0.215 \text{ nm}$ for nickel, we have

$$\lambda = \frac{1.226 \text{ nm} \cdot \text{V}^{1/2}}{\sqrt{48 \text{ V}}} = 0.177 \text{ nm}; \quad \text{and} \quad \phi = \sin^{-1}\left(\frac{\lambda}{D}\right) = \sin^{-1}\left(\frac{0.177 \text{ nm}}{0.215 \text{ nm}}\right) = 55.4^\circ.$$

$$\text{At } 64 \text{ eV: } \lambda = \frac{1.226 \text{ nm} \cdot \text{V}^{1/2}}{\sqrt{64 \text{ V}}} = 0.153 \text{ nm}; \quad \text{and} \quad \phi = \sin^{-1}\left(\frac{\lambda}{D}\right) = \sin^{-1}\left(\frac{0.153 \text{ nm}}{0.215 \text{ nm}}\right) = 45.4^\circ.$$

21. First we compute the wavelength of the electrons:

$$\lambda = \frac{h}{p} = \frac{hc}{\sqrt{E^2 - E_0^2}} = \frac{1240 \text{ eV} \cdot \text{nm}}{\sqrt{(513 \text{ keV})^2 - (511 \text{ keV})^2}} = 2.74 \times 10^{-2} \text{ nm}.$$

From Figure 5.7(a) we see that $2\theta_1 = \tan^{-1}(2.1 \text{ cm}/35 \text{ cm}) = 3.434^\circ$ or $\theta_1 = 1.717^\circ$. Now

$$\text{because } \lambda = 2d \sin \theta \text{ we have: } d_1 = \frac{\lambda}{2 \sin \theta_1} = \frac{2.74 \times 10^{-2} \text{ nm}}{2 \sin(1.717^\circ)} = 0.457 \text{ nm};$$

$$\theta_2 = \frac{1}{2} \tan^{-1}(2.3 \text{ cm}/35 \text{ cm}) = 1.880^\circ; \quad d_2 = \frac{\lambda}{2 \sin \theta_2} = \frac{2.74 \times 10^{-2} \text{ nm}}{2 \sin(1.880^\circ)} = 0.412 \text{ nm}.$$

$$\theta_3 = \frac{1}{2} \tan^{-1}(3.2 \text{ cm}/35 \text{ cm}) = 2.612^\circ; \quad d_3 = \frac{\lambda}{2 \sin \theta_3} = \frac{2.74 \times 10^{-2} \text{ nm}}{2 \sin(2.612^\circ)} = 0.301 \text{ nm}$$

$$22. \quad \lambda = \frac{h}{\sqrt{2mK}} = \frac{hc}{\sqrt{2mc^2K}} = \frac{1240 \text{ eV} \cdot \text{nm}}{\sqrt{2(939 \times 10^6 \text{ eV})(0.025 \text{ eV})}} = 0.181 \text{ nm};$$

$$\lambda = D \sin \phi; \quad \phi = \sin^{-1}\left(\frac{\lambda}{D}\right) = \sin^{-1}\left(\frac{0.181 \text{ nm}}{0.45 \text{ nm}}\right) = 23.7^\circ$$

23. We begin with Equation (5.23) $\Delta \omega \Delta t = 2\pi$. Since $\omega = 2\pi f$ the first equation is

$$\text{equivalent to } \Delta f \Delta t = 1. \text{ Then } \Delta t = \frac{1}{\Delta f} = \frac{1}{0.3 \text{ Hz}} = 3.33 \text{ s}. \text{ If we want the time to be one-}$$

half of that found from the bandwidth relation, then we must monitor the frequency of the system every 1.67 s.

24. Refer to Problem 11. The 6-eV electrons are nonrelativistic. We must treat the 600 keV electrons relativistically.

$$\text{For the 6 eV electrons, } \lambda = \frac{hc}{\sqrt{2mc^2K}} = \frac{1240 \text{ eV} \cdot \text{nm}}{\sqrt{2(0.511 \times 10^6 \text{ eV})(6 \text{ eV})}} = 0.501 \text{ nm}. \text{ The}$$

electron speed can be found using $p = mv = \frac{h}{\lambda}$. Therefore

$$v = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{(9.11 \times 10^{-31} \text{ kg})(0.501 \times 10^{-9} \text{ m})} = 1.45 \times 10^6 \text{ m/s} \text{ and } \beta = \frac{v}{c} = 4.83 \times 10^{-3}.$$

Alternatively for this nonrelativistic case you could use $K = 6 \text{ eV} = \frac{1}{2}mv^2$ to find v .

The phase velocity is found using a modification of Equation (5.32). We must note that E represents the total energy of the particle, so

$$v_{\text{ph}} = \frac{\omega}{k} = \frac{E/\hbar}{p/\hbar} = \frac{E}{p} = \frac{0.511 \times 10^6 \text{ eV}(1.602 \times 10^{-19} \text{ J/eV})}{(9.11 \times 10^{-31} \text{ kg})(1.45 \times 10^6 \text{ m/s})} = 6.20 \times 10^{10} \text{ m/s}.$$

This speed exceeds the speed of light but is not associated with the transmission of information. In Problem 28 we will show that a simpler approach to find the phase

velocity is to use $v_{\text{ph}} = \frac{c}{\beta}$ but either approach yields the same answer. We find the group

velocity from Equation (5.31), namely, $u_{\text{gr}} = \frac{pc^2}{E}$. In problem 73 from chapter 2, we

showed that $\beta = \frac{pc}{E}$ so $u_{gr} = \beta c = 1.45 \times 10^6 \text{ m/s}$ which is the same as the particle speed.

For the 600 keV electrons,

$$\lambda = \frac{hc}{\sqrt{K^2 + 2(mc^2)K}} = \frac{1240 \text{ eV} \cdot \text{nm}}{\sqrt{(600 \times 10^3 \text{ eV})^2 + 2(0.511 \times 10^6 \text{ eV})(600 \times 10^3 \text{ eV})}} \text{ so}$$

$$\lambda = 1.26 \times 10^{-3} \text{ nm}. \text{ Using the same approach as before } p = \frac{h}{\lambda} = 5.26 \times 10^{-22} \text{ kg} \cdot \text{m/s}.$$

However, this is a relativistic momentum so $p = \gamma mv$. We find $\gamma = \frac{K + E_0}{E_0} = 2.174$ and

therefore $v = 2.66 \times 10^8 \text{ m/s} = 0.885c$. The phase velocity is $v_{ph} = \frac{c}{\beta} = 1.13c$ and again

the group velocity is the same as the particle velocity, $u_{gr} = 0.885c$.

$$25. (a) f = \frac{v}{\lambda} = \frac{5.0 \text{ cm/s}}{7.0 \text{ cm}} = 0.714 \text{ Hz}$$

(b) From the initial conditions given, the phase in Equation (5.18) is $\phi = \pi/2 \text{ rad} = 90^\circ$. This is equivalent to representing the wave with a cosine function and zero phase:

$$\Psi = A \cos \left[\left(\frac{2\pi}{\lambda} \right) (x - vt) \right] = (4.0 \text{ cm}) \cos \left[\left(\frac{2\pi}{7.0 \text{ cm}} \right) (10 \text{ cm} - (5.0 \text{ cm/s})(13 \text{ s})) \right] = 2.49 \text{ cm}$$

$$26. (a) f = \frac{v}{\lambda} = \frac{4.2 \text{ cm/s}}{4.0 \text{ cm}} = 1.05 \text{ Hz}$$

$$b) T = 1/f = 0.95 \text{ s}$$

$$c) k = 2\pi/\lambda = \pi/2 \text{ cm}^{-1}$$

$$d) \omega = 2\pi/T = 2.1\pi \text{ rad/s}$$

$$27. (a) \Psi = \Psi_1 + \Psi_2 = 0.003 [\sin(6.0x - 300t) + \sin(7.0x - 250t)] \text{ We can use a trig identity}$$

$$\sin A + \sin B = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right) \text{ so } \Psi = 0.006 \sin(6.5x - 275t) \cos(-0.5x - 25t)$$

or because cosine is an even function $\Psi = 0.006 \sin(6.5x - 275t) \cos(0.5x + 25t)$.

$$(b) v_{ph} = \frac{\omega_1 + \omega_2}{k_1 + k_2} = \frac{550 \text{ rad/s}}{13 \text{ m}^{-1}} = 42.3 \text{ m/s} \text{ and } u_{gr} = \frac{\Delta\omega}{\Delta k} = \frac{50 \text{ rad/s}}{1 \text{ m}^{-1}} = 50 \text{ m/s}$$

(c) As in Equation (5.22) $\Delta x = (2\pi/\Delta k) = 2\pi \text{ m}$ is the separation between zeros.

$$(d) \Delta k \Delta x = (1 \text{ m}^{-1})(2\pi \text{ m}) = 2\pi$$

$$28. u_{gr} = \frac{d\omega}{dk} = \frac{dE}{dp} = \frac{d}{dp} \left(p^2 c^2 + E_0^2 \right)^{1/2} = \frac{pc^2}{\sqrt{p^2 c^2 + E_0^2}} = \frac{pc^2}{E} = \beta c;$$

$v_{ph} = \lambda v = \frac{h}{p} \frac{\omega}{2\pi} = \frac{E}{p} = \frac{pc^2 / v}{p} = \frac{c^2}{v} = \frac{c}{\beta}$. The particle and its “signal” are associated with the group velocity, not the phase velocity.

29. As in Example 5.6 we start with $v_{ph} = c\lambda^n$ where c is a constant. We also know from

Equation (5.33) that $u_g = v_{ph} + k \frac{dv_{ph}}{dk}$. We note further that $\frac{dv_{ph}}{dk} = \frac{dv_{ph}}{d\lambda} \left(\frac{d\lambda}{dk} \right)$. Since

$$\lambda = \frac{2\pi}{k} \text{ then } \frac{d\lambda}{dk} = \frac{-2\pi}{k^2} = \frac{-\lambda^2}{2\pi}. \text{ Therefore } u_g = v_{ph} + \left(\frac{2\pi}{\lambda} \right) \frac{dv_{ph}}{d\lambda} \left(\frac{-\lambda^2}{2\pi} \right) = v_{ph} - (\lambda) \frac{dv_{ph}}{d\lambda}.$$

Setting $u_g = v_{ph} = c\lambda^n$, we find $c\lambda^n = c\lambda^n - c n \lambda^{n-1}$. This can be satisfied only if $n = 0$, so v_{ph} is independent of λ . This is consistent with the idea that when a medium is non-dispersive, the phase and group velocities are equal and the speed independent of wavelength.

$$30. \text{ Protons: } \gamma = \frac{K + E_0}{E_0} = \frac{948.27 \text{ MeV}}{938.27 \text{ MeV}} = 1.01066; \quad \beta = \sqrt{1 - \frac{1}{1.01066^2}} = 0.145;$$

$$u_{gr} = \beta c = 0.145c \text{ and } v_{ph} = \frac{c}{\beta} = 6.90c.$$

$$\text{Electrons: } \gamma = \frac{K + E_0}{E_0} = \frac{10.511 \text{ MeV}}{0.511 \text{ MeV}} = 20.57; \quad \beta = \sqrt{1 - \frac{1}{20.57^2}} = 0.9988;$$

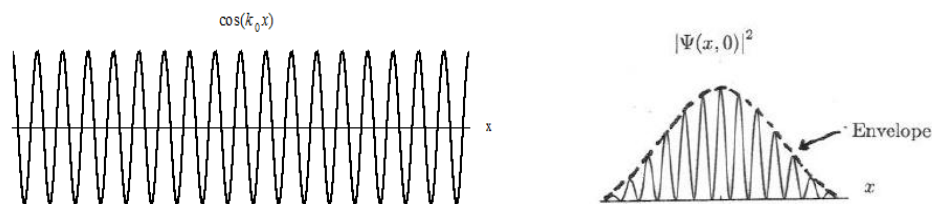
$$u_{gr} = \beta c = 0.9988c \text{ and } v_{ph} = \frac{c}{\beta} = 1.001c$$

31.

$$\begin{aligned} \Psi(x, 0) &= \int \tilde{A}(k) \cos(kx) dk = A_0 \int_{k_0 - \Delta k/2}^{k_0 + \Delta k/2} \cos(kx) dk \\ &= A_0 \frac{\sin(kx)}{x} \Big|_{k_0 - \Delta k/2}^{k_0 + \Delta k/2} = \frac{A_0}{x} \left(\sin(k_0 + \Delta k/2)x - \sin(k_0 - \Delta k/2)x \right) \\ &= \frac{2A_0}{x} \sin\left(\frac{\Delta kx}{2}\right) \cos(k_0 x) \end{aligned}$$

See the diagrams below. At the half-width of $|\Psi(x, 0)|^2$, we have $\sin^2\left(\frac{\Delta kx}{2}\right) = \frac{1}{2}$, so

$$\frac{\Delta k x}{2} = \frac{\pi}{4} \text{ and } x = \frac{\pi}{(2\Delta k)}. \text{ Then } \Delta x = 2x = \frac{\pi}{\Delta k}, \text{ and } \Delta k \Delta x = \pi.$$



32. Relativistically: $u = \frac{dE}{dp} = \frac{d}{dp} (p^2 c^2 + E_0^2)^{1/2} = \frac{pc^2}{\sqrt{p^2 c^2 + E_0^2}} = \frac{pc^2}{E};$

Classically: $u = \frac{dE}{dp} = \frac{d}{dp} \left(\frac{p^2}{2m} \right) = \frac{p}{m}$

33. For a double slit, the amplitude of E is doubled, and hence the intensity (proportional to E^2) is higher by a factor of four for the double slit.

34. $d = \frac{\lambda}{\sin \theta} = \frac{h/p}{\sin \theta} = \frac{hc}{(\sqrt{E^2 - E_0^2}) \sin \theta} = \frac{1240 \text{ eV} \cdot \text{nm}}{(\sqrt{(512 \text{ keV})^2 - (511 \text{ keV})^2}) \sin 1^\circ} = 2.22 \text{ nm}$

35. We make use of the small angle approximations: $\sin \theta \approx \tan \theta$ and $\sin \theta \approx \theta$.

$$\sin \theta \approx \tan \theta = \frac{0.3 \text{ mm}}{0.8 \text{ m}} = 3.75 \times 10^{-4};$$

$$\lambda = d \sin \theta \approx d \theta = (2000 \text{ nm})(3.75 \times 10^{-4}) = 0.75 \text{ nm};$$

$$p = \frac{h}{\lambda} = \frac{hc}{\lambda c} = \frac{1240 \text{ eV} \cdot \text{nm}}{(0.75 \text{ nm})c} = 1.653 \text{ keV}/c;$$

$$K = E - E_0 = \sqrt{p^2 c^2 + E_0^2} - E_0 = \sqrt{(1.653 \text{ keV})^2 + (511 \text{ keV})^2} - 511 \text{ keV} = 2.67 \text{ eV}.$$

Such low energies will present problems, because low-energy electrons take longer to move through the region of the field and thus the deflection by stray electric fields is greater.

36. By the uncertainty principle $\Delta p \Delta x = \frac{\hbar}{2}$ at a minimum. Non-relativistically (with $\Delta x = d$)

$$E_{\min} = \frac{(\Delta p)^2}{2m} = \frac{\hbar^2}{8md^2} = \frac{(\hbar c)^2}{8mc^2 d^2} = \frac{(197.3 \text{ eV} \cdot \text{nm})^2}{8(938.27 \times 10^6 \text{ eV})(1.44 \times 10^{-5} \text{ nm})^2} = 25.0 \text{ keV}$$

37. For the $n = 1$ energy level:

$$E = \frac{\hbar^2}{8md^2} = \frac{h^2 c^2}{8mc^2 d^2} = \frac{(1240 \text{ eV} \cdot \text{nm})^2}{8(939.57 \times 10^6 \text{ eV})(3.1 \times 10^{-6} \text{ nm})^2} = 21.3 \text{ MeV}.$$

By the uncertainty principle $\Delta p \Delta x = \frac{\hbar}{2}$ at a minimum. Non-relativistically (with $\Delta x = d$)

$$E_{\min} = \frac{(\Delta p)^2}{2m} = \frac{\hbar^2}{8md^2} = 0.539 \text{ MeV}.$$

The two answers differ by the factor of $(2\pi)^2$ in using $\hbar$ or h in the energy formula.

38. The uncertainty ratio is the same for any mass and independent of the box length.

$$\Delta p \Delta x = m \Delta v \Delta x \geq \frac{\hbar}{2} \text{ or } \Delta v \geq \frac{\hbar}{2m \Delta x} = \frac{\hbar}{2mL}$$

$$E = \frac{1}{2} m v^2 = \frac{\hbar^2}{8mL^2} \text{ or } v = \sqrt{\frac{\hbar^2}{4m^2 L^2}} = \frac{\hbar}{2mL}$$

$$\frac{\Delta v}{v} = \frac{\hbar / 2mL}{\hbar / 2mL} = \frac{1}{2\pi}$$

39. For circular motion $L = rp$ and so $\Delta L = r \Delta p$. Along the circle $x = r\theta$ and $\Delta x = r\Delta\theta$.

Thus $\Delta p \Delta x = \frac{\Delta L}{r} (r\Delta\theta) = \Delta L \Delta\theta \geq \frac{\hbar}{2}$. For complete uncertainty $\Delta\theta = 2\pi$ and

$$\Delta L = \frac{\hbar / 2}{2\pi} = \frac{\hbar}{4\pi}.$$

$$40. \Delta E \Delta t \geq \frac{\hbar}{2} \text{ so } \Delta E \geq \frac{\hbar}{2\Delta t} = \frac{1.054 \times 10^{-34} \text{ J} \cdot \text{s}}{2(1 \times 10^{36} \text{ y})(3.16 \times 10^7 \text{ s/y})} = 1.67 \times 10^{-78} \text{ J}$$

$$41. \text{ If we use Equation (5.44), } \Delta \omega \Delta t = \frac{1}{2} \text{ we find } \Delta \omega = \frac{1}{2\Delta t} = \frac{1}{2(2.4 \mu\text{s})} = 2.1 \times 10^5 \text{ Hz}.$$

$$42. \Delta p \Delta x = m \Delta v \Delta x \geq \frac{\hbar}{2} \text{ so at minimum uncertainty}$$

$$\Delta v = \frac{\hbar}{2m \Delta x} = \frac{1.054 \times 10^{-34} \text{ J} \cdot \text{s}}{2(3.0 \times 10^{-15} \text{ kg})(1.0 \times 10^{-6} \text{ m})} = 1.76 \times 10^{-14} \text{ m/s}$$

$$43. (a) \Delta E \Delta t \geq \frac{\hbar}{2} \text{ so } \Delta E \geq \frac{\hbar}{2\Delta t} = \frac{6.582 \times 10^{-16} \text{ eV} \cdot \text{s}}{2(1.0 \times 10^{-13} \text{ s})} = 3.29 \times 10^{-3} \text{ eV}$$

$$(b) \text{ Using the photon relation } E = \frac{hc}{\lambda} \text{ and taking a derivative } dE = -\frac{hc}{\lambda^2} d\lambda = -\frac{E^2}{hc} d\lambda$$

Then letting $\Delta\lambda = d\lambda$ and $\Delta E = dE$ we have

$$|\Delta\lambda| = hc \frac{\Delta E}{E^2} = (1240 \text{ eV} \cdot \text{nm}) \frac{3.29 \times 10^{-3} \text{ eV}}{(4.7 \text{ eV})^2} = 0.185 \text{ nm}.$$

44. The wavelength of the electrons should be 0.14 nm or less. For this wavelength

$$p = \frac{h}{\lambda} = \frac{hc}{\lambda c} = \frac{1240 \text{ eV} \cdot \text{nm}}{(0.14 \text{ nm})c} = 8.86 \text{ keV}/c;$$

$$K = E - E_0 = \sqrt{p^2 c^2 + E_0^2} - E_0 = \sqrt{(8.86 \text{ keV})^2 + (511 \text{ keV})^2} - 511 \text{ keV} = 77 \text{ eV}.$$

45. For the angle θ_R we find $\theta_R \approx \tan \theta_R = \frac{4000 \text{ nm}}{20 \text{ cm}} = 2 \times 10^{-5}$. The wavelength is

$$\lambda = \frac{d \theta_R}{1.22} = \frac{(0.05 \text{ m})(2 \times 10^{-5})}{1.22} = 820 \text{ nm}.$$

$$(a) E = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{820 \text{ nm}} = 1.51 \text{ eV}$$

(b) For non-relativistic electrons

$$K = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2} = \frac{h^2 c^2}{2mc^2 \lambda^2} = \frac{(1240 \text{ eV} \cdot \text{nm})^2}{2(511 \times 10^3 \text{ eV})(820 \text{ nm})^2} = 2.24 \times 10^{-6} \text{ eV}.$$

Check with the uncertainty principle:

$$\Delta p \geq \frac{\hbar}{2\Delta x} = \frac{1.055 \times 10^{-34} \text{ J} \cdot \text{s}}{2(4000 \times 10^{-9} \text{ m})} = 1.32 \times 10^{-29} \text{ kg} \cdot \text{m/s}.$$
 The actual momentum is

$$p = \frac{h}{\lambda} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{820 \times 10^{-9} \text{ m}} = 8.08 \times 10^{-28} \text{ kg} \cdot \text{m/s}.$$
 This is allowed because $p > \Delta p$.

$$46. \lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{1240 \text{ eV} \cdot \text{nm}}{\sqrt{2(3727 \text{ MeV})(5.5 \text{ MeV})}} = 6.12 \text{ fm}.$$
 The minimum kinetic energy

according to the uncertainty principle is

$$K = \frac{(\Delta p)^2}{2m} = \frac{(\hbar c)^2}{8mc^2 (\Delta x)^2} = \frac{(197.33 \text{ eV} \cdot \text{nm})^2}{8(3727 \text{ MeV})(16 \times 10^{-6} \text{ nm})^2} = 5.10 \text{ keV}.$$
 Because the kinetic

energy exceeds the minimum, it is allowed.

47. The proof is completed in Example 6.11. With $\omega = \sqrt{k/m}$ we have a minimum energy

$$E = \frac{\hbar \omega}{2} = \frac{\hbar}{2} \sqrt{\frac{k}{m}} = \frac{1.055 \times 10^{-34} \text{ J} \cdot \text{s}}{2} \sqrt{\frac{8.2 \text{ N/m}}{0.028 \text{ kg}}} = 9.03 \times 10^{-34} \text{ J}$$

48. We can determine the value using the normalization condition:

$$\int_0^L A^2 \sin^2\left(\frac{\pi x}{L}\right) dx = 1 = A^2 \frac{L}{2} \text{ or } A = \sqrt{\frac{2}{L}}.$$

No answers are provided for Problems 49 through 51

$$52. \psi_1 = A \sin\left(\frac{\pi x}{L}\right); \quad \psi_2 = A \sin\left(\frac{2\pi x}{L}\right); \quad \psi_3 = A \sin\left(\frac{3\pi x}{L}\right);$$

where $A = \sqrt{2/L}$. Refer to problem 48.

53. As before $\lambda = \frac{2L}{n}$ so $p_n = \frac{nh}{2L}$. At high energies we must use relativistic formulas, so

$$E = \sqrt{p^2 c^2 + E_0^2} = E_0 \sqrt{\frac{p^2 c^2}{E_0^2} + 1}. \text{ The ratios follow:}$$

$$\frac{E_2}{E_1} = \left[\frac{1 + h^2 c^2 / (L^2 E_0^2)}{1 + h^2 c^2 / (4L^2 E_0^2)} \right]^{1/2}; \quad \frac{E_3}{E_1} = \left[\frac{1 + 9h^2 c^2 / (4L^2 E_0^2)}{1 + h^2 c^2 / (4L^2 E_0^2)} \right]^{1/2}; \text{ and}$$

$$\frac{E_4}{E_1} = \left[\frac{1 + 4h^2 c^2 / (L^2 E_0^2)}{1 + h^2 c^2 / (4L^2 E_0^2)} \right]^{1/2}.$$

These are quite different from the non-relativistic results, as one might expect. They do reduce to the non-relativistic results in the low-energy limit.

54. At time $t = 0$ the velocity is uncertain by at least $\Delta v_0 = \frac{\Delta p}{m} = \frac{\hbar}{2m\Delta x}$. After a time

$t = \frac{m(\Delta x)^2}{\hbar}$ we have $\Delta x' = (\Delta v_0)t = \left(\frac{\hbar}{2m\Delta x}\right) \frac{m(\Delta x)^2}{\hbar} = \frac{\Delta x}{2}$ which means the uncertainty equals half the distance of travel.

55. Both the spatial distribution $\psi(x)$ and the wavenumber distribution $\phi(k)$ should have the

same Gaussian form: $\psi(x) \propto \exp\left[-\frac{x^2}{(2\Delta x)^2}\right]$ and $\phi(k) \propto \exp\left[-\frac{k^2}{(2\Delta k)^2}\right]$. For

conjugate variables (x, k) it is possible to obtain one distribution by taking a Fourier transform of the other. Letting A be a normalization constant for $\phi(k)$ we have

$$\psi(x) = \frac{A}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk \exp\left[-\frac{k^2}{(2\Delta k)^2}\right] \exp(ikx).$$

The integral is evaluated by completing the square:

$$\psi(x) = \frac{A}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk \exp\left[-\frac{k^2}{4\Delta k^2} + ikx - x^2 \Delta k^2 + x^2 \Delta k^2\right]$$

$$\psi(x) = \frac{A}{\sqrt{2\pi}} \exp(-x^2 \Delta k^2) \int_{-\infty}^{\infty} dk \exp\left[\left(-\frac{1}{4\Delta k^2}\right)(k - 2ix\Delta k^2)^2\right]$$

Letting $u = \frac{(k - 2ix\Delta k^2)}{2\Delta k}$ we have $\psi(x) = A\sqrt{\frac{2}{\pi}}\Delta k \exp(-x^2\Delta k^2) \int_{-\infty}^{\infty} \exp(-u^2) du$. The integral has a value $\sqrt{\pi}$ so $\psi(x) = \sqrt{2}A\Delta k \exp(-x^2\Delta k^2)$. Now comparing with the Gaussian form $\psi(x) \propto \exp\left[-\frac{x^2}{(2\Delta x)^2}\right]$ we see that $(\Delta k)^2 = \frac{1}{(2\Delta x)^2}$ or $\Delta k\Delta x = \frac{1}{2}$.

56. (a) These electrons have a kinetic energy that is a significant fraction of the rest energy, so they must be considered relativistically. These electrons have a total energy $E = K + E_0 = 390 \text{ keV} + 511 \text{ keV} = 901 \text{ keV}$.

$$\lambda = \frac{h}{p} = \frac{hc}{pc} = \frac{hc}{\sqrt{E^2 - E_0^2}} = \frac{1.240 \text{ keV} \cdot \text{nm}}{\sqrt{(901 \text{ keV})^2 - (511 \text{ keV})^2}} = 0.00167 \text{ nm} = 1.67 \text{ pm}.$$

- (b) Using the Davisson-Germer result Equation (5.5),

$$\sin \phi = \frac{n\lambda}{D} = \frac{(1)(0.00167 \text{ nm})}{0.215 \text{ nm}} = 0.00777 \text{ which gives an angle } \phi = 0.45^\circ. \text{ This small angle is quite difficult to measure, which is why less energetic electrons are better for this type of scattering experiment.}$$

57. For such highly relativistic electrons $p \approx E/c$, so the wavelength is

$$\lambda = \frac{h}{p} \approx \frac{h}{E/c} = \frac{hc}{E} = \frac{1240 \text{ eV} \cdot \text{nm}}{6.0 \times 10^9 \text{ eV}} = 2.1 \times 10^{-7} \text{ nm} = 2.1 \times 10^{-16} \text{ m}. \text{ Typically only gamma ray photons have such high energies.}$$

58. (a) This is a fairly low energy, but to be safe we should use relativistic calculations. The electron's total energy is $E = K + E_0 = 5.7 \text{ keV} + 511.0 \text{ keV} = 516.7 \text{ keV}$.

$$\lambda = \frac{h}{p} = \frac{hc}{pc} = \frac{hc}{\sqrt{E^2 - E_0^2}} = \frac{1.24 \text{ keV} \cdot \text{nm}}{\sqrt{(516.7 \text{ keV})^2 - (511.0 \text{ keV})^2}} = 0.0162 \text{ nm} = 16.2 \text{ pm}.$$

- (b) For this analysis we will find the minimum energy of an electron confined in a one-dimensional box of length 3.4 fm. We can use either the uncertainty principle result Equation (5.43) or the exact particle-in-a-box result Equation (5.51). The two results are the same order of magnitude. Using the latter,

$$E_1 = \frac{h^2}{8m\ell^2} = \frac{h^2 c^2}{8mc^2 \ell^2} = \frac{(1240 \text{ eV} \cdot \text{nm})^2}{8(5.11 \times 10^5 \text{ eV})(3.4 \times 10^{-6} \text{ nm})^2} = 3.3 \times 10^{10} \text{ eV} = 33 \text{ GeV}.$$

This is much larger than the energy of the emitted electron, which implies that it did not exist in the nucleus prior to emission. Further, note that the electron's wavelength in part (a) is much larger than the nucleus, also implying that it could not have been confined there.

59. From Chapter 4, $r = n^2 a_0$ where $a_0 = 5.29 \times 10^{-11}$ m is the Bohr radius. In this case we have $n = 10$, so $r = 100a_0 = 5.29 \times 10^{-9}$ m. Using Equation (5.51), the electron's energy is

$$E_1 = \frac{h^2}{8m\ell^2} = \frac{h^2 c^2}{8mc^2 \ell^2} = \frac{(1240 \text{ eV} \cdot \text{nm})^2}{8(5.11 \times 10^5 \text{ eV})(2 \times 5.29 \text{ nm})^2} = 3.36 \times 10^{-3} \text{ eV}.$$

This is a very small energy compared with the electron's actual kinetic energy, even in such a high orbital state.

60. Using Equation (5.23), the range of angular frequencies is

$$\Delta\omega = 2\pi / \Delta t = \frac{2\pi}{2.5 \text{ s}} = 2.51 \text{ s}^{-1}. \text{ Because } \omega = 2\pi f, \text{ the range of frequencies is}$$

$$\Delta f = \Delta\omega / 2\pi = \frac{2.51 \text{ s}^{-1}}{2\pi} = 0.40 \text{ Hz. Thus if the central frequency is 440 Hz, the range is from about 439.8 Hz to 440.2 Hz.}$$

61. (a) Because of conservation of energy and momentum, each photon gets an energy equal to the electron's rest energy, so its wavelength is given by $E = hc/\lambda$, or

$$\lambda = \frac{hc}{E} = \frac{1240 \text{ eV} \cdot \text{nm}}{5.11 \times 10^5 \text{ eV}} = 2.43 \times 10^{-3} \text{ nm} = 2.43 \text{ pm}.$$

- (b) Now the total energy of each electron is (Equation 2.65):

$$E = \frac{mc^2}{\sqrt{1-\beta^2}} = \frac{5.11 \times 10^5 \text{ eV}}{\sqrt{1-0.30^2}} = 5.357 \times 10^5 \text{ eV. This is the new energy of each photon, so}$$

$$\text{the photon's wavelength is: } \lambda = \frac{hc}{E} = \frac{1240 \text{ eV} \cdot \text{nm}}{5.357 \times 10^5 \text{ eV}} = 2.31 \times 10^{-3} \text{ nm} = 2.31 \text{ pm}.$$

Because of the additional kinetic energy in the initial configuration, the photons generated in the process have more energy and hence shorter wavelength.

62. (a) The uncertainty principle (Equation 5.45) provides the relationship between uncertainty in energy and time as, $\Delta E \Delta t \geq \hbar / 2$ or

$$\Delta E \geq \hbar / 2(\Delta t) = (6.5821 \times 10^{-16} \text{ eV} \cdot \text{s}) / (2 \cdot 8.4 \times 10^{-17} \text{ s}) = 3.92 \text{ eV}.$$

- (b) The uncertainty in mass is given by the energy-mass relation as $\Delta m = \Delta E / c^2$. With a mass of $135 \text{ MeV}/c^2$, the relative uncertainty is

$$\frac{\Delta m}{m} = \frac{\Delta E}{mc^2} = \frac{3.92 \text{ eV}}{135 \times 10^6 \text{ eV}} = 2.9 \times 10^{-8}.$$

63. $\Delta t = \frac{d}{c} = \frac{1.2 \times 10^{-15} \text{ m}}{3.0 \times 10^8 \text{ m/s}} = 4.0 \times 10^{-24} \text{ s}$ and $\Delta E \geq \frac{\hbar}{2\Delta t} = \frac{6.5821 \times 10^{-16} \text{ eV} \cdot \text{s}}{2(4.0 \times 10^{-24} \text{ s})} = 82 \text{ MeV}.$

This "lower bound" estimate of the rest mass is $\Delta E / c^2$ which is within a factor of two of the rest energy.

64. (a) In general $n\lambda = 2d \sin \theta$ so $\sin \theta = \frac{n\lambda}{2d} = n \frac{0.5 \text{ nm}}{2(0.8 \text{ nm})} = 0.3125n$. It is required that

$\sin \theta \leq 1$ so the allowed values of n are 1, 2, 3. Substituting we find $\theta = 18.2^\circ$ for $n = 1$, $\theta = 38.7^\circ$ for $n = 2$, and $\theta = 69.6^\circ$ for $n = 3$.

- (b) For electrons $p = \frac{h}{\lambda} = \frac{hc}{\lambda c} = \frac{1240 \text{ eV} \cdot \text{nm}}{(0.5 \text{ nm})c} = 2480 \text{ eV}/c$ and

$$K = E - E_0 = \sqrt{p^2 c^2 + E_0^2} - E_0 = \sqrt{(2.480 \text{ keV})^2 + (511 \text{ keV})^2} - 511 \text{ keV} = 6.0 \text{ eV}.$$

65. The uncertainty of the strike zone width is $\Delta x < 0.38 \text{ m}$. There is a second uncertainty, in the x -component of the ball's velocity, given by $\Delta p_x = m\Delta v_x \geq \frac{\hbar}{2\Delta x}$. Due to the

uncertainty in velocity the ball's x position will be uncertain by the time it reaches home plate in an amount $\Delta v_x t = \frac{\hbar t}{2m\Delta x} \leq 0.38 \text{ m}$ where we have added the inequality because

this is the condition for a strike. Since this inequality must be satisfied simultaneously with the first one $\Delta x < 0.38 \text{ m}$, we multiply the two inequalities together to find

$$\frac{\hbar t}{2m} \leq (0.38 \text{ m})^2 = 0.144 \text{ m}^2. \text{ Rearranging we find}$$

$$\hbar \leq \frac{2m(0.144 \text{ m}^2)}{t} = \frac{2(0.145 \text{ kg})(0.144 \text{ m}^2)}{(18 \text{ m})/(35 \text{ m/s})} = 0.081 \text{ J} \cdot \text{s}.$$

66. Assume that the given angle corresponds to the first order reflection. We have:

$$\lambda = 2d \sin \theta = 2(0.156 \text{ nm}) \sin 26^\circ = 0.1368 \text{ nm}. \text{ Next we find the energy:}$$

$$p = \frac{h}{\lambda} = \frac{hc}{\lambda c} = \frac{1240 \text{ eV} \cdot \text{nm}}{(0.1368 \text{ nm})c} = 9.06 \times 10^3 \text{ eV}/c.$$

$$K = E - E_0 = \sqrt{p^2 c^2 + E_0^2} - E_0 = \sqrt{(9.06 \text{ keV})^2 + (939.57 \text{ MeV})^2} - 939.57 \text{ MeV} \\ = 4.36 \times 10^{-2} \text{ eV}.$$

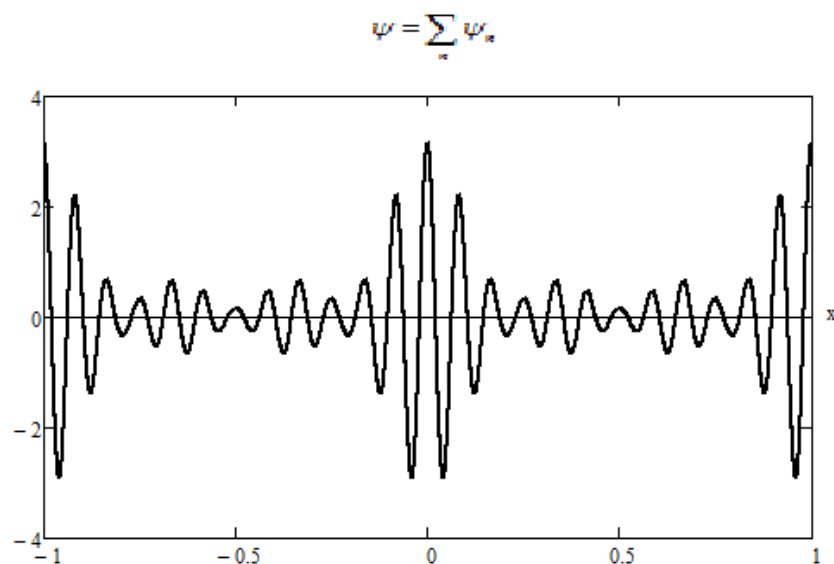
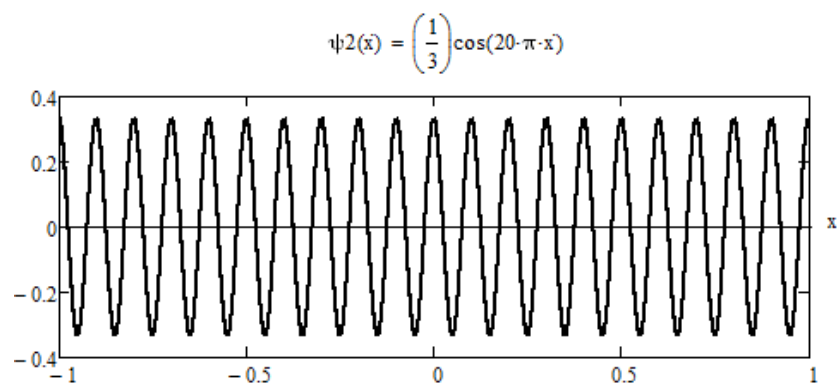
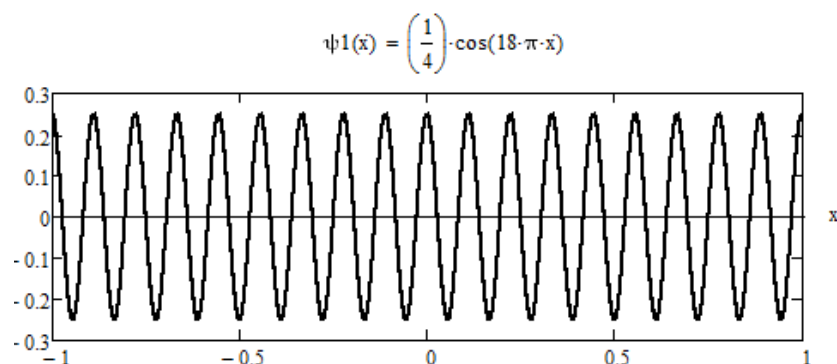
The Oak Ridge Electron Linear Accelerator Pulsed Neutron Source (ORELA) produces intense, nanosecond bursts of neutrons, each burst containing neutrons with energies from 10^{-3} to 10^8 eV .

67. (a) Using the given information, the wave packet will equal:

$$\psi = \sum_{n=9}^{15} \psi_n = \frac{1}{4} \cos(18\pi x) + \frac{1}{3} \cos(20\pi x) + \frac{1}{2} \cos(22\pi x) + \cos(24\pi x) + \frac{1}{2} \cos(26\pi x) \\ + \frac{1}{3} \cos(28\pi x) + \frac{1}{4} \cos(30\pi x).$$

A graph of the first two terms and a graph of the entire packet are shown below. ψ_9 has a frequency of 9 and an amplitude of 0.25. ψ_{10} has a frequency of 10 and an amplitude of 0.33. Other terms would have a frequency of 11 and amplitude of 0.5, etc. The peak value of the packet is approximately 3.16.

(b) The packet is centered about $x = 0$ but extends to $\pm\infty$. The packet repeats every unit along the x axis.



68. (a) Starting from Equation (5.45), $\Delta E \Delta t \geq \frac{\hbar}{2}$ and substituting, we have $\left(\frac{\Gamma}{2}\right) \tau \geq \frac{\hbar}{2}$.

Therefore $\Gamma \tau \geq \hbar$.

(b) We can find the minimum value for Γ from the equation above. Using the data for the

neutron, for example, $\Gamma_{\text{neutron}} = \frac{6.5821 \times 10^{-16} \text{ eV} \cdot \text{s}}{887 \text{ s}} = 7.42 \times 10^{-19} \text{ eV}$. The other values

follow in a similar fashion. $\Gamma_{\text{pion}} = 2.53 \times 10^{-8} \text{ eV}$ and $\Gamma_{\text{upsilon}} = 65.8 \text{ keV}$.

69. (a) With $E = hf$, the uncertainty in energy is $\Delta E = h\Delta f$. With this the uncertainty relation $\Delta E \Delta t \geq \hbar/2$ becomes $h\Delta f \Delta t \geq \hbar/2$. This simplifies to $\Delta f \Delta t \geq 1/4\pi$, so the minimum uncertainty in frequency can be computed:

$$\Delta f = \frac{1}{4\pi\Delta t} = \frac{1}{4\pi(10 \times 10^{-15} \text{ s})} = 7.96 \times 10^{12} \text{ Hz}.$$

The original frequency is $f = c/\lambda$, so the relative uncertainty is

$$\frac{\Delta f}{f} = \frac{\lambda \Delta f}{c} = \frac{(532 \times 10^{-9} \text{ m})(7.96 \times 10^{12} \text{ Hz})}{3.00 \times 10^8 \text{ m/s}} = 0.014. \text{ That's a fairly large uncertainty, over 1\%!}$$

(b) With $f = c/\lambda$, $\Delta f / \Delta \lambda \approx -c / \lambda^2$, so the absolute value of the wavelength range is

$$\Delta \lambda = \frac{\lambda^2 \Delta f}{c} = \frac{(532 \times 10^{-9} \text{ m})^2 (7.96 \times 10^{12} \text{ Hz})}{3.00 \times 10^8 \text{ m/s}} = 7.5 \times 10^{-9} \text{ m} = 7.5 \text{ nm}.$$

(c) This is an appreciable fraction of the wavelength (also 0.014) and of the original pulse, which has a length $L = ct = (3.00 \times 10^8 \text{ m/s})(10 \times 10^{-15} \text{ s}) = 3.0 \times 10^{-6} \text{ m} = 3000 \text{ nm}$. The wavelength spread is 0.0025 of the pulse length.

70. As in the preceding problem, the uncertainty principle leads to a frequency spread $\Delta f = 1/(4\pi \Delta t)$. In this case the wavelengths (and frequencies) differ by enough that it would be risky to use the approximation $\Delta f / \Delta \lambda = c / \lambda^2$. Instead we will compute the value of Δf directly:

$$\Delta f = f_2 - f_1 = \frac{c}{\lambda_2} - \frac{c}{\lambda_1} = \frac{3.00 \times 10^8 \text{ m/s}}{400 \times 10^{-9} \text{ m}} - \frac{3.00 \times 10^8 \text{ m/s}}{700 \times 10^{-9} \text{ m}} = 3.21 \times 10^{14} \text{ Hz}.$$

Solving for the pulse duration Δt :

$$\Delta t = \frac{1}{4\pi \Delta f} = \frac{1}{4\pi(3.21 \times 10^{14} \text{ Hz})} = 2.5 \times 10^{-16} \text{ s} = 0.25 \text{ fs}.$$

This is just possible with today's technology.

Chapter 6

1. The function's output does not approach zero in the limits $-\infty$ and $+\infty$, so it cannot be normalized over these limits.
2. (a) Given the placement of the $-$ sign, the wave moves in the $+x$ -direction. (One always assumes t increases starting from 0. For the argument of the sine function to remain constant x must increase as t increases. Therefore a value of constant phase will have increasing x as t increases.)
 (b) By the same reasoning as in (a), it moves in the $-x$ -direction.
 (c) It is a complex number. Euler's formula $e^{ix} = \cos x + i \sin x$ shows that an expression written as $e^{i(kx - \omega t)}$ will have a real and complex part. Therefore the expression is complex.
 (d) It moves in the $+x$ -direction. Looking at a particular phase $kx - \omega t$, x must increase as t increases in order to keep the phase constant.
3. The derivatives are $\partial\Psi / \partial t = -i\omega\Psi$ and $\partial^2\Psi / \partial x^2 = -k^2\Psi$. Substituting in the time-dependent Schrödinger equation, we find: $i\hbar(-i\omega\Psi) = -\frac{\hbar^2}{2m}(-k^2\Psi) + V\Psi$. The term on the left reduces to $\hbar\omega\Psi = E\Psi$ and the first term on the right is $\frac{\hbar^2 k^2}{2m}\Psi = \frac{p^2}{2m}\Psi = K\Psi$ where K is the kinetic energy. The result is $E = K + V$, which is a statement of conservation of mechanical energy.
4. $\Psi^*\Psi = A^2 \exp[-i(kx - \omega t) + i(kx - \omega t)] = A^2$. The condition for normalization becomes $\int_0^a \Psi^*\Psi dx = A^2 \int_0^a dx = A^2 a = 1$ so $A = \frac{1}{\sqrt{a}}$ and $\Psi = \frac{1}{\sqrt{a}} \exp[i(kx - \omega t)]$.
5. $\Psi^*\Psi = A^2 r^2 \exp\left(\frac{-2r}{\alpha}\right)$. The condition for normalization becomes $\int_0^\infty \Psi^*\Psi dr = A^2 \int_0^\infty r^2 \exp\left(\frac{-2r}{\alpha}\right) dr = A^2 \left[\frac{2}{(2/\alpha)^3} \right] = \frac{A^2 \alpha^3}{4} = 1$. Therefore $A = \sqrt{\frac{4}{\alpha^3}} = 2\alpha^{-3/2}$.
6. In order for a particle to have a greater probability of being at a given point than at an adjacent point, it would need to have infinite speed. This is seen as $p = i\hbar \left(\frac{\partial\Psi}{\partial x} \right) \rightarrow \infty$ at a discontinuity. Another problem is that the second derivative must exist in order to satisfy the Schrödinger equation.

7. (a) The wave function does not satisfy condition 3. The derivative of the wave function is not continuous at $x = 0$.
 (b) Based on (a) the wave function cannot be realized physically.
 (c) Very close to $x = 0$ we could modify the function so that its derivative is continuous. If we do so just in the neighborhood of $x = 0$, we need not change the function elsewhere.

8.

$$\bar{x} = \frac{3.4 + 3.9 + 5.2 + 4.7 + 4.1 + 3.8 + 3.9 + 4.7 + 4.1 + 4.5 + 3.8 + 4.5 + 4.8 + 3.9 + 4.4}{15}$$

$$= 4.247$$

$$\langle x^2 \rangle =$$

$$\frac{3.4^2 + 3.9^2 + 5.2^2 + 4.7^2 + 4.1^2 + 3.8^2 + 3.9^2 + 4.7^2 + 4.1^2 + 4.5^2 + 3.8^2 + 4.5^2 + 4.8^2 + 3.9^2 + 4.4^2}{15}$$

$$\langle x^2 \rangle = 18.254$$

The standard deviation is

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N}} = \sqrt{\frac{\sum (x_i^2 - 2x_i\bar{x} + \bar{x}^2)}{N}} = \sqrt{\frac{\sum x_i^2}{N} - 2\bar{x} \frac{\sum x_i}{N} + \frac{\sum \bar{x}^2}{N}}. \text{ Look at the three}$$

terms in the sum. The first is just $\langle x^2 \rangle$. The second is $-2\bar{x}(\bar{x}) = -2\bar{x}^2$. The third term is

$$\frac{\sum \bar{x}^2}{N} = \frac{N\bar{x}^2}{N} = \bar{x}^2. \text{ Putting the results together } \sigma = \sqrt{\langle x^2 \rangle - 2\bar{x}^2 + \bar{x}^2} = \sqrt{\langle x^2 \rangle - \bar{x}^2}. \text{ For}$$

$$\text{the data given we have } \sigma = \sqrt{\langle x^2 \rangle - \bar{x}^2} = \sqrt{18.254 - (4.247)^2} = 0.466.$$

9. If V is independent of time, then we can use the time-independent Schrödinger equation.

Then by Equation (6.15) $\Psi^* \Psi = \psi^*(x) \psi(x) e^{-i\omega t} e^{i\omega t} = \psi^*(x) \psi(x)$. Then

$$\int \Psi^* \Psi dx = \int \psi^*(x) \psi(x) dx \text{ which is independent of time.}$$

10. Using the Euler relations between exponential and trig functions, we find

$$\psi = A(e^{ix} + e^{-ix}) = 2A \cos(x).$$

$$\text{Normalization: } \int_{-\pi}^{\pi} \psi^* \psi dx = 4A^2 \int_{-\pi}^{\pi} \cos^2(x) dx = 4A^2 \pi = 1. \text{ Thus } A = \frac{1}{2\sqrt{\pi}}.$$

- (a) The probability of being in the interval $[0, \pi/8]$ is

$$P = \int_0^{\pi/8} \psi^* \psi dx = \frac{1}{\pi} \int_0^{\pi/8} \cos^2(x) dx = \frac{1}{\pi} \left(\frac{x}{2} + \frac{1}{4} \sin(2x) \right) \Bigg|_0^{\pi/8}$$

$$= \frac{1}{16} + \frac{1}{4\pi\sqrt{2}} = 0.119.$$

(b) The probability of being in the interval $[0, \pi/4]$ is

$$P = \int_0^{\pi/4} \psi^* \psi dx = \frac{1}{\pi} \int_0^{\pi/4} \cos^2(x) dx = \frac{1}{\pi} \left(\frac{x}{2} + \frac{1}{4} \sin(2x) \right) \Big|_0^{\pi/4}$$

$$= \frac{1}{8} + \frac{1}{4\pi} = 0.205.$$

11. $\int_0^{\pi} \psi^* \psi dx = A^2 \int_0^{\pi} \sin^2(x) dx = A^2 \frac{\pi}{2} = 1$ so $A = \sqrt{\frac{2}{\pi}}$.

(a) The probability of being in the interval $[0, \pi/4]$ is

$$P = \int_0^{\pi/4} \psi^* \psi dx = \frac{2}{\pi} \int_0^{\pi/4} \sin^2(x) dx = \frac{2}{\pi} \left(\frac{x}{2} - \frac{1}{4} \sin(2x) \right) \Big|_0^{\pi/4}$$

$$= \frac{2}{\pi} \left(\frac{\pi}{8} - \frac{1}{4} \right) = \frac{1}{4} - \frac{1}{2\pi} = 0.091$$

(b) The probability of being in the interval $[0, \pi/2]$ is

$$P = \int_0^{\pi/2} \psi^* \psi dx = \frac{2}{\pi} \int_0^{\pi/2} \sin^2(x) dx = \frac{2}{\pi} \left(\frac{x}{2} - \frac{1}{4} \sin(2x) \right) \Big|_0^{\pi/2}$$

$$= \frac{2}{\pi} \left(\frac{\pi}{4} - 0 \right) = \frac{1}{2} = 0.50.$$

12. $E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}; \quad E_{n+1} = \frac{(n+1)^2 \pi^2 \hbar^2}{2mL^2};$

$$\Delta E_n = E_{n+1} - E_n = \frac{\pi^2 \hbar^2}{2mL^2} [(n+1)^2 - n^2] = \frac{\pi^2 \hbar^2}{2mL^2} (2n+1).$$

Computing specific values:

$$\Delta E_1 = \frac{\pi^2 \hbar^2}{2mL^2} (3); \quad \Delta E_8 = \frac{\pi^2 \hbar^2}{2mL^2} (17); \quad \Delta E_{800} = \frac{\pi^2 \hbar^2}{2mL^2} (1601)$$

13. The wave function for the n th level is $\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$ so the average value of the square of the wave function is

$$\langle \psi_n^2(x) \rangle = \frac{\int_0^L \psi^* \psi dx}{\int_0^L dx} = \frac{1}{L} \int_0^L \psi^* \psi dx = \frac{2}{L^2} \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = \frac{2}{L^2} \frac{L}{2} = \frac{1}{L}.$$

This result for the

average value of the square of the wave function is independent of n and is the same as the classical probability. The classical probability is uniform throughout the box, but this is not so in the quantum mechanical case which is $\frac{2}{L} \sin^2(k_n x)$.

14. (a) We know the energy values from Equation (6.35). The energy value E_n is proportional to n^2 where n is the quantum number. If the ground state energy is 4.3 eV ,

then the next three levels correspond to: $4E_1 = 17.2 \text{ eV}$ for $n = 2$; $9E_1 = 38.7 \text{ eV}$ for $n = 3$; and $16E_1 = 68.8 \text{ eV}$ for $n = 4$.

(c) The wave functions and energy levels will be like those shown in Figure 6.3.

15. (a) Starting with Equation (6.35) and using the electron mass and the length given, we have:

$$E_n = n^2 \frac{\pi^2 \hbar^2}{2mL^2} = n^2 \frac{\pi^2 (\hbar c)^2}{2(mc^2)L^2} = n^2 \frac{\pi^2 (197.3 \text{ eV} \cdot \text{nm})^2}{2(5.11 \times 10^5 \text{ eV})(2000 \text{ nm})^2}$$

$$= n^2 (9.40 \times 10^{-8} \text{ eV})$$

Then the three lowest energy levels are: $E_1 = 9.40 \times 10^{-8} \text{ eV}$; $E_2 = 1.88 \times 10^{-7} \text{ eV}$; and $E_3 = 2.82 \times 10^{-7} \text{ eV}$.

(b) Average kinetic energy equals $\frac{3}{2}kT = \frac{3}{2} \left[(1.381 \times 10^{-23} \text{ J/K}) \left(\frac{1 \text{ eV}}{1.602 \times 10^{-19} \text{ J}} \right) \right] (13 \text{ K})$

which equals $1.68 \times 10^{-3} \text{ eV}$. Substitute this value into the equation above as E_n and solve for n . We find $n = 134$.

16. For this non-relativistic speed we have

$$E = \frac{1}{2}mv^2 = \frac{1}{2}(mc^2)\beta^2 = \frac{1}{2}(511000 \text{ eV}/c^2)(1.25 \times 10^{-4}c)^2 = 0.003992 \text{ eV}.$$

Using $E = \frac{n^2 \hbar^2}{8mL^2}$, we find

$$n^2 = \frac{8mL^2 E}{\hbar^2} = \frac{8mc^2 L^2 E}{\hbar^2 c^2} = \frac{8(511 \times 10^3 \text{ eV})(48.5 \text{ nm})^2 (0.003992 \text{ eV})}{(1240 \text{ eV} \cdot \text{nm})^2} = 24.97 \text{ and}$$

therefore $n = 5$.

17. The ground-state wave function is $\psi_1 = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$.

$$P_1 = \int_0^{L/3} \psi_1^2 dx = \frac{2}{L} \int_0^{L/3} \sin^2\left(\frac{\pi x}{L}\right) dx = \frac{2}{L} \left(\frac{x}{2} - \frac{L}{4\pi} \sin\left(\frac{2\pi x}{L}\right) \right) \Bigg|_0^{L/3}$$

$$= 2 \left(\frac{1}{6} - \frac{\sqrt{3}}{8\pi} \right) = 0.1955$$

$$P_2 = \int_{L/3}^{2L/3} \psi_1^2 dx = \frac{2}{L} \left(\frac{x}{2} - \frac{L}{4\pi} \sin\left(\frac{2\pi x}{L}\right) \right) \Bigg|_{L/3}^{2L/3}$$

$$= 2 \left(\frac{1}{6} + \frac{\sqrt{3}}{4\pi} \right) = 0.6090$$

$$P_3 = \int_{2L/3}^L \psi_1^2 dx = \frac{2}{L} \left(\frac{x}{2} - \frac{L}{4\pi} \sin\left(\frac{2\pi x}{L}\right) \right) \Big|_{2L/3}^L$$

$$= 2 \left(\frac{1}{6} - \frac{\sqrt{3}}{8\pi} \right) = 0.1955$$

Notice that $P_1 + P_2 + P_3 = 1$ as required.

18. The first excited state has a wave function $\psi_2 = \sqrt{\frac{2}{L}} \sin(kx)$ with $k = \frac{2\pi}{L}$.

$$P_1 = \int_0^{L/3} \psi_2^2 dx = \frac{2}{L} \left(\frac{x}{2} - \frac{1}{4k} \sin(2kx) \right) \Big|_0^{L/3} = 2 \left(\frac{1}{6} + \frac{\sqrt{3}}{16\pi} \right) = 0.402$$

Similarly $P_2 = 2 \left(\frac{1}{6} - \frac{\sqrt{3}}{8\pi} \right) = 0.196$ and $P_3 = P_1 = 0.402$. Notice that $P_1 + P_2 + P_3 = 1$.

$$19. E_1 = \frac{h^2}{8mL^2} = \frac{h^2 c^2}{8mc^2 L^2} = \frac{(1240 \text{ eV} \cdot \text{nm})^2}{8(511 \times 10^3 \text{ eV})(10^{-5} \text{ nm})^2} = 3.76 \text{ GeV}.$$

Then $E_2 = 4E_1 = 15.05 \text{ GeV}$ and $\Delta E = E_2 - E_1 = 11.3 \text{ GeV}$. This is much larger than any energy observed in nuclear process; the electron is not in the nucleus.

20. (a) As in the previous problem

$$E_1 = \frac{h^2 c^2}{8mc^2 L^2} = \frac{(1240 \text{ eV} \cdot \text{nm})^2}{8(938.27 \times 10^6 \text{ eV})(1.48 \times 10^{-5} \text{ nm})^2} = 935 \text{ keV}$$

$$(b) E_1 = \frac{h^2 c^2}{8mc^2 L^2} = \frac{(1240 \text{ eV} \cdot \text{nm})^2}{8(3727 \times 10^6 \text{ eV})(1.48 \times 10^{-5} \text{ nm})^2} = 235 \text{ keV}$$

21. As in previous problems the ground state energy is

$$E_1 = \frac{h^2}{8mL^2} = \frac{h^2 c^2}{8mc^2 L^2} = \frac{(1240 \text{ eV} \cdot \text{nm})^2}{8(511 \times 10^3 \text{ eV})(0.70 \text{ nm})^2} = 0.7676 \text{ eV}.$$

The other energy levels are $E_n = n^2 E_1$: $E_2 = 4E_1 = 3.07 \text{ eV}$; $E_3 = 9E_1 = 6.91 \text{ eV}$; $E_4 = 16E_1 = 12.28 \text{ eV}$. The allowed photon energies are:

$$E_4 - E_3 = 5.37 \text{ eV} \quad E_4 - E_2 = 9.21 \text{ eV} \quad E_4 - E_1 = 11.5 \text{ eV}$$

$$E_3 - E_2 = 3.84 \text{ eV} \quad E_3 - E_1 = 6.14 \text{ eV} \quad E_2 - E_1 = 2.30 \text{ eV}$$

22. Lacking an explicit equation for finite square well energies, we will approximate using the infinite square well formula. In order to contain three energy levels the depth of the

well must be at least $E = \frac{n^2 h^2}{8mL^2} = \frac{9h^2}{8mL^2}$. Evaluating numerically with the given mass

$$E = \frac{9h^2}{8mL^2} = \frac{9h^2 c^2}{8mc^2 L^2} = \frac{9(1240 \text{ eV} \cdot \text{nm})^2}{8(1.88 \times 10^9 \text{ eV})(3.00 \times 10^{-6} \text{ nm})^2} = 102 \text{ MeV}.$$

23. (a) The wavelengths are longer for the finite well, because the wave functions can leak outside the box.
 (b) Generally shorter wavelengths correspond to higher energies, so we expect energies to be lower for the finite well.
 (c) Generally the number of bound states is limited by the depth of the well. We expect no bound states for $E > V_0$.

24. Using the same notation as the text, from the boundary condition $\psi_I(x=0) = \psi_{II}(x=0)$ we have $Ae^0 = Ce^0 + De^0$ or $A = C + D$. From the condition $\psi'_I(x=0) = \psi'_{II}(x=0)$ we have $\alpha A = ikC - ikD$. Solving this last expression for A and combining with the first boundary condition gives $C + D = \frac{ik}{\alpha} C - \frac{ik}{\alpha} D$ or after rearranging $\frac{C}{D} = \frac{ik + \alpha}{ik - \alpha}$.

25. As in the previous problem matching the wave function at the boundary gives

$$Ce^{ikL} + De^{-ikL} = Be^{-\alpha L} \text{ and matching the first derivative gives}$$

$$ikCe^{ikL} - ikDe^{-ikL} = -\alpha Be^{-\alpha L}. \text{ Eliminating } B \text{ from these two equations gives}$$

$$Ce^{ikL} + De^{-ikL} = \frac{1}{\alpha} (ikDe^{-ikL} - ikCe^{ikL}). \text{ Thus } \frac{C}{D} = \frac{ik - \alpha}{ik + \alpha} e^{-2ikL}.$$

26. $E = \frac{\pi^2 \hbar^2}{2mL^2} (n_1^2 + n_2^2 + n_3^2) = E_0 (n_1^2 + n_2^2 + n_3^2)$ where $E_0 = \frac{\pi^2 \hbar^2}{2mL^2}$. Then the second, third, fourth, and fifth levels are:

$$E_2 = (2^2 + 1^2 + 1^2) E_0 = 6E_0 \quad (\text{degenerate})$$

$$E_3 = (2^2 + 2^2 + 1^2) E_0 = 9E_0 \quad (\text{degenerate})$$

$$E_4 = (3^2 + 1^2 + 1^2) E_0 = 11E_0 \quad (\text{degenerate})$$

$$E_5 = (2^2 + 2^2 + 2^2) E_0 = 12E_0 \quad (\text{not degenerate})$$

27. In general we have $\psi(x) = A \sin\left(\frac{n_1 \pi x}{L}\right) \sin\left(\frac{n_2 \pi y}{L}\right) \sin\left(\frac{n_3 \pi z}{L}\right)$.

For $\psi_2(x)$ we can have $(n_1, n_2, n_3) = (1, 1, 2)$ or $(1, 2, 1)$ or $(2, 1, 1)$.

For $\psi_3(x)$ we can have $(n_1, n_2, n_3) = (1, 2, 2)$ or $(2, 2, 1)$ or $(2, 1, 2)$.

For $\psi_4(x)$ we can have $(n_1, n_2, n_3) = (1, 1, 3)$ or $(1, 3, 1)$ or $(3, 1, 1)$.

For $\psi_5(x)$ we can have $(n_1, n_2, n_3) = (2, 2, 2)$.

28. We must normalize by evaluating the triple integral of $\psi^* \psi$: $\iiint \psi^* \psi dx dy dz = 1$ with $\psi(x, y, z)$ given by Equation (6.47) in the text. We can evaluate the iterated triple integral

$$A^2 \int_0^L \sin^2\left(\frac{\pi x}{L}\right) dx \int_0^L \sin^2\left(\frac{\pi y}{L}\right) dy \int_0^L \sin^2\left(\frac{\pi z}{L}\right) dz = A^2 \left(\frac{L}{2}\right)^3 = 1. \text{ Solving for } A \text{ we find}$$

$$A = \left(\frac{2}{L}\right)^{3/2}.$$

29. Taking the derivatives we find $\nabla^2 \psi = -(k_1^2 + k_2^2 + k_3^2) \psi$ so the Schrödinger equation

$$\text{becomes } -\frac{\hbar^2}{2m} \nabla^2 \psi = \frac{\hbar^2}{2m} (k_1^2 + k_2^2 + k_3^2) \psi = E \psi. \text{ From the boundary conditions } k_i = \frac{n_i \pi}{L_i}$$

so $E = \frac{\hbar^2 \pi^2}{2m} \left(\frac{n_1^2}{L_1^2} + \frac{n_2^2}{L_2^2} + \frac{n_3^2}{L_3^2} \right)$. The three quantum numbers come directly from the three boundary conditions.

30. One possible solution is $n_1 = 2, n_2 = 1, n_3 = 1$. Letting either of the other two quantum numbers be equal to 2 (with the remaining two equal to 1) gives the same result for A . For the choice given here, the wave function becomes

$$\psi = A \sin\left(\frac{2\pi x}{L}\right) \sin\left(\frac{\pi y}{L}\right) \sin\left(\frac{\pi z}{L}\right). \text{ To normalize, integrate } \psi^* \psi = \psi^2 \text{ over the entire}$$

$$\text{box } A^2 \int_0^L \sin^2\left(\frac{2\pi x}{L}\right) dx \int_0^L \sin^2\left(\frac{\pi y}{L}\right) dy \int_0^L \sin^2\left(\frac{\pi z}{L}\right) dz = A^2 \left(\frac{L}{2}\right)^3 = 1 \text{ Solving for } A, \text{ we}$$

find $A = \left(\frac{2}{L}\right)^{3/2}$. It is interesting that this is the same result as for the ground state (Problem 28).

31. From Equation (6.49), $E = \frac{\pi^2 \hbar^2}{2m} \left(\frac{n_1^2}{L^2} + \frac{n_2^2}{4L^2} + \frac{n_3^2}{16L^2} \right)$. For the ground state:

$$n_1 = n_2 = n_3 = 1. \text{ Thus } E_{\text{gs}} = \frac{\pi^2 \hbar^2}{2mL^2} \left(1 + \frac{1}{4} + \frac{1}{16} \right) = \frac{21\pi^2 \hbar^2}{32mL^2}. \text{ The least amount of energy}$$

$$\text{we can add is accomplished by making } n_3 = 2. \text{ Then } E_1 = \frac{\pi^2 \hbar^2}{2mL^2} \left(1 + \frac{1}{4} + \frac{1}{4} \right) = \frac{3\pi^2 \hbar^2}{4mL^2}. \text{ This}$$

state is not degenerate. For the next energy level, make $n_3 = 3$:

$$E_2 = \frac{\pi^2 \hbar^2}{2mL^2} \left(1 + \frac{1}{4} + \frac{9}{16} \right) = \frac{29\pi^2 \hbar^2}{32mL^2}, \text{ also non-degenerate. For the next level, take } n_1 = 1,$$

$$n_2 = 2, \text{ and } n_3 = 1: E_3 = \frac{\pi^2 \hbar^2}{2mL^2} \left(1 + 1 + \frac{1}{16} \right) = \frac{33\pi^2 \hbar^2}{32mL^2}. \text{ This state is also non-degenerate.}$$

Notice that this energy is less than we would obtain with $n_1 = 1, n_2 = 1$, and $n_3 = 4$.

32. $\psi_0(x)$ has these features: it is symmetric about $x = 0$; it has a maximum at $x = 0$ because the wave function must tend toward zero for $x \rightarrow \pm\infty$; there is no node in the ground state; the wave function decreases exponentially where $V > E$.

33. $\Delta E_n = E_{n+1} - E_n = \left(n + 1 + \frac{1}{2}\right)\hbar\omega - \left(n + \frac{1}{2}\right)\hbar\omega = \hbar\omega$ for all n . This is true for all n , and there is no restriction on the number of levels.

34. Normalization $1 = \int_{-\infty}^{\infty} \psi^* \psi dx = A^2 \int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} dx = 2A^2 \int_0^{\infty} x^2 e^{-\alpha x^2} dx = 2A^2 \frac{1}{4\alpha} \sqrt{\frac{\pi}{\alpha}}$. Solving

for A we find $A = \sqrt{2\pi^{-1/4} \alpha^{3/4}}$. $\langle x \rangle = A^2 \int_{-\infty}^{\infty} x^3 e^{-\alpha x^2} dx = 0$ because the integrand is odd

over symmetric limits. $\langle x^2 \rangle = A^2 \int_{-\infty}^{\infty} x^4 e^{-\alpha x^2} dx = \frac{3}{4} A^2 \pi^{1/2} \alpha^{-5/2} = \frac{3}{2\alpha}$;

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{3}{2\alpha}}$$

35. The energy is related to the frequency of oscillation by:

$$\begin{aligned} E &= \left(n + \frac{1}{2}\right)\hbar\omega = \left(n + \frac{1}{2}\right)hf = (4.136 \times 10^{-15} \text{ eV}\cdot\text{s})(1 \times 10^{13} \text{ s}^{-1})\left(n + \frac{1}{2}\right) \\ &= (4.136 \times 10^{-2} \text{ eV})\left(n + \frac{1}{2}\right) \end{aligned}$$

For the harmonic oscillator $\omega^2 = \frac{k}{m}$ so

$$k = \omega^2 m = 4\pi^2 f^2 m = 4\pi^2 (1 \times 10^{13} \text{ s}^{-1})^2 (2.32 \times 10^{-26} \text{ kg}) = 91.6 \text{ N/m}.$$

36. Taking the second derivative of ψ for the Schrödinger equation:

$$\frac{d\psi}{dx} = 5A\alpha x e^{-\alpha x^2/2} - 2\alpha^2 A x^3 e^{-\alpha x^2/2};$$

$$\frac{d^2\psi}{dx^2} = (5\alpha - 5\alpha^2 x^2 - 6\alpha^2 x^2 + 2\alpha^3 x^4) A e^{-\alpha x^2/2}.$$

Starting with Equation (6.56), and with the wave function given in the problem, we have:

$$\begin{aligned} \frac{d^2\psi}{dx^2} &= (\alpha^2 x^2 - \beta)\psi = (\alpha^2 x^2 - \beta)A(2\alpha x^2 - 1)e^{-\alpha x^2/2} \\ &= (2\alpha^3 x^4 + x^2(-2\alpha\beta - \alpha^2) + \beta)Ae^{-\alpha x^2/2} \end{aligned}$$

Matching our two values of $d^2\psi/dx^2$ we see that the Schrödinger equation can only be

satisfied if $\beta = 5\alpha$. Then $\frac{2mE}{\hbar^2} = 5\sqrt{\frac{mk}{\hbar^2}}$ or $E = \frac{5}{2}\hbar\omega$. This is the expected result,

because the wave function contains a second-order polynomial (in x), and with $n = 2$ we expect $E = \left(n + \frac{1}{2}\right)\hbar\omega = \frac{5}{2}\hbar\omega$.

37. By symmetry $\langle p \rangle = 0$. Setting the ground state energy equal to $\frac{\langle p^2 \rangle}{2m}$ we find

$\frac{\langle p^2 \rangle}{2m} = \frac{1}{2}\hbar\omega$ which gives $\langle p^2 \rangle = \hbar\omega m$. Note, however, that a detailed calculation gives $\langle p^2 \rangle = \frac{1}{2}\hbar\omega m$. The factor of one-half is evidently the difference between the kinetic energy and the total energy, which if taken into account does give the correct result.

38. Taking the derivatives for the Schrödinger equation:

$$\frac{d\psi}{dx} = Ae^{-\alpha x^2/2} - A\alpha x^2 e^{-\alpha x^2/2}; \quad \frac{d^2\psi}{dx^2} = -3A\alpha x e^{-\alpha x^2/2} + A\alpha^2 x^3 e^{-\alpha x^2/2} = (\alpha^2 x^2 - 3\alpha)\psi.$$

Combining Equation (6.56) with this, we see $\frac{d^2\psi}{dx^2} = (\alpha^2 x^2 - \beta)\psi = (\alpha^2 x^2 - 3\alpha)\psi$. Thus

we see that $\beta = 3\alpha$ or $\frac{2mE}{\hbar^2} = 3\sqrt{\frac{mk}{\hbar^2}}$. Solving for the energy, we find

$$E = \frac{3}{2}\hbar\sqrt{\frac{k}{m}} = \frac{3}{2}\hbar\omega.$$

39. The classical frequency for a two-particle oscillator is [see Chapter 10, Equation (10.4)]

$\omega = \sqrt{k/\mu} = \sqrt{k(m_1 + m_2)/m_1 m_2} = \sqrt{2k/m}$ since the masses are equal in this case. The energies of the ground state (E_0) and the first three excited states are given by

$E_n = \left(n + \frac{1}{2}\right)\hbar\omega$ so the possible transitions (from E_3 to E_2 , E_3 to E_1 , etc. are $\Delta E = \hbar\omega$, $2\hbar\omega$, and $3\hbar\omega$. Specifically these calculations give:

$$\hbar\omega = \hbar\sqrt{\frac{2k}{m}} = (6.582 \times 10^{-16} \text{ eV}\cdot\text{s})\sqrt{\frac{2(1.1 \times 10^3 \text{ N/m})}{1.673 \times 10^{-27} \text{ kg}}} = 0.755 \text{ eV with a wavelength,}$$

$$\lambda = \frac{hc}{E} = \frac{1240 \text{ eV}\cdot\text{nm}}{0.755 \text{ eV}} = 1640 \text{ nm}.$$

$$2\hbar\omega = 2(6.582 \times 10^{-16} \text{ eV}\cdot\text{s})\sqrt{\frac{2(1.1 \times 10^3 \text{ N/m})}{1.673 \times 10^{-27} \text{ kg}}} = 1.51 \text{ eV}$$

$$\lambda = \frac{hc}{E} = \frac{1240 \text{ eV}\cdot\text{nm}}{1.51 \text{ eV}} = 821 \text{ nm}$$

$$3\hbar\omega = 3(6.582 \times 10^{-16} \text{ eV} \cdot \text{s}) \sqrt{\frac{2(1.1 \times 10^3 \text{ N/m})}{1.673 \times 10^{-27} \text{ kg}}} = 2.26 \text{ eV}$$

$$\lambda = \frac{hc}{E} = \frac{1240 \text{ eV} \cdot \text{nm}}{2.26 \text{ eV}} = 549 \text{ nm}$$

40. The kinetic energy is (see Chapter 9)

$$K = \frac{3}{2}kT = \frac{3}{2}(8.617 \times 10^{-5} \text{ eV/K})(12000 \text{ K}) = 1.551 \text{ eV}. \text{ Assume a square-top potential of}$$

$$\text{height } V_0 = \frac{q_1 q_2}{4\pi\epsilon_0 r} = \frac{6e^2}{4\pi\epsilon_0 r} = \frac{6(1.440 \text{ eV} \cdot \text{nm})}{(1.2 \times 10^{-6} \text{ nm})(12)^{1/3}} = 3.145 \text{ MeV where we have used the}$$

fact that the radius of a nucleus is approximately $1.2A^{1/3} \text{ fm}$ (see Chapter 12). For the width of the potential barrier use twice the radius or

$$L = 2(1.2 \times 10^{-6} \text{ nm})(12)^{1/3} = 5.49 \times 10^{-6} \text{ nm}. \text{ Then}$$

$$\kappa = \frac{\sqrt{2mc^2(V_0 - E)}}{\hbar c} = \frac{\sqrt{2(938.27 \times 10^6 \text{ eV})[3.145 \times 10^6 \text{ eV} - 1.551 \text{ eV}]}}{197.3 \text{ eV} \cdot \text{nm}}$$

$$= 3.89 \times 10^5 \text{ nm}^{-1}$$

$$\kappa L = (3.89 \times 10^5 \text{ nm}^{-1})(5.49 \times 10^{-6} \text{ nm}) = 2.135$$

$$T = \left(1 + \frac{(3.145 \times 10^6 \text{ eV})^2 \sinh^2(2.135)}{4(1.551 \text{ eV})(3.145 \times 10^6 \text{ eV} - 1.551 \text{ eV})} \right)^{-1} = 1.14 \times 10^{-7}$$

$$41. (a) \quad p = \sqrt{2m(E - V_0)}; \quad \lambda = \frac{h}{p} = \frac{h}{\sqrt{2m(E - V_0)}}; \quad K = E - V_0$$

$$(b) \quad p = \sqrt{2m(E + V_0)}; \quad \lambda = \frac{h}{p} = \frac{h}{\sqrt{2m(E + V_0)}}; \quad K = E + V_0$$

$$42. \text{ In each case } \kappa L \gg 1 \text{ so we can use } T = 16 \frac{E}{V_0} \left(1 - \frac{E}{V_0} \right) e^{-2\kappa L} \text{ where}$$

$$\kappa = \frac{\sqrt{2mc^2(V_0 - E)}}{\hbar c} = \frac{(2(3727 \times 10^6 \text{ eV})(10 \times 10^6 \text{ eV}))^{1/2}}{197.4 \text{ eV} \cdot \text{nm}} = 1.38 \times 10^{15} \text{ m}^{-1}.$$

(a) With $L = 1.3 \times 10^{-14} \text{ m}$,

$$T_a = 16 \frac{5 \text{ MeV}}{15 \text{ MeV}} \left(1 - \frac{5 \text{ MeV}}{15 \text{ MeV}} \right) e^{-2(1.38 \times 10^{15} \text{ m}^{-1})(1.3 \times 10^{-14} \text{ m})} = 9.3 \times 10^{-16}.$$

(b) With $V_0 = 30 \text{ MeV}$,

$$\kappa = \frac{\sqrt{2mc^2(V_0 - E)}}{\hbar c} = \frac{(2(3727 \times 10^6 \text{ eV})(25 \times 10^6 \text{ eV}))^{1/2}}{197.4 \text{ eV} \cdot \text{nm}} = 2.19 \times 10^{15} \text{ m}^{-1}.$$

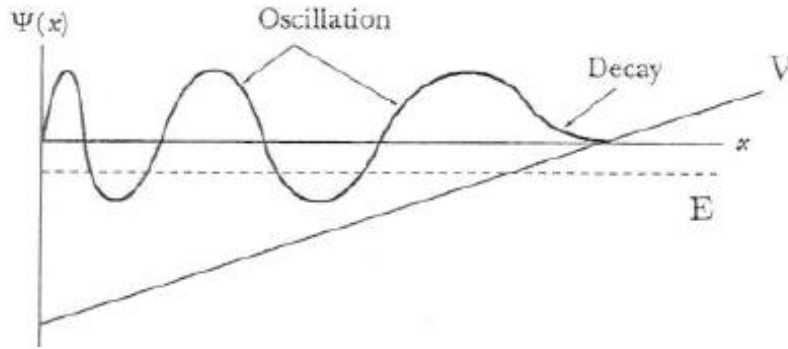
$$T_b = 16 \frac{5 \text{ MeV}}{30 \text{ MeV}} \left(1 - \frac{5 \text{ MeV}}{30 \text{ MeV}}\right) e^{-2(2.19 \times 10^{15} \text{ m}^{-1})(1.3 \times 10^{-14} \text{ m})} = 4.2 \times 10^{-25}.$$

(c) With $V_0 = 15 \text{ MeV}$ we return to the original value of κ , but now $L = 2.6 \times 10^{-14} \text{ m}$ and

$$T_c = 16 \frac{5 \text{ MeV}}{15 \text{ MeV}} \left(1 - \frac{5 \text{ MeV}}{15 \text{ MeV}}\right) e^{-2(1.38 \times 10^{15} \text{ m}^{-1})(2.6 \times 10^{-14} \text{ m})} = 2.4 \times 10^{-31}.$$

By comparison $T_a > T_b > T_c$.

43. When $E > V$ the wave function is oscillating, with a longer wavelength as $E - V$ decreases. Then when $E < V$ the wave function decays.



44. In general for $E > V_0$ we have $R = 1 - T = 1 - \left[1 + \frac{V_0^2 \sin^2(k_2 L)}{4E(E - V_0)}\right]^{-1}$. If $E \gg V_0$ then

$4E(E - V_0) \approx 4E^2$. From the binomial theorem $(1 + x)^{-1} \approx 1 - x$ for small x and

$$R \approx 1 - \left[1 - \frac{V_0^2 \sin^2(k_2 L)}{4E^2}\right] = \frac{V_0^2 \sin^2(k_2 L)}{4E^2} \text{ which can be written as } R \approx \left(\frac{V_0 \sin(k_2 L)}{2E}\right)^2.$$

$$45. \quad T = \left[1 + \frac{V_0^2 \sin^2(k_2 L)}{4E(E - V_0)}\right]^{-1}$$

(a) To obtain $T = 1$ we require $\sin(k_2 L) = 0$. Except for the trivial solution $L = 0$, this occurs whenever $k_2 L = n\pi$ with n an integer. Letting $n = 1$ we find

$$L = \frac{\pi}{k_2} = \frac{\pi \hbar}{\sqrt{2m(E - V_0)}} = \frac{1}{2} \frac{hc}{\sqrt{2mc^2(E - V_0)}} = \frac{1}{2} \frac{1240 \text{ eV} \cdot \text{nm}}{\sqrt{2(511 \times 10^3 \text{ eV})(7.8 \text{ eV})}} = 0.220 \text{ nm}$$

Any integer multiple of this value will work.

(b) For maximum reflection $\sin^2(k_2 L) = 1$ or $L = \frac{n\pi}{2k_2}$ for any odd integer n . From the result of (a) we see that the first maximum is with L equal to half the value of L for the first minimum, or $L = 0.110 \text{ nm}$.

$$46. \quad \kappa = \frac{\sqrt{2mc^2(V_0 - E)}}{\hbar c} = \frac{\left[2(511 \times 10^3 \text{ eV})(1.5 \text{ eV})\right]^{1/2}}{197.4 \text{ eV} \cdot \text{nm}} = 6.27 \text{ nm}^{-1}$$

With a probability of 2×10^{-4} we know $\kappa L \gg 1$ and we can use

$$T = 16 \frac{E}{V_0} \left(1 - \frac{E}{V_0}\right) e^{-2\kappa L} = 16 \frac{1}{2.5} \left(1 - \frac{1}{2.5}\right) e^{-2\kappa L} = 3.84 e^{-2\kappa L} = 2 \times 10^{-4}.$$

Solving for L :

$$L = \frac{\ln(1.92 \times 10^4)}{2(6.27 \times 10^9 \text{ m}^{-1})} = 7.86 \times 10^{-10} \text{ m}. \text{ Now using the proton mass:}$$

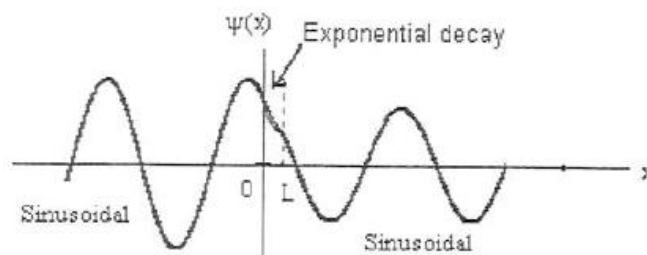
$$\kappa = \frac{\sqrt{2mc^2(V_0 - E)}}{\hbar c} = \frac{\left[2(938.27 \times 10^6 \text{ eV})(1.5 \text{ eV})\right]^{1/2}}{197.4 \text{ eV} \cdot \text{nm}} = 268.8 \text{ nm}^{-1}.$$

$$T = 3.84 e^{-2\kappa L} = 3.84 e^{-2(268.8 \times 10^9 \text{ m}^{-1})(7.86 \times 10^{-10} \text{ m})} = 1.2 \times 10^{-183}.$$

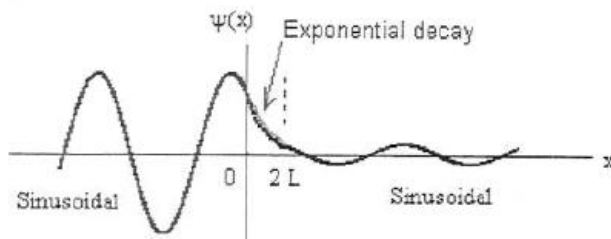
The proton's probability is much lower!

47. The sketch of the potential will be like Figure 6.12. Since the question mentions tunneling, the assumption is that the total energy is less than the potential. Therefore the wave function will be sinusoidal on either side of the barrier with an exponential decay in the barrier region. Each sketch will be similar to Figure 6.15. From Equation (6.70) we can see that in part (b) with a barrier twice as wide, the exponential factor will be markedly smaller (a ratio of $e^{-2} = 0.135$ while doubling the barrier height in c) will reduce the transmission coefficient as compared to (a) by less than $1/2$.

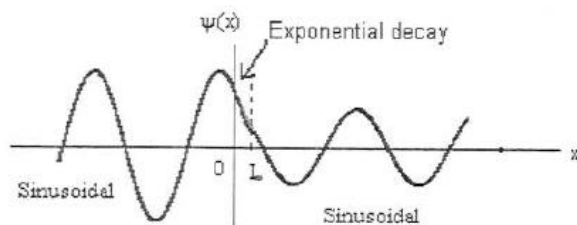
(a)



(b)



(c)



48. Starting with $T \approx 2e^{-2\kappa L}$ and $\kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar} = \frac{\sqrt{2mc^2(V_0 - E)}}{\hbar c}$, we find:

$$(a) \kappa = \frac{\sqrt{2(511 \times 10^3 \text{ eV})(6.4 \text{ eV} - 1.4 \text{ eV})}}{197.33 \text{ eV} \cdot \text{nm}} = 11.5 \text{ nm}^{-1}$$

$$\text{Then } T \approx e^{-2\kappa L} = 2e^{-2(11.5 \text{ nm}^{-1})(2.8 \text{ nm})} = 2e^{-64.4} = 2.1 \times 10^{-28}$$

$$(b) \kappa = \frac{\sqrt{2(3727 \times 10^6 \text{ eV})(19.2 \times 10^6 \text{ eV} - 4.4 \times 10^6 \text{ eV})}}{197.33 \text{ eV} \cdot \text{nm}} = 1.683 \times 10^6 \text{ nm}^{-1}$$

Then $T \approx e^{-2\kappa L} = 2e^{-2(1.683 \times 10^6 \text{ m}^{-1})(7.4 \times 10^{-15} \text{ m})} = 2e^{-22.6} = 3.04 \times 10^{-11}$. For values of $\kappa L \gg 1$, Equation (6.70) is a good approximation for Equation (6.67) and Equation (6.73) gives at least an estimate of the transmission that is to within an order of magnitude.

49. The second derivative of the wave function with respect to x is

$$\frac{d^2\psi}{dx^2} = -k^2 A \sin(kx) - k^2 B \sin(kx) = -k^2 \psi. \text{ Then the time-independent Schrödinger}$$

equation [Equation (6.13)] becomes $-\frac{\hbar^2 k^2}{2m} \psi = \frac{\hbar^2 k^2}{2m} \psi = E \psi$. We know that for a free particle $E = p^2 / 2m = \hbar^2 k^2 / 2m$, which verifies that this is a suitable wave function.

50. (a) We apply the time-dependent Schrödinger equation [Equation (6.1)] with

$V = 0$. The left side is $i\hbar \frac{\partial \Psi}{\partial t} = i\hbar [-\omega A \cos(kx - \omega t) + \omega B \sin(kx - \omega t)]$. The right side is

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} = -\frac{\hbar^2}{2m} [-k^2 A \sin(kx - \omega t) - k^2 B \cos(kx - \omega t)] = \frac{\hbar^2 k^2}{2m} \Psi. \text{ Because of the}$$

complex number i and the extra minus sign on the left side, there is no way the two expressions can be equal. This is not a valid wave function.

(b) Using the Euler relation $e^{i\theta} = \cos \theta + i \sin \theta$, this wave function is equivalent to $\Psi(x, t) = A \cos(kx - \omega t) - i A \sin(kx - \omega t) = A e^{i(kx - \omega t)}$. Example 6.2 verified that this is a valid wave function. You can also substitute the original wave function (containing sine and cosine) into Equation (6.1) to prove that it is valid.

51. As in the text we find $E = \frac{\hbar^2 \pi^2}{2m} \left(\frac{n_1^2}{L_1^2} + \frac{n_2^2}{L_2^2} + \frac{n_3^2}{L_3^2} \right)$ and substituting the given values of L we

find $E = \frac{\hbar^2 \pi^2}{2mL^2} \left(n_1^2 + 2n_2^2 + \frac{n_3^2}{4} \right)$. Letting $E_0 = \hbar^2 \pi^2 / 2mL^2$ we have:

$$E_1 = E_0 \left(1 + 2 + \frac{1}{4} \right) = \frac{13}{4} E_0; \quad E_2 = E_0 \left(1 + 2 + \frac{2^2}{4} \right) = 4E_0; \quad E_3 = E_0 \left(1 + 2 + \frac{3^2}{4} \right) = \frac{21}{4} E_0;$$

$$E_4 = E_0 \left(2^2 + 2 + \frac{1}{4} \right) = \frac{25}{4} E_0; \quad E_5 = E_0 \left(1 + 2 + \frac{4^2}{4} \right) = E_0 \left(2^2 + 2 + \frac{2^2}{4} \right) = 7E_0.$$

Of those listed, only E_5 is degenerate.

52. Recognizing this as the infinite square well wave function we see that $k = \frac{3\pi}{a}$ and

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m} = \frac{9\hbar^2 \pi^2}{2m a^2}.$$

53. (a) In general inside the box we have a superposition of sine and cosine functions, but only the sine function satisfies the boundary condition $\psi(0) = 0$, and thus $\psi = A \sin(kx)$.

With $V = 0$ inside the well, $E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$ or $k = \frac{\sqrt{2mE}}{\hbar}$. Outside the well the decaying

exponential is required as explained in section 6.4 of the text, with $E = \frac{\hbar^2 k^2}{2m} + V_0$ which

$$\text{reduces to } \kappa = ik = \frac{\sqrt{2m(V_0 - E)}}{\hbar}.$$

(b) Equating the wavefunctions and first derivatives at $x = L$:

$$A \sin(kL) = B e^{-\kappa L} \text{ and } k A \cos(kL) = -\kappa B e^{-\kappa L}.$$

Dividing the first equation by the second gives $\frac{\tan(kL)}{k} = -\frac{1}{\kappa}$ or $\kappa \tan(kL) = -k$.

54. From the previous problem $\kappa \tan(kL) = -k$ or $\kappa L \tan(kL) = -kL$. Let $\alpha = kL$ and $\beta = \kappa L$ so that

$$\beta \tan \alpha = -\alpha. \quad (1)$$

Now from the value of V_0 given in the problem and the definitions of α and β we have:

$$1 = \frac{2mV_0L^2}{\hbar^2} = \frac{2mEL^2}{\hbar^2} + \frac{2m(V_0 - E)L^2}{\hbar^2} = k^2L^2 + \kappa^2L^2 = \alpha^2 + \beta^2. \quad (2)$$

Solving Equations (1) and (2) numerically we find $\alpha = 0.20$ and $\beta = -0.98$. Then E is

$$\text{given by } E = \frac{\hbar^2 k^2}{2m} = \hbar^2 \frac{(\alpha/L)^2}{2m} = \frac{0.04\hbar^2}{2mL^2}.$$

55. Referring to the solution to the previous problem, we see that only a finite number of solutions to Equation (1) exist up to any particular (finite) value of V_0 . Therefore for any finite V_0 only a finite number of combinations of α and β will satisfy both equations, and the number of bound states is finite.

56. Using the nomenclature of Problem 53

$$\kappa L = \frac{\sqrt{2m(V_0 - E)} L}{\hbar} = \frac{\sqrt{2(939 \times 10^6 \text{ eV})(2.2 \times 10^6 \text{ eV})} (3.5 \times 10^{-6} \text{ nm})}{197.4 \text{ eV} \cdot \text{nm}} = 1.14 \text{ where we}$$

have used the mass of one nucleon, because one nucleon is “bound” by the other. Now $\kappa L = \beta = -\alpha / \tan \alpha$ so $\alpha \approx 2.07 = kL$. Then

$$E = \frac{\hbar^2 k^2}{2m} = \hbar^2 \frac{(\alpha/L)^2}{2m} = \hbar^2 \frac{(2.07/L)^2}{2m} = \frac{(2.07)^2 (197.4 \text{ eV} \cdot \text{nm})^2}{2(939 \times 10^6 \text{ eV})(3.5 \times 10^{-6} \text{ nm})^2} = 7.26 \text{ MeV}$$

This means that $V_0 = 2.2 \text{ MeV} + E = 9.46 \text{ MeV}$. The next solution of the equation

$\beta = -\alpha / \tan \alpha$ is at $\alpha \approx 4.94$, a value that will put $E > V_0$. Therefore there are no excited states.

57. For the one-dimensional well [Equation (6.35)],

$$E_1 = \frac{\pi^2 \hbar^2}{2mL^2} = \frac{\pi^2 (\hbar c)^2}{2mc^2 L^2} = \frac{\pi^2 (197.3 \text{ eV} \cdot \text{nm})^2}{2(5.11 \times 10^5 \text{ eV}) [2(5.292 \times 10^{-2} \text{ nm})]^2} = 33.6 \text{ eV}.$$

This is the same order of magnitude but larger than the electron’s ground-state kinetic energy. The one-dimensional model is a fair but inaccurate representation of the atom.

58. For the cubical well, the ground-state energy is given by Equation (6.51):

$$E = \frac{3\pi^2 \hbar^2}{2mL^2} = \frac{3\pi^2 (\hbar c)^2}{2mc^2 L^2} = \frac{3\pi^2 (197.3 \text{ eV} \cdot \text{nm})^2}{2(5.11 \times 10^5 \text{ eV}) [2(5.292 \times 10^{-2} \text{ nm})]^2} = 101 \text{ eV}.$$

This is about an order of magnitude higher than the electron’s kinetic energy in the Bohr atom. The cubical well is not a good approximation. Also, the electron’s spatial distribution goes beyond the Bohr radius, as you will see in Chapter 7. A less-confined electron would correspond to a larger well, leading to a lower ground-state energy.

59. (a) From introductory physics, the oscillator’s energy is $E = \frac{1}{2} kA^2$ where A is the amplitude. Equating this to the quantum oscillator energy [Equation (6.58)],

$$\left(n + \frac{1}{2}\right) \hbar \omega = \frac{1}{2} k A^2. \text{ The classical frequency is } \omega = \sqrt{k/m}. \text{ Solving for the quantum number, } \left(n + \frac{1}{2}\right) = \frac{k A^2}{2 \hbar \omega} = \frac{k A^2}{2 \hbar} \sqrt{\frac{m}{k}} = \frac{A^2 \sqrt{km}}{2 \hbar}$$

$$\left(n + \frac{1}{2}\right) = \frac{A^2 \sqrt{km}}{2 \hbar} = \frac{(0.10 \text{ m})^2 \sqrt{(12 \text{ N/m})(0.15 \text{ kg})}}{2(1.055 \times 10^{-34} \text{ J}\cdot\text{s})} = 6.4 \times 10^{31}.$$

Thus $n = 6.4 \times 10^{31}$. This is to be expected, because the classical system is exceedingly large as compared to a quantum oscillator.

(b) For the ground state $n = 0$, so $\left(n + \frac{1}{2}\right) \hbar \omega = \frac{1}{2} \hbar \omega = \frac{1}{2} k A^2$. Again using $\omega = \sqrt{k/m}$ and solving for amplitude A , $\hbar \sqrt{\frac{k}{m}} = k A^2$, so

$$A^2 = \sqrt{\frac{\hbar^2}{km}} = \sqrt{\frac{(1.055 \times 10^{-34} \text{ J}\cdot\text{s})^2}{(12 \text{ N/m})(0.15 \text{ kg})}} = 7.86 \times 10^{-35} \text{ m}^2, \text{ so } A = 8.9 \times 10^{-18} \text{ m}.$$

The first quantum state is too small to be observed.

(c) The difference between adjacent states is $\Delta E = \hbar \omega = \hbar \sqrt{k/m}$. Inserting numerical values, $\Delta E = \hbar \sqrt{k/m} = (1.055 \times 10^{-34} \text{ J}\cdot\text{s}) \sqrt{\frac{12 \text{ N/m}}{0.15 \text{ kg}}} = 9.4 \times 10^{-34} \text{ J}$. Again, this energy is too small to be observed.

60. (a) For the one-dimensional potential well [Equation (6.35)], the ground-state energy is

$$E_1 = \frac{\pi^2 \hbar^2}{2mL^2} = \frac{\pi^2 (1.055 \times 10^{-34} \text{ J}\cdot\text{s})^2}{2(2.0 \times 10^{-6} \text{ kg})(1.0 \times 10^{-3} \text{ m})^2} = 2.75 \times 10^{-56} \text{ J}. \text{ If this is the classical kinetic}$$

energy $K = \frac{1}{2} mv^2$, the resulting speed is $v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2(2.75 \times 10^{-56} \text{ J})}{2.0 \times 10^{-6} \text{ kg}}} = 1.7 \times 10^{-25} \text{ m/s}$, which is much too small to be measured.

(b) Now the kinetic energy is $K = \frac{1}{2} mv^2 = 6.25 \times 10^{-11} \text{ J}$. Solving Equation (6.35) for the quantum number n :

$$n = \left(\frac{2mL^2 E}{\pi^2 \hbar^2} \right)^{1/2} = \left(\frac{2(2.0 \times 10^{-6} \text{ kg})(1.0 \times 10^{-3} \text{ m})^2 (6.25 \times 10^{-11} \text{ J})}{\pi^2 (1.055 \times 10^{-34} \text{ J}\cdot\text{s})^2} \right)^{1/2} = 1.51 \times 10^{21}.$$

This extremely large value of n is an indication of how small the quantum states are.

61. (a) This was calculated in Problem 36.

(b) $\langle x \rangle = \int_{-\infty}^{\infty} x \psi^* \psi dx = 0$ because the integrand is odd over symmetric limits. To find $\langle x^2 \rangle$ we first need to normalize:

$$\begin{aligned} A^2 \int_{-\infty}^{\infty} (1 - 2\alpha x^2)^2 e^{-\alpha x^2} dx &= 2A^2 \int_0^{\infty} (1 - 2\alpha x^2)^2 e^{-\alpha x^2} dx = 1 \\ &= 2A^2 \int_0^{\infty} (1 - 4\alpha x^2 + 4\alpha^2 x^4) e^{-\alpha x^2} dx = 1 \\ &= 2A^2 \sqrt{\frac{\pi}{\alpha}} \left(\frac{1}{2} - 1 + \frac{3}{2} \right) = 1 \end{aligned}$$

Thus $A^2 = \frac{1}{2} \sqrt{\frac{\alpha}{\pi}}$ and

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} x^2 \psi^* \psi dx = 2A^2 \int_0^{\infty} x^2 (1 - 4\alpha x^2 + 4\alpha^2 x^4)^2 e^{-\alpha x^2} dx = \frac{1}{\alpha} \left(\frac{1}{4} - \frac{3}{2} + \frac{15}{4} \right) = \frac{5}{2\alpha}.$$

62. Using the known functions for ψ_1 and ψ_2 we see:

$$\begin{aligned} \psi &= \frac{1}{2} \psi_1 + \frac{\sqrt{3}}{2} \psi_2 = \frac{1}{2} \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) + \frac{\sqrt{3}}{2} \sqrt{\frac{2}{L}} \sin\left(\frac{2\pi x}{L}\right) \\ &= \sqrt{\frac{1}{2L}} \sin\left(\frac{\pi x}{L}\right) + \sqrt{\frac{3}{2L}} \sin\left(\frac{2\pi x}{L}\right) \end{aligned}$$

For normalization:

$$\int_0^L \psi^* \psi dx = \int_0^L \left(\frac{1}{2L} \sin^2\left(\frac{\pi x}{L}\right) + \frac{3}{2L} \sin^2\left(\frac{2\pi x}{L}\right) + \frac{\sqrt{3}}{2L} \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{2\pi x}{L}\right) \right) dx.$$

The third term vanishes because of the orthogonality of the trigonometric functions, leaving

$$\begin{aligned} \int_0^L \psi^* \psi dx &= \int_0^L \left(\frac{1}{2L} \sin^2\left(\frac{\pi x}{L}\right) + \frac{3}{2L} \sin^2\left(\frac{2\pi x}{L}\right) \right) dx \\ &= \frac{1}{2L} \int_0^L \sin^2\left(\frac{\pi x}{L}\right) dx + \frac{3}{2L} \int_0^L \sin^2\left(\frac{2\pi x}{L}\right) dx \\ &= \frac{1}{2L} \left(\frac{L}{2} \right) + \frac{3}{2L} \left(\frac{L}{2} \right) = 1 \quad \text{as required.} \end{aligned}$$

63. Using the Taylor approximation for the exponential $e^x \approx 1 + x$ for small x , we have

$$V(r) = D(1 - e^{-a(r-r_e)})^2 \approx D(1 - (1 - a(r-r_e)))^2 = D(a(r-r_e))^2 = Da^2(r-r_e)^2 \quad \text{which is quadratic in } (r-r_e).$$

64. We will solve for numerical values of the factors in front of the quantum numbers:

$$\begin{aligned}\hbar\omega &= \hbar a \sqrt{\frac{2D}{\mu}} = \hbar a \sqrt{\frac{2D(m_1 + m_2)}{m_1 m_2}} \\ \hbar\omega &= (6.582 \times 10^{-16} \text{ eV}\cdot\text{s}) (7.8 \times 10^9 \text{ m}^{-1}) \\ &\quad \times \sqrt{\frac{2(4.42 \text{ eV})(39.10 \text{ u} + 35.45 \text{ u})}{(39.10 \text{ u})(35.45 \text{ u})} \frac{1 \text{ u}}{931.5 \times 10^6 \text{ eV}/c^2} \left(\frac{2.998 \times 10^8 \text{ m/s}}{c} \right)}\end{aligned}$$

$$\hbar\omega = 0.03477 \text{ eV} \text{ and } \frac{\hbar^2 \omega^2}{4D} = \frac{(0.03477 \text{ eV})^2}{4(4.42 \text{ eV})} = 6.838 \times 10^{-5} \text{ eV}$$

Evaluating for specific energy levels:

$$E_0 = \left(0 + \frac{1}{2}\right)(0.03477 \text{ eV}) - \left(0 + \frac{1}{2}\right)^2 (6.838 \times 10^{-5} \text{ eV}) = 0.017 \text{ eV}$$

$$E_1 = \left(1 + \frac{1}{2}\right)(0.03477 \text{ eV}) - \left(1 + \frac{1}{2}\right)^2 (6.838 \times 10^{-5} \text{ eV}) = 0.052 \text{ eV}$$

$$E_2 = \left(2 + \frac{1}{2}\right)(0.03477 \text{ eV}) - \left(2 + \frac{1}{2}\right)^2 (6.838 \times 10^{-5} \text{ eV}) = 0.086 \text{ eV}$$

$$E_3 = \left(3 + \frac{1}{2}\right)(0.03477 \text{ eV}) - \left(3 + \frac{1}{2}\right)^2 (6.838 \times 10^{-5} \text{ eV}) = 0.121 \text{ eV}$$

Note that for these low quantum numbers the second-order correction is small.

65. The solution is identical to the presentation in the text for the three-dimensional box but without the z dimension. Briefly, we assume a trial function for the form

$\psi(x, y) = A \sin(k_1 x) \sin(k_2 y)$. Assuming that one corner is at the origin, applying the

boundary conditions leads to $k_1 = \frac{n_x \pi}{L}$ and $k_2 = \frac{n_y \pi}{L}$ and substituting into the Schrödinger

equation leads to $E = \frac{\pi^2 \hbar^2}{2mL^2} (n_x^2 + n_y^2)$. To normalize, solve the iterated double integral

$$\int_0^L \int_0^L \psi^* \psi dx dy = A^2 \int_0^L \int_0^L \sin^2\left(\frac{n_x \pi x}{L}\right) \sin^2\left(\frac{n_y \pi y}{L}\right) dx dy = A^2 \left(\frac{L}{2}\right) \left(\frac{L}{2}\right) = 1$$

so $A = \frac{2}{L}$. Now to find the energy levels use the energy equation with different values of

the quantum numbers. Letting $E_0 = \frac{\pi^2 \hbar^2}{2mL^2}$ we have: $E_1 = E_0 (1^2 + 1^2) = 2E_0$ with

$$n_1 = 1, n_2 = 1;$$

$$E_2 = E_0 (2^2 + 1^2) = 5E_0 \text{ with } n_1 = 2, n_2 = 1 \text{ or vice versa};$$

$$E_3 = E_0(2^2 + 2^2) = 8E_0 \text{ with } n_1 = 2, n_2 = 2;$$

$$E_4 = E_0(3^2 + 1^2) = 10E_0 \text{ with } n_1 = 3, n_2 = 1 \text{ or vice versa};$$

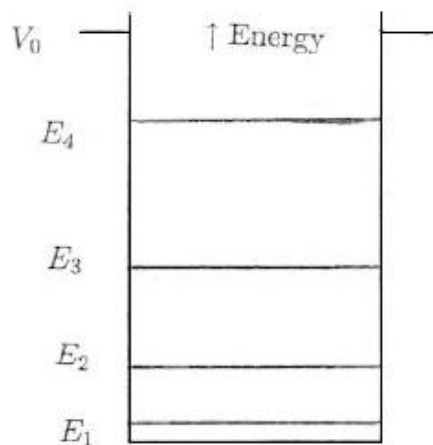
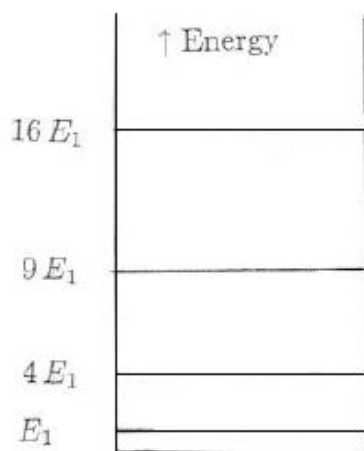
$$E_5 = E_0(3^2 + 2^2) = 13E_0 \text{ with } n_1 = 3, n_2 = 2 \text{ or vice versa};$$

$$E_6 = E_0(4^2 + 1^2) = 17E_0 \text{ with } n_1 = 4, n_2 = 1 \text{ or vice versa}.$$

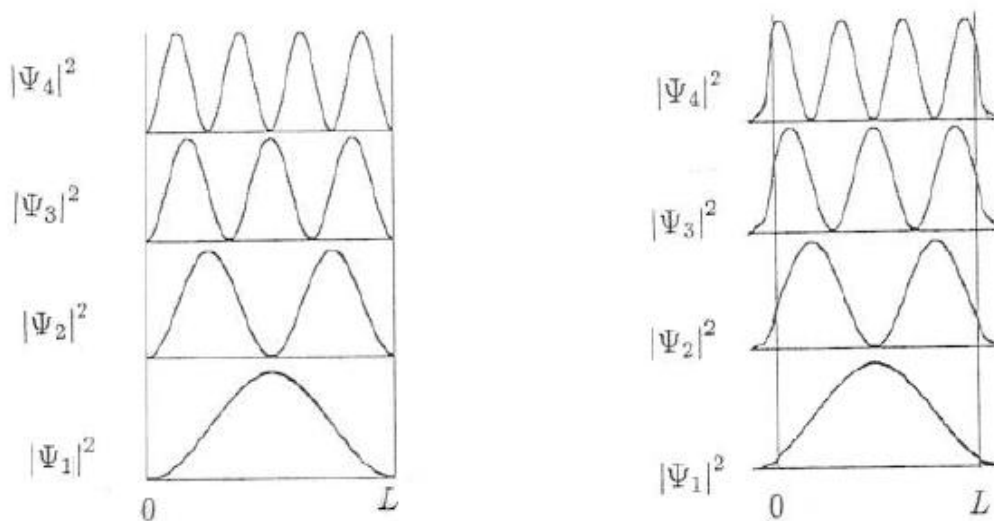
66. (a) For the infinite square well, the energies are given by Equation (6.35) and increase as n^2 . The energies for the finite square well are expressed in a transcendental equation of

the form $\tan\left(\frac{\sqrt{2m(V_0 - |E|)}L}{\hbar}\right) = -\frac{\sqrt{V_0 - |E|}}{\sqrt{|E|}}$ where L is the width of the well. (See a

Quantum Mechanics text such as Griffiths.) This equation always has at least one solution, no matter how weak the well.



- (b) For the infinite square well, the wave functions are given by Equation (6.34) and the $|\psi|^2$ can be determined easily. Note that for the finite well, the wave functions extend beyond the boundaries.



(c) For the infinite square well, the wave functions equal zero exactly at the boundaries. This is not true for the finite square well. The determination of the exact energies is rather difficult and is often treated in quantum mechanics texts. The number of available energies depends on the value of the potential. We assume that four states exist in this problem.

67. This tunneling problem is similar to Example 6.14. Because of the information given in the problem we will use Equation (6.73) to approximate the transmission probability. We

$$\text{know } \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar} = \frac{\sqrt{2mc^2(V_0 - E)}}{\hbar c} \text{ so } \kappa = \frac{\sqrt{2(5.11 \times 10^5)(0.9 \text{ eV})}}{197.33 \text{ eV} \cdot \text{nm}} = 4.86 \text{ nm}^{-1}.$$

$$\text{Then from Equation (6.73), } T \approx 2e^{-2\kappa L} = 2e^{-2(4.86 \text{ nm}^{-1})(1.3 \text{ nm})} = 6.51 \times 10^{-6}.$$

68. (a) Note that this is an approximate procedure for one-dimensional problems with a gradually varying potential, $V(x)$. We begin with Equation (6.62b) which was derived

for a scenario where $E > V$ but with V constant. We found $k = \frac{\sqrt{2m(E - V_0)}}{\hbar}$ for a constant V_0 . Because our potential varies slowly, we approximate the wave number by

$$k = \frac{\sqrt{2m[E - V(x)]}}{\hbar}. \text{ We also know that } p = \hbar k \text{ and } \lambda = \frac{h}{p}. \text{ Combining all of the}$$

above, we find a position-dependent wavelength $\lambda(x) = \frac{h}{\sqrt{2m[E - V(x)]}}.$

- (b) If we neglect barrier penetration, then the wave function must be zero at the turning points. From the particle in a box example, we know that the number of wavelengths that

fit between the turning points is $\frac{1}{2}$, or 1, or $\frac{3}{2}$, etc., which equals the distance divided by the wavelength. By analogy, the number of wavelengths that can fit inside our potential well with a slowly varying wavelength is $\int \frac{dx}{\lambda(x)} = \frac{n}{2}$ where n is an integer. Substituting

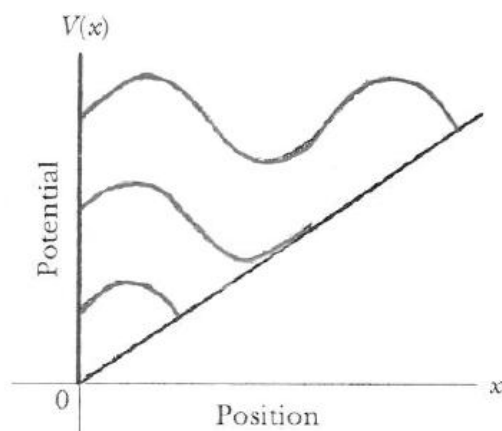
from above and rearranging, we have $2 \int \sqrt{2m[E - V(x)]} dx = nh$ where n is an integer.

69. (a) Strictly speaking this approach is valid only when the potential varies slowly. From the previous problem, we know $2 \int \sqrt{2m[E - V(x)]} dx = nh$, where n is an integer. For the potential energy $V(x) = \infty$ for $x \leq 0$ and $V(x) = Ax$ for $x > 0$, we know the classical turning points occur at $x = 0$ and where $E - V(x) = 0 = E - Ax$. Let us call this point b ; then $E - Ab = 0$ so $b = E/A$. Now we must evaluate $2 \int_0^b \sqrt{2m[E - Ax]} dx = nh$ with $b = E/A$. We know that $\int (a - qx)^{1/2} dx = \frac{-2}{3q} \sqrt{(a - qx)^3}$. Therefore

$$2\sqrt{2m} \int_0^b \sqrt{E - Ax} dx = 2\sqrt{2m} \left[\left(\frac{-2}{3A} \right) (E - Ax)^{3/2} \right]_0^b = nh. \text{ Simplifying, we find}$$

$$\frac{-4\sqrt{2m}}{3A} [-E^{3/2}] = nh \text{ so } E^{3/2} = \frac{3Anh}{4\sqrt{2m}} \text{ or } E = \frac{(3Anh)^{2/3}}{2^{5/3} m^{1/3}}.$$

(b) A sketch is shown below. The wave functions are oscillating with the lowest state being one-half a cycle, the second state a full cycle, etc. As mentioned in the answer for previous problem, part (b), these sketches ignore barrier penetration in the region where the potential is finite. No barrier penetration occurs where $x = 0$ where the potential is infinite.



70. Equation (6.70) gives the transmission coefficient for tunneling: $T = 16 \frac{E}{V_0} \left(1 - \frac{E}{V_0} \right) e^{-2\kappa L}$.

The tunneling current is proportional to the transmission coefficient. Suppose there is initially a tunneling current I_0 with a barrier width L_0 . Then the barrier width is increased by 0.4 nm to L , resulting in a current decrease to I . The ratio of currents is the ratio of the transmission coefficients. All the factors in the ratio cancel except the exponential,

$$\text{leaving } \frac{I}{I_0} = \frac{e^{-2\kappa L}}{e^{-2\kappa L_0}} = e^{-2\kappa \Delta L}$$

where $\Delta L = 0.4$ nm. We wish to solve for $V_0 - E$, the difference between the particle energy and barrier height. This is contained in the factor κ in the exponential. Taking the natural logarithm:

$$\ln\left(\frac{I}{I_0}\right) = -2\kappa \Delta L. \text{ Notice that in this case } I < I_0, \text{ so the logarithm is negative, and the}$$

equation is correct with $\Delta L > 0$. Thus $\kappa = -\frac{\ln(I/I_0)}{2\Delta L} = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$. Solving for $V_0 - E$,

$$V_0 - E = \frac{\hbar^2 (\ln(I/I_0))^2}{8m(\Delta L)^2} = \frac{(\hbar c)^2 (\ln(I/I_0))^2}{8(mc^2)(\Delta L)^2}$$

$$V_0 - E = \frac{(197.3 \text{ eV} \cdot \text{nm})^2 (\ln(10^{-4}))^2}{8(5.11 \times 10^5 \text{ eV})(0.4 \text{ nm})^2} = 5.0 \text{ eV}.$$

This is entirely reasonable, depending on the materials used for the sample and probe.

Chapter 7

- Starting with Equation (7.7), let the electron move in a circle of radius a in the xy plane, so $\sin \theta = \sin(\pi/2) = 1$. With both r and θ constant, R and f are also constant. Let

$R = f = 1$. Then $g = \psi$ and the derivatives of R and f are zero. With this Equation (7.7)

reduces to $-\frac{2\mu}{\hbar^2} a^2 (E - V) = \frac{1}{\psi} \frac{d^2 \psi}{d\phi^2}$. In uniform circular motion with an inverse-square

force, we know from the planetary model that $E = V/2$, and $E - V = \frac{V}{2} - V = -\frac{V}{2} = |E|$.

Thus $-\frac{2\mu}{\hbar^2} a^2 |E| = \frac{1}{\psi} \frac{d^2 \psi}{d\phi^2}$ or $\frac{1}{a^2} \frac{d^2 \psi}{d\phi^2} + \frac{2\mu}{\hbar^2} |E| \psi = 0$.

- This is a simple harmonic oscillator equation. Assume a standard trial solution $\psi = A \exp(iB\phi)$. With this trial solution $d^2 \psi / d\phi^2 = -B^2 \psi$. Substituting this into the equation from the previous problem we find: $\frac{1}{a^2} (-B^2) \psi + \frac{2\mu}{\hbar^2} |E| \psi = 0$. Solving for B ,

$B = \frac{\sqrt{2\mu|E|}a}{\hbar}$. To find A , normalize $\int_0^{2\pi} \psi^* \psi d\phi = 1 = A^2 \int_0^{2\pi} d\phi = 2\pi A^2$ so $A = \sqrt{1/2\pi}$.

Note that B must be an integer (let $B = n$) so that ψ will be single-valued

$[\psi(0) = \psi(2\pi)]$. With $B = n$ we have $n^2 = \frac{2\mu}{\hbar^2} |E| a^2$ and $|E| = \frac{n^2 \hbar^2}{2\mu a^2}$. For circular

motion $|E| = \frac{L^2}{2I}$ where rotational inertia $I = \mu a^2$ for a particle of mass μ . Thus

$L^2 = 2I |E| = 2\mu a^2 \frac{n^2 \hbar^2}{2\mu a^2} = n^2 \hbar^2$ or $L = n\hbar$, which is the Bohr condition.

- Assuming a trial solution $g(\phi) = Ae^{ik\phi}$ (which is easily verified by direct substitution), and using the boundary condition $g(0) = g(2\pi)$, we find $Ae^0 = Ae^{2\pi ik}$ which is only true if k is an integer.
- Using the transformations it can be shown that for any vector $\vec{A}$

$$\vec{\nabla} \psi = \hat{r} \frac{\partial \psi}{\partial r} + \theta \frac{1}{r} \frac{\partial \psi}{\partial \theta} + \hat{\phi} \frac{1}{r \sin \theta} \frac{\partial \psi}{\partial \phi} \text{ and}$$

$$\vec{\nabla} \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}.$$

Because $\nabla^2 \psi = \vec{\nabla} \cdot \vec{\nabla} \psi$ we can combine our results to find

$\nabla^2\psi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2}$ and from this a simple rearrangement gives the desired result.

5. Letting the constants in the front of R be called A we have $R = A \left(2 - \frac{r}{a_0} \right) e^{-r/2a_0}$,

$$\frac{dR}{dr} = A \left(-\frac{2}{a_0} + \frac{r}{2a_0^2} \right) e^{-r/2a_0} \text{ and } \frac{d^2R}{dr^2} = A \left(\frac{3}{2a_0^2} - \frac{r}{4a_0^3} \right) e^{-r/2a_0}.$$

Substituting these into Equation (7.13) we have

$$\left(-\frac{1}{4a_0^3} - \frac{2\mu E}{a_0 \hbar^2} \right) r + \left(\frac{5}{2a_0^2} + \frac{4\mu E}{\hbar^2} - \frac{2\mu e^2}{4\pi\epsilon_0 a_0 \hbar^2} \right) + \left(-\frac{4}{a_0} + \frac{4\mu e^2}{4\pi\epsilon_0 \hbar^2} \right) \frac{1}{r} = 0.$$

To satisfy the equation, each of the expressions in parentheses must equal zero. From the $1/r$ term we

find $a_0 = \frac{4\pi\epsilon_0 \hbar^2}{\mu e^2}$ which is correct. From the r term we obtain $E = -\frac{\hbar^2}{8\mu a_0^2} = -\frac{E_0}{4}$ which is

consistent with the Bohr result. The other expression in parentheses also leads directly to $E = -E_0/4$, so the solution is verified.

6. As in the previous problem $R = A r e^{-r/2a_0}$, $\frac{dR}{dr} = A \left(1 - \frac{r}{2a_0} \right) e^{-r/2a_0}$,

$$r^2 \frac{dR}{dr} = A \left(r^2 - \frac{r^3}{2a_0} \right) e^{-r/2a_0}, \text{ and } \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = A \left(2r - \frac{2r^2}{a_0} + \frac{r^3}{4a_0^2} \right) e^{-r/2a_0}.$$

Substituting these into Equation (7.10) with $\ell=1$ and after substituting the Coulomb potential, we

have $\left(\frac{1}{4a_0^2} + \frac{2\mu E}{\hbar^2} \right) r + \left(-\frac{1}{2a_0} + \frac{2\mu e^2}{4\pi\epsilon_0 \hbar^2} \right) + (2-2) \frac{1}{r} = 0$. The $1/r$ term vanishes, and the

middle expression (without r) reduces to $a_0 = \frac{4\pi\epsilon_0 \hbar^2}{\mu e^2}$ which is correct. From the r term

we get $E = -\frac{\hbar^2}{8\mu a_0^2} = -\frac{E_0}{4}$ which is consistent with the Bohr result.

7. $R = \frac{e^{-r/2a_0}}{\sqrt{3}(2a_0)^{3/2}} \frac{r}{a_0}$ so $R^* R = \frac{e^{-r/a_0}}{3(2a_0)^3} \left(\frac{r}{a_0} \right)^2$

To normalize, we integrate over all r space:

$$\int_0^\infty r^2 R^* R dr = \frac{1}{24a_0^5} \int_0^\infty r^4 e^{-r/a_0} dr = \frac{1}{24a_0^5} \frac{4!}{(1/a_0)^5} = 1, \text{ so the wave function } R_{21} \text{ was}$$

normalized.

8. The wave function given is $\psi_{100}(r, \theta, \phi) = Ae^{-r/a_0}$ so $\psi^* \psi$ is given by $\psi_{100}^* \psi_{100} = A^2 e^{-2r/a_0}$.

To normalize the wave function, compute the triple integral over all space

$$\iiint \psi^* \psi dV = A^2 \int_0^{2\pi} \int_0^\pi \int_0^\infty r^2 \sin \theta e^{-2r/a_0} dr d\theta d\phi. \text{ The } \phi \text{ integral yields } 2\pi, \text{ and the } \theta$$

$$\text{integral yields } 2. \text{ This leaves } \iiint \psi^* \psi dV = 4\pi A^2 \int_0^\infty r^2 e^{-2r/a_0} dr = 4\pi A^2 \frac{2}{(2/a_0)^3} = \pi a_0^3 A^2$$

This integral must equal 1 due to normalization which leads to $\pi a_0^3 A^2 = 1$ so $A = \frac{1}{\sqrt{\pi a_0^3}}$.

9. It is required that $\ell < 5$ and $|m_\ell| \leq \ell$.

$$\ell = 4: m_\ell = 0, \pm 1, \pm 2, \pm 3, \pm 4; \quad \ell = 3: m_\ell = 0, \pm 1, \pm 2, \pm 3;$$

$$\ell = 2: m_\ell = 0, \pm 1, \pm 2 \quad \ell = 1: m_\ell = 0, \pm 1 \quad \ell = 0: m_\ell = 0$$

10. $n = 3$ and $\ell = 1$, so $m_\ell = 0$ or ± 1 . Thus $L_z = 0$ or $\pm \hbar$. $L = \sqrt{\ell(\ell+1)} \hbar = \sqrt{2} \hbar$

L_y and L_x are unrestricted except for the constraint $L_x^2 + L_y^2 = L^2 - L_z^2$.

11. For $3p$ state $n = 3$ and $\ell = 1$ and $m_\ell = 0, \pm 1$.

$$\psi_{310} = R_{31} Y_{10} = \frac{1}{81} \sqrt{\frac{2}{\pi}} a_0^{-3/2} \left(6 - \frac{r}{a_0}\right) \left(\frac{r}{a_0}\right) e^{-r/3a_0} \cos \theta$$

$$\psi_{31\pm 1} = R_{31} Y_{1\pm 1} = \frac{1}{81\sqrt{\pi}} a_0^{-3/2} \left(6 - \frac{r}{a_0}\right) \left(\frac{r}{a_0}\right) e^{-r/3a_0} \sin \theta e^{\pm i\phi}$$

12. The sum is of the form $\sum_{y=-x}^x y^2$ which by symmetry is equivalent to $2 \sum_{y=1}^x y^2$. Let us first

consider (as a lemma) the sum:

$$\begin{aligned} \sum_{y=1}^x [(y+1)^3 - y^3] &= \sum_{y=1}^x [3y^2 + 3y + 1] \\ &= (2^3 - 1^3) + (3^3 - 2^3) + \dots + (x+1)^3 - x^3 \\ &= (x+1)^3 - 1^3 = x^3 + 3x^2 + 3x \end{aligned}$$

Now let us write $3 \sum_{y=1}^x y^2 = \sum_{y=1}^x [(1+y)^3 - y^3] - 3 \sum_{y=1}^x y - \sum_{y=1}^x 1$. The first of these sums is

given by our lemma above. The others are $\sum_{y=1}^x y = \frac{1}{2} x(x+1)$ and $\sum_{y=1}^x 1 = x$. Combining

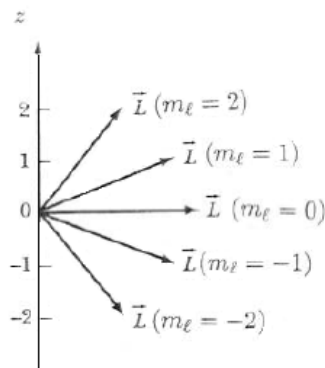
these results $3 \sum_{y=1}^x y^2 = x^3 + 3x^2 + 3x - \frac{3}{2} x(x+1) - x = \frac{1}{2} x(2x+1)(x+1)$. Therefore

$$\sum_{y=1}^x y^2 = \frac{1}{6}x(2x+1)(x+1) \text{ and } \sum_{y=-x}^x y^2 = \frac{1}{3}x(2x+1)(x+1). \text{ Then}$$

$$\langle L^2 \rangle = 3\langle L_z^2 \rangle = \frac{3}{2\ell+1} \sum_{m_\ell=-\ell}^{\ell} m_\ell^2 \hbar^2 = \ell(\ell+1)\hbar^2.$$

13. As in Example 7.4 the degeneracy is $n^2 = 36$.

14. There are five possible orientations, corresponding to the five different values of $m_\ell = 0, \pm 1, \pm 2$.



For the $m_\ell = -1$ component we have (with $\ell = 2$), $L = \sqrt{\ell(\ell+1)} \hbar = \sqrt{6}\hbar$ and

$$L_z = m_\ell \hbar = -\hbar \text{ with } L_x^2 + L_y^2 = L^2 - L_z^2 = 6\hbar^2 - \hbar^2 = 5\hbar^2.$$

15. $\cos \theta = \frac{L_z}{L} = \frac{m_\ell}{\sqrt{\ell(\ell+1)}}$ (See diagram in the preceding problem.) For this extreme case we

could have $\ell = m_\ell$ so $\cos(10^\circ) = \frac{\ell}{\sqrt{\ell(\ell+1)}}$ or $\cos^2(10^\circ) = \frac{\ell^2}{\ell(\ell+1)} = \frac{\ell^2}{\ell^2 + \ell}$. Rearranging

we find $\ell = \left(\frac{1}{\cos^2(10^\circ)} - 1 \right)^{-1} = 32.16$ and we have to round up in order to get within 10° ,

so $\ell = 33$.

16. There is one possible m_ℓ value for $\ell = 0$, three values of m_ℓ for $\ell = 1$, five values of m_ℓ for $\ell = 2$, and so on, so that the degeneracy of the n th level is $1+3+5+\dots(2\ell+1) = 1+3+5+\dots(2(n-1)+1) = n^2$.

$$17. \psi_{21-1} = R_{21}Y_{1-1} = \frac{1}{8\sqrt{\pi}a_0^{3/2}} \left(\frac{r}{a_0} \right) e^{-r/2a_0} \sin \theta e^{-i\phi};$$

$$\psi_{210} = R_{21}Y_{10} = \frac{1}{4\sqrt{2\pi}a_0^{3/2}}\left(\frac{r}{a_0}\right)e^{-r/2a_0}\cos\theta;$$

$$\psi_{32-1} = R_{32}Y_{2-1} = \frac{1}{81\sqrt{\pi}a_0^{3/2}}\left(\frac{r^2}{a_0^2}\right)e^{-r/3a_0}\sin\theta\cos\theta e^{-i\phi}$$

18. We must calculate the triple integral over all space. The definite integrals can be evaluated from a handbook or from Appendix 3.

$$\iiint \psi_{200}^* \psi_{200} dV = \frac{1}{32\pi a_0^3} \int_0^{2\pi} \int_0^\pi \int_0^\infty \left(2 - \frac{r}{a_0}\right)^2 e^{-r/a_0} r^2 \sin\theta dr d\theta d\phi$$

The ϕ integral yields 2π , and the θ integral yields 2. This leaves

$$\begin{aligned} \iiint \psi^* \psi dV &= \frac{1}{8a_0^3} \int_0^\infty \left(2 - \frac{r}{a_0}\right)^2 r^2 e^{-r/a_0} dr = \frac{1}{8a_0^3} \int_0^\infty \left(4r^2 - \frac{4r^3}{a_0} + \frac{r^4}{a_0^2}\right) e^{-r/a_0} dr \\ &= \frac{1}{8a_0^3} \left[4(2!a_0^3) - \frac{4}{a_0}(3!a_0^4) + \frac{1}{a_0^2}(4!a_0^5) \right] \\ &= 1 \end{aligned}$$

as required.

Repeat the process with the second wave function:

$$\iiint \psi_{21-1}^* \psi_{21-1} dV = \frac{1}{64\pi a_0^3} \int_0^{2\pi} \int_0^\pi \int_0^\infty \left(\frac{r}{a_0}\right)^2 r^2 \sin^3\theta e^{-r/a_0} dr d\theta d\phi.$$

The ϕ integral yields 2π as before. The θ integral can be found in integral tables and

$$\text{equals } 4/3. \text{ This leaves } \iiint \psi^* \psi dV = \frac{1}{24a_0^3} \int_0^\infty \left(\frac{r}{a_0}\right)^2 r^2 e^{-r/a_0} dr = \frac{1}{24a_0^3} \left[\frac{1}{a_0^2}(4!a_0^5) \right] = 1$$

as required.

19. The value of n sets the maximum value of ℓ . However the letter p identifies the value of ℓ for this problem. With $\ell=1$ we have $m_\ell=0, \pm 1$ and $L_z = m_\ell \hbar = 0, \pm \hbar$.

20. The maximum difference is between the $m_\ell = -2$ and $m_\ell = +2$ levels, so $\Delta m_\ell = 4$. Then

$$\Delta V = \mu_B (\Delta m_\ell) B = (5.788 \times 10^{-5} \text{ eV/T})(4)(3.5 \text{ T}) = 8.10 \times 10^{-4} \text{ eV}.$$

21. Differentiating $E = \frac{hc}{\lambda}$ we find: $dE = -\frac{hc}{\lambda^2} d\lambda$ or $|\Delta E| = \frac{hc}{\lambda^2} |\Delta \lambda|$. In the normal Zeeman

$$\text{effect, between adjacent } m_\ell \text{ states } |\Delta E| = \mu_B B \text{ so } \mu_B B = \left(\frac{hc}{\lambda_0^2}\right) |\Delta \lambda| \text{ or } \Delta \lambda = \frac{\lambda_0^2 \mu_B B}{hc}.$$

22. See the solution to problem 14 for the sketch. To compute the angles with $\ell = 2$:

$$\cos \theta = \frac{L_z}{L} = \frac{m_\ell}{\sqrt{\ell(\ell+1)}} = \frac{m_\ell}{\sqrt{6}}. \text{ There are five different values of } \theta, \text{ corresponding to the}$$

different m_ℓ values $0, \pm 1, \pm 2$:

$$\theta = \cos^{-1}\left(\frac{2}{\sqrt{6}}\right) = 35.3^\circ \quad \theta = \cos^{-1}\left(\frac{1}{\sqrt{6}}\right) = 65.9^\circ \quad \theta = \cos^{-1}(0) = 90^\circ$$

$$\theta = \cos^{-1}\left(\frac{-1}{\sqrt{6}}\right) = 114.1^\circ \quad \theta = \cos^{-1}\left(\frac{-2}{\sqrt{6}}\right) = 144.7^\circ$$

23. With $\ell = 3$ we have (as in the previous problem) $\cos \theta = \frac{L_z}{L} = \frac{m_\ell}{\sqrt{\ell(\ell+1)}} = \frac{m_\ell}{\sqrt{12}}$. For the

$$\text{minimum angle } m_\ell = \ell = 3 \text{ and } \theta = \cos^{-1}\left(\frac{3}{\sqrt{12}}\right) = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = 30^\circ.$$

24. From Problem 21, $\Delta\lambda = \frac{\lambda_0^2 \mu_B B}{hc}$ so the magnetic field is

$$B = \frac{hc\Delta\lambda}{\lambda_0^2 \mu_B} = \frac{(1240 \text{ eV} \cdot \text{nm})(0.04 \text{ nm})}{(656.5 \text{ nm})^2 (5.788 \times 10^{-5} \text{ eV/T})} = 1.99 \text{ T}.$$

25. There are seven different states, corresponding to $m_\ell = 0, \pm 1, \pm 2, \pm 3$. In the absence of a

magnetic field $E = -\frac{E_0}{n^2} = -\frac{13.606 \text{ eV}}{25} = -0.544 \text{ eV}$. The Zeeman splitting is given by

$$\Delta E = \mu_B B (\Delta m_\ell) = (5.788 \times 10^{-5} \text{ eV/T})(3.00 \text{ T})m_\ell = (1.7364 \times 10^{-4} \text{ eV})m_\ell.$$

For $m_\ell = 0$ we have $\Delta E = 0$. For the other m_ℓ states:

$$m_\ell = \pm 1: \Delta E = (1.7364 \times 10^{-4} \text{ eV})(\pm 1) = \pm 1.74 \times 10^{-4} \text{ eV};$$

$$m_\ell = \pm 2: \Delta E = (1.7364 \times 10^{-4} \text{ eV})(\pm 2) = \pm 3.47 \times 10^{-4} \text{ eV};$$

$$m_\ell = \pm 3: \Delta E = (1.7364 \times 10^{-4} \text{ eV})(\pm 3) = \pm 5.21 \times 10^{-4} \text{ eV}.$$

26. From the text the magnitude of the spin magnetic moment is

$$\mu_s = \frac{2\mu_B \|S\|}{\hbar} = \frac{2\mu_B}{\hbar} \frac{\sqrt{3}\hbar}{2} = \sqrt{3}\mu_B.$$

The z-component of the magnetic moment is (see Figure 7.9)

$$\mu_z = \mu_s \cos \theta = \mu_s \frac{1/2}{\sqrt{3}/2} = \frac{\mu_s}{\sqrt{3}} = \mu_B.$$

The potential energy is $V = -\vec{\mu} \cdot \vec{B} = -\mu_z B_z$ and so the vertical component of force is

$F_z = -dV / dz = \mu_z (dB_z / dz)$. From mechanics the acceleration is $a_z = \frac{F_z}{m} = \frac{\mu_z}{m} \frac{dB_z}{dz}$ and

with constant acceleration the vertical deflection of each beam is $z = \frac{1}{2} a_z t^2$. With the time equal to the horizontal distance divided by incoming speed, or $t = x / v_x$, we have

$$z = \frac{1}{2} a_z t^2 = \frac{1}{2} \left(\frac{\mu_z}{m} \frac{dB_z}{dz} \right) \left(\frac{x}{v_x} \right)^2 = \frac{1}{2} \left(\frac{9.27 \times 10^{-24} \text{ J/T}}{1.81 \times 10^{-25} \text{ kg}} \right) (2000 \text{ T/m}) \left(\frac{0.071 \text{ m}}{925 \text{ m/s}} \right)^2 = 3.017 \times 10^{-4} \text{ m}$$

The separation between the two silver beams is twice this amount, $6.035 \times 10^{-4} \text{ m}$.

27. The kinetic energy of the atoms is

$$K = \frac{3}{2} kT = \frac{3}{2} (1.38 \times 10^{-23} \text{ J/K}) (1273 \text{ K}) = 2.64 \times 10^{-20} \text{ J} . \text{ From the previous problem,}$$

we see that the separation of the beams is (remember $\mu_z = \mu_B$) $s = 2z = \left(\frac{\mu_B}{m} \frac{dB_z}{dz} \right) \left(\frac{x}{v_x} \right)^2$.

$$\text{Rearranging we see that } x^2 \frac{dB_z}{dz} = \frac{smv^2}{\mu_B} = \frac{2sK}{\mu_B} = \frac{2(0.01 \text{ m})(2.64 \times 10^{-20} \text{ J})}{9.27 \times 10^{-24} \text{ J/T}} = 57.0 \text{ T} \cdot \text{m} .$$

The magnet should be designed so that the product of its length squared and its vertical magnetic field gradient is $57 \text{ T} \cdot \text{m}$.

28. As shown in Figure 7.9 the electron spin vector cannot point in the direction of B ,

because its magnitude is $S = \sqrt{s(s+1)} = \sqrt{3/4} \hbar$ and its z-component is $S_z = m_s \hbar = \hbar / 2$.

If the z-component of a vector is less than the vector's magnitude, the vector does not lie along the z-axis.

29. For the $4f$ state $n = 4$ and $\ell = 3$. The possible m_ℓ values are $0, \pm 1, \pm 2$, and ± 3 with

$m_s = \pm 1/2$ for each possible m_ℓ value. The degeneracy of the $4f$ state is then (with 2 spin states per m_ℓ) equal to $2(7) = 14$.

30. For the $5d$ state $n = 5$ and $\ell = 2$. The possible m_ℓ values are $0, \pm 1$, and ± 2 , with

$m_s = \pm 1/2$ for each possible m_ℓ value. The degeneracy of the $5d$ state is then (with 2 spin states per m_ℓ) equal to $2(5) = 10$.

31. If we determine the thermal energy that equals the energy required for the spin-flip transition, we have $5.9 \times 10^{-6} \text{ eV} = \frac{3}{2} kT = \frac{3}{2} (8.617 \times 10^{-5} \text{ eV/K}) T$. This gives

$$T = 0.0456 \text{ K}.$$

32. The spin degeneracy is 2 and the n^2 is shown in Problem 16.

33. The selection rule $\Delta m_\ell = 0, \pm 1$ gives three lines in each case.

34. (a) $\Delta \ell = 0$ is forbidden

(b) allowed but with $\Delta n = 0$ there is no energy difference unless an external magnetic field is present

(c) $\Delta \ell = -2$ is forbidden

(d) allowed with absorbed photon of energy $\Delta E = E_0 \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = 2.55 \text{ eV}$

35. We must find the maxima and minima of the following function.

$$P(r) = r^2 |R(r)|^2 = A^2 e^{-r/a_0} \left(2 - \frac{r}{a_0} \right)^2 r^2 = A^2 \left(4r^2 - \frac{4r^3}{a_0} + \frac{r^4}{a_0^2} \right) e^{-r/a_0}$$

To find the extremes set $\frac{dP}{dr} = 0$:

$$0 = -\frac{1}{a_0} \left(4r^2 - \frac{4r^3}{a_0} + \frac{r^4}{a_0^2} \right) e^{-r/a_0} + \left(8r - \frac{12r^2}{a_0} + \frac{4r^3}{a_0^2} \right) e^{-r/a_0} \text{ which simplifies to}$$

$$0 = -\frac{r^3}{a_0^3} + \frac{8r^2}{a_0^2} - \frac{16r}{a_0} + 8$$

Letting $x = \frac{r}{a_0}$ the equation above can be factored into $(x-2)(x^2-6x+4)=0$. From the

first factor we obtain $x = 2$ (or $r = 2a_0$), which from Figure 7.12 we can see is a

minimum. The second parenthesis gives a quadratic equation with solutions $x = 3 \pm \sqrt{5}$,

so $r = (3 \pm \sqrt{5})a_0$. These are both maxima.

36. In the previous problem we found that the two maxima are at $r = (3 \pm \sqrt{5})a_0$. From

Figure 7.12 it is clear that the peak at $r = (3 + \sqrt{5})a_0$ is higher. This can be verified by substituting the two values for r and computing $P(r)$. The most probable location is

therefore at $r = (3 + \sqrt{5})a_0 = 5.24a_0$, which is significantly further from the nucleus than the $2p$ peak at $r = 4a_0$.

37. From the solution to Problem 35 we see that $P(r) = 0$ at $r = 2a_0$. Note that $P(0) = 0$ also. A sketch is found in Figure 7.12.

38. The radial probability distribution for the ground state is $P(r) = r^2 |R(r)|^2 = \frac{4}{a_0^3} r^2 e^{-2r/a_0}$.

With $r \ll a_0$ throughout this interval we can approximate the exponential term as $e^{-2r/a_0} \approx 1$. Therefore the probability of being inside a radius 1.2×10^{-15} m is

$$\int_0^{1.2 \times 10^{-15}} P(r) dr \approx \frac{4}{a_0^3} \int_0^{1.2 \times 10^{-15}} r^2 dr = \frac{4r^3}{3a_0^3} \bigg|_0^{1.2 \times 10^{-15}} = 1.55 \times 10^{-14}$$

39. As in the previous problem, $P(r) = r^2 |R(r)|^2 = \frac{4}{a_0^3} r^2 e^{-2r/a_0}$. To find the desired

probability, integrate $P(r)$ over the appropriate limits: $\int_{0.95a_0}^{1.05a_0} P(r) dr = \frac{4}{a_0^3} \int_{0.95a_0}^{1.05a_0} r^2 e^{-2r/a_0} dr$.

Letting $x = r/a_0$ which implies $dx = dr/a_0$. Thus $\int_{0.95a_0}^{1.05a_0} P(r) dr = 4 \int_{0.95}^{1.05} x^2 e^{-2x} dx = 0.056$ where the definite integral was evaluated using Mathcad.

40. In general $\langle r \rangle = \int_0^\infty r P(r) dr = \int_0^\infty r^3 |R(r)|^2 dr$. For the $2s$ state:

$$\langle r \rangle = \frac{1}{8a_0^3} \int_0^\infty r^3 \left(4 - \frac{4r}{a_0} + \frac{r^2}{a_0^2} \right) e^{-r/a_0} dr. \quad \text{Using integral tables } \int_0^\infty r^n e^{-r/a_0} dr = n!(a_0)^{n+1}$$

$$\langle r \rangle = \frac{1}{8a_0^3} \left[4(3!)a_0^4 - \frac{4}{a_0} (4!)(a_0^5) + \frac{1}{a_0^2} (5!)a_0^6 \right] = \frac{a_0}{8} (24 - 96 + 120) = 6a_0$$

For the $2p$ state:

$$\begin{aligned} \langle r \rangle &= \frac{1}{24a_0^3} \int_0^\infty r^3 \left(\frac{r}{a_0} \right)^2 e^{-r/a_0} dr = \frac{1}{24a_0^5} \int_0^\infty r^5 e^{-r/a_0} dr \\ &= \frac{1}{24a_0^5} (5!)(a_0^6) = \frac{120a_0}{24} = 5a_0 \end{aligned}$$

41. For the 2s state: $P(r) = r^2 |R(r)|^2 = \frac{1}{8a_0^3} r^2 \left(2 - \frac{r}{a_0}\right)^2 e^{-r/a_0}$. As in Problem 38 for $r \ll a_0$

we can say $e^{-r/a_0} \approx 1$, so the probability is given by the integral

$$\begin{aligned} \int_0^{1.2 \times 10^{-15}} P(r) dr &\approx \frac{1}{8a_0^3} \int_0^{1.2 \times 10^{-15}} r^2 \left(2 - \frac{r}{a_0}\right)^2 dr = \frac{1}{8a_0^3} \int_0^{1.2 \times 10^{-15}} \left(4r^2 - \frac{4r^3}{a_0} + \frac{r^4}{a_0^2}\right) dr \\ &= \frac{1}{8} \left(\frac{4}{3} \frac{r^3}{a_0^3} - \frac{r^4}{a_0^4} + \frac{r^5}{5a_0^5} \right) \bigg|_0^{1.2 \times 10^{-15}} = 1.9 \times 10^{-15} \end{aligned}$$

Similarly for the 2p state: $P(r) = r^2 |R(r)|^2 = \frac{1}{24a_0^5} r^4 e^{-r/a_0}$.

$$\int_0^{1.2 \times 10^{-15}} P(r) dr \approx \frac{1}{24a_0^5} \int_0^{1.2 \times 10^{-15}} r^4 dr = \frac{r^5}{120a_0^5} \bigg|_0^{1.2 \times 10^{-15}} = 5.0 \times 10^{-26}$$

42. To find the most probable radial position in the 3d state:

$$P(r) = r^2 |R(r)|^2 = A^2 e^{-2r/3a_0} \left(\frac{r^2}{a_0^2}\right)^2 r^2 = A^2 \left(\frac{r^6}{a_0^4}\right) e^{-2r/3a_0}$$

To find the extrema, set $\frac{dP}{dr} = 0$: $0 = \frac{1}{a_0} \left(\frac{6r^5}{a_0^3} - \frac{2r^6}{3a_0^4}\right) e^{-2r/3a_0}$ which is equivalent to

$$0 = \frac{6r^5}{a_0^3} - \frac{2r^6}{3a_0^4}. \quad r = 9a_0 \text{ and } r = 0 \text{ are the two solutions. From Figure (7.12) you can see}$$

that $r = 0$ is a minimum where $P(r) = 0$ and $r = 9a_0$ is a maximum.

43. For the 3d state:

$$P(r) = r^2 |R(r)|^2 = r^2 \left(\frac{1}{a_0}\right)^3 \left(\frac{16}{(81)^2 (30)}\right) \left(\frac{r^2}{a_0^2}\right)^2 e^{-2r/3a_0} = \left(\frac{8}{(81)^2 (15)}\right) \left(\frac{r^6}{a_0^7}\right) e^{-2r/3a_0}$$

To find that probability that the electron in the 3d state is location at a position greater

$$\text{than } a_0, \text{ we must evaluate } \int_{a_0}^{\infty} P(r) dr = \int_{a_0}^{\infty} \left(\frac{8}{(81)^2 (15)}\right) \left(\frac{r^6}{a_0^7}\right) e^{-2r/3a_0} dr.$$

(Alternatively, we could evaluate the integral from 0 to a_0 and subtract this answer from 1 since we know that the wave functions are normalized.) From integral tables, we find

integrals of the form $\int x^m e^{bx} dx = e^{bx} \sum_{i=0}^m (-1)^i \frac{m! x^{m-i}}{(m-i)! b^{i+1}}$ where $m = 6$ and $b = \frac{-2}{3a_0}$ for our example. Evaluation of the summation gives a probability of 0.9999935 that the $3d$ electron will be at a position greater than a_0 . That means it is almost certain!

44. $R = \frac{e^2}{4\pi\epsilon_0 mc^2} = \frac{1.44 \times 10^{-9} \text{ eV} \cdot \text{m}}{511 \times 10^3 \text{ eV}} = 2.82 \times 10^{-15} \text{ m}$. From the angular momentum

equation $v = \frac{3\hbar}{4mR} = \frac{3\hbar c}{4mc^2 R} = \frac{3(197.33 \text{ eV} \cdot \text{nm})}{4(511 \times 10^3 \text{ eV})(2.82 \times 10^{-6} \text{ nm})} c = 103c$. A speed of $103c$ is prohibited by the postulates of relativity.

45. The electron radius would be $\frac{\lambda_c}{2} = 1.21 \times 10^{-12} \text{ m}$. As in the previous problem

$$v = \frac{3\hbar}{4mR} = \frac{3\hbar c}{4mc^2 R} = \frac{3(197.33 \text{ eV} \cdot \text{nm})}{4(511 \times 10^3 \text{ eV})(1.21 \times 10^{-3} \text{ nm})} c = 0.24c$$

This result is allowed by relativity. However, in order to obtain this allowed result, we had to assume an unreasonably large size for the electron (one thousand times larger in radius than a proton!).

46. (a) The only change in Equation (7.3) is in the potential energy term, with $V = -\frac{Ze^2}{4\pi\epsilon_0 r}$ so

the Schrödinger equation becomes:

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{2\mu}{\hbar^2} \left(E + \frac{Ze^2}{4\pi\epsilon_0 r} \right) \psi = 0.$$

(b) Because V occurs only in the radial part, there is no change in the separation of variables.

(c) Yes, from Equation (7.10) it is clear that the radial wave functions will change.

(d) No, there is no change in the θ or ϕ dependence.

47. Carrying Z through the derivation in the text [Equations (7.12) through (7.14)] we find

$$\psi_{100} = \frac{1}{\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} e^{-Zr/a_0}.$$

48. Use the wave function $R_{31} = Ar \left(1 - \frac{r}{6a_0} \right) e^{-r/3a_0}$ where A is constant. Then

$$P(r) = r^2 |R(r)|^2 = A^2 \left(r^2 - \frac{r^3}{3a_0} + \frac{r^4}{36a_0^2} \right) e^{-2r/3a_0}. \text{ To find the extrema, set } \frac{dP}{dr} = 0. \text{ Doing}$$

so and factoring out $A^2 r e^{-2r/3a_0}$ gives $2 - \frac{5r}{3a_0} + \frac{r^2}{3a_0^2} - \frac{r^3}{54a_0^3} = 0$. Letting $x = \frac{r}{a_0}$ and

multiplying both sides by -54 we obtain

$$x^3 - 18x^2 + 90x - 108 = 0 = (x - 6)(x^2 - 12x + 18).$$

(a) The minimum is at $x = 6$, or $r = 6a_0$, and we find $P(6a_0) = 0$. Clearly $P(0) = 0$ also.

(b) The two maxima come from the quadratic equation in parentheses, with $x = 6 \pm 3\sqrt{2}$ or $r = (6 \pm 3\sqrt{2})a_0$.

(c) $Y_{1\pm1} = (\text{constant}) \sin \theta e^{\pm i\phi}$. Then Y^*Y is proportional to $\sin^2 \theta$, and the probability is zero at $\theta = 0$ and $\theta = 180^\circ$.

49. The ground state energy can be obtained using the standard Rydberg formula with the

reduced mass, μ , of the muonic atom.
$$E_0 = \frac{e^2}{8\pi\epsilon_0 a_0} = \frac{\mu e^4}{2(4\pi\epsilon_0)^2 \hbar^2}$$

Computing the reduced mass:

$$\mu = \frac{m_p m_\mu}{m_p + m_\mu} = \frac{1}{c^2} \frac{(938.27 \text{ MeV})(105.66 \text{ MeV})}{(938.27 \text{ MeV} + 105.66 \text{ MeV})} = 94.966 \text{ MeV}/c^2$$

Thus

$$E_0 = \frac{\mu e^4}{2(4\pi\epsilon_0)^2 \hbar^2} = \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \frac{\mu c^2}{2(\hbar^2 c^2)} = \frac{(1.44 \text{ eV} \cdot \text{nm})^2 (94.966 \times 10^6 \text{ eV})}{2(197.33 \text{ eV} \cdot \text{nm})^2} = 2.53 \text{ keV}$$

50. The interaction between the magnetic moment of the proton and magnetic moment of the electron causes hyperfine splitting. The transition between the two states causes emission

of a photon with energy of $E = \frac{hc}{\lambda} = \frac{(1240 \text{ eV} \cdot \text{nm})}{21 \times 10^7 \text{ nm}} = 5.9 \times 10^{-6} \text{ eV}$. From the uncertainty

principle, we know $\Delta E \Delta t \geq \frac{\hbar}{2}$. With a lifetime of $\Delta t = 1 \times 10^7 \text{ y}$ then

$$\Delta E \geq \frac{6.5821 \times 10^{-16} \text{ eV}}{2(1 \times 10^7 \text{ y})(3.16 \times 10^7 \text{ s/y})} = 1.041 \times 10^{-30} \text{ eV}.$$

51. Break the vector $\vec{r}$ into its components $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$. Because the two factors R and Y in the hydrogen wave functions are given in spherical coordinates, it is best to express each of the Cartesian components in spherical coordinates using the standard transformations $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, and $z = r \cos \theta$ as shown in Figure 7.1.

We know that in spherical coordinates, the volume element $d\tau = r^2 \sin \theta dr d\theta d\phi$.

In this case we have $Y = Y_{00} = 1/2\sqrt{\pi}$ for both the initial and final states, so $Y_f^* Y_i = \frac{1}{4\pi}$.

For simplicity we will write each component of the integral separately. Ignoring constant factors, the x component is $\int \psi_f^* x \psi_i d\tau = \int_0^\infty r^3 R_f^* R_i dr \int_0^\pi \sin^2 \theta d\theta \int_0^{2\pi} \cos \phi d\phi$.

The integral over ϕ is zero, so the product is zero. Similarly, the y and z components of the integral are zero, and so the transition probability is zero. (For the z component, it is the integral over θ that vanishes.)

52. The solution is similar to the preceding problem, but now the spherical harmonics give

$$Y_f^* Y_i = \frac{1}{2\sqrt{\pi}} \frac{1}{4} \sqrt{\frac{5}{\pi}} (3\cos^2 \theta - 1) = \frac{\sqrt{5}}{8\pi} (3\cos^2 \theta - 1)$$

Ignoring constant factors, the x component of the integral is

$$\int \psi_f^* x \psi_i d\tau = \int_0^\infty r^3 R_f^* R_i dr \int_0^\pi \sin^2 \theta (3\cos^2 \theta - 1) d\theta \int_0^{2\pi} \cos \phi d\phi.$$

As in the preceding problem, the integral over ϕ is zero, so the product is zero. Similarly, the y and z components of the integral are zero, and so the transition probability is zero. (For the z component, it's the integral over θ that vanishes.)

53. In this case we want to show that the transition integral is not zero. From Table 7.2,

$$Y_f^* Y_i = \frac{1}{2\sqrt{\pi}} \frac{1}{2} \sqrt{\frac{3}{\pi}} \cos \theta = \frac{\sqrt{3}}{4\pi} \cos \theta. \text{ Ignoring constant factors, the x component of the}$$

integral is $\int \psi_f^* x \psi_i d\tau = \int_0^\infty r^3 R_f^* R_i dr \int_0^\pi \sin \theta \cos \theta d\theta \int_0^{2\pi} \cos \phi d\phi$ which is still zero, as is the y component. However, in this case the z component is

$$\int \psi_f^* z \psi_i d\tau = \int_0^\infty r^3 R_f^* R_i dr \int_0^\pi \sin \theta \cos^2 \theta d\theta \int_0^{2\pi} d\phi.$$

All three factors in this integral are nonzero, so the net result is nonzero, implying that the transition may occur. This is consistent with the selection rules described in Section 7.6.

54. The probability density is

$$P = r^2 |R_{32}(r)|^2 = \frac{1}{a_0^7} \left(\frac{4}{81\sqrt{30}} \right)^2 r^6 e^{-2r/3a_0}$$

To find the peak, set the derivative of this function equal to zero:

$$\frac{dP}{dr} = \frac{1}{a_0^7} \left(\frac{4}{81\sqrt{30}} \right)^2 \left[6r^5 e^{-2r/3a_0} + r^6 \left(-\frac{2r}{3a_0} \right) e^{-2r/3a_0} \right] = 0$$

Therefore $0 = \left[6 - \frac{2r}{3a_0} \right]$ which leads to $r = 9a_0$. This matches the Bohr result.

55. (a) As in Example 7.14, $\langle r_{3p} \rangle = \int_0^\infty rP(r)dr = \int_0^\infty r^3 |R_{3p}(r)|^2 dr$. Using the value from

Table 7.1 $R_{3p} = \frac{1}{a_0^{3/2}} \frac{4}{81\sqrt{6}} \left(6 - \frac{r}{a_0} \right) \frac{r}{a_0} e^{-r/3a_0}$, so

$$\langle r_{3p} \rangle = \frac{1}{a_0^3} \left(\frac{4}{81\sqrt{6}} \right)^2 \int_0^\infty r^3 \left(6 - \frac{r}{a_0} \right)^2 \frac{r^2}{a_0^2} e^{-2r/3a_0} dr$$

Integration using a computer program yields $\langle r_{3p} \rangle = \frac{25a_0}{2}$.

This is $12.5a_0$, which is substantially larger than the Bohr result $9a_0$. Our answer seems reasonable, given the shape of the graph of $P(r)$ in Figure 7.12.

(b) The probability is

$$P = \int_{9a_0}^\infty P(r)dr = \int_{9a_0}^\infty r^2 |R_{3p}(r)|^2 dr = \frac{1}{a_0^3} \left(\frac{4}{81\sqrt{6}} \right)^2 \int_{9a_0}^\infty r^2 \left(6 - \frac{r}{a_0} \right)^2 \frac{r^2}{a_0^2} e^{-2r/3a_0} dr.$$

Because the lower limit is no longer zero, we will use numerical integration on a computer. The result to three significant figures is $P = 0.820$. That may seem rather high, but again it is reasonable, considering the shape of the graph of $P(r)$ in Figure 7.12.

Chapter 8

1. The first two electrons are in the 1s subshell and have $\ell = 0$, with $m_s = \pm 1/2$. The third electron is in the 2s subshell and has $\ell = 0$, with either $m_s = 1/2$ or $-1/2$. With four particles there are six possible interactions: the nucleus with electrons 1, 2, 3; electron 1 with electron 2; electron 1 with electron 3; or electron 2 with electron 3. In each case it is possible to have a Coulomb interaction and a magnetic moment interaction.
2. H: $1s^1$, He: $1s^2$, Li: $1s^2 2s^1$, Be: $1s^2 2s^2$, B: $1s^2 2s^2 2p^1$, C: $1s^2 2s^2 2p^2$, N: $1s^2 2s^2 2p^3$, O: $1s^2 2s^2 2p^4$, F: $1s^2 2s^2 2p^5$, Ne: $1s^2 2s^2 2p^6$
3. L: $n = 2$, so there are two (2s and 2p)
 N: $n = 4$, so there are four (4s, 4p, 4d, and 4f)
 O: $n = 5$, so there are five (5s, 5p, 5d, 5f, and 5g)
4. In the first excited state, go to the next higher level. In neon one of the 2p electrons is promoted to 3s, so the configuration is $2p^5 3s^1$. By the same reasoning the first excited state of xenon is $5p^5 6s^1$.
5. K: $[\text{Ar}]4s^1$, As: $[\text{Ar}]4s^2 3d^{10} 4p^3$, Nb: $[\text{Kr}]5s^1 4d^4$, Pd: $[\text{Kr}]4d^{10}$, Sm: $[\text{Xe}]6s^2 4f^6$, Po: $[\text{Xe}]6s^2 4f^{14} 6s^2 5d^{10} 6p^4$ U: $[\text{Rn}]7s^2 6d^1 5f^3$ where the bracket represents a closed inner shell. For example, $[\text{Ar}]$ represents $1s^2 2s^2 2p^6 3s^2 3p^6$.
6.
 - (a) He, Ne, Ar, Kr, Xe, Rn
 - (b) Li, Na, K, Rb, Cs, Fr
 - (c) F, Cl, Br, I, At
 - (d) Be, Mg, Ca, Sr, Ba, Ra

7. From Figure 8.4 we see that the radius of Na is about 0.16 nm. We know that for single-electron atoms $E = -\frac{Ze^2}{8\pi\epsilon_0 r}$. Therefore

$$Ze = -\frac{8\pi\epsilon_0 r E}{e^2} e = -2 \frac{4\pi\epsilon_0}{e^2} r E e = -\frac{2(0.16 \text{ nm})(-5.14 \text{ eV})}{1.44 \text{ eV} \cdot \text{nm}} e = 1.14e.$$

8. Using the (n, ℓ, m_ℓ, m_s) notation, the first four electrons (in the 1s and 2s orbitals) have quantum numbers $(1, 0, 0, \pm 1/2)$ and $(2, 0, 0, \pm 1/2)$. The remaining two electrons are in

the $2p$ orbital with $n = 2$ and $\ell = 1$. By Hund's rules, we want the total spin angular momentum, S , to be maximized, so $m_s = +1/2$ or $m_s = -1/2$, and the total angular momentum, L , should be maximized, so the possibilities are $(2, 1, 1, +1/2)$, $(2, 1, 0, +1/2)$ or $(2, 1, 1, -1/2)$, $(2, 1, 0, -1/2)$.

9. (a) count the number of electrons, 9; F, or use Table 8.2
(b) count the number of electrons, 12; Mg
(c) filled $3p$ shell: Ar
10. Ag: $[\text{Kr}]4d^{10}5s^1$, Hf: $[\text{Xe}]4f^{14}5d^26s^2$, Sb: $[\text{Kr}]4d^{10}5s^25p^3$ where the bracket represents a closed inner shell. For example, $[\text{Kr}]$ represents $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$.
11. (a) filled $3d$: Se
(b) filled $4p$ and $4d$: Ag
(c) The $4f$ shell fills in the lanthanides: Er
12. J ranges from $|L - S|$ to $|L + S|$ or 2, 3, 4. Then in spectroscopic notation $^{2S+1}L_J$, we have three possibilities: 3F_2 , 3F_3 , or 3F_4 . The ground state has the lowest J value, or 3F_2 . With $n = 4$ the full notation is 4^3F_2 .
13. For indium (In) there is a single electron in the $5p$ subshell. Therefore $S = \frac{1}{2}$ and $2S + 1 = 2$. Because that electron is in the p subshell we have $L = 1$. The possible J values range from $|L - S|$ to $|L + S|$, which in this case gives $J = 1/2$ or $J = 3/2$. Therefore the possible states are $^2P_{1/2}$ and $^2P_{3/2}$, with $^2P_{1/2}$ being the ground state. The full notation is $5^2P_{1/2}$.
14. No spin-orbit interaction requires *either* a closed subshell ($S = 0$), which occur in He, Be, Ne, Mg, Ar, and Ca *or* $L = 0$ which we have in H, Li, Na, and K. (It is the 1st excited state of the alkali metals which accounts for the splitting in the alkali metal spectra with the visible transition begin from the 2nd excited state to the first in Na for example.)
15. In the $4d$ state $\ell = 2$ and $s = 1/2$, so $j = 5/2$ or $3/2$. As usual $m_\ell = 0, \pm 1, \pm 2$. The value of m_j ranges from $-j$ to j so its possible values are $\pm 1/2$, $\pm 3/2$, and $\pm 5/2$. As always $m_s = \pm 1/2$. The two possible term notations are $4D_{5/2}$ and $4D_{3/2}$.
16. $^1S_0 : S = 0, L = 0, J = 0$ $^2D_{5/2} : S = 1/2, L = 2, J = 5/2$
 $^5F_1 : S = 2, L = 3, J = 1$ $^3F_4 : S = 1, L = 3, J = 4$

17. The quantum number m_j ranges from $-J$ to J , or $-7/2$ to $+7/2$. Then $J_z = m_j \hbar = \pm \hbar/2, \pm 3\hbar/2, \pm 5\hbar/2, \pm 7\hbar/2$.

18. (a) The quantum number m_j ranges from $-J$ to J , or $-7/2$ to $+7/2$. Then

$$J_z = m_j \hbar = \pm \hbar/2, \pm 3\hbar/2, \pm 5\hbar/2, \pm 7\hbar/2.$$

(b) The minimum angle occurs when m_j is at its maximum value, which is $+7/2$. Then

$$J_z = 7\hbar/2 \text{ and } \cos \theta = \frac{J_z}{J} = \frac{m_j \hbar}{\sqrt{J(J+1)} \hbar} = \frac{7/2}{\sqrt{(7/2)(9/2)}} = \frac{\sqrt{7}}{3} \text{ so } \theta = 28.1^\circ.$$

19. The 1s subshell is filled, and there is one 2s electron. Therefore $L = 0$ (which gives the S in the middle of the symbol). The single unpaired electron gives $S = 1/2$ or $2S + 1 = 2$. With $L = 0$ and $S = 1/2$ the only possible value for J is $J = 1/2$.

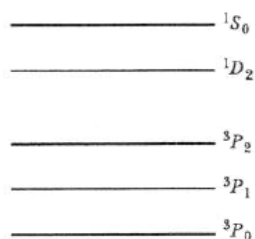
20. Ga has a ground state configuration with a filled 3d shell and $4s^2 4p^1$. The single unpaired electron gives $S = 1/2$, so $2S + 1 = 2$. The unpaired electron is in a p subshell, so $L = 1$, for which the symbol is P. Then $J = L \pm S$ which is $1/2$ or $3/2$, but the $1/2$ state has a slightly lower energy. Therefore the symbol is $4^2P_{1/2}$.

$$21. \Delta E = \frac{hc}{\lambda_1} - \frac{hc}{\lambda_2} = (1240 \text{ eV} \cdot \text{nm}) \left(\frac{1}{766.41 \text{ nm}} - \frac{1}{769.90 \text{ nm}} \right) = 7.334 \times 10^{-3} \text{ eV}.$$

As in Example 8.8 the internal magnetic field is

$$B = \frac{m\Delta E}{e\hbar} = \frac{(9.109 \times 10^{-31} \text{ kg})(7.334 \times 10^{-3} \text{ eV})}{(1.602 \times 10^{-19} \text{ C})(6.582 \times 10^{-16} \text{ eV} \cdot \text{s})} = 63.4 \text{ T}$$

22. By the selection rules $\Delta L = \pm 1$, $\Delta S = 0$, $\Delta J = 0, \pm 1$, no transitions are allowed between the pictured levels. The selection rule $\Delta S = 0$ prohibits singlet $\rightarrow$ triplet transitions.



23. The 2s to 1s transition is forbidden by the $\Delta L = \pm 1$ selection rule. The two lines result from the transitions from the two 2p levels to the 1s level.

24. As in Example 8.8

$$\Delta E = \frac{e\hbar B}{m} = \frac{(1.602 \times 10^{-19} \text{ C})(6.582 \times 10^{-16} \text{ eV} \cdot \text{s})(2.55 \text{ T})}{9.109 \times 10^{-31} \text{ kg}} = 2.95 \times 10^{-4} \text{ eV}.$$

25. The ground state in He is a singlet, so an excited state may be a singlet or triplet, and the behavior of Ca ($4s^2$) and Sr ($5s^2$) must be the same. Al has a $3s^2 3p^1$ configuration. The single unpaired electron gives $S = 1/2$ and $2S + 1 = 2$, which is a doublet.

26. The minimum angle corresponds to the maximum value of J_z and hence the maximum

value of m_j , which is $m_j = j$. Then $\cos \theta = \frac{J_z}{J} = \frac{m_j \hbar}{\sqrt{j(j+1)} \hbar} = \frac{j}{\sqrt{j(j+1)}}$. Solving for

$$j \text{ we find } j = \frac{1}{\frac{1}{\cos^2 \theta} - 1} = 2.50 = 5/2.$$

27. $I = \frac{dq}{dt} = \frac{Ze}{2\pi r / v} = \frac{Zev}{2\pi r}$ From the Biot-Savart Law, $B = \frac{\mu_0 I}{2r} = \frac{\mu_0 Zev}{4\pi r^2}$. With the

assumption that the orbit is circular, the angular momentum is $L = mvr$, so $v = \frac{L}{mr}$ and

$$B = \frac{\mu_0 ZeL}{4\pi mr^3} = \frac{ZeL}{4\pi \epsilon_0 mc^2 r^3} \text{ where we have used the fact that } \mu_0 = \frac{1}{c^2 \epsilon_0}.$$

The directions of the vectors follow from the right-hand rule.

28. Using the fact that $g = 2$ we have $\vec{\mu}_s = g \frac{e\vec{S}}{2m} = \frac{e\vec{S}}{m}$ and thus $V_{s\ell} = \vec{\mu}_s \cdot \vec{B} = \frac{Ze^2}{4\pi \epsilon_0} \frac{\vec{S} \cdot \vec{L}}{m^2 c^2 r^3}$.

29. (a) In order to use the result of the previous problem, we need to know the directions of $\vec{S}$ and $\vec{L}$. The electron has $S = \pm 1/2$, so $\|\vec{S}\| = \sqrt{3/4} \hbar$ and the angle $\vec{S}$ makes with the

$$+z\text{-axis is } \theta = \cos^{-1} \left(\frac{\pm 1/2}{\sqrt{3/4}} \right) = 54.7^\circ \text{ or } 125.3^\circ.$$

With $L = 1$ we have $\|\vec{L}\| = \sqrt{2} \hbar$ and the vector $\vec{L}$ can have three possible orientations,

corresponding to $m_\ell = 0, \pm 1$. If we choose $m_\ell = 0$, then the orientation of the $\vec{L}$ vector is in the xy plane. Therefore for either spin state the angle between $\vec{L}$ and $\vec{S}$ is 35.3° and

$\vec{S} \cdot \vec{L} = \sqrt{\frac{3}{4}} \hbar \sqrt{2} \hbar \cos(35.3^\circ) = \hbar^2$. Then using $r = 5a_0$ for a $2p$ electron (see Chapter 7

Problem 40), we have

$$V = \frac{e^2 \hbar^2}{4\pi\epsilon_0 m^2 c^2 r^3} = \frac{(1.44 \text{ eV} \cdot \text{nm})(1240 \text{ eV} \cdot \text{nm})^2}{4\pi^2 (5.11 \times 10^5 \text{ eV})^2 (5 \times 0.0529 \text{ nm})^3} = 1.2 \times 10^{-5} \text{ eV}.$$

The difference between spin-up and spin-down states is twice this amount, or $2.4 \times 10^{-5} \text{ eV}$, which is just over half the measured value.

(b) The two possibilities are $j = 1/2$ and $j = 3/2$. The difference between these two is

$$\begin{aligned} \Delta V &= \frac{Z^4 \alpha^4}{2n^3} mc^2 \left(\frac{2}{2\left(\frac{1}{2}\right)+1} - \frac{2}{2\left(\frac{3}{2}\right)+1} \right) \\ &= \frac{\left(\frac{1}{137}\right)^4}{2(2)^3} (5.11 \times 10^5 \text{ eV}) \left(\frac{2}{2\left(\frac{1}{2}\right)+1} - \frac{2}{2\left(\frac{3}{2}\right)+1} \right) = 4.53 \times 10^{-5} \text{ eV} \end{aligned}$$

which gives a more accurate result.

30. In Example 8.10, it was shown that the normal Zeeman effect is expected when the total spin $S = 0$. The normal Zeeman effect is due to the interaction of the external magnetic field and the orbital angular momentum, which is true for all levels. If S is not zero, then the anomalous effect is expected. Thus the two effects are not expected in the same atom.

31. $\vec{\mu} = -\frac{\mu_B}{\hbar}(\vec{L} + 2\vec{S})$ and $\vec{J} = \vec{L} + \vec{S}$ Using the equation in the problem, we have

$$\vec{\mu}_{\text{av}} = -\frac{\mu_B}{\hbar} \frac{(\vec{L} + 2\vec{S}) \cdot (\vec{L} + \vec{S})}{J^2} \vec{J}$$

Now $(\vec{L} + 2\vec{S}) \cdot (\vec{L} + \vec{S}) = L^2 + 2S^2 + 3\vec{L} \cdot \vec{S}$ and using $\vec{J} = \vec{L} + \vec{S}$ we have

$$\vec{J} \cdot \vec{J} = L^2 + S^2 + 2\vec{L} \cdot \vec{S} \text{ so } \vec{L} \cdot \vec{S} = \frac{J^2 - L^2 - S^2}{2} \text{ and } (\vec{L} + 2\vec{S}) \cdot (\vec{L} + \vec{S}) = \frac{3J^2 + S^2 - L^2}{2}.$$

Therefore $\vec{\mu}_{\text{av}} = -\frac{\mu_B}{\hbar} \frac{3J^2 + S^2 - L^2}{2J^2} \vec{J}$. With $\vec{B}$ defined to be in the $+z$ -direction,

$$V = -\vec{\mu}_{\text{av}} \cdot \vec{B} = \frac{\mu_B B}{\hbar} \frac{3J^2 + S^2 - L^2}{2J^2} J_z. \text{ As usual } J_z = m_J \hbar, \text{ so } V = \mu_B B m_J \frac{3J^2 + S^2 - L^2}{2J^2}.$$

Now the vector magnitudes are $J^2 = j(j+1)\hbar^2$, $S^2 = s(s+1)\hbar^2$, and $L^2 = \ell(\ell+1)\hbar^2$ so we

obtain $V = \mu_B B m_j g$ where

$$g = \frac{3j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)} = 1 + \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)}.$$

$$32. \quad \Delta E = \frac{hc}{\lambda_1} - \frac{hc}{\lambda_2} = \frac{1240 \text{ eV} \cdot \text{nm}}{422.7 \text{ nm}} - \frac{1240 \text{ eV} \cdot \text{nm}}{422.7168 \text{ nm}} = 1.166 \times 10^{-4} \text{ eV}$$

[Note: The same answer to four significant figures is obtained using the approximation $|\Delta E| = hc |\Delta \lambda| / \lambda^2$ as in Example 8.3.] Also $\Delta E = \mu_B B \Delta m_\ell = (e\hbar / 2m) B$ with $\Delta m_\ell = 1$.

$$\text{Thus } \frac{e\hbar}{m} = \frac{2\Delta E}{B} = \frac{2(1.166 \times 10^{-4} \text{ eV})}{2.00 \text{ T}} \frac{1.602 \times 10^{-19} \text{ J}}{\text{eV}} = 1.868 \times 10^{-23} \text{ J/T}. \text{ The accepted}$$

value is $\frac{e\hbar}{m} = 2\mu_B = 2(9.274 \times 10^{-24} \text{ J/T}) = 1.855 \times 10^{-23} \text{ J/T}$ which is less than 0.2% lower

than the experimental value. Using the experimental data with the known value of $\hbar$,

$$\frac{e}{m} = \left(\frac{e\hbar}{m} \right) \frac{1}{\hbar} = \frac{1.868 \times 10^{-23} \text{ J/T}}{1.055 \times 10^{-34} \text{ J} \cdot \text{s}} = 1.77 \times 10^{11} \text{ C/kg}, \text{ which compares favorably to the known}$$

$$\text{value of } \frac{e}{m} = \frac{1.6022 \times 10^{-19} \text{ C}}{9.1094 \times 10^{-31} \text{ kg}} = 1.76 \times 10^{11} \text{ C/kg}.$$

$$33. \text{ (a) } L = 0, S = J = 1/2; \quad g = 1 + \frac{(1/2)(3/2) + (1/2)(3/2)}{2(1/2)(3/2)} = 1 + 1 = 2$$

$$\text{(b) } L = 1, S = 1/2, \text{ and } J = 3/2; \quad g = 1 + \frac{(3/2)(5/2) + (1/2)(3/2) - 1(2)}{2(3/2)(5/2)} = 1 + \frac{1}{3} = \frac{4}{3}$$

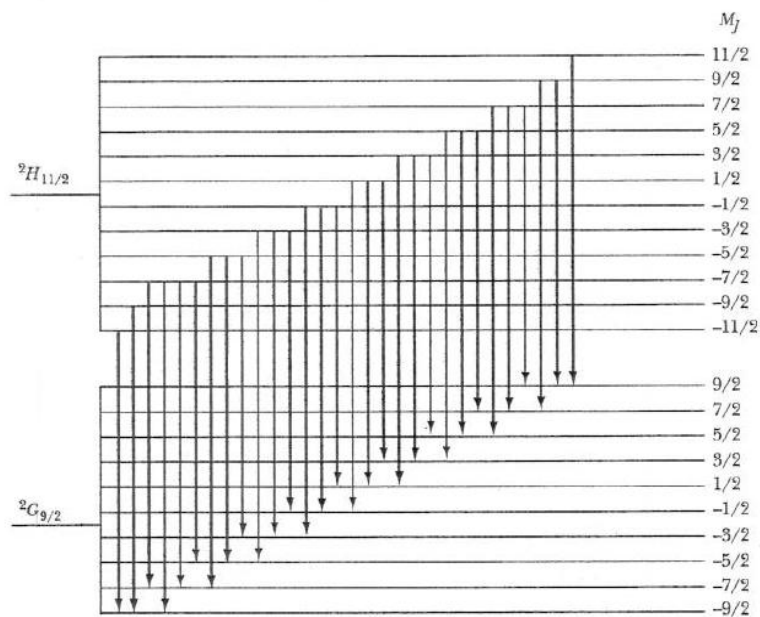
$$\text{(c) } L = 2, S = 1/2, \text{ and } J = 5/2; \quad g = 1 + \frac{(5/2)(7/2) + (1/2)(3/2) - 2(3)}{2(5/2)(7/2)} = 1 + \frac{1}{5} = \frac{6}{5}$$

34. Computing the g factors: ${}^2G_{9/2}$: $L = 4$, $S = 1/2$, and $J = 9/2$;

$$g = 1 + \frac{(9/2)(11/2) + (1/2)(3/2) - 4(5)}{2(9/2)(11/2)} = 1 + \frac{1}{9} = \frac{10}{9}$$

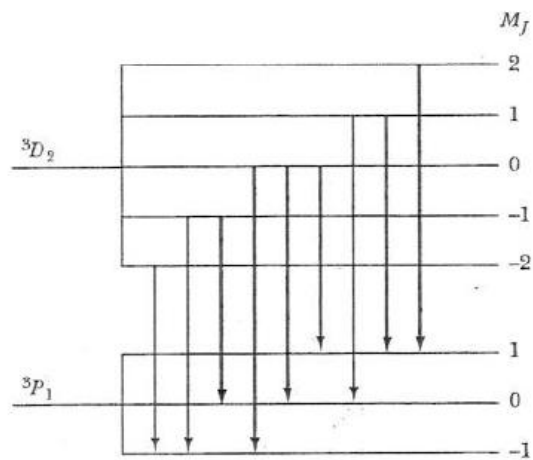
${}^2H_{11/2}$: $L = 5$, $S = 1/2$, and $J = 11/2$;

$$g = 1 + \frac{(11/2)(13/2) + (1/2)(3/2) - 5(6)}{2(11/2)(13/2)} = 1 + \frac{1}{11} = \frac{12}{11}$$



35. 3P_1 : $L=1$, $S=1$, and $J=1$; $g = 1 + \frac{(1)(2) + (1)(2) - 1(2)}{2(1)(2)} = 1 + \frac{1}{2} = \frac{3}{2}$

3D_2 : $L=2$, $S=1$, and $J=2$; $g = 1 + \frac{(2)(3) + (1)(2) - 2(3)}{2(2)(3)} = 1 + \frac{1}{6} = \frac{7}{6}$



36. From Equation (8.22) we have an energy splitting of $V = \mu_B B g m_j$.

For the $^2S_{1/2}$ state $g = 2$, so $V = 2\mu_B B m_j$. Now m_j can vary from $-J$ to $+J$, so $m_j = \pm 1/2$ and we find $V = \pm \mu_B B$.

For the $^2P_{1/2}$ state $g = 2/3$, so $V = \frac{2}{3}\mu_B B m_j$. Now m_j can vary from $-J$ to $+J$, so $m_j = \pm 1/2$ and we find $V = \pm \frac{1}{3}\mu_B B$.

The maximum difference is from $+\frac{1}{3}\mu_B B$ to $-\mu_B B$ or from $-\frac{1}{3}\mu_B B$ to $+\mu_B B$. In each case the magnitude of the energy difference is $\frac{4}{3}\mu_B B$ for a total difference of $\frac{8}{3}\mu_B B$. Calling this difference ΔE , it will show up as the usual wavelength difference $\Delta\lambda$ with $\Delta E = (hc / \lambda^2) \Delta\lambda$. Thus

$$\Delta\lambda = \frac{\lambda^2 \Delta E}{hc} = \frac{\lambda^2}{hc} \frac{8}{3} \mu_B B = \frac{8}{3} \frac{(5.8976 \times 10^{-7} \text{ m})^2 (9.274 \times 10^{-24} \text{ J/T})(2.50 \text{ T})}{(6.626 \times 10^{-34} \text{ J}\cdot\text{s})(2.997 \times 10^8 \text{ m/s})}$$

$$= 1.083 \times 10^{-10} \text{ m}$$

37. As in the preceding problem we have $g = 4/3$ for the $^2P_{3/2}$ state and $V = \frac{4}{3}\mu_B B m_j$. The states are split by $\Delta m_j = 1$ and so the energy difference between two adjacent states is $\Delta E = \frac{4}{3}\mu_B B$.

$$\text{Then } \Delta E = \frac{4}{3}\mu_B B = \frac{4}{3}(9.274 \times 10^{-24} \text{ J/T})(1.20 \text{ T}) = 6.18 \times 10^{-24} \text{ J} = 9.26 \times 10^{-5} \text{ eV}.$$

38. (a) Calcium has two $4s$ subshell electrons outside a closed $3p^6$ subshell. So the electronic configuration is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$. Aluminum has one $3p$ electron outside the closed $3s$ level of magnesium, so its electronic configuration is $1s^2 2s^2 2p^6 3s^2 3p^1$.

(b) The LS coupling for calcium can be determined since both outer shell electrons have $\ell = 0$, then $L = 0$ and $S = 0$ since one electron will have spin up and one will have spin down. Then $J = L + S$ will also be 0. Therefore in spectroscopic notation, $n^{2S+1}L_J$, calcium will be 4^1S_0 . The $3p$ electron for aluminum will have $\ell = 1$. Therefore $L = 1$ and $S = 1/2$. Now $J = L \pm S$, so $J = 3/2$ or $J = 1/2$. The state with lower J has lower energy, so in spectroscopic notation, aluminum is $3^2P_{1/2}$.

39. (a) Yttrium (Y) has one $4d$ and two $5s$ electrons outside a closed $4p^6$ subshell. So Y^- would have one additional electron in the $4d$ subshell. So the electronic configuration is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^2$. Aluminum has one $3p$ electron outside the closed

3s subshell. So Al^- would have one additional electron in the 3p subshell. So its electronic configuration is $1s^2 2s^2 2p^6 3s^2 3p^2$.

(b) We will use the LS coupling model and Hund's rules to find the lowest energy state. For Y^- we must consider the two 4d electrons. According to Hund's rules, the triplet state $S = 1$ will be lowest in energy. Additionally, we attempt to maximize L as long as we do not violate the exclusion principle. Both 4d electrons have $\ell = 2$, so we can have $L = 0, 1, 2, 3$, or 4. As discussed in Example 8.9 of the text, the antisymmetrization of the wave function requires that the m_ℓ values for the electrons forming the $S = 1$ state must be unequal. The largest allowed L is thus 3. Finally, J should be minimized, so we have $J = 2$. Therefore in spectroscopic notation, $n^{2S+1}L_J$, we have 5^3F_2 .

For Al^- we consider the two 3p electrons outside the closed subshell. As above, the triplet state $S = 1$ will have lowest energy. The 3p electrons have $\ell = 1$ so we can have $L = 0, 1$, or 2. Again using the antisymmetrization of the wave function requires the m_ℓ values for the electrons forming the $S = 1$ state to be unequal. The largest allowed L is therefore 1. Finally the smallest J will be 0. So in spectroscopic notation we have 3^3P_0 .

40. (a) $^{2S+1}L_J$ is standard spectroscopic notation. For $^2L_{3/2}$, reading from the notation, $J = 3/2$ and $2S + 1 = 2$, so $S = 1/2$. The “D” implies that $L = 2$. This is permitted, because in this case $J = L - S$. With $J = 3/2$, the possible values of J_z are $\pm\hbar/2$ and $\pm 3\hbar/2$.
 (b) With $J = 3/2$, the quantum number m_J ranges from $-3/2$ to $+3/2$. The minimum angle is achieved when $m_J = 3/2$. In this case

$$\cos \theta = \frac{J_z}{J} = \frac{m_J \hbar}{\sqrt{J(J+1)} \hbar} = \frac{3/2}{\sqrt{(3/2)(5/2)}} = \sqrt{3/5}$$

Thus $\theta = 39.2^\circ$.

41. Co has a ground state configuration $3d^7 4s^2$. By Hund's rules there are two pairs of d -electrons and three unpaired ones, so $S = 3/2$ and $2S + 1 = 4$. The unpaired electrons have $m_\ell = 0, 1, 2$, so the total angular momentum is $L = 3$, indicating the symbol F . The maximum value of J is $L + S = 3 + 3/2 = 9/2$. Therefore the full term symbol is $^4F_{9/2}$.

Chapter 9

1. (a)

$$\begin{aligned}\overline{v_x^2} &= \int_{-\infty}^{\infty} v_x^2 g(v_x) dv_x = \left(\frac{\beta m}{2\pi}\right)^{1/2} \int_{-\infty}^{\infty} v_x^2 \exp\left(-\frac{1}{2}\beta m v_x^2\right) dv_x \\ &= 2\left(\frac{\beta m}{2\pi}\right)^{1/2} \int_0^{\infty} v_x^2 \exp\left(-\frac{1}{2}\beta m v_x^2\right) dv_x = 2\left(\frac{\beta m}{2\pi}\right)^{1/2} \frac{\sqrt{\pi}}{4} \left(\frac{2}{\beta m}\right)^{3/2} = \frac{1}{\beta m}\end{aligned}$$

Therefore

$$v_{x\text{rms}} = \left(\overline{v_x^2}\right)^{1/2} = \left(\frac{1}{\beta m}\right)^{1/2} = \left(\frac{kT}{m}\right)^{1/2}$$

$$(b) \quad g(v_x) = \left(\frac{\beta m}{2\pi}\right)^{1/2} \exp\left(-\frac{1}{2}\beta m v_x^2\right)$$

and from (a) we see that $(\beta m)^{1/2} = v_{x\text{rms}}^{-1}$, so

$$g(v_x) dv_x = \frac{1}{\sqrt{2\pi}} v_{x\text{rms}}^{-1} \exp\left(-\frac{1}{2} \frac{v_x^2}{v_{x\text{rms}}^2}\right) dv_x$$

$$2. (a) \text{ With } v_x = 0.01 v_{x\text{rms}} \text{ we have } \exp\left(-\frac{1}{2} \frac{v_x^2}{v_{x\text{rms}}^2}\right) \approx 1.$$

$$g(v_x) dv_x = \frac{1}{\sqrt{2\pi}} v_{x\text{rms}}^{-1} \exp\left(-\frac{1}{2} \frac{v_x^2}{v_{x\text{rms}}^2}\right) dv_x = \frac{1}{\sqrt{2\pi}} v_{x\text{rms}}^{-1} (1) (0.002 v_{x\text{rms}}) = 7.98 \times 10^{-4}$$

This is the probability that a given molecule will be in this range, so in one mole the number is

$$N = (7.98 \times 10^{-4}) N_A = (7.98 \times 10^{-4}) (6.022 \times 10^{23}) = 4.81 \times 10^{20}$$

$$(b) \text{ With } v_x = 0.20 v_{x\text{rms}} \text{ we have } \exp\left[-\frac{1}{2} \frac{(0.20 v_{x\text{rms}})^2}{(v_{x\text{rms}})^2}\right] = 0.980. \text{ Continuing as in (a) we find}$$

$$g(v_x) dv_x = \frac{1}{\sqrt{2\pi}} v_{x\text{rms}}^{-1} (0.98) (0.002 v_{x\text{rms}}) = 7.82 \times 10^{-4} \text{ and therefore } N = 4.71 \times 10^{20}.$$

$$(c) N = 2.91 \times 10^{20}$$

(d) $N = 1.79 \times 10^{15}$

(e) In this case $g(v_x)dv_x = (7.98 \times 10^{-4}) \exp(-5 \times 10^3)$ which is on the order of 10^{-2175} .

Therefore we conclude that no molecules travel at that speed.

3. (a) $\bar{f} = f_0 \left(1 + \frac{v}{c}\right) = f_0 \left(1 + \frac{v_x}{c}\right) = f_0 (1 + 0) = f_0$

(b) $\sigma = \left(\overline{(f - f_0)^2}\right)^{1/2} = \left(\overline{\left(\frac{f_0 v_x}{c}\right)^2}\right)^{1/2} = \left(f_0^2 \frac{\overline{v_x^2}}{c^2}\right)^{1/2}$

But we know that $\overline{v_x^2} = kT / m$, so $\sigma = \left(\frac{f_0^2}{c^2} \frac{kT}{m}\right)^{1/2} = \frac{f_0}{c} \sqrt{\frac{kT}{m}}$.

(c) From (b) we have $\frac{\sigma}{f_0} = \frac{1}{c} \sqrt{\frac{kT}{m}}$.

H₂ at $T = 293$ K: $\frac{\sigma}{f_0} = \frac{1}{3.00 \times 10^8 \text{ m/s}} \sqrt{\frac{1.381 \times 10^{-23} \text{ J/K} (293 \text{ K})}{2(1.674 \times 10^{-27} \text{ kg})}} = 3.66 \times 10^{-6}$

H at $T = 5500$ K: $\frac{\sigma}{f_0} = \frac{1}{3.00 \times 10^8 \text{ m/s}} \sqrt{\frac{1.381 \times 10^{-23} \text{ J/K} (5500 \text{ K})}{(1.674 \times 10^{-27} \text{ kg})}} = 2.25 \times 10^{-5}$

This is how we could deduce the surface temperature of a star.

4. (a) Letting d be the distance between the two atoms, we have

$$I_x = 2(mr^2) = 2m\left(\frac{d}{2}\right)^2 = \frac{md^2}{2} = \frac{16(1.66 \times 10^{-27} \text{ kg})(1.21 \times 10^{-10} \text{ m})^2}{2}$$

$$= 1.94 \times 10^{-46} \text{ kg} \cdot \text{m}^2$$

(b)

$$I_z = 2\left(\frac{2}{5}mR^2\right) = \frac{4}{5}mR^2 = 0.8(16)(1.66 \times 10^{-27} \text{ kg})(3.0 \times 10^{-15} \text{ m})^2$$

$$= 1.91 \times 10^{-55} \text{ kg} \cdot \text{m}^2$$

(c) The rigid rotor is quantized (see Chapter 10) with an energy

$$E = \frac{\hbar^2 \ell(\ell+1)}{2I} = \frac{(1.055 \times 10^{-34} \text{ J} \cdot \text{s})^2 (1)(2)}{2(1.94 \times 10^{-46} \text{ kg} \cdot \text{m}^2)} = 5.74 \times 10^{-23} \text{ J} = 3.58 \times 10^{-4} \text{ eV}$$

(d) Rearranging the energy equation in (c) and using the value of I_z to find the ℓ value, we find

$$\ell(\ell+1) = \frac{2IE}{\hbar^2} = \frac{2(1.91 \times 10^{-55} \text{ kg} \cdot \text{m}^2)(5.74 \times 10^{-23} \text{ J})}{(1.055 \times 10^{-34} \text{ J} \cdot \text{s})^2} = 1.97 \times 10^{-9}$$

This shows that a much larger energy (larger E_{rot}) is required to have $\ell=1$ for the rotation about the z axis. Therefore the diatomic molecule acts as if there were only two degrees of rotational freedom.

5. (a) $\int_c^\infty F(v) dv = 4\pi C \int_c^\infty v^2 \exp\left(-\frac{1}{2}\beta m v^2\right) dv$ with $T = 293 \text{ K}$ and $C = \left(\frac{\beta m}{2\pi}\right)^{3/2}$.

(b) For example H_2 gas at $T = 293 \text{ K}$ we have

$$\frac{1}{2}\beta m c^2 = \frac{(1)(2)(938 \times 10^6 \text{ eV})}{2(8.62 \times 10^{-5} \text{ eV/K})(293 \text{ K})} = 3.7 \times 10^{10}.$$

The exponential of the negative of this value, $\exp(-3.7 \times 10^{10})$, is almost 0.

6. Computations depend on the software but should yield numbers very close to zero.

7. (a) $\bar{v} = \frac{4}{\sqrt{2\pi}} \sqrt{\frac{kT}{m}} = \frac{4}{\sqrt{2\pi}} \sqrt{\frac{(1.381 \times 10^{-23} \text{ J/K})(300 \text{ K})}{(1.675 \times 10^{-27} \text{ kg})}} = 2510 \text{ m/s}$

$$v^* = \sqrt{\frac{2kT}{m}} = \sqrt{\frac{2(1.381 \times 10^{-23} \text{ J/K})(300 \text{ K})}{(1.675 \times 10^{-27} \text{ kg})}} = 2220 \text{ m/s}$$

(b) $\bar{v} = \frac{4}{\sqrt{2\pi}} \sqrt{\frac{kT}{m}} = \frac{4}{\sqrt{2\pi}} \sqrt{\frac{(1.381 \times 10^{-23} \text{ J/K})(630 \text{ K})}{(1.675 \times 10^{-27} \text{ kg})}} = 3640 \text{ m/s}$

$$v^* = \sqrt{\frac{2kT}{m}} = \sqrt{\frac{2(1.381 \times 10^{-23} \text{ J/K})(630 \text{ K})}{(1.675 \times 10^{-27} \text{ kg})}} = 3220 \text{ m/s}$$

8. $F(v) = 4\pi C \exp\left(-\frac{1}{2}\beta m v^2\right) v^2$. In the limit as $v \rightarrow 0$ the exponential is approximately $e^0 = 1$ and v^2 approaches 0, so $\lim_{v \rightarrow 0} F(v) = 0$.

The other limit is $\lim_{v \rightarrow \infty} F(v) = 4\pi C \lim_{v \rightarrow \infty} \frac{v^2}{\exp\left(\frac{1}{2}\beta m v^2\right)}$.

Applying L'Hopital's rule,

$$\lim_{v \rightarrow \infty} F(v) = \lim_{v \rightarrow \infty} \frac{2v}{\beta m v \exp\left(\frac{1}{2}\beta m v^2\right)} = \lim_{v \rightarrow \infty} \frac{2}{\beta m} \exp\left(-\frac{1}{2}\beta m v^2\right) = 0.$$

9. (a) $v^* = \sqrt{\frac{2kT}{m}} = \sqrt{\frac{2(1.381 \times 10^{-23} \text{ J/K})(258 \text{ K})}{28(1.6605 \times 10^{-27} \text{ kg})}} = 391 \text{ m/s}$

(b) $v^* = \sqrt{\frac{2kT}{m}} = \sqrt{\frac{2(1.381 \times 10^{-23} \text{ J/K})(308 \text{ K})}{28(1.6605 \times 10^{-27} \text{ kg})}} = 428 \text{ m/s}$

10. The equation to be satisfied is $2v^2 \exp\left(-\frac{1}{2}\beta m v^2\right) = v^{*2} \exp\left(-\frac{1}{2}\beta m v^{*2}\right) = \frac{2kT}{m} e^{-1}$

where we have used the fact that $v^* = \sqrt{\frac{2kT}{m}}$. Thus $v^2 \exp\left(-\frac{1}{2}\beta m v^2\right) = \frac{kT}{m} e^{-1} \approx 28000$

which can be solved graphically to yield $v = 188 \text{ m/s}$ and $v = 639 \text{ m/s}$. The lower of these is closer to $v^* = 390 \text{ m/s}$, which follows from the shape of the distribution curve.

11. Various software packages should all give results very close to 1.

12. The calculations start from Equation (9.14) and are of the form:

$$I = 4\pi \left(\frac{\beta m}{2\pi}\right)^{3/2} \int_a^b v^2 \exp\left(-\frac{1}{2}\beta m v^2\right) dv$$

The limits a and b are given in each part. Values are, as a fraction of the total number of molecules,

(a) 2.0×10^{-7} (b) 2.0×10^{-4} (c) 0.156 (d) 0.494 (e) 0.350 (f) 0.99987

13. (a) From Equation (9.20) we have $v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$. The mass of $\text{H}_2 = 2(1.008 \text{ u}) = 2.02 \text{ u}$ and the mass of $\text{N}_2 = 2(14.003 \text{ u}) = 28.0 \text{ u}$.

$$\text{H}_2 : \quad v_{\text{rms}} = \sqrt{\frac{(3)(1.38 \times 10^{-23} \text{ J/K})(293 \text{ K})}{(2.02 \text{ u})(1.66 \times 10^{-27} \text{ kg/u})}} = 1902 \text{ m/s}$$

$$\text{N}_2 : \quad v_{\text{rms}} = \sqrt{\frac{(3)(1.38 \times 10^{-23} \text{ J/K})(293 \text{ K})}{(28.0 \text{ u})(1.66 \times 10^{-27} \text{ kg/u})}} = 511 \text{ m/s}$$

- (b) From classical physics we know the escape velocity $v_{\text{esc}} = \sqrt{\frac{2GM}{R}}$. Using the mass and radius of the earth we find $v_{\text{esc}} = 1.12 \times 10^4 \text{ m/s}$ if the object starts at the surface. Neither H_2 nor N_2 has $v_{\text{rms}} > v_{\text{esc}}$; however, a very small percentage of molecules in the exponential tail of the distribution may have speeds greater than v_{esc} and will escape. Since H_2 has a larger v_{rms} , a larger fraction will eventually escape.

14. (a) Use Equation (9.8) for the translational kinetic energy of one atom and multiply by Avogadro's number for one mole.

$$K = N_A \left(\frac{3}{2} kT \right) = (6.022 \times 10^{23}) \left(\frac{3}{2} (1.381 \times 10^{-23} \text{ J/K})(273 \text{ K}) \right) = 3406 \text{ J}.$$

- (b) Since the translational kinetic energy depends only on temperature and one mole of anything contains the same number of objects, the answer is the same for argon or oxygen.

$$15. (a) \quad \bar{E} = \int_0^\infty EF(E)dE = \frac{8\pi C}{\sqrt{2}m^{3/2}} \int_0^\infty E^{3/2} \exp(-\beta E)dE = \frac{8\pi C}{\sqrt{2}m^{3/2}} \frac{\Gamma(5/2)}{\beta^{5/2}}$$

$$\text{Using } \Gamma(5/2) = \frac{3}{2} \Gamma(3/2) = \frac{3\sqrt{\pi}}{4} \text{ and } C = (\beta m / 2\pi)^{3/2} \text{ we find}$$

$$\bar{E} = \frac{8\pi}{\sqrt{2}m^{3/2}} \left(\frac{\beta m}{2\pi} \right)^{3/2} \frac{3\sqrt{\pi}}{4\beta^{5/2}} = \frac{3}{2\beta} = \frac{3}{2} kT.$$

(b) As we know from the text $\bar{E} = \frac{1}{2} m \bar{v}^2$ and by Equation (9.17) $\frac{1}{2} m \bar{v}^2 = \frac{4}{\pi} kT \approx 1.27kT$ which is a bit less than $\frac{3}{2} kT$.

16. Starting with the distribution $F(E) = \frac{8\pi C}{\sqrt{2m^{3/2}}} E^{1/2} \exp(-\beta E)$ and setting $\frac{dF}{dE} = 0$, we obtain

$$0 = \frac{d}{dE} [E^{1/2} \exp(-\beta E)] = \frac{1}{2} E^{-1/2} \exp(-\beta E) - \beta E^{1/2} \exp(-\beta E).$$

Therefore $0 = E^{-1/2} - 2\beta E^{1/2}$ and solving for E gives the desired result $E^* = \frac{kT}{2}$.

17. The ratio of the numbers on the two levels is

$$\frac{n_2(E)}{n_1(E)} = \frac{8 \exp(-\beta E_2)}{2 \exp(-\beta E_1)} = 4 \exp(-\beta(E_2 - E_1)) = 10^{-5}$$

Therefore $\exp(-\beta(E_2 - E_1)) = 2.5 \times 10^{-6}$.

Taking the logarithm of each side gives:

$$-\beta(E_2 - E_1) = -\frac{E_2 - E_1}{kT} = \ln(2.5 \times 10^{-6}) = -12.90. \text{ For atomic hydrogen}$$

$$E_2 - E_1 = \frac{3}{4} E_0 = 10.20 \text{ eV. Finally}$$

$$T = -\frac{E_2 - E_1}{k(-12.90)} = -\frac{10.20 \text{ eV}}{(8.618 \times 10^{-5} \text{ eV/K})(-12.90)} = 9175 \text{ K.}$$

18. (a) With $E = \frac{p^2}{2m}$ and the mean energy $E = \frac{3}{2} kT$ we obtain $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{3mkT}}$.

(b) We have $\lambda \ll d$. Using λ from part (a) and $d = (V/N)^{1/3}$ we obtain

$$\frac{h}{\sqrt{3mkT}} \ll \left(\frac{V}{N}\right)^{1/3} \text{ If we cube both sides and rearrange } \frac{N}{V} \frac{h^3}{(3mkT)^{3/2}} \ll 1.$$

(c) For any ideal gas $\frac{N}{V} = \frac{6.022 \times 10^{23}}{22.4 \times 10^{-3} \text{ m}^3} = 2.69 \times 10^{25} \text{ m}^{-3}$.

For argon gas (a monatomic gas) at room temperature

$$\frac{N}{V} \frac{h^3}{(3mkT)^{3/2}} = (2.69 \times 10^{25} \text{ m}^{-3}) \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})^3}{(3(40)(1.66 \times 10^{-27} \text{ kg})(1.38 \times 10^{-23} \text{ J/K})(293 \text{ K}))^{3/2}}$$

$$\approx 3 \times 10^{-7}$$

so Maxwell-Boltzmann statistics are fine. However, for electrons in silver,

$$N/V = 5.86 \times 10^{28} \text{ m}^{-3} \text{ and}$$

$$\frac{N}{V} \frac{h^3}{(3mkT)^{3/2}} = (5.86 \times 10^{28} \text{ m}^{-3}) \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})^3}{(3(9.11 \times 10^{-31} \text{ kg})(1.38 \times 10^{-23} \text{ J/K})(293 \text{ K}))^{3/2}}$$

$$\approx 1.5 \times 10^4$$

so in this case Maxwell-Boltzmann statistics fail.

$$19. \quad F(v)dv = 4\pi C v^2 \exp\left(-\frac{1}{2}\beta m v^2\right) dv = F(E)dE$$

With $E = \frac{1}{2}mv^2$ we differentiate to obtain $dE = mv dv$ or $dv = \frac{dE}{mv} = \frac{dE}{\sqrt{2mE}}$. Then

$$F(E)dE = 4\pi C \frac{2E}{m} \exp(-\beta E) \frac{dE}{\sqrt{2mE}} = 8\pi C \frac{E^{1/2}}{\sqrt{2m^{3/2}}} \exp(-\beta E) dE$$

$$= \frac{8\pi C}{\sqrt{2m^{3/2}}} E^{1/2} \exp(-\beta E) dE.$$

20. (a) We assume that the magnetic moment is due to spin alone. In general

$n(E) = g(E) F_{\text{MB}}$. There is no reason to prefer one spin state or the other, so the two

$g(E)$ are the same. Thus the ratio of the numbers in the two spin states is governed by the

$$\text{Maxwell-Boltzmann distribution: } \frac{n(E_2)}{n(E_1)} = \frac{F_{\text{MB}}(E_2)}{F_{\text{MB}}(E_1)} = \frac{\exp(-\beta E_2)}{\exp(-\beta E_1)} = \exp(\beta(E_1 - E_2)).$$

The energy of a magnetic moment $\vec{\mu}$ in a magnetic field $\vec{B}$ is $E = -\vec{\mu} \cdot \vec{B}$. We know from

Chapter 7 that this is equal to $E = \frac{e}{m} \vec{S} \cdot \vec{B} = \frac{e}{m} S_z B = \pm \frac{e\hbar}{2m} B = \pm \mu_B B$.

Then $E_1 = -\mu_B B$ is the energy of an electron aligned with the field, and $E_2 = +\mu_B B$ is the energy of the spin opposed to the field. Therefore

$$\frac{n(E_2)}{n(E_1)} = \exp(\beta(E_1 - E_2)) = \exp\left(\frac{-\mu_B B - \mu_B B}{kT}\right) = \exp\left(\frac{-2\mu_B B}{kT}\right).$$

$$(b) \text{ At } T = 77 \text{ K: } \frac{n(E_2)}{n(E_1)} = \exp\left(\frac{-2\mu_B B}{kT}\right) = \exp\left(\frac{-2(9.274 \times 10^{-24} \text{ J/T})(5.0 \text{ T})}{(1.381 \times 10^{-23} \text{ J/K})(77 \text{ K})}\right) = 0.916$$

$$\text{At } T = 273 \text{ K: } \frac{n(E_2)}{n(E_1)} = \exp\left(\frac{-2\mu_B B}{kT}\right) = \exp\left(\frac{-2(9.274 \times 10^{-24} \text{ J/T})(5.0 \text{ T})}{(1.381 \times 10^{-23} \text{ J/K})(273 \text{ K})}\right) = 0.976$$

$$\text{At } T = 600 \text{ K: } \frac{n(E_2)}{n(E_1)} = \exp\left(\frac{-2\mu_B B}{kT}\right) = \exp\left(\frac{-2(9.274 \times 10^{-24} \text{ J/T})(5.0 \text{ T})}{(1.381 \times 10^{-23} \text{ J/K})(600 \text{ K})}\right) = 0.993$$

As temperature is increased, the probabilities for spin aligned with the magnetic field (lower energy) and anti-aligned (higher energy) become essentially equal, as the energy difference between them becomes much less than kT .

21. Starting with Equation (9.30) and setting $F_{FD} = 0.5$ when $E = E_F$, we have

$$0.5 = \frac{1}{B_{FD} \exp(\beta E_F) + 1}. \text{ Solving for } B_{FD} \text{ we find } B_{FD} \exp(\beta E_F) + 1 = 2 \text{ so}$$

$$B_{FD} \exp(\beta E_F) = 1 \text{ and } B_{FD} = \exp(-\beta E_F). \text{ Therefore in general}$$

$$F_{FD} = \frac{1}{B_{FD} \exp(\beta E) + 1} = \frac{1}{\exp(-\beta E_F) \exp(\beta E) + 1} = \frac{1}{\exp(\beta(E - E_F)) + 1}.$$

22. At first one might think it should be 0.5, but this is not quite true due to the asymmetric shape of the distribution. Starting with Equation (9.43) for $g(E)$ and using the fact that

$F_{FD} \approx 1$ in this range, we have

$$N(E < E_F) = \int_0^{\bar{E}} g(E) dE = \frac{3}{2} N E_F^{-3/2} \int_0^{\bar{E}} E^{1/2} dE = N E_F^{-3/2} \bar{E}^{3/2}. \text{ Recall that } \bar{E} = \frac{3}{5} E_F, \text{ and}$$

$$\text{we see that } N(E < E_F) = N \left(\frac{3}{5} \right)^{3/2} = 0.465 N.$$

23. (a) From dimensional analysis

$$1.05 \times 10^4 \text{ kg/m}^3 \left(\frac{1 \text{ mol}}{0.10787 \text{ kg}} \right) \left(\frac{6.022 \times 10^{23}}{\text{mol}} \right) = 5.86 \times 10^{28} \text{ m}^{-3}.$$

(b) For electrons an extra factor of 2 is required due to the Pauli principle:

$$\frac{N}{V} = \frac{2A}{h^3} (2\pi mkT)^{3/2} \quad \text{so}$$

$$T = \frac{\left(\frac{N}{2AV}\right)^{2/3} h^2}{2\pi mk} = \frac{\left(\frac{5.86 \times 10^{28} \text{ m}^{-3}}{2(1)}\right)^{2/3} (6.626 \times 10^{-34} \text{ J}\cdot\text{s})^2}{2\pi (9.109 \times 10^{-31} \text{ kg})(1.381 \times 10^{-23} \text{ J/K})} = 5.28 \times 10^4 \text{ K}.$$

$$(c) T = \frac{\left(\frac{N}{2AV}\right)^{2/3} h^2}{2\pi mk} = \frac{\left(\frac{5.86 \times 10^{28} \text{ m}^{-3}}{2(0.001)}\right)^{2/3} (6.626 \times 10^{-34} \text{ J}\cdot\text{s})^2}{2\pi (9.109 \times 10^{-31} \text{ kg})(1.381 \times 10^{-23} \text{ J/K})} = 5.28 \times 10^6 \text{ K}.$$

24. The number is given by $\int_{0.90E_F}^{E_F} g(E) dE = \frac{3}{2} N E_F^{-3/2} \int_{0.90E_F}^{E_F} E^{1/2} dE = N E_F^{-3/2} E^{3/2} \Big|_{0.90E_F}^{E_F}$

$$= N (1.00^{3/2} - 0.90^{3/2}) = 0.146N$$

We see that about 14.6% of the electrons are in this range, which is about what one would expect from the shape of the distribution.

25. (a) As in Problem 23, $N/V = 5.86 \times 10^{28} \text{ m}^{-3}$. Then

$$E_F = \frac{h^2}{8m} \left(\frac{3N}{\pi V}\right)^{2/3} = \frac{(6.626 \times 10^{-34} \text{ J}\cdot\text{s})^2}{8(9.109 \times 10^{-31} \text{ kg})} \left(\frac{3}{\pi} (5.86 \times 10^{28} \text{ m}^{-3})\right)^{2/3}$$

$$= 8.81 \times 10^{-19} \text{ J} = 5.50 \text{ eV}.$$

$$(b) u_F = \sqrt{\frac{2E_F}{m}} = \sqrt{\frac{2(8.81 \times 10^{-19} \text{ J})}{9.109 \times 10^{-31} \text{ kg}}} = 1.39 \times 10^6 \text{ m/s}.$$

26. (a) Note: the term $\alpha(kT)^2/E_F$ is a small fraction of one eV and can be ignored. Then

$$\bar{E} = \frac{3}{5} E_F = \frac{3}{5} (5.51 \text{ eV}) = 3.31 \text{ eV}$$

$$(b) \text{ With } \bar{E} = \frac{3}{2} kT \text{ we have } T = \frac{2\bar{E}}{3k} = \frac{2(3.31 \text{ eV})}{3(8.617 \times 10^{-5} \text{ eV/K})} = 2.56 \times 10^4 \text{ K}$$

(c) As discussed in the text, thermal energies are small compared with the Fermi energy, except at high temperatures.

$$27. 8.92 \times 10^3 \text{ kg/m}^3 \left(\frac{1 \text{ mol}}{0.063546 \text{ kg}} \right) \left(\frac{6.022 \times 10^{23}}{\text{mol}} \right) = 8.45 \times 10^{28} \text{ m}^{-3}.$$

This represents a difference of 0.2% from the experimental value so within rounding errors there is one conduction electron per atom.

$$28. (a) 2.70 \times 10^3 \text{ kg/m}^3 \left(\frac{1 \text{ mol}}{0.02698 \text{ kg}} \right) \left(\frac{6.022 \times 10^{23}}{\text{mol}} \right) = 6.03 \times 10^{28} \text{ m}^{-3}$$

$$(b) E_F = \frac{h^2}{8m} \left(\frac{3N}{\pi V} \right)^{2/3}$$

$$\begin{aligned} \frac{N}{V} &= \frac{\pi}{3} \left(\frac{8mE_F}{h^2} \right)^{3/2} = \frac{\pi}{3} \left(\frac{8(9.109 \times 10^{-31} \text{ kg})(11.63\% \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})}{(6.626 \times 10^{-34} \text{ J}\cdot\text{s})^2} \right)^{3/2} \\ &= 1.80 \times 10^{29} \text{ m}^{-3} \end{aligned}$$

(c) Dividing the conduction electron density by the number density we obtain almost exactly 3; from this we conclude that the valence number is three.

$$29. \text{ In general } E_F = \frac{1}{2} m u_F^2 \text{ so } u_F = \sqrt{2E_F / m}.$$

$$(a) u_F = \sqrt{\frac{2E_F}{m}} = \sqrt{\frac{2(4.69 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})}{9.109 \times 10^{-31} \text{ kg}}} = 1.28 \times 10^6 \text{ m/s}$$

$$(b) u_F = \sqrt{\frac{2E_F}{m}} = \sqrt{\frac{2(14.3 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})}{9.109 \times 10^{-31} \text{ kg}}} = 2.24 \times 10^6 \text{ m/s}$$

30. Beginning with Equation (9.34) consider the following cases as $T \rightarrow 0$:

$$E > E_F: \frac{E - E_F}{kT} \rightarrow \infty \text{ so } F_{FD} \rightarrow 0$$

$$E < E_F: \frac{E - E_F}{kT} \rightarrow -\infty \text{ so } F_{FD} \rightarrow 1$$

$$E = E_F: \frac{E - E_F}{kT} \rightarrow 0 \text{ so } F_{FD} \rightarrow \frac{1}{2}$$

31. In general $n(E) = g(E) F_{FD}$. Using Equation (9.43) for $g(E)$ and the result of Problem 21

$$\text{for } F_{FD}, \text{ we can substitute to find } n(E) = \frac{3N}{2} E_F^{-3/2} \frac{E^{1/2}}{\exp(\beta(E - E_F)) + 1}.$$

32. The graphs will resemble those of Figure 9.11 (b). The $T = 0$ K line will match the dashed line and the $T = 300$ K line will match the solid line. The $T = 1500$ K line will deviate a bit more from the dashed line.
33. Numerical integration should yield accurate results with $kT = 0.02586$ eV. However, in Mathcad, for example, the upper limit cannot exceed 20 as the engine calculates e raised to the power before taking the reciprocal and thus rejects the problem. As you adjust the upper limit above 10, the integral equals 1 within rounding error.

$$1.5(7)^{-3/2} \int_0^{\infty} \frac{E^{1/2}}{\exp((E-7)/(0.02586))+1)} dE \approx 1$$

34. Setting up the numerical integration in Mathcad we have with $kT = 0.02525$ eV,

$$1.5(7)^{-3/2} \int_6^7 \frac{E^{1/2}}{\exp((E-7)/(0.02525))+1)} dE = 0.203.$$

We see that about one-fifth of the electrons are within 1 eV of the Fermi energy, which makes sense given the shape of the distribution.

35. We can use Equation (9.42) $E_F = \frac{h^2}{8m} \left(\frac{3N}{\pi L^3} \right)^{2/3}$. We use the neutron mass and from

$$\text{dimensional analysis } \frac{N}{L^3} = \frac{4.50 \times 10^{30} \text{ kg}}{\frac{4}{3}\pi(10^4 \text{ m})^3} \frac{1 \text{ (neutron)}}{1.675 \times 10^{-27} \text{ kg}} = 6.41 \times 10^{44} \text{ m}^{-3}. \text{ Then}$$

$$E_F = \frac{h^2}{8m} \left(\frac{3N}{\pi L^3} \right)^{2/3} = \frac{(6.626 \times 10^{-34} \text{ J}\cdot\text{s})^2}{8(1.675 \times 10^{-27} \text{ kg})} \left[\frac{3}{\pi} (6.41 \times 10^{44} \text{ m}^{-3}) \right]^{2/3}$$

$$= 2.36 \times 10^{-11} \text{ J} = 147 \text{ MeV}$$

The close packing of the neutrons makes the Fermi energy large compared with Fermi energies in normal matter.

36. The probability that a state will be occupied can be determined from the Fermi-Dirac factor. With $kT = 0.02525$ eV:

$$(a) F_{FD} = \frac{1}{\exp\left[\frac{(E-E_F)}{kT}\right]+1} = \frac{1}{\exp\left[\frac{(-0.1)}{0.02525}\right]+1} = 0.981 = 98.1\%$$

$$(b) \text{ When } E = E_F, \text{ then } F_{FD} = \frac{1}{2} = 50\%.$$

$$(c) F_{FD} = \frac{1}{\exp\left[\frac{(E - E_F)}{kT}\right] + 1} = \frac{1}{\exp\left[\frac{(0.1)}{0.02525}\right] + 1} = 0.0187 = 1.9\%$$

Therefore a state with energy less than the Fermi energy is almost certainly occupied and one above the Fermi energy has a very small probability of being occupied.

37. We assume that the collection of fermions behaves like an ideal gas. Using Maxwell-Boltzmann statistics, we know that $E = \frac{3}{2} kT$ or $\beta E = 3/2$. We note that $\exp(\beta E) = \exp(3/2) = 4.4817$. Now the MB factor is $\exp(-\beta E)$ and we want this to be within 1 % of the FD factor: $F_{FD} = \frac{1}{\exp\left[\frac{(E - E_F)}{kT}\right] + 1}$. So we want $\exp[\beta(E - E_F)] + 1 = (1.01) \cdot (4.4817) = 4.5265$ or $[\beta(E - E_F)] = \ln(3.5265)$. At room temperature $(\beta)^{-1} = 2.525 \times 10^{-2}$ eV so the expression above can be solved to give $(E - E_F) = 3.2 \times 10^{-2}$ eV. This small value is possible at room temperature.

38. (a) Computing carefully:

$$N/V = \frac{1.93 \times 10^4 \text{ kg/m}^3}{197.0 \times 1.66 \times 10^{-27} \text{ kg}} = 5.90 \times 10^{28} / \text{m}^3$$

$$E_F = \frac{h^2}{8m} \left(\frac{3N}{\pi L^3} \right)^{2/3} = 5.53 \text{ eV. This is an exact match with Table 9.4.}$$

(b) Thermal energy is $1.5kT = 0.04$ eV. The result shows that electrons aren't classical and the energy levels stack up in the Fermi distribution, forcing the electrons into higher energy states.

39. (a) Use the same method as in the text for helium. The liquid density of 1200 kg/m^3 translates to a number density of about $3.6 \times 10^{28} / \text{m}^3$. Plugging this into the usual formula (9.65) gives $T > 0.87$ K.
(b) Neon is not a liquid at that temperature, so it cannot be a superfluid.

40. (a) To find N/V integrate $n(E)dE$ over the whole range of energies:

$$\frac{N}{V} = \int_0^\infty n(E)dE = \frac{8\pi}{h^3 c^3} \int_0^\infty \frac{E^2}{\exp(E/kT) - 1} dE. \text{ From integral tables we have the following:}$$

$\int_0^\infty \frac{x^{n-1}}{e^{mx}-1} dx = m^{-n} \Gamma(n) \zeta(n)$. For our integral $m = \frac{1}{kT}$, $\Gamma(3) = 2! = 2$, and from numerical tables $\zeta(3) \approx 1.20$. Therefore $\frac{N}{V} = \frac{8\pi}{h^3 c^3} (kT)^3 (2)(1.20) = \frac{8\pi k^3 T^3}{h^3 c^3} (2.40)$.

(b) With $T = 500$ K:

$$\frac{N}{V} = \frac{8\pi k^3 T^3}{h^3 c^3} (2.40) = 8\pi (2.40) \left(\frac{(1.381 \times 10^{-23} \text{ J/K})(500 \text{ K})}{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(2.998 \times 10^8 \text{ m/s})} \right)^3$$

$$= 2.53 \times 10^{15} \text{ m}^{-3}.$$

(c) With $T = 5500$ K :

$$\frac{N}{V} = \frac{8\pi k^3 T^3}{h^3 c^3} (2.40) = 8\pi (2.40) \left(\frac{(1.381 \times 10^{-23} \text{ J/K})(5500 \text{ K})}{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(2.998 \times 10^8 \text{ m/s})} \right)^3$$

$$= 3.37 \times 10^{18} \text{ m}^{-3}.$$

41. Evaluating the following integral in Mathcad we find $\int_0^\infty \frac{u^{1/2}}{e^u - 1} du = 2.315$.

42. Take the expressions in Equation (9.67) to be in the “normal” order (1,2). That is,

$$\Psi_s(1,2) = \frac{1}{\sqrt{2}} [\Psi_a(1)\Psi_b(2) + \Psi_a(2)\Psi_b(1)].$$

Reversing the order for the symmetric wave function yields $\Psi_s(2,1) = \frac{1}{\sqrt{2}} [\Psi_a(2)\Psi_b(1) + \Psi_a(1)\Psi_b(2)]$, which is the same as

$$\Psi_s(1,2) \text{ with the order of terms reversed. Thus, } \Psi_s(1,2) = \Psi_s(2,1).$$

Following the same procedure for the antisymmetric wave function,

$$\Psi_A(1,2) = \frac{1}{\sqrt{2}} [\Psi_a(1)\Psi_b(2) - \Psi_a(2)\Psi_b(1)] \text{ and}$$

$$\Psi_A(2,1) = \frac{1}{\sqrt{2}} [\Psi_a(2)\Psi_b(1) - \Psi_a(1)\Psi_b(2)].$$

Comparison of these two expressions verifies that $\Psi_A(1,2) = -\Psi_A(2,1)$.

43. $E = K + V = \frac{p^2}{2m} + mgz :$

$$\exp(-\beta E) = \exp\left(-\beta\left(\frac{p^2}{2m} + mgz\right)\right) = \exp\left(-\frac{\beta p^2}{2m}\right) \exp(-\beta mgz)$$

Absorbing the (assumed constant) first exponential factor into the normalization constant C_z , $f(z)dz = C_z \exp(-\beta mgz)dz$. To find C_z normalize the integral:

$$\int_0^\infty f(z)dz = C_z \int_0^\infty \exp(-\beta mgz)dz = C_z (\beta mg) \text{ and therefore } C_z = \frac{1}{\beta mg} = \frac{kT}{mg}.$$

44. For air we will use an average $m = 29 \text{ u} = 4.82 \times 10^{-26} \text{ kg}$ and $T = 273 \text{ K}$. In general

$$\frac{\rho(h)}{\rho(0)} = \frac{\exp(-\beta mgh)}{\exp(-\beta mg(0))} = \exp(-\beta mgh).$$

$$\text{For Chicago: } \rho(h) = \exp\left(-\frac{(4.82 \times 10^{-26} \text{ kg})(9.80 \text{ m/s}^2)(176 \text{ m})}{(1.381 \times 10^{-23} \text{ J/K})(273 \text{ K})}\right) \rho(0) = 0.978 \rho(0)$$

$$\text{For Denver: } \rho(h) = \exp\left(-\frac{(4.82 \times 10^{-26} \text{ kg})(9.80 \text{ m/s}^2)(1610 \text{ m})}{(1.381 \times 10^{-23} \text{ J/K})(273 \text{ K})}\right) \rho(0) = 0.817 \rho(0)$$

$$\text{For Mt. Rainier: } \rho(h) = \exp\left(-\frac{(4.82 \times 10^{-26} \text{ kg})(9.80 \text{ m/s}^2)(4390 \text{ m})}{(1.381 \times 10^{-23} \text{ J/K})(273 \text{ K})}\right) \rho(0) = 0.577 \rho(0)$$

45. In equilibrium a fluid layer of density ρ , mass M , thickness h , and surface area A has a force $F_2 = P_2 A$ acting downward on its upper surface and a force $F_1 = P_1 A$ acting upward on its lower surface. The difference between these forces equals the weight of the fluid layer. $F_2 - F_1 = (P_1 - P_2)A = Mg = \rho gAh$. With $dP \approx \Delta P = P_2 - P_1$ and $h = \Delta z \approx dz$, we have $dP = -\rho g dz$. With N particles of mass m , the mass density is $\rho = Nm/V$. Putting these together: $dP = -\rho g dz = -\frac{Nmg}{V} dz$. From the ideal gas law $N/V = P/kT$, so

$$dP = -\frac{mgP}{kT} dz. \text{ Using separation of variables we can solve this differential equation for } P$$

$$\text{as a function of } z: \frac{dP}{P} = -\frac{mg}{kT} dz \quad \ln P = -\frac{mgz}{kT} + \text{constant} = -\beta mgz + \text{constant}.$$

$$P = (\text{constant}) \exp(-\beta mgz) = P_0 \exp(-\beta mgz)$$

46. (a) $\frac{dN}{dt} = -\frac{n\bar{v}}{4} A = -\frac{N\bar{v}A}{4V}$; Rearranging this expression we find the following

$$\text{differential equation: } \frac{dN}{N} = -\frac{\bar{v}A}{4V} dt \text{ or } \ln N = -\frac{\bar{v}A}{4V} t + \text{constant}. \text{ This can be rewritten as}$$

$$N = (\text{constant}) \exp\left(-\frac{\bar{v}A}{4V} t\right) = N_0 \exp\left(-\frac{\bar{v}A}{4V} t\right). \text{ Setting } \frac{N}{N_0} = \frac{1}{2} \text{ at } t = t_{1/2}, \text{ we find}$$

$$\frac{1}{2} = \exp\left(-\frac{\bar{v}A}{4V} t_{1/2}\right) \text{ or } t_{1/2} = \frac{4V}{\bar{v}A} \ln 2.$$

$$(b) V = \frac{\pi D^3}{6} = \frac{\pi (0.4 \text{ m})^3}{6} = 0.0335 \text{ m}^3; \quad A = \frac{\pi d^2}{4} = \frac{\pi (0.001 \text{ m})^2}{4} = 7.85 \times 10^{-7} \text{ m}^2;$$

$$\bar{v} = \frac{4}{\sqrt{2\pi}} \sqrt{\frac{kT}{m}} = \frac{4}{\sqrt{2\pi}} \sqrt{\frac{(1.381 \times 10^{-23} \text{ J/K})(293 \text{ K})}{29(1.66 \times 10^{-27} \text{ kg})}} = 462.6 \text{ m/s}$$

$$t_{1/2} = \frac{4V}{\bar{v}A} \ln 2 = \frac{4(0.0335 \text{ m}^3)}{(462.6 \text{ m/s})(7.85 \times 10^{-7} \text{ m}^2)} \ln 2 = 256 \text{ s}$$

47. The number of molecules with speed v that hit the wall per unit time is proportional to v and $F(v)$, so that the distribution $W(v)$ of the escaping molecules is by proportion

$$W(v) \sim vF(v) \sim v^3 \exp\left(-\frac{1}{2}\beta mv^2\right). \text{ Let the normalization constant for } W(v) \text{ be } C', \text{ so}$$

$$C' \int_0^\infty v^3 \exp\left(-\frac{1}{2}\beta mv^2\right) dv = 1 = C' \left(\frac{1}{2}\right) \left(\frac{\beta m}{2}\right)^{-2} \text{ or } C' = \beta^2 m^2 / 2. \text{ The mean kinetic energy of the escaping molecules is}$$

$$\bar{E} = \frac{1}{2} m \bar{v}^2 = \frac{1}{2} m C' \int_0^\infty v^5 \exp\left(-\frac{1}{2}\beta mv^2\right) dv = \frac{1}{2} m \left(\frac{\beta^2 m^2}{2}\right) \left(\frac{\beta m}{2}\right)^{-3} = \frac{2}{\beta} = 2kT$$

$$48. (a) v_{\text{rms}} = \sqrt{\frac{3kT}{m}} \text{ so } T = \frac{mv_{\text{rms}}^2}{3k} = 19,500 \text{ K}$$

- (b) With the shape of the Maxwell-Boltzmann distribution, there are always some atoms that are moving fast enough to escape. Whenever equilibrium is reestablished, more helium atoms fill that part of the distribution, so more will continue to escape.

$$49. v_{\text{rms}} = \sqrt{\frac{3kT}{m}} = 1.20 \times 10^4 \text{ m/s. This is much less than escape speed, so we do not expect many hydrogen atoms to escape from the sun.}$$

$$50. \text{ From Example 9.9 we have } \frac{N}{V} = \frac{A}{h^3} [2\pi mkT]^{3/2}.$$

- (a) Letting m be the electron mass and inserting a factor of 2 for the Pauli principle,

$$\begin{aligned} \frac{N}{V} &= \frac{2A}{h^3} [2\pi mkT]^{3/2} \\ &= \frac{2(1)}{(6.626 \times 10^{-34} \text{ J}\cdot\text{s})^3} \left[2\pi (9.109 \times 10^{-31} \text{ kg})(1.381 \times 10^{-23} \text{ J/K})(293 \text{ K}) \right]^{3/2} \\ &= 2.42 \times 10^{25} \text{ m}^{-3} \end{aligned}$$

This is quite a bit less than the density of conduction electrons in a metal (such as copper), which indicates that Fermi-Dirac statistics should be used.

$$\begin{aligned} \text{(b) For He gas the Pauli principle does not apply, so } \frac{N}{V} &= \frac{A}{h^3} [2\pi mkT]^{3/2} \\ &= \frac{1}{(6.626 \times 10^{-34} \text{ J}\cdot\text{s})^3} \left[2\pi(4)(1.66 \times 10^{-27} \text{ kg})(1.381 \times 10^{-23} \text{ J/K})(293 \text{ K}) \right]^{3/2} \\ &= 7.54 \times 10^{30} \text{ m}^{-3}. \end{aligned}$$

This is much larger than the density of He gas, which explains why MB statistics work for helium.

$$\begin{aligned} \text{(c) For the neutron star } \frac{N}{V} &= \frac{2A}{h^3} [2\pi mkT]^{3/2} \\ \frac{N}{V} &= \frac{2(1)}{(6.626 \times 10^{-34} \text{ J}\cdot\text{s})^3} \left[2\pi(1.6749 \times 10^{-27} \text{ kg})(1.381 \times 10^{-23} \text{ J/K})(10^6 \text{ K}) \right]^{3/2} = 1.90 \times 10^{35} \text{ m}^{-3} \end{aligned}$$

(d) With mass density on the order of 10^{17} kg/m^3 , the number density of the neutron star is on the order of $10^{43}/\text{m}^3$, which is why Fermi-Dirac statistics are required there.

$$51. E_F = \frac{h^2}{8m} \left(\frac{3N}{\pi L^3} \right)^{2/3} \quad \text{Rearrange to solve for } N/L^3:$$

$$E_F = \frac{h^2}{8m} \left(\frac{3N}{\pi L^3} \right)^{2/3} \quad \frac{N}{L^3} = \frac{\pi}{3h^2} (8mE_F)^{3/2} = 1.81 \times 10^{29}$$

where the numerical value comes from the Fermi energy in the table. We need to compare this with the number density for aluminum atoms, which is

$$N/V = \frac{2.70 \times 10^3 \text{ kg/m}^3}{27.0 \times 1.66 \times 10^{-27} \text{ kg}} = 6.0 \times 10^{28} / \text{m}^3$$

Note that the conduction electron density is exactly three times the number density, so there are three conduction electrons per atom. This makes sense, because aluminum has two 3s electrons and one 3p electron.

52. Expressions for I_1 and I_2 are known from Appendix 6:

$$\begin{aligned} I_3 &= -\frac{dI_1}{da} = -\left(-\frac{1}{2a^2}\right) = \frac{1}{2a^2} \\ I_4 &= -\frac{dI_2}{da} = -\frac{\sqrt{\pi}}{4} \left(-\frac{3}{2}\right) a^{-5/2} = \frac{3\sqrt{\pi}}{8} a^{-5/2} \\ I_5 &= -\frac{dI_3}{da} = -\frac{1}{2} (-2a^{-3}) = a^{-3} \end{aligned}$$

53. For the harmonic oscillator the position and velocity are $x = x_0 \cos(\omega t)$ and

$$v = \frac{dx}{dt} = -\omega x_0 \sin(\omega t) \text{ respectively. } V = \frac{1}{2} kx^2 = \frac{1}{2} k x_0^2 \cos^2(\omega t);$$

$K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2 x_0^2 \sin^2(\omega t) = \frac{1}{2}k x_0^2 \sin^2(\omega t)$ where we have used the fact that $\omega^2 m = k$. Over one cycle the average of the square of the sine or cosine function is one-half. Also the total energy is $E = \frac{1}{2}k x_0^2$. Therefore $\bar{K} = \bar{V} = \frac{1}{2}k x_0^2 \left(\frac{1}{2}\right) = \frac{E}{2}$.

54. (a) For these elements the number of electrons is half the number of nucleons. The number of nucleons is

$$\frac{m(\text{sun})}{m(\text{nucleon})} = 1.1916 \times 10^{57}$$

The number of electrons is half this, or 5.96×10^{56} .

$$(b) \text{ Number density} = \text{number/volume} = \frac{5.96 \times 10^{56}}{\frac{4}{3}\pi (6.96 \times 10^6 \text{ m})^3} = 4.22 \times 10^{35} \text{ m}^{-3}.$$

$$(c) \text{ From text } E_F = \frac{h^2}{8m} \left(\frac{3N}{\pi L^3} \right)^{2/3} = 3.28 \times 10^{-14} \text{ J} = 205 \text{ keV}$$

This is huge! The high density of electrons gives them an enormous Fermi energy.

55. (a) They have different wave functions, so when you stack up the energies the protons and neutrons go in separate categories. Because they are fundamentally different particles, which must have different wave functions, it's not a violation of Pauli to have two of them in the same energy state.

(b) i) For protons $m = 1.673 \times 10^{-27} \text{ kg}$, and the number density is

$$\frac{N}{V} = \frac{N}{\frac{4}{3}\pi r^3} = \frac{92}{\frac{4}{3}\pi (7.4 \times 10^{-15} \text{ m})^3} = 5.42 \times 10^{43} \text{ m}^{-3}$$

Then the Fermi energy is

$$E_F = \frac{h^2}{8m} \left(\frac{3N}{\pi L^3} \right)^{2/3} = \frac{(6.626 \times 10^{-34} \text{ J}\cdot\text{s})^2}{8(1.673 \times 10^{-27} \text{ kg})} \left(\frac{3}{\pi} (5.42 \times 10^{43} \text{ m}^{-3}) \right)^{2/3}$$

$$\text{or } E_F = 4.53 \times 10^{-12} \text{ J} = 28.3 \text{ MeV}.$$

ii) For neutrons $m = 1.675 \times 10^{-27} \text{ kg}$, and the number density is

$$\frac{N}{V} = \frac{N}{\frac{4}{3}\pi r^3} = \frac{146}{\frac{4}{3}\pi (7.4 \times 10^{-15} \text{ m})^3} = 8.60 \times 10^{43} \text{ m}^{-3}$$

The Fermi energy is $E_F = \frac{h^2}{8m} \left(\frac{3N}{\pi L^3} \right)^{2/3} = \frac{(6.626 \times 10^{-34} \text{ J}\cdot\text{s})^2}{8(1.675 \times 10^{-27} \text{ kg})} \left[\frac{3}{\pi} (8.60 \times 10^{43} \text{ m}^{-3}) \right]^{2/3}$

or $E_F = 6.19 \times 10^{-12} \text{ J} = 38.7 \text{ MeV}$.

These energies are enormous, but still small compared with the binding energy of the uranium nucleus.

56. The mass of gas molecules would differ slightly depending on the isotope of uranium. $^{235}\text{UF}_6$ would have a molecular mass of $235 + 6(19) = 349$ while $^{238}\text{UF}_6$ would have a molecular mass of 352. Find the rms velocity for each molecule.

$$^{235}\text{UF}_6: \quad v_{\text{rms}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.3807 \times 10^{-23} \text{ J/K})(293 \text{ K})}{349(1.6605 \times 10^{-27} \text{ kg})}} = 1.447 \times 10^2 \text{ m/s}$$

$$^{238}\text{UF}_6 \quad v_{\text{rms}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.3807 \times 10^{-23} \text{ J/K})(293 \text{ K})}{352(1.6605 \times 10^{-27} \text{ kg})}} = 1.441 \times 10^2 \text{ m/s}$$

The difference is about 0.6 m/s which represents a 0.4% change from one isotope to the other.

57. Rearranging Equation (9.64) and with $m = 4 \text{ u}$, we have

$$\begin{aligned} \frac{N}{V} &\leq \frac{2\pi(2.315)}{h^3} [2mkT]^{3/2} \\ &\leq \frac{2\pi(2.315)}{(6.626 \times 10^{-34} \text{ J}\cdot\text{s})^3} \left[2(4)(1.6605 \times 10^{-27} \text{ kg})(1.381 \times 10^{-23} \text{ J/K})(293) \right]^{3/2} \\ &\leq 1.97 \times 10^{31} \text{ m}^{-3}. \end{aligned}$$

The number density of an ideal gas at STP is $\frac{N}{V} = \frac{P}{kT} = \frac{1.0135 \times 10^5 \text{ Pa}}{(1.381 \times 10^{-23} \text{ J/K})(293 \text{ K})}$.

Therefore $\frac{N}{V} = 2.50 \times 10^{25} \text{ m}^{-3}$. As you would expect, the condensate has a number density nearly one million times greater than the ideal gas.

58. From Appendix 8, the ^{85}Rb isotope has atomic mass of 84.9 u:

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38 \times 10^{-23} \text{ J/K})(20 \times 10^{-9} \text{ K})}{84.9(1.66 \times 10^{-27} \text{ kg})}} = 2.42 \times 10^{-3} \text{ m/s}$$

For sodium, the mass is 23.0 u, so

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38 \times 10^{-23} \text{ J/K})(450 \times 10^{-12} \text{ K})}{23.0(1.66 \times 10^{-27} \text{ kg})}} = 6.99 \times 10^{-4} \text{ m/s}$$

which is even slower!

59. (a) Beginning with Equation (9.65) and with $\frac{N}{V} = 2.5 \times 10^{28} \text{ m}^{-3}$ we have

$$\begin{aligned} T &\geq \frac{h^2}{2mk} \left(\frac{1}{2\pi(2.315)} \right)^{2/3} \left(\frac{N}{V} \right)^{2/3} \\ &\geq \frac{(6.626 \times 10^{-34} \text{ J/K})^2}{2(40)(1.6605 \times 10^{-27} \text{ kg})(1.381 \times 10^{-23} \text{ J/K})} \left(\frac{1}{2\pi(2.315)} \right)^{2/3} (2.5 \times 10^{28} \text{ m}^{-3})^{2/3} \\ &\geq 3.43 \times 10^{-1} \text{ K} = 0.343 \text{ K}. \end{aligned}$$

(b) The temperature found in (a) is far below the freezing point of 84 K so no condensate is possible.

60. Beginning with Equation (9.65)

$$\begin{aligned} T &\geq \frac{h^2}{2mk} \left(\frac{N}{2\pi V(2.315)} \right)^{2/3} \\ &\geq \frac{(6.626 \times 10^{-34} \text{ J/K})^2}{2(87)(1.6605 \times 10^{-27} \text{ kg})(1.381 \times 10^{-23} \text{ J/K})} \left(\frac{2000}{2\pi(1 \times 10^{-15} \text{ m}^3)(2.315)} \right)^{2/3} \\ &\geq 2.93 \times 10^{-8} \text{ K} = 29.3 \text{ nK}. \end{aligned}$$

Chapter 10

1.

(a) For each state the energy is given by $E_{\text{rot}} = \frac{\hbar^2 \ell(\ell+1)}{2I}$, so the transition energy is

$$\Delta E = \frac{\hbar^2}{2I} [2(3) - 1(2)] = \frac{2\hbar^2}{I} = \frac{2(1.055 \times 10^{-34} \text{ J}\cdot\text{s})^2}{10^{-46} \text{ kg}\cdot\text{m}^2} = 2.2 \times 10^{-22} \text{ J} = 1.4 \times 10^{-3} \text{ eV}.$$

(b) As in part (a), $\Delta E = \frac{\hbar^2}{2I} (10(11) - 9(10)) = \frac{31\hbar^2}{2I} = 1.73 \times 10^{-21} \text{ J} = 1.08 \times 10^{-2} \text{ eV}.$

This is still in the infrared part of the spectrum.

2. From Table 10.1 $\kappa = 1860 \text{ N/m}$ and $f = 6.42 \times 10^{13} \text{ Hz}.$

(a) $\Delta E = hf = (4.136 \times 10^{-15} \text{ eV}\cdot\text{s})(6.42 \times 10^{13} \text{ Hz}) = 0.266 \text{ eV}$

(b) Set $\Delta E = kT$ with two degrees of freedom in the vibrational mode. Then

$$T = \frac{\Delta E}{k} = \frac{0.266 \text{ eV}}{8.617 \times 10^{-5} \text{ eV/K}} = 3090 \text{ K}$$

3. In the ground state $E = \left(n + \frac{1}{2}\right)\hbar\omega = \frac{1}{2}\hbar\omega = \frac{1}{2}kA^2 = \frac{1}{2}\mu\omega^2 A^2$. Solving for A:

$$A = \sqrt{\frac{\hbar}{\mu\omega}}. \text{ Using the } {}^{35}\text{Cl } \mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{35}{36} \text{ u} = 1.614 \times 10^{-27} \text{ kg}.$$

From Table 10.1 $f = 8.66 \times 10^{13} \text{ Hz}.$ With $\omega = 2\pi f$ we have

$$A = \sqrt{\frac{\hbar}{\mu\omega}} = \sqrt{\frac{1.055 \times 10^{-34} \text{ J}\cdot\text{s}}{(1.614 \times 10^{-27} \text{ kg})(2\pi)(8.66 \times 10^{13} \text{ s}^{-1})}} = 1.10 \times 10^{-11} \text{ m}.$$

4.

(a) Using the result of Problem 8 for the rotational inertia, we have

$$I = \mu R^2 = (1.614 \times 10^{-27} \text{ kg})(1.28 \times 10^{-10} \text{ m})^2 = 2.64 \times 10^{-47} \text{ kg}\cdot\text{m}^2.$$

For $\ell = 1$ we have $E_{\text{rot}} = \frac{\ell(\ell+1)\hbar^2}{2I} = \frac{\hbar^2}{I} = \frac{1}{2}I\omega^2$. Therefore

$$\omega = \sqrt{\frac{2\hbar^2}{I^2}} = \sqrt{\frac{2(1.055 \times 10^{-34} \text{ J}\cdot\text{s})^2}{(2.64 \times 10^{-47} \text{ kg}\cdot\text{m}^2)^2}} = 5.65 \times 10^{12} \text{ rad/s}.$$

For $\ell = 5$ we have $E_{\text{rot}} = \frac{\ell(\ell+1)\hbar^2}{2I} = \frac{15\hbar^2}{I} = \frac{1}{2}I\omega^2$. Therefore

$$\omega = \sqrt{\frac{110\hbar^2}{I^2}} = \sqrt{\frac{30(1.055 \times 10^{-34} \text{ J}\cdot\text{s})^2}{(2.64 \times 10^{-47} \text{ kg}\cdot\text{m}^2)^2}} = 2.19 \times 10^{13} \text{ rad/s}.$$

(b) The distance of the center of mass from the H atom is

$$x_{\text{cm}} = \frac{m_{\text{Cl}}x_{\text{Cl}}}{M} = \frac{35}{36}R = 1.24 \times 10^{-10} \text{ m}. \text{ Then from the rotational kinematics we have}$$

for $\ell = 1$: $v = \omega x_{\text{cm}} = (5.65 \times 10^{12} \text{ rad/s})(1.24 \times 10^{-10} \text{ m}) = 700 \text{ m/s}$ and for $\ell = 5$:

$$v = \omega x_{\text{cm}} = (2.19 \times 10^{13} \text{ rad/s})(1.24 \times 10^{-10} \text{ m}) = 2720 \text{ m/s}.$$

$$(c) \quad \omega = \frac{v}{x_{\text{cm}}} = \frac{0.1c}{x_{\text{cm}}} = \frac{2.998 \times 10^7 \text{ m/s}}{1.24 \times 10^{-10} \text{ m}} = 2.42 \times 10^{17} \text{ rad/s}. \text{ Then } \omega = \sqrt{\frac{\ell(\ell+1)\hbar^2}{I^2}} \text{ and}$$

$$\ell(\ell+1) = \frac{\omega^2 I^2}{\hbar^2} = \left(\frac{(2.42 \times 10^{17} \text{ rad/s})(2.64 \times 10^{-47} \text{ kg}\cdot\text{m}^2)}{1.055 \times 10^{-34} \text{ J}\cdot\text{s}} \right)^2 = 3.67 \times 10^9$$

from which it follows that $\ell \approx 6.1 \times 10^4$.

$$(d) \quad E_{\text{rot}} = kT = \frac{\hbar^2 \ell(\ell+1)}{2I}. \text{ Thus}$$

$$T = \frac{\hbar^2 \ell(\ell+1)}{2Ik} = \frac{(1.055 \times 10^{-34} \text{ J}\cdot\text{s})^2 (3.67 \times 10^9)}{2(2.64 \times 10^{-47} \text{ kg}\cdot\text{m}^2)(1.38 \times 10^{-23} \text{ J/K})} = 5.61 \times 10^{10} \text{ K}.$$

5. With Bohr's condition $L = n\hbar$ we find $E_{\text{rot}} = \frac{L^2}{2I} = \frac{n^2\hbar^2}{2I}$. The Bohr version and the correct version become similar for large values of quantum number n or ℓ , but they are quite different for small ℓ .

$$6. \quad \Delta E = E_1 - E_0 = E_1 = \frac{\hbar^2}{I} = \frac{hc}{\lambda}.$$

$$(a) \quad I = \frac{\hbar^2 \lambda}{hc} = \frac{\hbar \lambda}{2\pi c} = \frac{(1.055 \times 10^{-34} \text{ J}\cdot\text{s})(2.60 \times 10^{-3} \text{ m})}{2\pi(2.998 \times 10^8 \text{ m/s})} = 1.46 \times 10^{-46} \text{ kg}\cdot\text{m}^2$$

(b) The minimum energy in a vibrational transition $\Delta E = hf$. From Table 10.1

$f = 6.42 \times 10^{13} \text{ Hz}$, which corresponds to a photon of wavelength $\lambda = c/f = 4.67 \mu\text{m}$. A photon of this wavelength or less is required to excite the vibrational mode, so the 2.60-mm photon is too weak.

7. $\Delta E = hf = (6.626 \times 10^{-34} \text{ J}\cdot\text{s})(6.42 \times 10^{13} \text{ Hz}) = 4.25 \times 10^{-20} \text{ J}$. Using this energy for a rotational transition from the ground state to the ℓ th state we have $\Delta E = \frac{\hbar^2 \ell(\ell+1)}{2I}$ so
- $$\ell(\ell+1) = \frac{2I\Delta E}{\hbar^2} = \frac{2(7.28 \times 10^{-47} \text{ kg}\cdot\text{m}^2)(4.25 \times 10^{-20} \text{ J})}{(1.055 \times 10^{-34} \text{ J}\cdot\text{s})^2} = 556 \text{ so } \ell \approx 24.$$

This is prohibited by the $\Delta\ell = \pm 1$ selection rule.

8. Combining $R = r_1 + r_2$ with $m_1 r_1 = m_2 r_2$ we find that $r_1 = \frac{m_2}{m_1 + m_2} R$ and $r_2 = \frac{m_1}{m_1 + m_2} R$.

Therefore

$$\begin{aligned} I &= m_1 r_1^2 + m_2 r_2^2 = m_1 \left(\frac{m_2}{m_1 + m_2} R \right)^2 + m_2 \left(\frac{m_1}{m_1 + m_2} R \right)^2 \\ &= \frac{m_1 m_2 (m_1 + m_2)}{(m_1 + m_2)^2} R^2 = \mu R^2 \end{aligned}$$

9. (a) $\Delta E = \frac{\hbar^2}{2I} (3(4) - 2(3)) = \frac{3\hbar^2}{I}$ so

$$I = \frac{3\hbar^2}{\Delta E} = \frac{3(1.055 \times 10^{-34} \text{ J}\cdot\text{s})^2}{(1.43 \times 10^{-3} \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})} = 1.46 \times 10^{-46} \text{ kg}\cdot\text{m}^2$$

- (b) $\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{(12)(16)}{28} \text{ u} = 1.139 \times 10^{-26} \text{ kg}$. Then $I = \mu R^2$ becomes

$$R = \sqrt{\frac{I}{\mu}} = \sqrt{\frac{1.46 \times 10^{-46} \text{ kg}\cdot\text{m}^2}{1.139 \times 10^{-26} \text{ kg}}} = 1.13 \times 10^{-10} \text{ m}, \text{ which is a reasonable answer.}$$

10.

- (a) The distance of each H atom from the line is

$$d = (0.0958 \text{ nm})(\sin 52.5^\circ) = 7.60 \times 10^{-2} \text{ nm}.$$

$$\text{Then } I = 2m_{\text{H}}d^2 = 2(1.67 \times 10^{-27} \text{ kg})(7.60 \times 10^{-11} \text{ m})^2 = 1.93 \times 10^{-47} \text{ kg}\cdot\text{m}^2.$$

- (b) $E_1 = \frac{\hbar^2}{I} = \frac{(1.055 \times 10^{-34} \text{ J}\cdot\text{s})^2}{1.93 \times 10^{-47} \text{ kg}\cdot\text{m}^2} = 5.77 \times 10^{-22} \text{ J} = 3.61 \text{ MeV}$

$$E_2 = \frac{3\hbar^2}{I} = \frac{3(1.055 \times 10^{-34} \text{ J}\cdot\text{s})^2}{1.93 \times 10^{-47} \text{ kg}\cdot\text{m}^2} = 1.73 \times 10^{-21} \text{ J} = 10.81 \text{ MeV}$$

$$(c) \lambda = \frac{hc}{E_1} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(2.998 \times 10^8 \text{ m/s})}{5.77 \times 10^{-22} \text{ J}} = 344 \mu\text{m}$$

11. First we need to compute the rotational inertia. Including both the nucleus and electrons:

$$\begin{aligned} I &= \frac{2}{5} m_\alpha r_{\text{nuc}}^2 + \frac{2}{5} (2m_e) a_0^2 \\ &= \frac{2}{5} \left((6.64 \times 10^{-27} \text{ kg})(1.9 \times 10^{-15} \text{ m})^2 + 2(9.109 \times 10^{-31} \text{ kg})(5.29 \times 10^{-11} \text{ m})^2 \right) \\ &= 2.04 \times 10^{-51} \text{ kg} \cdot \text{m}^2 \end{aligned}$$

Notice that the nuclear contribution is negligible.

$$(a) E_{\text{rot}} = \frac{\hbar^2}{I} = \frac{(1.055 \times 10^{-34} \text{ J} \cdot \text{s})^2}{2.04 \times 10^{-51} \text{ kg} \cdot \text{m}^2} = 5.46 \times 10^{-18} \text{ J} = 34.1 \text{ eV}.$$

(b) This is greater than the ionization energy for helium and note likely to be observed.

$$12. f' = f \pm \frac{\hbar}{2\pi I} (2\ell + 3) \text{ with } f = c/\lambda = 1.00 \times 10^{13} \text{ Hz. Thus}$$

$$f' = 1.00 \times 10^{13} \text{ Hz} \pm \frac{1.055 \times 10^{-34} \text{ J} \cdot \text{s}}{2\pi (1.30 \times 10^{-45} \text{ kg} \cdot \text{m}^2)} (2\ell + 3) \text{ or}$$

$$f' = 1.00 \times 10^{13} \text{ Hz} \pm \left((2.58 \times 10^{10} \text{ Hz})\ell + 3.87 \times 10^{10} \text{ Hz} \right) \text{ where } \ell \text{ is an integer and as}$$

$$\text{usual } \lambda = \frac{c}{f'}.$$

$$13. \Delta E = hf = \frac{\hbar^2}{2I} [(\ell + 1)(\ell + 2) - \ell(\ell + 1)]. \text{ Therefore we can say that for a particular}$$

transition $f = C/\mu$ where C is a constant, because $I = \mu R^2$. Then $\frac{df}{d\mu} = -\frac{C}{\mu^2}$; taking

absolute values $\frac{d f}{d \mu} = C / \mu^2 = \frac{f}{\mu}$. Thus $\frac{\Delta f}{f} = \frac{\Delta \mu}{\mu}$ as required.

14. (a) The Maxwell-Boltzmann factor is $F_{\text{MB}} = A \exp(-E/kT)$. The energies of the three

rotational levels are $E_0 = 0$; $E_1 = \frac{\hbar^2}{I}$; $E_2 = \frac{3\hbar^2}{I}$. Thus the Maxwell-Boltzmann factors

are $\ell = 0$: $F_{\text{MB}} = A$.

For $\ell = 1$:

$$F_{\text{MB}} = A \exp\left(-\frac{\hbar^2}{IkT}\right) = A \exp\left(-\frac{(1.055 \times 10^{-34} \text{ J} \cdot \text{s})^2}{(1.4 \times 10^{-46} \text{ kg} \cdot \text{m}^2)(1.381 \times 10^{-23} \text{ J/K})(293\text{K})}\right) = 0.981A.$$

For $\ell = 2$:

$$F_{\text{MB}} = A \exp\left(-\frac{3\hbar^2}{IkT}\right) = A \exp\left(-\frac{3(1.055 \times 10^{-34} \text{ J}\cdot\text{s})^2}{(1.4 \times 10^{-46} \text{ kg}\cdot\text{m}^2)(1.381 \times 10^{-23} \text{ J/K})(293 \text{ K})}\right) = 0.943A.$$

(b) The degeneracy factor $g(E)$ is 1 for $\ell = 0$, 3 for $\ell = 1$, and 5 for $\ell = 2$. Therefore the level populations $n(E)$

$$\ell = 0: \quad n(E) = g(E)F_{\text{MB}} = A$$

$$\ell = 1: \quad n(E) = g(E)F_{\text{MB}} = 3(0.981A) = 2.94A$$

$$\ell = 2: \quad n(E) = g(E)F_{\text{MB}} = 5(0.943A) = 4.72A$$

(c) For the lower rotational states the degeneracy factor causes the state population to increase with increasing ℓ . However as ℓ increases and the rotational energy increases, the exponential factor begins to take over and decrease the state populations.

15. The gap between adjacent lines is $h(\Delta f) = \hbar^2 / I$.

$$(a) \quad I = \frac{\hbar^2}{h(\Delta f)} = \frac{(1.055 \times 10^{-34} \text{ J}\cdot\text{s})^2}{(6.626 \times 10^{-34} \text{ J}\cdot\text{s})(7 \times 10^{11} \text{ Hz})} = 2.4 \times 10^{-47} \text{ kg}\cdot\text{m}^2$$

(b) $\omega = 2\pi f = \sqrt{\kappa / \mu}$ with $f = 8.65 \times 10^{13} \text{ Hz}$. The reduced mass is (using the ^{35}Cl

isotope) $\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{35}{36} \text{ u} = 1.614 \times 10^{-27} \text{ kg}$. Solving for κ we find

$$\kappa = 4\pi^2 f^2 \mu = 4\pi^2 (8.65 \times 10^{13} \text{ Hz})^2 (1.614 \times 10^{-27} \text{ kg}) = 478 \text{ N/m}$$

in good agreement with Table 10.1.

$$16. (a) \quad P = q\chi = (1.602 \times 10^{-19} \text{ C})(2.67 \times 10^{-10} \text{ m}) = 4.28 \times 10^{-29} \text{ C}\cdot\text{m}$$

$$(b) \quad \text{The fractional ionic character is the ratio } \frac{4.28 \times 10^{-29} \text{ C}\cdot\text{m}}{5.41 \times 10^{-29} \text{ C}\cdot\text{m}} = 0.79$$

17. $\Delta E = hf = \frac{hc}{\lambda}$. Using the data from Table 10.1, the wavelength is

$$\lambda = \frac{c}{f} = \frac{2.998 \times 10^8 \text{ m/s}}{6.69 \times 10^{13} \text{ Hz}} = 4.48 \text{ }\mu\text{m}$$

which corresponds to an energy of 0.277 eV. This photon is in the infrared.

18. (a) Using dimensional analysis and the fact that the energy of each photon is

$$hc / \lambda = 3.14 \times 10^{-19} \text{ J}, \text{ then } \frac{N}{t} = \frac{5 \times 10^{-3} \text{ J/s}}{3.14 \times 10^{-19} \text{ J/photon}} = 1.59 \times 10^{16} \text{ photon/s}.$$

- (b) 0.02 mole is equivalent to $0.02 N_A = 1.20 \times 10^{22}$ atoms. The fraction participating is

$$\frac{1.59 \times 10^{16}}{1.20 \times 10^{22}} = 1.33 \times 10^{-6}.$$

- (c) The transitions involved have a fairly low probability, even with stimulated emission. We are saved by the large number of atoms available.

19. (a) The energy of each photon is $E_2 - E_1 = \frac{1.15 \text{ eV}}{\text{photon}} \cdot \frac{1.602 \times 10^{-19} \text{ J}}{1 \text{ eV}} = 1.84 \times 10^{-19} \text{ J/photon}.$

$$\text{Since } P = E / t \text{ then we have } P = (1.84 \times 10^{-19} \text{ J/photon}) (5.50 \times 10^{18} \text{ photons/s}) = 1.00 \text{ W}.$$

$$(b) \lambda = \frac{hc}{\Delta E} = \frac{1.986 \times 10^{-25} \text{ J} \cdot \text{m}}{1.84 \times 10^{-19} \text{ J}} = 1.08 \times 10^{-6} \text{ m} = 1.08 \mu\text{m}$$

20. (a) With the given wavelength, we know that each photon has an energy of

$$E = \frac{hc}{\lambda} = \frac{1.986 \times 10^{-25} \text{ J} \cdot \text{m}}{351 \times 10^{-9} \text{ m}} = 5.658 \times 10^{-19} \text{ J}.$$

Therefore the number of transitions required is

$$N = \frac{40 \times 10^3 \text{ J}}{5.658 \times 10^{-19} \text{ J/photon}} = 7.07 \times 10^{22} \text{ photons}.$$

$$(b) \text{ Average power is energy divided by time: } P = \frac{40 \times 10^3 \text{ J}}{3.5 \times 10^{-9} \text{ s}} = 1.14 \times 10^{13} \text{ W}$$

or about 11 TW!

21. (a) From Chapter 9 we have $\Delta f = \frac{f_0}{c} \sqrt{\frac{kT}{m}}$. We also have, in general, $f = c / \lambda$ so,

calculating the differential we have $\Delta f = (c / \lambda^2) \Delta \lambda$. Combining the two expressions we

$$\text{have } \Delta \lambda = \frac{\lambda^2 \Delta f}{c} = \frac{\lambda^2 f_0}{c^2} \sqrt{\frac{kT}{m}} = \frac{\lambda}{c} \sqrt{\frac{kT}{m}}. \text{ Using the mass of neon as } 3.32 \times 10^{-26} \text{ kg we}$$

$$\text{calculate } \Delta \lambda = \frac{6.328 \times 10^{-7} \text{ m}}{2.998 \times 10^8 \text{ m/s}} \sqrt{\frac{(1.381 \times 10^{-23} \text{ J/K})(293 \text{ K})}{3.32 \times 10^{-26} \text{ kg}}} = 7.37 \times 10^{-13} \text{ m}.$$

$$(b) \Delta E = \frac{hc}{\lambda^2} \Delta \lambda = \frac{\hbar}{2\tau}$$

$$\Delta \lambda = \frac{\hbar \lambda^2}{2hc\tau} = \frac{\lambda^2}{4\pi c\tau} = \frac{(6.328 \times 10^{-7} \text{ m})^2}{4\pi(2.998 \times 10^8 \text{ m/s})(10^{-3} \text{ s})} = 1.06 \times 10^{-19} \text{ m}.$$

The Doppler broadening is much more significant than the Heisenberg broadening.

$$22. (a) \Delta t = \frac{2(1 \text{ m})}{2.998 \times 10^8 \text{ m/s}} = 6.67 \times 10^{-9} \text{ s}$$

(b) For the 16-km round trip

$$\Delta t = \frac{16 \times 10^3 \text{ m}}{2.998 \times 10^8 \text{ m/s}} - \frac{16 \times 10^3 \text{ m}}{(1 - 3 \times 10^{-4}) 2.998 \times 10^8 \text{ m/s}} = -1.6 \times 10^{-8} \text{ s}.$$

Because this result is larger than the desired uncertainty in timing, it is important to take atmospheric effects into account.

23. In the three-level system, the population of the upper-level must exceed the population of the ground state. This is not necessary in a four-level system.

24. Because the 3s state has two possible configurations and the $n = 2$ level has eight, let us assign a density of states $g = 2$ to the excited state and $g = 8$ to the ground state.

$$\frac{f_3}{f_2} = \frac{2 \exp(-\beta E_3)}{8 \exp(-\beta E_2)} = 0.25 \exp\left(-\frac{E_3 - E_2}{kT}\right)$$

$$(a) = 0.25 \exp\left(-\frac{16.6 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(293 \text{ K})}\right) = 7.2 \times 10^{-287}$$

$$(b) \frac{f_3}{f_2} = 0.25 \exp\left(-\frac{16.6 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(150 \text{ K})}\right) = 4.4 \times 10^{-559}$$

$$(c) \frac{f_3}{f_2} = 0.25 \exp\left(-\frac{16.6 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(600 \text{ K})}\right) = 9.1 \times 10^{-141}$$

(d) Thermal excitations can be neglected.

25. Using dimensional analysis the number density is

$$1980 \text{ kg/m}^3 \frac{1 \text{ mol}}{0.07455 \text{ kg}} \frac{2(6.022 \times 10^{23})}{\text{mol}} = 3.20 \times 10^{28} \text{ m}^{-3}. \text{ Therefore the distance is}$$

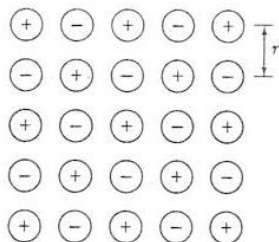
$$d = (3.20 \times 10^{28} \text{ m}^{-3})^{-1/3} = 3.15 \times 10^{-10} \text{ m} = 0.315 \text{ nm}.$$

26. Each charge has two unlike charges a distance r away, two like charges a distance $2r$

away, and so on: $V = -\frac{2e^2}{4\pi\epsilon_0 r} \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \right)$. The bracketed expression is the Taylor

series expansion for $\ln 2$, so $V = -\frac{2e^2}{4\pi\epsilon_0 r} \ln 2 = -\frac{\alpha e^2}{4\pi\epsilon_0 r}$ and we see that $\alpha = 2 \ln 2$.

27. Using the central positive charge as a guide, we find:



$$V = \frac{e^2}{4\pi\epsilon_0 r} \left(-\frac{4}{r} + \frac{4}{\sqrt{2}r} + \frac{4}{2r} - \frac{8}{\sqrt{5}r} + \frac{4}{\sqrt{8}r} - \dots \right),$$

so by definition of α we have $\alpha = 4 - \frac{4}{\sqrt{2}} - 2 + \frac{8}{\sqrt{5}} - \frac{4}{\sqrt{8}} + \dots$

28. $F = -\frac{dV}{dr} = -\frac{\alpha e^2}{4\pi\epsilon_0 r^2} + \frac{\lambda}{\rho} e^{-r/\rho}$. From Equation 10.20a we have $1 = e^{r_0/\rho} \frac{\rho \alpha e^2}{4\pi\epsilon_0 r_0^2 \lambda}$.

Multiplying this factor of 1 by the last term in the force equation, we obtain

$$\begin{aligned} F &= -\frac{\alpha e^2}{4\pi\epsilon_0 r^2} + \frac{\lambda}{\rho} e^{-r/\rho} e^{r_0/\rho} \frac{\rho \alpha e^2}{4\pi\epsilon_0 r_0^2 \lambda} \\ &= \frac{\alpha e^2}{4\pi\epsilon_0 r_0^2} \left(-\frac{r_0^2}{r^2} + e^{-(r-r_0)/\rho} \right) \end{aligned}$$

29. Inserting $r = r_0 + \delta r$ into the force equation from Problem 28:

$$F = \frac{\alpha e^2}{4\pi\epsilon_0 r_0^2} \left(-\frac{r_0^2}{(r_0 + \delta r)^2} + e^{-\delta r/\rho} \right). \text{ Factoring the first term in parentheses leaves}$$

$$F = \frac{\alpha e^2}{4\pi\epsilon_0 r_0^2} \left(-\frac{1}{\left(1 + \frac{\delta r}{r_0} \right)^2} + e^{-\delta r/\rho} \right). \text{ Applying the binomial theorem}$$

$$\left(1 + \frac{\delta r}{r_0} \right)^{-2} = 1 - \frac{2}{r_0} \delta r + \frac{3}{r_0^2} (\delta r)^2 - \dots \text{ and we end the series at that point for small } \delta r. \text{ The}$$

Taylor series for the exponential term is $e^{-\delta r/\rho} = 1 - \frac{\delta r}{\rho} + \frac{(\delta r)^2}{2\rho^2} - \dots$ Combining these

two series approximations gives: $F \approx \frac{\alpha e^2}{4\pi\epsilon_0 r_0^2} \left(-1 + \frac{2}{r_0} \delta r - \frac{3}{r_0^2} (\delta r)^2 + 1 - \frac{\delta r}{\rho} + \frac{(\delta r)^2}{2\rho^2} \right)$.

Collecting terms we have:

$$F \approx \frac{\alpha e^2}{4\pi\epsilon_0 r_0^2} \left(\left(\frac{2}{r_0} - \frac{1}{\rho} \right) (\delta r) + \left(-\frac{3}{r_0^2} + \frac{1}{2\rho^2} \right) (\delta r)^2 \right)$$

$$\approx K_1 (\delta r) + K_2 (\delta r)^2$$

where $K_1 = \frac{\alpha e^2}{4\pi\epsilon_0 r_0^2} \left(\frac{2}{r_0} - \frac{1}{\rho} \right)$ and $K_2 = \frac{\alpha e^2}{4\pi\epsilon_0 r_0^2} \left(-\frac{3}{r_0^2} + \frac{1}{2\rho^2} \right)$

30. (a) Looking at the result of the preceding problem we see that the spring constant is $\kappa = -K_1$, and we know that for the harmonic oscillator $\omega = \sqrt{\kappa/\mu}$. For NaCl

$\mu = \frac{m_1 m_2}{m_1 + m_2} = 2.32 \times 10^{-26}$ kg. Recall that for NaCl we know $\alpha = 1.7476$,

$r_0 = 0.282$ nm, and $\rho = 0.0316$ nm. Substituting these values gives

$-K_1 = \kappa = -\frac{\alpha e^2}{4\pi\epsilon_0 r_0^2} \left(\frac{2}{r_0} - \frac{1}{\rho} \right) = 124.7$ N/m. Then the oscillation frequency is

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{\kappa}{\mu}} = \frac{1}{2\pi} \sqrt{\frac{124.7 \text{ N/m}}{2.32 \times 10^{-26} \text{ kg}}} = 1.17 \times 10^{13} \text{ Hz}.$$

(b) $\lambda = \frac{c}{f} = \frac{2.998 \times 10^8 \text{ m/s}}{1.17 \times 10^{13} \text{ Hz}} = 25.6 \mu\text{m}$ which is about one-half the observed value.

31. (a) $\bar{F} = 0 = K_1 \bar{\delta r} + K_2 \overline{(\delta r)^2}$ and therefore $\bar{\delta r} = -\frac{K_2}{K_1} \overline{(\delta r)^2}$.

(b) From the equipartition theorem $K_1 \overline{(\delta r)^2} = kT$ so with a) $\bar{\delta r} = -\frac{K_2}{K_1^2} kT$. The

coefficient of thermal expansion α comes from $\Delta L = L \alpha \Delta T$, or $\alpha = \frac{1}{L} \frac{\Delta L}{\Delta T}$. Therefore in

our nomenclature $\alpha = \frac{1}{r_0} \frac{d(\bar{\delta r})}{dT} = \frac{1}{r_0} \frac{k K_2}{K_1^2}$.

(c) Evaluating with $K_1 = -124.7$ N/m and $K_2 = 2.35 \times 10^{12}$ N/m², we find

$\alpha = 7.4 \times 10^{-6} \text{ K}^{-1}$, which is the right order of magnitude.

32. We begin with Equation (10.32): $L \equiv \frac{K}{\sigma T} = \frac{4k^2}{\pi e^2}$. Solving for the thermal conductivity,

K , we find

$$K = \frac{4\sigma k^2 T}{\pi e^2} = \frac{4(6.30 \times 10^7 \Omega^{-1} \cdot \text{m}^{-1})(1.38 \times 10^{-23} \text{ J} \cdot \text{K}^{-1})^2 293 \text{ K}}{\pi(1.60 \times 10^{-19} \text{ C})^2} = 175 \text{ W} \cdot \text{K}^{-1} \cdot \text{m}^{-1}.$$

As mentioned in the text, the Wiedemann-Franz Law was derived using classical expressions for the mean speed and the molar heat capacity. Quantum mechanical

corrections give an additional factor of $\frac{\pi^3}{12} = 2.58$. With this correction, our answer

would be $451 \text{ W} \cdot \text{K}^{-1} \cdot \text{m}^{-1}$ which is much closer to the measured value.

33. We begin with Equation (10.26) and compare that equation to the graph given in Figure 10.23. We notice that in the figure the mean displacement is plotted on the vertical axis and the absolute temperature is on the horizontal axis. Equation (10.26) is of the form $y = mx + b$ where the mean displacement represents the vertical or y quantity and the absolute temperature is the horizontal or x quantity. Thus the slope, m is the constant multiplying the temperature, or in this case, $m = \frac{3bk}{4a^2}$. We can estimate the slope from the graph. In the region between temperatures of about 40 K and 90 K, the graph is nearly linear. The slope is approximately $m = \frac{0.546 \text{ nm} - 0.534 \text{ nm}}{85 \text{ K} - 40 \text{ K}} = 2.67 \times 10^{-4} \text{ nm} \cdot \text{K}^{-1}$.

From above we also know that $\frac{b}{a^2} = \frac{4m}{3k} = \frac{4(2.67 \times 10^{-4} \text{ nm} \cdot \text{K}^{-1})}{3(1.3807 \times 10^{-23} \text{ J} \cdot \text{K}^{-1})} = 2.58 \times 10^{10} \text{ N}^{-1}$.

34. (a) Substituting the results from Equations (10.24) and (10.25) into Equation (10.23), we

$$\text{have } \frac{\frac{3\sqrt{\pi}}{4} b a^{-5/2} \beta^{-3/2}}{\pi^{1/2} a^{-1/2} \beta^{-1/2}} = \frac{3}{4} b a^{-2} \beta^{-1} = \frac{3bkT}{4a^2}.$$

(b) Let $\frac{3bk}{4a^2} = C_0$. Then $\langle x \rangle = C_0 T$.

$C_0 = \frac{\Delta \langle x \rangle}{\Delta T} = \alpha x = (1.67 \times 10^{-5} \text{ K}^{-1})(8.47 \times 10^{28} \text{ m}^{-3})^{-1/3} = 3.80 \times 10^{-15} \text{ m/K}$ where we used the number density of copper from Table 9.3.

35. The potential energy of a harmonic oscillator would have only the quadratic term and be of the form $\frac{1}{2}\kappa x^2$. Examine Equation (10.22) and notice that a is the multiplicative constant for the x^2 term. Thus $a = (1/2)\kappa$. From Table 10.1 we see that an average value might be $\kappa \approx 1000 \text{ N/m}$. Using one-half this value for α along with the definition of C_0 from the preceding problem we find

$$b = \frac{4a^2 C_0}{3k} = \frac{4(5 \times 10^2 \text{ N/m})^2 (3.80 \times 10^{-15} \text{ m/K})}{3(1.381 \times 10^{-23} \text{ J/K})} = 9.2 \times 10^{13} \text{ N/m}^2.$$

36. (a) $\bar{E} = \frac{3}{2}kT$ so $T = \frac{2\bar{E}}{3k}$ and the ideal gas law becomes $PV = NkT = Nk \frac{2\bar{E}}{3k} = \frac{2N\bar{E}}{3}$.

(b) From Chapter 9 we know $\frac{N}{V} = 8.47 \times 10^{28} \text{ m}^{-3}$ and we know $E = \frac{3}{5}E_F = 6.76 \times 10^{-19} \text{ J}$.

Thus in SI units we have

$$P = \frac{2N\bar{E}}{3V} = \frac{2}{3}(8.47 \times 10^{28} \text{ m}^{-3})(6.76 \times 10^{-19} \text{ J}) = 3.82 \times 10^{10} \text{ N/m}^2 \text{ which is quite high. The ideal gas law may not be the best assumption for conduction electrons.}$$

37. From the previous problem $P = \frac{2N\bar{E}}{3V} = \frac{2N}{3V} \frac{3}{5}E_F = \frac{2NE_F}{5V}$. We must be careful because

$$E_F \text{ depends on the volume as: } E_F = \frac{h^2}{8m} \left(\frac{3N}{\pi V} \right)^{2/3} \text{ so}$$

$$P = \frac{2N}{5V} \frac{h^2}{8m} \left(\frac{3N}{\pi V} \right)^{2/3} = \left(\frac{3}{\pi} \right)^{2/3} \frac{N^{5/3} h^2}{20m} V^{-5/3}.$$

The Bulk modulus is:

$$\begin{aligned} B &= -V \frac{\partial P}{\partial V} = -V \left(\frac{3}{\pi} \right)^{2/3} \frac{N^{5/3} h^2}{20m} \left(-\frac{5}{3} \right) V^{-8/3} \\ &= \frac{5}{3} \left(\frac{3}{\pi} \right)^{2/3} \frac{N^{5/3} h^2}{20m} V^{-5/3} = \frac{5P}{3} \end{aligned}$$

Using the fact from above that $P = \frac{2NE_F}{5V}$ we find $B = \frac{5}{3} \frac{2NE_F}{5V} = \frac{2NE_F}{3V}$.

38. For silver $N/V = 5.86 \times 10^{28} \text{ m}^{-3}$ and $E_F = 5.49 \text{ eV}$.

(a) $B = \frac{2NE_F}{3V} = \frac{2}{3}(5.86 \times 10^{28} \text{ m}^{-3})(5.49 \text{ eV})(1.602 \times 10^{-19} \text{ J/eV}) = 3.44 \times 10^{10} \text{ N/m}^2$.

(b) The computed result is about one-third of the measured value.

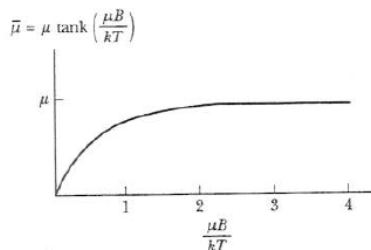
39. (a) Following the arguments in the text, regardless of the sense of rotation of the electron, as the magnetic field increases from zero, the flux upward through the loop increases. Then, as in the text, the tangential electric field is still directed clockwise. The torque does not depend on the sense of rotation either, so the torque is directed out of the page. Thus the vector $\Delta \vec{L}$ is out of the page which means that $\Delta \mu = -\frac{e}{2m} \Delta L = \frac{e^2 r^2 B}{4m}$, and $\Delta \vec{\mu}$ is directed *into* the page and thus opposite the direction of B as before. (If the reader is uncertain that the sense of rotation is irrelevant, recall that $\Phi_B = \int \vec{B} \cdot d\vec{a}$. You use the right-hand rule to determine the sense of $d\vec{a}$. While the direction of $d\vec{a}$ will depend on the sense of rotation, the other side of the equation for Faraday's law contains the expression $\oint \vec{E} \cdot d\vec{\ell}$. The sense of transit along the increment $d\vec{\ell}$ will also change.)

(b) With the field directed into the paper, now the magnetic flux increases downward through the loop. Thus Faraday's law will indicate that the electric field is tangent to the orbit but now directed counterclockwise. Thus the torque will be directed into the page and thus $\Delta \vec{L}$ will be directed into the page. Finally the negative sign in $\Delta \vec{\mu} = -\frac{e}{2m} \Delta \vec{L}$ will mean that $\Delta \mu = \frac{e^2 r^2 B}{4m}$, with $\Delta \vec{\mu}$ directed *out of* the page.

40. We use the result of the previous problem and the semi-classical model that m_ℓ represents a projection of the angular momentum vector on the z-axis and thus represents a sense of rotation. If we say that $m_\ell = +1$ represents a clockwise rotation, then $m_\ell = -1$ represents a counterclockwise rotation. Regardless of the sense of rotation, there is a diamagnetic effect for both. The electron with $m_\ell = 0$ will not contribute to the change in the magnetic moment. The angular momentum of this electron is perpendicular to the magnetic field. There is no magnetic flux through the orbit so Faraday's law indicates there will be no tangential electric field. So the total induced magnetic moment is twice the effect of the previous problem. That is $\Delta \mu = 2 \left(\frac{e^2 r^2 B}{4m} \right) = \frac{e^2 r^2 B}{2m}$.

41. This is the same as the high-field limit. With $\mu B / kT \gg 1$ we have $\tanh(\mu B / kT) \approx 1$ so $\bar{\mu} \approx \mu$.

42. (a) See graph:



(b) $\bar{\mu} = \mu \tanh(5) = 0.99991\mu$ and the approximate result of Problem 41 is off by only 0.009%.

(c) $\bar{\mu} = \mu \tanh(0.10) = 0.0997\mu$. The approximate result is off by just 0.3%.

43. The magnetic dipole moment has units of $A \cdot m$, so M has units of $A \cdot m^2 / m^3 = A/m$.

μ_0 has units of $T \cdot m/A$ and B has units T , so $\chi = \frac{\mu_0 M}{B}$ has units $\frac{(T \cdot m/A)(A/m)}{T}$ which reduces to no unit.

44. (a) If we assume that every atom's magnetic moment is a Bohr magneton aligned in the same direction, $M = n\mu_B$ where n is the number density, we have

$$n = \frac{7.87 \times 10^3 \text{ kg}}{\text{m}^3} \frac{1 \text{ mol}}{0.05585 \text{ kg}} \left(\frac{6.022 \times 10^{23}}{1 \text{ mol}} \right) = 8.49 \times 10^{28} \text{ m}^{-3}. \text{ Thus}$$

$$M = n\mu_B = (8.49 \times 10^{28} \text{ m}^{-3}) (9.274 \times 10^{-24} \text{ J/T}) = 7.87 \times 10^5 \text{ A/m}.$$

(b) The computed value is almost exactly one-half the measured value.

(c) This implies that there are two unpaired spins per atom.

$$45. B_c = B_c(0) \left(1 - \left(\frac{T}{T_c} \right)^2 \right) = 0.25 B_c(0). \text{ Thus } (T/T_c)^2 = 0.75 \text{ and } T = \sqrt{0.75} T_c \approx 0.87 T_c.$$

Similarly for a ratio of 0.50 we find $T = \sqrt{0.50} T_c \approx 0.71 T_c$; for a ratio of 0.75 we find

$$T = \sqrt{0.25} T_c \approx 0.50 T_c.$$

46. The energy gap at $T = 2 \text{ K}$ is $E_g(2 \text{ K}) = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{5.68 \times 10^5 \text{ nm}} = 2.18 \times 10^{-3} \text{ eV}$. Inserting

Equation (10.46) into Equation (10.47) gives $E_g(T) = 1.74(3.54 k T_c) \left(1 - \frac{T}{T_c} \right)^{1/2}$. Using

$E_g(2 \text{ K}) = 2.18 \times 10^{-3} \text{ eV}$ and $k = 8.62 \times 10^{-5} \text{ eV/K}$ we obtain

$$4.11 \text{ K} = T_c \left(1 - \frac{T}{T_c} \right)^{1/2} = T_c \left(1 - \frac{2}{T_c} \right)^{1/2}. \text{ Solving this equation using MathCAD we obtain}$$

$T_c = 5.23 \text{ K}$ which is closest to vanadium.

47. Using the value given in the text just below Equation (10.43), $T_c = 4.146$ K for a mass of 203.4 u, we find $M^{0.5}T_c = \text{constant} = 59.1296 \text{ u}^{0.5} \cdot \text{K}$. For a mass of 201 u we find $T_c = 4.171$ K and for mass of 204 u we find $T_c = 4.140$ K.

48. With ^{16}O the molar mass in grams is $88.906 + 2(137.34) + 3(63.546) + 7(16.00) = 666.224$. Replacing all of the ^{16}O atoms with ^{18}O atoms adds 14 grams per mole, changing the mass to 680.224. Using the BCS formula for the isotope effect $M_1^{0.5}T_{c1} = M_2^{0.5}T_{c2}$ and assuming $T_c = 93$ K (exactly) for the first sample $T_{c2} = \left(\frac{M_1}{M_2}\right)^{1/2} T_{c1} = \left(\frac{666.224}{680.224}\right)^{1/2} (93 \text{ K}) = 92.0 \text{ K}$, a change of 1.0 K.

49. Extrapolating on the graph in Figure 10.37, it could be at about 130 K.

50. $B = \mu_0 I n = (4\pi \times 10^{-7} \text{ N/A}^2)(4.5 \text{ A})(2500 \text{ m}^{-1}) = 14.1 \text{ mT}$

$$\Phi = BA = \frac{B \pi d^2}{4} = (14.1 \times 10^{-3} \text{ T}) \frac{\pi (0.032 \text{ m})^2}{4} = 1.13 \times 10^{-5} \text{ T} \cdot \text{m}^2$$

$$\frac{\Phi}{\Phi_0} = \frac{1.13 \times 10^{-5} \text{ T} \cdot \text{m}^2}{2.068 \times 10^{-15} \text{ T} \cdot \text{m}^2} = 5.5 \times 10^9 \text{ flux quanta}.$$

This large number shows how small the flux quantum is.

51. We know that for niobium $B_c = 0.206$ T. Then the diameter (twice the radius) is

$$D = 2R = \frac{\mu_0 I}{\pi B} = \frac{(4\pi \times 10^{-7} \text{ N/A}^2)(2.5 \text{ A})}{\pi (0.206 \text{ T})} = 4.9 \times 10^{-6} \text{ m which is quite small.}$$

52. $P = I^2 R = \frac{I^2 \rho L}{A}$ where ρ is resistivity, L is length, and A is area. Since $A = \pi r^2$ we have

$$P = \frac{I^2 \rho L}{\pi r^2}. \text{ The surface area is } 2\pi r L \text{ so the power per unit area is}$$

$$\frac{P}{\text{area}} = \frac{I^2 \rho L}{\pi r^2 (2\pi r L)} = \frac{I^2 \rho}{2\pi^2 r^3} = 80 \text{ W/m}^2. \text{ Using } r = 3.75 \times 10^{-4} \text{ m we find}$$

$$I^2 = \frac{2\pi^2 (3.75 \times 10^{-4} \text{ m})^3 (80 \text{ W/m}^2)}{1.72 \times 10^{-8} \Omega \cdot \text{m}} = 4.842 \text{ A}^2 \text{ so } I = 2.2 \text{ A.}$$

53. (a) From the BCS theory we have

$$B_c(4.2 \text{ K}) = B_c(0) \left(1 - \left(\frac{T}{T_c} \right)^2 \right) = (0.206 \text{ T}) \left(1 - \left(\frac{4.2}{9.25} \right)^2 \right) = 0.1635 \text{ T} . \text{ From the result of}$$

Problem 51 we know that we know that $I = \frac{\pi B_c D}{\mu_0} = \frac{\pi (0.1635 \text{ T}) (1.29 \times 10^{-3} \text{ m})}{4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}} = 527 \text{ A}.$

- (b) This current is more than 200 times the current that the copper wire can carry.

54. Because f_j is directly proportional to V , the frequency is known to within one part in 10^{10} or $(10^{-10})(483.6 \times 10^9 \text{ Hz}) = 48.36 \text{ Hz}$, which is pretty fine tuning.

55. By conversion $420 \text{ km/h} = 119.44 \text{ m/s}$. Then from kinematics $v^2 = 2ax$, so

$$a = \frac{v^2}{2x} = \frac{(119.44 \text{ m/s})^2}{2(3000 \text{ m})} = 2.38 \text{ m/s}^2. \text{ This is about } g/4 \text{ which would certainly be noticeable.}$$

56. (a) To compute the escape speed use conservation of energy with $\frac{1}{2}mv_{\text{esc}}^2 = \frac{GMm}{R_e}$:

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R_e}} = \sqrt{\frac{2(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(5.98 \times 10^{24} \text{ kg})}{6.378 \times 10^6 \text{ m}}} = 11.1 \text{ km/s}.$$

- (b) From Chapter 9 we know

$$\bar{v} = \frac{4}{\sqrt{2\pi}} \sqrt{\frac{kT}{m}} = \frac{4}{\sqrt{2\pi}} \sqrt{\frac{(1.381 \times 10^{-23} \text{ J/K})(293 \text{ K})}{4(1.661 \times 10^{-27} \text{ kg})}} = 1245 \text{ m/s}.$$

- (c) There are always enough helium atoms on the (high-speed) tail of the Maxwell-Boltzmann distribution that a significant number can escape, given enough time.

57. Equating the centripetal force with the Lorentz (magnetic) force we have $\frac{mv^2}{R} = qvB$ or

$mv = p = qBR$. The formula $p = qBR$ is also correct relativistically and note that for these extremely high energies $E \approx pc = qBRc$. Therefore the energy is

$$E \approx qBRc = (1.602 \times 10^{-19} \text{ C})(13.5 \text{ T}) \left(\frac{27000 \text{ m}}{2\pi} \right) (2.998 \times 10^8 \text{ m/s}) \\ = 2.786 \times 10^{-6} \text{ J} = 17.4 \text{ TeV}.$$

58. (a) Let the atoms' distances from the molecule's center of mass be r_1 and r_2 , respectively, for the atoms with masses m_1 and m_2 . Thus $R = r_1 + r_2$. The rotational inertia is

$$I = m_1 r_1^2 + m_2 r_2^2, \text{ which can also be written}$$

$$I = \frac{(m_1 + m_2)m_1r_1^2 + (m_1 + m_2)m_2r_2^2}{m_1 + m_2} = \frac{m_1^2r_1^2 + m_2^2r_2^2 + m_1m_2r_1^2 + m_1m_2r_2^2}{m_1 + m_2}$$

From the definition of center of mass, $m_1r_1 = m_2r_2$. Applying this to the last expression above,

$$I = \frac{2m_1m_2r_1r_2 + m_1m_2r_1^2 + m_1m_2r_2^2}{m_1 + m_2} = \frac{m_1m_2(r_1 + r_2)^2}{m_1 + m_2}$$

which reduces to $I = \mu R^2$.

(b) Looking up the masses in Appendix 8, $m_1 = 23.0$ u and $m_2 = 35.0$ u. Then

$$I = \mu R^2 = \frac{(23 \text{ u})(35 \text{ u})}{23 \text{ u} + 35 \text{ u}} \frac{1.66 \times 10^{-27} \text{ kg}}{\text{u}} (2.36 \times 10^{-10} \text{ m})^2 = 1.28 \times 10^{-45} \text{ kg} \cdot \text{m}^2$$

This answer is in agreement with published values.

59. (a) Using the result of the preceding problem, the rotational inertias of the two isotopes are:

$$I = \mu R^2 = \frac{(1 \text{ u})(35 \text{ u})}{1 \text{ u} + 35 \text{ u}} \frac{1.66 \times 10^{-27} \text{ kg}}{\text{u}} (1.27 \times 10^{-10} \text{ m})^2 = 2.603 \times 10^{-47} \text{ kg} \cdot \text{m}^2$$

$$I = \mu R^2 = \frac{(1 \text{ u})(37 \text{ u})}{1 \text{ u} + 37 \text{ u}} \frac{1.66 \times 10^{-27} \text{ kg}}{\text{u}} (1.27 \times 10^{-10} \text{ m})^2 = 2.607 \times 10^{-47} \text{ kg} \cdot \text{m}^2$$

Thus the change is only about $4 \times 10^{-50} \text{ kg} \cdot \text{m}^2$, a fractional difference of 1.5×10^{-3} .

(b) Using the two different rotational inertia values from (a),

$$\Delta E_{\text{rot}} = \frac{\hbar^2}{I} = \frac{(1.0546 \times 10^{-34} \text{ J} \cdot \text{s})^2}{2.603 \times 10^{-47} \text{ kg} \cdot \text{m}^2} = 4.2727 \times 10^{-22} \text{ J}$$

$$\Delta E_{\text{rot}} = \frac{\hbar^2}{I} = \frac{(1.0546 \times 10^{-34} \text{ J} \cdot \text{s})^2}{2.607 \times 10^{-47} \text{ kg} \cdot \text{m}^2} = 4.2661 \times 10^{-22} \text{ J}$$

The difference is $6.6 \times 10^{-25} \text{ J} = 4.1 \times 10^{-6} \text{ eV}$, a very small difference.

60. (a) With $E_{\text{rot}} = \frac{\hbar^2 \ell(\ell+1)}{2I}$, for this transition we have $\Delta E_{\text{rot}} = \frac{\hbar^2(6-2)}{2I} = \frac{2\hbar^2}{I}$. Solving for I ,

$$I = \frac{2\hbar^2}{\Delta E_{\text{rot}}} = \frac{2(1.055 \times 10^{-34} \text{ J} \cdot \text{s})^2}{(9.55 \times 10^{-4} \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})} = 1.46 \times 10^{-46} \text{ kg} \cdot \text{m}^2$$

(b) From Problem 58 $I = \mu R^2$, The reduced mass is $\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{(12 \text{ u})(16 \text{ u})}{12 \text{ u} + 16 \text{ u}} = 6.86 \text{ u}$,
so the bond length is

$$R = \left(\frac{I}{\mu} \right)^{1/2} = \left(\frac{1.46 \times 10^{-46} \text{ kg} \cdot \text{m}^2}{(6.86 \text{ u})(1.66 \times 10^{-27} \text{ kg/u})} \right)^{1/2} = 1.13 \times 10^{-10} \text{ m}$$

61. (a) $P = \frac{\Delta E}{\Delta t} = \frac{1.8 \times 10^6 \text{ J}}{4.0 \times 10^{-9} \text{ s}} = 4.5 \times 10^{14} \text{ W}$ or 150 times the average US power consumption.

(b) The energy of each photon is $E = hc/\lambda = 5.68 \times 10^{-18} \text{ J}$, so the number of photons is

$$\frac{1.8 \times 10^6 \text{ J}}{5.68 \times 10^{-18} \text{ J/photon}} = 3.2 \times 10^{23} \text{ photons.}$$

62. According to BCS theory, $\frac{B_c(T)}{B_c(0)} = 1 - \left(\frac{T}{T_c} \right)^2$, so

$$\frac{T}{T_c} = \left[1 - \frac{B_c(T)}{B_c(0)} \right]^{1/2} = \left(1 - \frac{5 \times 10^{-5} \text{ T}}{4.11 \times 10^{-2} \text{ T}} \right)^{1/2} = 0.9994$$

For mercury $T_c = 4.15 \text{ K}$, so $T \approx 4.148 \text{ K}$, which is probably not noticeable.

63. From Table 10.5, $T_c = 3.72 \text{ K}$. Take this to be the transition temperature associated with the weighted atomic mass from the periodic table, 118.7 u. The BCS theory gives the isotope effect for the two isotopes, relative to this average.

$$\text{For } m = 112 \text{ u, } M_{112}^{0.5} T_{c112} = M_{\text{av}}^{0.5} T_{c,\text{av}}, \text{ so } T_{c112} = \frac{M_{\text{av}}^{0.5} T_{c,\text{av}}}{M_{112}^{0.5}} = \left(\frac{118.7}{112} \right)^{0.5} (3.72 \text{ K}) = 3.83 \text{ K}$$

$$\text{Similarly, for } m = 124 \text{ u, } T_{c124} = \frac{M_{\text{av}}^{0.5} T_{c,\text{av}}}{M_{124}^{0.5}} = \left(\frac{118.7}{124} \right)^{0.5} (3.72 \text{ K}) = 3.64 \text{ K}$$

The transition temperatures differ by 0.19 K.

64. (a) In an RL circuit the current is $I = I_0 e^{-Rt/L}$. For small values of R let us approximate the exponential with the Taylor expansion $1 - Rt/L$. Then $10^{-9} = 1 - \frac{I}{I_0} = 1 - e^{-Rt/L} \approx \frac{Rt}{L}$;

$$R \leq 10^{-9} \frac{L}{t} = 10^{-9} \left(\frac{3.14 \times 10^{-8} \text{ H}}{2.5 \text{ y} (3.16 \times 10^7 \text{ s/y})} \right) = 4.0 \times 10^{-25} \Omega.$$

- (b) For a 10% loss $t = \frac{0.1}{10^{-9}} (2.5 \text{ y}) = 2.5 \times 10^8 \text{ y}$.

65. From the BCS theory $B = B_c(0) \left(1 - \left(\frac{T}{T_c} \right)^2 \right)$. Then

$$\frac{\Delta S}{V} = - \frac{\partial}{\partial T} \left(\frac{B^2}{2\mu_0} \right) = \frac{2B_c^2(0)}{\mu_0 T_c} \left(\frac{T}{T_c} - \left(\frac{T}{T_c} \right)^3 \right). \text{ For numerical values use } T = 6 \text{ K},$$

$T_c = 9.25 \text{ K}$, and $B_c(0) = 0.206 \text{ T}$. Substituting, we have

$$\frac{\Delta S}{V} = \frac{2(0.206 \text{ T})^2}{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(9.25 \text{ K})} \left(\frac{6}{9.25} - \left(\frac{6}{9.25} \right)^3 \right) = 2743 \text{ J} \cdot \text{m}^{-3} \cdot \text{K}^{-1}.$$

The volume of one mole of niobium is $V = \frac{92.91 \text{ g}}{8.57 \text{ g/cm}^3} = 10.84 \text{ cm}^3 = 1.084 \times 10^{-5} \text{ m}^3$ and

thus $\Delta S = (2743 \text{ J} \cdot \text{m}^{-3} \cdot \text{K}^{-1})(1.084 \times 10^{-5} \text{ m}^3) = 2.97 \times 10^{-2} \text{ J/K}$ for one mole of niobium.

The superconducting state has a lower entropy than the normal state.

66. Provided the supercomputing wire is in its superconducting state and remains in that state, the resistance is zero. Therefore, the resistance of the circuit is determined by the copper wire.

- (a) The current, from Ohm's law is $I = \frac{V}{R} = \frac{12 \text{ V}}{1.5 \Omega} = 8.0 \text{ A}$.

- (b) The potential difference across the copper is 12 V. The superconducting wire will carry the current but will have zero potential difference across it.

Chapter 11

1. The Clement-Quinnell equation, $\log R + \frac{K}{\log R} = A + \frac{B}{T}$, can be rearranged to find

$$(\log R)^2 - \left(A + \frac{B}{T}\right)(\log R) + K = 0$$
 which is a quadratic equation that can be solved numerically at any given temperature.
 - a) $T = 77 \text{ K} : \log R = 2.372$ so $R = 236 \Omega$
 - b) $T = 20 \text{ K} : \log R = 2.786$ so $R = 610 \Omega$
 - c) $T = 1 \text{ K} : \log R = 7.739$ so $R = 5.48 \times 10^7 \Omega$
2. $\log R + \frac{K}{\log R} = A + \frac{B}{T}$; inserting each of the R and T values, we have three equations in three unknowns (A, B, K) which can be solved to yield $A = 2.09$, $B = 1.96$, and $K = 1.10$.
3. Positive charges drift to the right, so the right side of the strip is at a higher potential and the voltmeter reads a positive value.
4. (a) Starting from Equation (11.6) and with $A = yz$, we have

$$n = \frac{IB}{eV_H z} = \frac{(0.10 \text{ A})(0.050 \text{ T})}{(1.602 \times 10^{-19} \text{ C})(11.5 \times 10^{-3} \text{ V})(1.5 \times 10^{-4} \text{ m})} = 1.81 \times 10^{22} \text{ m}^{-3}$$
 (b) Graphing B versus V_H we find a slope of approximately 4.56 T/V. Algebraically we see that $B = \frac{nez}{I} V_H$. Thus the slope, m is equal to $\frac{nez}{I}$. So

$$n = \frac{mI}{ez} = \frac{(4.56 \text{ T/V})(0.10 \text{ A})}{(1.602 \times 10^{-19} \text{ C})(1.5 \times 10^{-4} \text{ m})} = 1.90 \times 10^{22} \text{ m}^{-3}$$
5. $E = V / L = QdT / dx$ so $Q = \frac{V}{L \frac{dT}{dx}} = \frac{12.8 \times 10^{-6} \text{ V}}{(0.10 \text{ m})\left(\frac{10 \text{ K}}{0.1 \text{ m}}\right)} = 1.28 \times 10^{-6} \text{ V/K}$.
6. From the table 3.03 mV corresponds to 58.4°C .
7. Over most of the table, 0.05 mV corresponds to a temperature change of 1°C . Therefore for 0.01°C there is a corresponding voltage difference of $(0.01)(0.05 \text{ mV}) = 5 \times 10^{-7} \text{ V}$.
8. (a) P is in Group V; Ge is in Group IV; so adding phosphorous will create an n -type semiconductor.
 (b) Ga is in Group III; Ge is in Group IV; so adding gallium will create a p -type semiconductor.

9. The relation between energy and photon wavelength is $E = \frac{hc}{\lambda}$. The photon must provide enough energy to excite the electron across the gaps, so $\lambda = \frac{hc}{\Delta E}$.

$$(a) \quad \lambda = \frac{1240 \text{ eV} \cdot \text{nm}}{0.67 \text{ eV}} = 1850 \text{ nm}$$

$$(b) \quad \lambda = \frac{1240 \text{ eV} \cdot \text{nm}}{1.10 \text{ eV}} = 1130 \text{ nm}$$

$$(c) \quad \lambda = \frac{1240 \text{ eV} \cdot \text{nm}}{0.36 \text{ eV}} = 3440 \text{ nm}$$

$$(d) \quad \lambda = \frac{1240 \text{ eV} \cdot \text{nm}}{3.6 \text{ eV}} = 344 \text{ nm}$$

10. The relation between energy and photon wavelength is $E = \frac{hc}{\lambda}$. A photon with wavelength 574 nm corresponds to $E = \frac{1240 \text{ eV} \cdot \text{nm}}{574 \text{ nm}} = 2.16 \text{ eV}$.

11. Using Equation (11.9), the ratio of currents in forward and reverse is

$$\frac{I_f}{I_r} = \frac{e^{eV_f/kT} - 1}{e^{eV_r/kT} - 1} = \frac{\exp\left[(1.60 \times 10^{-19} \text{ C})(1.5 \text{ V}) / (1.38 \times 10^{-23} \text{ J/K})(293 \text{ K})\right] - 1}{\exp\left[(1.60 \times 10^{-19} \text{ C})(-1.5 \text{ V}) / (1.38 \times 10^{-23} \text{ J/K})(293 \text{ K})\right] - 1}$$

which simplifies to $\frac{I_f}{I_r} = -6.0 \times 10^{25}$.

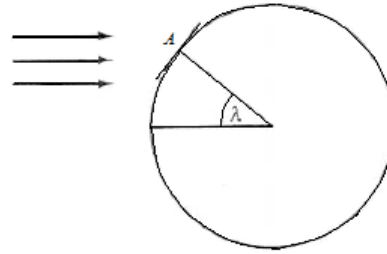
12. From Equation (11.9), the ratio of the two currents is $\frac{I_f}{I_r} = \frac{e^{eV_f/kT} - 1}{e^{eV_r/kT} - 1}$.

When the ratio of forward-bias to reverse-bias current is this high, the first term in the denominator is nearly zero, and the first term in the numerator is much greater than one,

so to a good approximation $\frac{I_f}{I_r} = -e^{eV_f/kT}$. Solving for V_f ,

$$V_f = \frac{kT}{e} \ln\left(\frac{I_f}{I_r}\right) = \frac{(1.38 \times 10^{-23} \text{ J/K})(293 \text{ K})}{1.6 \times 10^{-19} \text{ C}} \ln(10^6) = 0.35 \text{ V}.$$

13. The angle between the solar rays and a vector normal to the surface A is λ . Therefore $\Phi = \Phi_0 \cos \lambda$.



- (a) On the equinox $\lambda = 26^\circ$ so $\Phi = \Phi_0 \cos(26^\circ) = 0.900 \Phi_0$;
 On the winter solstice $\lambda = 26^\circ + 23^\circ = 49^\circ$ and $\Phi = \Phi_0 \cos(49^\circ) = 0.656 \Phi_0$;
 On the summer solstice $\lambda = 26^\circ - 23^\circ = 3^\circ$ and $\Phi = \Phi_0 \cos(3^\circ) = 0.999 \Phi_0$
- (b) On the equinox $\lambda = 50^\circ$ so $\Phi = \Phi_0 \cos(50^\circ) = 0.643 \Phi_0$;
 On the winter solstice $\lambda = 50^\circ + 23^\circ = 73^\circ$ and $\Phi = \Phi_0 \cos(73^\circ) = 0.292 \Phi_0$;
 On the summer solstice $\lambda = 50^\circ - 23^\circ = 27^\circ$ and $\Phi = \Phi_0 \cos(27^\circ) = 0.891 \Phi_0$
- (c) On the equinox $\lambda = 60^\circ$ so $\Phi = \Phi_0 \cos(60^\circ) = 0.500 \Phi_0$;
 On the winter solstice $\lambda = 60^\circ + 23^\circ = 83^\circ$ and $\Phi = \Phi_0 \cos(83^\circ) = 0.122 \Phi_0$;
 On the summer solstice $\lambda = 60^\circ - 23^\circ = 37^\circ$ and $\Phi = \Phi_0 \cos(37^\circ) = 0.799 \Phi_0$

14. We require $10^9 \text{ W} = 0.3(200 \text{ W/m}^2) A$, so rearranging

$$A = \frac{10^9 \text{ W}}{0.3(200 \text{ W/m}^2)} = 1.67 \times 10^7 \text{ m}^2$$

which corresponds to a square array 4.1 km on a side.

15. In general $I = I_0 (\exp(eV/kT) - 1)$ and in Example 11.4

$$I_0 = \frac{I}{\exp(eV/kT) - 1} \approx 18.16 \text{ } \mu\text{A}.$$

- (a) $I = (18.16 \times 10^{-6} \text{ A}) \left(\exp \left(\frac{0.250 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(250 \text{ K})} \right) - 1 \right) = 1.99 \text{ A}$
- (b) $I = (18.16 \times 10^{-6} \text{ A}) \left(\exp \left(\frac{0.250 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(300 \text{ K})} \right) - 1 \right) = 0.288 \text{ A} = 288 \text{ mA}$
- (c) $I = (18.16 \times 10^{-6} \text{ A}) \left(\exp \left(\frac{0.250 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(500 \text{ K})} \right) - 1 \right) = 0.006 \text{ A} = 6.00 \text{ mA}$

16. Because the tube is single-walled, we can find the surface area density, σ in SI units. Each atom has mass 12 u, with $u = 1.6605 \times 10^{-27}$ kg.

$$\sigma = (2.3 \times 10^{19} \text{ atoms/m}^2)(12 \text{ u})(1.6605 \times 10^{-27} \text{ kg/u}) = 4.58 \times 10^{-7} \text{ kg/m}^2$$

- (a) To determine the density of the material, we need the mass per unit volume. The mass will equal the mass density, σ from above, times the area A of the cylindrical wall. If the tube's length is L with radius R , then $m = 2\pi RL\sigma$. Therefore the

$$\text{density will be } \rho = \frac{m}{V} = \frac{2\pi RL\sigma}{\pi R^2 L} = \frac{2\sigma}{R} = \frac{2(4.58 \times 10^{-7} \text{ kg/m}^2)}{0.7 \times 10^{-9} \text{ m}} = 1300 \text{ kg/m}^3.$$

- (b) The material is less dense than steel and has a greater tensile strength.

17. We know the density of the single-walled nanotube from the previous problem. Consider a tube that is 1 nm long. The tube will contain one buckyball. The total mass will be the mass of the tube plus the mass of the buckyball which is

$m = \rho V + (60)(12 \text{ u})(1.6605 \times 10^{-27} \text{ kg/u}) = \rho(\pi R^2 L) + 1.20 \times 10^{-24} \text{ kg}$. We find the term ρV equal to $2.0 \times 10^{-24} \text{ kg}$, so the total mass equals $3.2 \times 10^{-24} \text{ kg}$. Therefore the density of the nanotube peapod is

$$\rho = \frac{m}{V} = \frac{3.2 \times 10^{-24} \text{ kg}}{\pi R^2 L} = \frac{3.2 \times 10^{-24} \text{ kg}}{\pi (0.7 \times 10^{-9} \text{ m})^2 (1 \times 10^{-9} \text{ m})} = 2080 \text{ kg/m}^3.$$

18. (a) Using $E = E_g$ for the conduction band we have $E - E_F = E_g - E_g / 2 = E_g / 2$. Then $\exp[(E - E_F) / kT] = \exp[E_g / 2kT]$. With $E_g \approx 1 \text{ eV}$ for a semiconductor (and larger for an insulator) and $kT \approx \frac{1}{40} \text{ eV}$ at room temperature, we can see that $E_g \gg 2kT$ and the exponential term will be much greater than 1, so we can neglect the +1 term in the Fermi-Dirac factor. This leaves $F_{FD} \approx \frac{1}{\exp[E_g / 2kT]} = \exp[-E_g / 2kT]$.

$$(b) F_{FD} \approx \exp[-E_g / 2kT] = \exp\left[-\frac{8.0 \text{ eV}}{2(8.617 \times 10^{-5} \text{ eV/K})(300 \text{ K})}\right] = 6.3 \times 10^{-68}$$

$$(c) F_{FD} \approx \exp[-E_g / 2kT] = \exp\left[-\frac{1.11 \text{ eV}}{2(8.617 \times 10^{-5} \text{ eV/K})(293 \text{ K})}\right] = 2.8 \times 10^{-10}$$

- (d) For one mole of silicon there are 6×10^{23} atoms, so there are still many conduction electrons available.

19. (a) Replacing ϵ_0 with $\kappa\epsilon_0$ we have for the new Bohr radius

$$a'_0 = \frac{4\pi\epsilon_0\kappa\hbar^2}{me^2} = 11.7a_0 = 11.7(5.29 \times 10^{-2} \text{ nm}) = 0.619 \text{ nm}.$$

(b) This value is about 2.6 times the lattice spacing. This is consistent with the fact that the electron is very weakly bound, and hence the doped silicon should have a higher electrical conductivity than pure silicon.

20. From the Bohr theory $E_0 = \frac{me^4}{2\hbar^2(4\pi\epsilon_0)^2}$. Replacing ϵ_0 with $\kappa\epsilon_0$ we have a new Rydberg

$$\text{energy } E'_0 = \frac{me^4}{2\hbar^2(4\pi\epsilon_0^2)\kappa^2} = \frac{E_0}{\kappa^2} = \frac{13.6 \text{ eV}}{(11.7)^2} = 0.099 \text{ eV}.$$

This is more than a factor of ten less than the band gap for pure silicon which is consistent with the idea that the doped version has a higher electrical conductivity.

21. Answers will vary depending on the algorithm used, but one should find that using a second-order method results in some improvement.

22. (a) $I = I_0(\exp(eV/kT) - 1)$. To find the value of V for the diode, use the loop rule:

$V + IR = 6 \text{ V}$ so $V = 6 \text{ V} - IR$. We are given that $I_0 = 1.05 \mu\text{A}$ and $I = 140 \text{ mA}$ with

$$T = 293 \text{ K so } \frac{I}{I_0} = \frac{140 \times 10^{-3}}{1.05 \times 10^{-6}} = 1.33 \times 10^5 = \exp(eV/kT) \text{ so}$$

$$\ln(1.33 \times 10^5) = \frac{eV}{kT} = \frac{e(6 \text{ V} - IR)}{kT}. \text{ Solving for } R \text{ we find}$$

$$R = \frac{6 \text{ V} - (kT \ln(1.33 \times 10^5))/e}{I} = \frac{6 \text{ V} - (8.617 \times 10^{-5} \text{ eV/K})[293 \text{ K}(\ln(1.33 \times 10^5))]/e}{0.140 \text{ A}} = 40.7 \Omega$$

$$(b) V_R = IR = (0.140 \text{ A})(40.7 \Omega) = 5.70 \text{ V}$$

23. (a) $7 = \exp(eV/kT) - 1$ so $V = \frac{kT \ln 8}{e} = \frac{(8.617 \times 10^{-5} \text{ eV/K})(293 \text{ K})(\ln 8)}{e} = 52.5 \text{ mV}$

$$(b) -0.7 = \exp(eV/kT) - 1 \text{ so}$$

$$V = \frac{kT \ln 0.3}{e} = \frac{(8.617 \times 10^{-5} \text{ eV/K})(293 \text{ K})(\ln 0.3)}{e} = -30.4 \text{ mV}$$

24. $E_g = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{650 \text{ nm}} = 1.91 \text{ eV}$

25. The GaN laser's wavelength is 405 nm, so $E_g = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{405 \text{ nm}} = 3.06 \text{ eV}$.
26. (a) The total area is $(580 \times 10^6)(45 \times 10^{-9} \text{ m})^2 = 1.12 \times 10^{-6} \text{ m}^2$ so each side is the square root of this which equals 1.08 mm.
- (b) The area of each transistor is now $(32 \times 10^{-9} \text{ m})^2 = 1.024 \times 10^{-15} \text{ m}^2$ so the number is
- $$N = \frac{1.12 \times 10^{-6} \text{ m}^2}{1.024 \times 10^{-15} \text{ m}^2} = 1.09 \times 10^9 \text{ or nearly a factor of 2 improvement.}$$
27. From Problem 18(a) we have $F_{\text{FD}} \approx \exp[-E_g / 2kT]$.

$$\text{Si at } T = 0^\circ \text{C: } F_{\text{FD}} \approx \exp\left(-\frac{1.11 \text{ eV}}{2(8.617 \times 10^{-5} \text{ eV/K})(273 \text{ K})}\right) = 5.67 \times 10^{-11}$$

$$\text{Si at } T = 75^\circ \text{C: } F_{\text{FD}} \approx \exp\left(-\frac{1.11 \text{ eV}}{2(8.617 \times 10^{-5} \text{ eV/K})(348 \text{ K})}\right) = 9.16 \times 10^{-9}$$

$$\text{Ge at } T = 0^\circ \text{C: } F_{\text{FD}} \approx \exp\left(-\frac{0.67 \text{ eV}}{2(8.617 \times 10^{-5} \text{ eV/K})(273 \text{ K})}\right) = 6.54 \times 10^{-7}$$

$$\text{Ge at } T = 75^\circ \text{C: } F_{\text{FD}} \approx \exp\left(-\frac{0.67 \text{ eV}}{2(8.617 \times 10^{-5} \text{ eV/K})(348 \text{ K})}\right) = 1.41 \times 10^{-5}$$

The Fermi-Dirac factor is orders of magnitude higher in germanium, making the conduction electron density too high. The result is a reverse-bias current that is too large. See Physics Today December 1997 p. 38.

28. (a) Energy is power multiplied by time:

$$E = Pt = (0.15)(200 \text{ W})(3.156 \times 10^7 \text{ s}) = 9.47 \times 10^8 \text{ J.}$$

- (b) Converting we find $2.8 \times 10^{12} \text{ kW} \cdot \text{h} = 1.01 \times 10^{19} \text{ J}$. Then the area is

$$\frac{1.01 \times 10^{19} \text{ J}}{9.47 \times 10^8 \text{ J/m}^2} = 1.06 \times 10^{10} \text{ m}^2$$

- c) Using 2.5 times the area found in (b) the fraction of the U.S. covered is

$$\frac{2.5(1.06 \times 10^{10} \text{ m}^2)}{9.6 \times 10^{12} \text{ m}^2} = 0.0028 \text{ or about one-fourth of one percent. See American Scientist}$$

July-August 1993, p. 368.

29. From the diode equation the current is $I = I_0 \exp(eV / kT) - 1$. Then the ratio is

$$\frac{I_f}{I_r} = \frac{I_0 \exp(eV_f / kT) - 1}{I_0 \exp(eV_r / kT) - 1}.$$

$$(a) \text{ At } T = 77 \text{ K: } \frac{I_f}{I_r} = \frac{\exp\left(\frac{1.50 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(77 \text{ K})}\right) - 1}{\exp\left(\frac{-1.50 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(77 \text{ K})}\right) - 1} = -1.5 \times 10^{98}$$

$$\text{At } T = 273 \text{ K: } \frac{I_f}{I_r} = \frac{\exp\left(\frac{1.50 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(273 \text{ K})}\right) - 1}{\exp\left(\frac{-1.50 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(273 \text{ K})}\right) - 1} = -4.9 \times 10^{27}$$

$$\text{At } T = 350 \text{ K: } \frac{I_f}{I_r} = \frac{\exp\left(\frac{1.50 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(350 \text{ K})}\right) - 1}{\exp\left(\frac{-1.50 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(350 \text{ K})}\right) - 1} = -4.0 \times 10^{21}$$

$$\text{At } T = 600 \text{ K: } \frac{I_f}{I_r} = \frac{\exp\left(\frac{1.50 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(600 \text{ K})}\right) - 1}{\exp\left(\frac{-1.50 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(600 \text{ K})}\right) - 1} = -4.0 \times 10^{12}$$

(b) The ratio changes significantly as a function of temperature which is something diode designers must keep in mind.

30.

(a) An electron can be produced if the 1.11 eV band gap can be overcome. If we divide the total energy available by the band gap energy, the maximum number of electrons

$$\text{that can be produced is } N = \frac{1.04 \times 10^6 \text{ eV}}{1.11 \text{ eV}} = 9.37 \times 10^5.$$

(b) If the silicon is cooled well below room temperature, very few electrons will be in the conduction band. At room temperature, however, enough electrons are in the conduction band so that additional current will be measured, tending to mask the gamma-ray signal.

31. From the diode equation the current is $I = I_0 \exp(eV / kT) - 1$. Then the ratio is

$$\frac{I_f}{I_r} = \frac{I_0 \exp(eV_f / kT) - 1}{I_0 \exp(eV_r / kT) - 1} = \frac{\exp(eV_f / kT) - 1}{\exp(eV_r / kT) - 1}. \text{ Substituting the given values we find}$$

$$\frac{I_f}{I_r} = \frac{\exp\left(\frac{0.25 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(293 \text{ K})}\right) - 1}{\exp\left(\frac{-0.25 \text{ eV}}{(8.617 \times 10^{-5} \text{ eV/K})(293 \text{ K})}\right) - 1} = -20000. \quad \text{So the ratio of the currents in the forward- and reverse-bias is 20,000.}$$

32. Young's Modulus relates the elongation of an object due to an applied force. The formula is $Y = \frac{F}{A} \left(\frac{L}{\Delta L} \right)$. Rearranging this formula to solve for F and substituting, we find

$$F = Y A \left(\frac{\Delta L}{L} \right) = 1050 \times 10^9 \text{ Pa} \left(\frac{\pi (1.6 \times 10^{-9} \text{ m})^2}{4} \right) (0.01) = 2.1 \times 10^{-8} \text{ N}.$$

33. The number of bits stored is $(4.7 \times 10^9)(8) = 3.76 \times 10^{10}$ bits. To find the area we use $A = \pi(r_2^2 - r_1^2) = \pi(0.058^2 - 0.023^2) = 8.91 \times 10^{-3} \text{ m}^2$. The number of bits stored per square meter is then $\frac{3.76 \times 10^{10} \text{ bits}}{8.91 \times 10^{-3} \text{ m}^2} = 4.22 \times 10^{12} \text{ bits/m}^2$.

34. $25 \text{ GB} \times 8 \text{ bits/byte} = 200 \text{ Gbits}$ With an inner radius of 2.3 cm and outer radius 5.8 cm, the effective useful area is $\pi(0.058 \text{ m})^2 - \pi(0.023 \text{ m})^2 = 8.91 \times 10^{-3} \text{ m}^2$. The average area per bit is $\frac{8.91 \times 10^{-3} \text{ m}^2}{200 \times 10^9} = 4.46 \times 10^{-14} \text{ m}^2$. For an approximately rectangular bit, the area is length times width, so the length is area divided by width, or $\frac{4.46 \times 10^{-14} \text{ m}^2}{3.2 \times 10^{-7} \text{ m}} = 1.4 \times 10^{-7} \text{ m} = 0.14 \text{ } \mu\text{m}$.

35. 1 hectare = 10^4 m^2 , so 140 hectares = $1.40 \times 10^6 \text{ m}^2$.

$$\text{The facility's power output per unit area is } \frac{64 \times 10^6 \text{ W}}{1.40 \times 10^6 \text{ m}^2} = 45.7 \text{ W/m}^2.$$

$$\text{The efficiency is this output divided by the sun's input: } \frac{45.7 \text{ W/m}^2}{630 \text{ W/m}^2} = 0.073 = 7.3\%.$$

This seems low compared with solar cells. However, the NSO facility does not use every square meter of land surface, so the comparison is not an exact one.

- 36.

$$(a) \text{ The mass of the dot is } m = \rho V = (4820 \text{ kg/m}^3) \frac{4}{3} \pi (2.50 \times 10^{-9} \text{ m})^3 = 3.15 \times 10^{-22} \text{ kg}.$$

From the periodic table, the average mass of a CdS molecule is

$112.4 \text{ u} + 32.1 \text{ u} = 144.5 \text{ u} = 2.40 \times 10^{-25} \text{ kg}$. The number of molecules is

$$\frac{3.15 \times 10^{-22} \text{ kg}}{2.40 \times 10^{-25} \text{ kg}} = 1312.5. \text{ The number of atoms is twice this or } 2625.$$

(b) From Chapter 6, the energy levels are given by $E = \frac{n^2 h^2}{8mL^2}$. For $n = 1$, we find

$$E = \frac{h^2}{8mL^2} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})^2}{8(9.11 \times 10^{-31} \text{ kg})(2.0 \times 10^{-9} \text{ m})^2} = 1.51 \times 10^{-20} \text{ J} = 0.094 \text{ eV}.$$

(c) The difference in energy levels is $\frac{n_u^2 h^2}{8mL^2} - \frac{n_l^2 h^2}{8mL^2}$. With $n_u = 6$ and $n_l = 5$,

$$\Delta E = (6^2 - 5^2) 0.094 \text{ eV} = 1.0 \text{ eV}.$$

Chapter 12

1. From Compton scattering $\Delta\lambda = \frac{h}{mc}(1 - \cos\theta)$, so $\Delta\lambda_{\max} = \frac{2h}{mc}$. From Chapter 3 Problem

51 we know that the kinetic energy of the recoiling particle is

$$K = \left[\frac{\Delta\lambda / \lambda}{1 + \Delta\lambda / \lambda} \right] \frac{hc}{\lambda} = 5.7 \text{ MeV. Using } \Delta\lambda = \frac{2h}{mc} \text{ from above and for convenience letting}$$

$$x = \frac{2h}{mc\lambda}, \text{ we have } K = \frac{2h/(mc\lambda)}{1 + 2h/(mc\lambda)} \frac{hc}{\lambda} = \frac{x}{1+x} \frac{mc^2}{2} x; K(1+x) = x^2 \frac{mc^2}{2};$$

$$\frac{mc^2}{2} x^2 - Kx - K = 0. \text{ This is a quadratic equation that can be solved numerically to find}$$

$$x = 0.117 = 2h/mc\lambda, \text{ so } \lambda = \frac{2h}{mcx} = \frac{2(1240 \text{ eV} \cdot \text{nm})}{(938.27 \times 10^6 \text{ eV})(0.117)} = 2.26 \times 10^{-5} \text{ nm. The}$$

$$\text{photon's energy is } E = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{2.26 \times 10^{-5} \text{ nm}} = 54.9 \text{ MeV.}$$

2. (a) integral: ${}^6\text{Li}$, ${}^{18}\text{F}$; (b) half-integral: ${}^3\text{He}$, ${}^7\text{Li}$, ${}^{19}\text{F}$
3. In each case the atomic number equals the number of protons (Z), and the atomic charge is Ze .

$${}^3_2\text{He} : Z = 2, N = 1, A = 3, m = 3.02 \text{ u}$$

$${}^4_2\text{He} : Z = 2, N = 2, A = 4, m = 4.00 \text{ u}$$

$${}^{18}_8\text{O} : Z = 8, N = 10, A = 18, m = 18.00 \text{ u}$$

$${}^{44}_{20}\text{Ca} : Z = 20, N = 24, A = 44, m = 43.96 \text{ u}$$

$${}^{209}_{83}\text{Bi} : Z = 83, N = 126, A = 209, m = 208.98 \text{ u}$$

$${}^{235}_{92}\text{U} : Z = 92, N = 143, A = 235, m = 235.04 \text{ u}$$

4. ${}^6\text{Li} : Z = 3, N = 3$

$${}^{13}\text{C} : Z = 6, N = 7$$

$${}^{40}\text{K} : Z = 19, N = 21$$

$${}^{102}\text{Pd} : Z = 46, N = 56$$

5. Isotopes ${}^{38}\text{Ca}$ through ${}^{52}\text{Ca}$; Isobars ${}^{40}\text{S}$, ${}^{40}\text{Cl}$, ${}^{40}\text{Ar}$, ${}^{40}\text{K}$, ${}^{40}\text{Ca}$; Isotones ${}^{32}\text{Mg}$, ${}^{34}\text{Si}$, ${}^{35}\text{P}$, ${}^{36}\text{S}$, ${}^{37}\text{Cl}$, ${}^{38}\text{Ar}$, ${}^{39}\text{K}$, ${}^{40}\text{Ca}$

6. ${}^{14}\text{N}$ (99.63%), ${}^{15}\text{N}$ (0.37%); ${}^{50}\text{V}$ (0.250%), ${}^{51}\text{V}$ (99.75%);

$${}^{84}\text{Sr}$$
 (0.56%), ${}^{86}\text{Sr}$ (9.86%), ${}^{87}\text{Sr}$ (7.00%), ${}^{88}\text{Sr}$ (82.58%)

7. ^{39}Ar (269 y), ^{41}Ar (1.822h), ^{42}Ar (32.9 y), ^{43}Ar (5.37 m), ^{44}Ar (11.87 m), ^{45}Ar (21.48 s), ^{46}Ar (8.4 s).

^{250}No (0.25 ms), ^{251}No (0.8 s), ^{252}No (2.30 s), ^{254}No (55 s), ^{256}No (2.91 s),
 ^{258}No (1.2 ms), ^{259}No (58 m), ^{260}No (106 ms).

8. From the text nuclear density is $2.3 \times 10^{17} \text{ kg/m}^3$, which is 2.3×10^{14} times the density of water.

$$9. \quad \frac{\mu_p}{\mu_e} = \frac{2.79\mu_N}{-1.00116\mu_B} = -2.786 \frac{\mu_N}{\mu_B} = -2.787 \frac{m_e}{m_p} = -2.786 \frac{0.51100}{938.27} = -1.52 \times 10^{-3}$$

10. From Appendix 8 the mass of the nuclide is 55.935 u or $9.29 \times 10^{-26} \text{ kg}$.

$$\rho = \frac{m}{\frac{4}{3}\pi r^3} = \frac{m}{\frac{4}{3}\pi r_0^3 A} = \frac{9.29 \times 10^{-26} \text{ kg}}{\frac{4}{3}\pi (1.2 \times 10^{-15} \text{ m})^3 (56)} = 2.29 \times 10^{17} \text{ kg/m}^3$$

11. The electron binding energy is virtually the same as in hydrogen, or 13.6 eV. The deuteron rest energy is $mc^2 = 1876 \text{ MeV}$. The ratio of these is

$$\frac{13.6 \text{ eV}}{1876 \times 10^6 \text{ eV}} = 7.25 \times 10^{-9}.$$

“Nuclear calculations” might also refer to nuclear binding energies. For the deuteron the binding energy is 2.2 MeV, so the ratio of the electronic to nuclear binding energy is on the order of 10^{-5} . Therefore it is generally safe to ignore electronic binding energies.

12. Using Equation (12.14), the values of B_d from Equation (12.9) and the atomic deuterium mass in order to evaluate the second term inside the parenthesis, we find

$$E_{\min} = B_d \left(1 + \frac{B_d}{2M(^2\text{H})c^2} \right) = 1.00059B_d \approx 2.22 \text{ MeV}.$$

From this calculation we can see that the error is 0.00059 or 0.059%.

13. The incoming photon's momentum is p . Its energy is $hf = pc$. By conservation of momentum, the total momentum of the two particles after the reaction is also p , and the total mass is $m_p + m_n$. (We will ignore electron masses here.) Conservation of relativistic energy gives

$$pc + M_d c^2 = \sqrt{p^2 c^2 + ((m_p + m_n) c^2)^2}.$$

Squaring both sides of the equation:

$$(pc)^2 + 2pM_d c^3 + M_d^2 c^4 = (pc)^2 + ((m_p + m_n) c^2)^2$$

The first term on each side cancels. The resulting equation can be solved for $2pc$:

$$2pc = \frac{\left((m_p + m_n)c^2\right)^2 - M_d^2c^4}{M_dc^2} = \left[\frac{(m_p + m_n)^2 - M_d^2c^4}{M_dc^2} \right]$$

Now $B_d = [m_p + m_n - M_d]c^2$, so we can substitute $(m_p + m_n)c^2 = B_d + M_dc^2$, giving

$$2pc = \left[\frac{(B_d + M_dc^2)^2 - M_d^2c^4}{M_dc^2} \right] = \left[\frac{B_d^2 + M_d^2c^4 + 2B_dM_dc^2 - M_d^2c^4}{M_dc^2} \right]$$

$$2pc = \left[2B_d + \frac{B_d^2}{M_dc^2} \right] = B_d \left[2 + \frac{B_d}{M_dc^2} \right]$$

Now for the photon $pc = hf$, and so

$$hf = B_d \left[1 + \frac{B_d}{2M_dc^2} \right]$$

which is equivalent to Equation (12.14).

14. (a) Think of the nucleus as the composite of deuteron and 1_0n , so that

$$B = [M({}^2_1\text{H}) + m_n - M({}^3_1\text{H})]c^2. \text{ Use the atomic masses from Appendix 8:}$$

$$B = (2.014102 \text{ u} + 1.008665 \text{ u} - 3.016049 \text{ u})c^2 (931.49 \text{ MeV}/(\text{u} \cdot c^2)) = 6.26 \text{ MeV}$$

- (b) With the nucleus as two neutrons and a proton, we have:

$$B = [2m_n + M({}^1_1\text{H}) - M({}^3_1\text{H})]c^2$$

$$B = (2(1.008665) \text{ u} + 1.00728 \text{ u} - 3.016049 \text{ u})c^2 (931.49 \text{ MeV}/(\text{u} \cdot c^2)) = 8.48 \text{ MeV}$$

- (c) The difference is just the binding energy of the deuteron, 2.22 MeV. This makes sense, because the first step (a) separates the triton into a neutron and deuteron, and a second step of separating the deuteron into a proton and neutron completes the disintegration of the original nucleus.

15. The distance equals the nuclear radius: $r = r_0 A^{1/3} = (1.2 \text{ fm})(3^{1/3}) = 1.73 \text{ fm}$

$$|F_g| = \frac{GMm}{r^2} = \frac{(6.673 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2)(1.673 \times 10^{-27} \text{ kg})^2}{(1.73 \times 10^{-15} \text{ m})^2} = 6.24 \times 10^{-35} \text{ N}$$

$$|F_e| = \frac{ke^2}{r^2} = \frac{(8.988 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(1.602 \times 10^{-19} \text{ C})^2}{(1.73 \times 10^{-15} \text{ m})^2} = 77.1 \text{ N}$$

To compare with the strong force, we need the potential energy:

$$|V_g| = \frac{GMm}{r} = \frac{(6.673 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2)(1.673 \times 10^{-27} \text{ kg})^2}{(1.73 \times 10^{-15} \text{ m})(1.602 \times 10^{-13} \text{ J/MeV})} = 6.7 \times 10^{-37} \text{ MeV}$$

$$|V_e| = \frac{ke^2}{r} = \frac{(8.988 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(1.602 \times 10^{-19} \text{ C})^2}{(1.73 \times 10^{-15} \text{ m})(1.602 \times 10^{-13} \text{ J/MeV})} = 0.83 \text{ MeV}$$

The electrostatic force is about 50 times weaker than the strong force. The gravitational force is almost 10^{38} times weaker than the strong force.

16. The required nuclear force must be at least of this magnitude:

$$|F_e| = \frac{ke^2}{r^2} = \frac{(8.988 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(1.602 \times 10^{-19} \text{ C})^2}{(2.4 \times 10^{-15} \text{ m})^2} = 40.0 \text{ N. We can also compare the}$$

$$\text{potential energy: } |V| = \frac{ke^2}{r} = \frac{(8.988 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(1.602 \times 10^{-19} \text{ C})^2}{(2.4 \times 10^{-15} \text{ m})(1.602 \times 10^{-13} \text{ J/MeV})} = 0.60 \text{ MeV}$$

17. (a) Think of the nucleus as the composite of ${}^{A-1}_Z X$ and ${}^1_0 n$, so that

$$B = [M({}^{A-1}_Z X) + m_n - M({}^A_Z X)]c^2$$

(b) Details for ${}^6\text{Li}$ are shown. The other examples are similar. Use the atomic masses from Appendix 8:

$$B = [M({}^5\text{Li}) + m_n - M({}^6\text{Li})]c^2 \\ = (5.012540 \text{ u} + 1.008665 \text{ u} - 6.015122 \text{ u})c^2 (931.49 \text{ MeV}/(\text{u} \cdot c^2)) = 5.67 \text{ MeV}$$

$${}^{16}\text{O}: B = [M({}^{15}\text{O}) + m_n - M({}^{16}\text{O})]c^2 = 15.7 \text{ MeV}$$

$${}^{207}\text{Pb}: B = [M({}^{206}\text{Pb}) + m_n - M({}^{207}\text{Pb})]c^2 = 6.74 \text{ MeV}$$

18. (a) As in the previous problem, consider the nucleus as the composite of ${}^{A-1}_{Z-1} Y$ and ${}^1_1 \text{H}$, so that $B = [M({}^{Z-1}_{Z-1} Y) + M({}^1_1 \text{H}) - M({}^A_Z X)]c^2$.

(b) ${}^8\text{Be}$:

$$B = [M({}^7\text{Li}) + M({}^1_1 \text{H}) - M({}^8\text{Be})]c^2 \\ = (7.016004 \text{ u} + 1.007825 \text{ u} - 8.005305 \text{ u})c^2 (931.49 \text{ MeV}/(\text{u} \cdot c^2)) = 17.3 \text{ MeV}$$

$${}^{15}\text{O}: B = [M({}^{14}\text{N}) + M({}^1_1 \text{H}) - M({}^{15}\text{O})]c^2 = 7.3 \text{ MeV}$$

$$^{32}\text{S}: B = \left[M(^{31}\text{P}) + M(^1\text{H}) - M(^{32}\text{S}) \right] c^2 = 8.9 \text{ MeV}$$

19. The energy release comes from the mass difference:

$$\begin{aligned} \Delta E = \Delta mc^2 &= \left[3 M(^4\text{He}) - M(^{12}\text{C}) \right] c^2 \\ &= \left[3(4.002603 \text{ u}) - 12.000 \text{ u} \right] c^2 \left(931.49 \text{ MeV}/(\text{u} \cdot c^2) \right) = 7.27 \text{ MeV} \end{aligned}$$

20. With an even number of nucleons the ^4He nucleus has integer spin. Because alignment is unlikely, the net spin should be zero. The ^3He nucleus has three nucleons, so its spin is half-integer. Because alignment is unlikely, the net spin should be one-half.

21. As in Problem 18, $B = \left[M(^{106}\text{Pd}) + M(^1\text{H}) - M(^{107}\text{Ag}) \right] c^2 = 5.8 \text{ MeV}$. The electron binding energy is less by a factor of $5.8 \text{ MeV} / 25.6 \text{ keV} = 227$.

22. For ^4He the radius is $r = r_0 A^{1/3} = (1.2 \text{ fm})(4^{1/3}) = 1.90 \text{ fm}$.

$$V = \frac{e^2}{4\pi\epsilon_0} \frac{1}{r} = \frac{1.44 \times 10^{-9} \text{ eV} \cdot \text{m}}{1.90 \times 10^{-15} \text{ m}} = 0.76 \text{ MeV}$$

For ^{40}Ca :

$$\begin{aligned} \Delta E_{\text{Coul}} &= \frac{3}{5} \frac{Z(Z-1)e^2}{4\pi\epsilon_0 R} = 0.72 Z(Z-1) A^{-1/3} \text{ MeV} \\ &= 0.72(20)(19) 40^{-1/3} \text{ MeV} = 80 \text{ MeV} \end{aligned}$$

For ^{208}Pb : $\Delta E_{\text{Coul}} = 0.72(82)(81) 208^{-1/3} \text{ MeV} = 807 \text{ MeV}$

There is roughly a factor of ten between each of these three nuclides.

23. Refer to Equation (12.20), the vonWeizsäcker semi-empirical mass formula. Adding another neutron to ^{18}O reduces δ from $+\Delta$ to 0, makes the symmetry term more negative, and makes the surface term more negative. The net effect of these three terms more than offsets the increase in the volume term.

24. Refer to Equation (12.20), the von Weizsäcker semi-empirical mass formula. δ drops from $+\Delta$ in ^{42}Ca to 0 in ^{41}Ca . The volume term is substantially higher in ^{42}Ca , more than offsetting the difference in the surface effect. The symmetry term is only slightly higher for ^{42}Ca . Comparing ^{42}Ca and ^{42}Ti , we see that the coulomb term is higher in ^{42}Ti , because it has two more protons.

25. We begin with Equation (12.20) and substitute Equation (12.19) for the Coulomb term.

$$\text{Therefore, we have: } B\left({}_Z^AX\right) = a_v A - a_s A^{2/3} - 0.72 \left[Z(Z-1) \right] A^{-1/3} - a_s \frac{(N-Z)^2}{A} + \delta$$

Evaluating this expression for ^{48}Ca , we have:

$$B(^{48}_{20}\text{Ca}) = (14\text{MeV})48 - (13\text{MeV})48^{2/3} - 0.72 \frac{[20 \cdot 19]}{48^{1/3}} - (19\text{MeV}) \frac{(28-20)^2}{48} + \frac{33\text{MeV}}{48^{3/4}}.$$

This gives $B(^{48}_{20}\text{Ca}) = 401.492\text{ MeV}$ which equals

$$401.492\text{ MeV} \left(\frac{c^2 \cdot \text{u}}{931.5\text{MeV}} \right) = 0.43102\text{ u} \cdot c^2. \text{ From Equation (12.10), we can calculate the}$$

mass: $M(^{48}_{20}\text{Ca}) = 28 \cdot m_n + 20 \cdot M(^1\text{H}) - [B(^{48}_{20}\text{Ca})/c^2]$. Substituting, we find

$$M(^{48}_{20}\text{Ca}) = 28 \cdot (1.008665\text{u}) + 20 \cdot (1.007825\text{u}) - [0.43102\text{ u} \cdot c^2 / c^2] = 47.9681\text{u, or}$$

$$47.9681\text{u} \cdot \left(\frac{931.5\text{MeV}}{c^2 \cdot \text{u}} \right) = 4.468 \times 10^4\text{ MeV} / c^2. \text{ The atomic mass given in Appendix 8 is}$$

47.952534 u. Our calculation based on a semi-empirical formula is different by about 0.03%.

26. (a) We use Equation (12.20) to calculate the binding energy with Equation (12.19) for the Coulomb term. We find that $\Delta = (33\text{MeV}) \cdot 18^{-3/4} = 3.776\text{MeV}$ for the pairing term, so $\delta = -3.776\text{MeV}$ for ^{18}N but $+3.776\text{MeV}$ for the others, which are even-even nuclei. Once we determine the binding energy, we must divide by 18 to find the binding energy per nucleon. Evaluating this expression for $^{18}_6\text{C}$, we have:

$$\begin{aligned} B(^{18}_6\text{C}) &= (14\text{MeV})18 - (13\text{MeV})18^{2/3} - 0.72 \frac{[6 \cdot 5]}{18^{1/3}} - (19\text{MeV}) \frac{(12-6)^2}{18} + 3.776\text{MeV} \\ &= 120.2466\text{MeV} \end{aligned}$$

which gives a binding energy per nucleon for $^{18}_6\text{C}$ of 6.68 MeV. The values for the other elements are 7.25 MeV, 8.16 MeV, and 7.64 MeV in the order given.

(b) The nuclide ^{18}O is most stable, which is expected because it is the closest to having the same number of neutrons and protons and it is even-even.

(c) Using the information in Appendix 8 and Equation (12.10), the binding energy of ^{18}O is: $B(^{18}_8\text{O}) = [10 \cdot m_n + 8M(^1\text{H}) - M(^{18}_8\text{O})]c^2 = 0.1501\text{u} \cdot c^2 = 139.8\text{ MeV}$ which gives a binding energy of 7.77 MeV per nucleon.

$$27. \lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{(5.271\text{ y})(3.156 \times 10^7\text{ s/y})} = 4.167 \times 10^{-9}\text{ s}^{-1}$$

$$N = \frac{R}{\lambda} = \frac{4.4 \times 10^7\text{ s}^{-1}}{4.167 \times 10^{-9}\text{ s}^{-1}} = 1.06 \times 10^{16};$$

$$m = (1.06 \times 10^{16}) \frac{1\text{ mol}}{6.022 \times 10^{23}} \left(\frac{60\text{ g}}{\text{mol}} \right) = 1.05\text{ }\mu\text{g}$$

$$28. R = R_0 e^{-\lambda t} = \frac{R_0}{5} \text{ at } t = T = 3600 \text{ s}; \quad \lambda = \frac{\ln 5}{T}$$

$$t_{1/2} = \frac{\ln 2}{\lambda} = \frac{\ln 2}{\ln 5} T = \frac{\ln 2}{\ln 5} (3600 \text{ s}) = 1550 \text{ s} \approx 26 \text{ minutes}$$

29. In general (using the definition of the mean value of a function)

$$\tau = \frac{\int_0^\infty t R(t) dt}{\int_0^\infty R(t) dt} = \frac{1}{N_0} \int_0^\infty t R(t) dt$$

because all nuclei must decay between $t = 0$ and $t = \infty$. Using $R = R_0 e^{-\lambda t}$ we have

$$\tau = \frac{R_0}{N_0} \int_0^\infty t e^{-\lambda t} dt = \frac{R_0}{N_0} \cdot \frac{1}{\lambda^2} = \frac{\lambda N_0}{\lambda^2 N_0} = \frac{1}{\lambda} = \frac{t_{1/2}}{\ln 2}.$$

30. With 96% ^{238}U and 4% ^{235}U the number N of ^{238}U atoms is

$$N = (10 \text{ kg}) \frac{1 \text{ mol}}{0.235 \text{ kg}} \frac{6.022 \times 10^{23}}{\text{mol}} \left(\frac{0.96}{0.04} \right) = 6.15 \times 10^{26}. \text{ [A careful reading of example}$$

12.11 indicates that the fuel rod has 10-kg of ^{235}U . The final ratio in the equation is to determine the amount of ^{238}U present in the sample.] The half-life for ^{238}U alpha emission is found in Appendix 8 as $4.47 \times 10^9 \text{ y}$ which equals $1.41 \times 10^{17} \text{ s}$. Therefore,

$$R = \lambda N = \frac{\ln 2}{t_{1/2}} N = \frac{\ln 2}{1.41 \times 10^{17} \text{ s}} (6.15 \times 10^{26}) = 3.02 \times 10^9 \text{ Bq}.$$

31. $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{(1.277 \times 10^9 \text{ y})(3.156 \times 10^7 \text{ s/y})} = 1.720 \times 10^{-17} \text{ s}^{-1}$ using the $t_{1/2}$ for ^{40}K from Appendix 8.

$$N = (60 \text{ kg}) \frac{1 \text{ mol}}{0.040 \text{ kg}} \frac{6.022 \times 10^{23}}{\text{mol}} (0.00012)(0.003) = 3.252 \times 10^{20}$$

$$R = \lambda N = (1.720 \times 10^{-17} \text{ s}^{-1})(3.252 \times 10^{20}) = 5.6 \times 10^3 \text{ Bq}$$

32. $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{(109.8 \text{ min})(60 \text{ s/min})} = 1.052 \times 10^{-4} \text{ s}^{-1}$

$$R = R_0 e^{-\lambda t} = (1.2 \times 10^7 \text{ Bq}) \exp\left(- (1.052 \times 10^{-4} \text{ s}^{-1})(48)(3600 \text{ s})\right) = 0.153 \text{ Bq}$$

33. $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{12.33 \text{ y}} = 5.622 \times 10^{-2} \text{ y}^{-1}$. The mass decreases by the same exponential factor as N , so $m = m_0 e^{-\lambda t} = (75 \text{ kg}) \exp\left[-(5.622 \times 10^{-2} \text{ y}^{-1})(7 \text{ y})\right] = 51 \text{ kg}$

34. For convenience assume a mass of 1 kg of each material.

For ${}^3\text{H}$: $N = (1 \text{ kg}) \frac{1 \text{ mol}}{0.003 \text{ kg}} \frac{6.022 \times 10^{23}}{\text{mol}} = 2.01 \times 10^{26}$

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{(12.33 \text{ y})} = 5.622 \times 10^{-2} \text{ y}^{-1}$$

$$R = \lambda N = (5.622 \times 10^{-2} \text{ y}^{-1})(2.01 \times 10^{26}) = 1.13 \times 10^{25} \text{ y}^{-1}$$

For ${}^{222}\text{Rn}$: $N = (1 \text{ kg}) \frac{1 \text{ mol}}{0.222 \text{ kg}} \frac{6.022 \times 10^{23}}{\text{mol}} = 2.71 \times 10^{24}$

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{(3.82 \text{ d})(1 \text{ y}/365.25 \text{ d})} = 66.28 \text{ y}^{-1}$$

$$R = \lambda N = (66.28 \text{ y}^{-1})(2.71 \times 10^{24}) = 1.80 \times 10^{26} \text{ y}^{-1}$$

For ${}^{239}\text{Pu}$: $N = (1 \text{ kg}) \frac{1 \text{ mol}}{0.239 \text{ kg}} \frac{6.022 \times 10^{23}}{\text{mol}} = 2.52 \times 10^{24}$

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{(24110 \text{ y})} = 2.875 \times 10^{-5} \text{ y}^{-1}$$

$$R = \lambda N = (2.875 \times 10^{-5} \text{ y}^{-1})(2.52 \times 10^{24}) = 7.245 \times 10^{19} \text{ y}^{-1}$$

The order of activity is ${}^{222}\text{Rn} > {}^3\text{H} > {}^{239}\text{Pu}$.

35. The decays in question are ${}^{52}\text{Fe} \rightarrow n + {}^{51}\text{Fe}$ and ${}^{52}\text{Fe} \rightarrow p + {}^{51}\text{Mn}$.

neutron decay: $Q = (51.948117 - 50.956825 - 1.008665) \text{ u} \cdot c^2 = -16.2 \text{ MeV}$

which is not allowed because $Q < 0$.

proton decay: $Q = (51.948117 - 50.948216 - 1.007825) \text{ u} \cdot c^2 = -7.4 \text{ MeV}$

which is not allowed because $Q < 0$. Because ${}^{40}\text{Fe}$ has an excess of protons, it should decay.

36. The *separation energy* is the energy required to “extract” a nucleon from the nucleus and is equal to the absolute value of the Q values found: 7.7 MeV for the neutron and 5.6 MeV for the proton.

37. We need not consider β^- decay, because there is no physical object with mass 1 and charge 2. The β^+ decay is $p \rightarrow n + \beta^+$, with

$$Q = (m_p - m_n - m_e)c^2 = (1.0072876 - 1.008665 - 0.00054858) \text{ u} \cdot c^2 = -1.8 \text{ MeV}, \text{ which is not allowed with } Q < 0.$$

For electron capture $p + \beta^- \rightarrow n$, with

$$Q = (m_p + m_e - m_n)c^2 = (1.0072876 - 1.008665 + 0.00054858) \text{ u} \cdot c^2 = -0.79 \text{ MeV},$$

which is not allowed with $Q < 0$.

38. For ^{144}Sm :

$$Q = [M(^{144}\text{Sm}) - M(^{140}\text{Nd}) - M(^4\text{He})]c^2 \\ = [143.911995 - 139.909310 - 4.002603] \text{ u} \cdot c^2 = +0.076 \text{ MeV}$$

For ^{147}Sm :

$$Q = [M(^{147}\text{Sm}) - M(^{143}\text{Nd}) - M(^4\text{He})]c^2 \\ = [146.914893 - 142.909810 - 4.002603] \text{ u} \cdot c^2 = +2.31 \text{ MeV}$$

With $Q > 0$ both isotopes may decay, though the lighter one is barely stable. The ^{147}Sm abundance is greater because its decay has an extremely long half-life of 10^{11} years.

39. The decay is $^{241}\text{Am} \rightarrow ^{237}\text{Np} + ^4\text{He}$.

$$Q = [M(^{241}\text{Am}) - M(^{237}\text{Np}) - M(^4\text{He})]c^2 \\ = [241.056823 - 237.048167 - 4.002603] \text{ u} \cdot c^2 = 5.64 \text{ MeV}$$

By Equation (12.32) the emitted alpha particle's energy is

$$K_\alpha = \frac{A-4}{A} Q = \frac{237}{241} (5.64 \text{ MeV}) = 5.55 \text{ MeV}.$$

Then by conservation of energy the daughter's kinetic energy is 0.09 MeV.

40. Let M be the mass of the decaying nucleus, E_1 and p_1 the energy and momentum of the recoiling nucleus, and let E_2 and p_2 be the energy and momentum of the β^- . Then by conservation of momentum $p_1 + p_2 = 0$, and by conservation of energy $Mc^2 = E_1 + E_2$. Then using the energy-momentum invariant $E^2 = p^2c^2 + E_0^2$ one can rewrite the energy conservation equation in terms of the two momenta. Then we would have two equations in two unknowns (p_1 and p_2), and by the laws of algebra there are unique solutions for p_1 and p_2 . Hence there are unique solutions for E_1 and E_2 .

41. $^{80}\text{Br} \rightarrow ^{76}\text{As} + ^4\text{He} :$

$$Q = [79.918530 - 75.922394 - 4.002603] \text{u} \cdot c^2 = -6.0 \text{ MeV (not allowed)}$$

$$^{80}\text{Br} \rightarrow ^{80}\text{Kr} + \beta^- : Q = (79.918530 - 79.916378) \text{u} \cdot c^2 = 2.0 \text{ MeV (allowed)} ;$$

$$^{80}\text{Br} \rightarrow ^{80}\text{Se} + \beta^+ :$$

$$Q = (79.918530 - 79.916522 - 2(0.000549)) \text{u} \cdot c^2 = 0.85 \text{ MeV (allowed)}$$

$$^{80}\text{Br} + \beta^- \rightarrow ^{80}\text{Se} :$$

$$Q = (79.918530 - 79.916522) \text{u} \cdot c^2 = 1.9 \text{ MeV (allowed)}$$

42. $^{227}\text{Ac} \rightarrow ^{223}\text{Fr} + ^4\text{He} :$

$$Q = [227.027747 - 223.019731 - 4.002603] \text{u} \cdot c^2 = 5.0 \text{ MeV (allowed)}$$

$$^{227}\text{Ac} \rightarrow ^{227}\text{Th} + \beta^- :$$

$$Q = (227.027747 - 227.027699) \text{u} \cdot c^2 = 0.045 \text{ MeV (allowed, barely)}$$

$$^{227}\text{Ac} \rightarrow ^{227}\text{Ra} + \beta^+ :$$

$$Q = (227.027747 - 227.029171 - 2(0.000549)) \text{u} \cdot c^2 = -2.3 \text{ MeV (not allowed)}$$

$$^{227}\text{Ac} + \beta^- \rightarrow ^{227}\text{Ra} :$$

$$Q = (227.027747 - 227.029171) \text{u} \cdot c^2 = -1.3 \text{ MeV (not allowed)}$$

43. $^{230}\text{Pa} \rightarrow ^{226}\text{Ac} + ^4\text{He} :$

$$Q = [230.034533 - 226.026090 - 4.002603] \text{u} \cdot c^2 = 5.4 \text{ MeV (allowed)}$$

$$^{230}\text{Pa} \rightarrow ^{230}\text{U} + \beta^- :$$

$$Q = (230.034533 - 230.033927) \text{u} \cdot c^2 = 0.56 \text{ MeV (allowed)}$$

$$^{230}\text{Pa} \rightarrow ^{230}\text{Th} + \beta^+ :$$

$$Q = (230.034533 - 230.033127 - 2(0.000549)) \text{u} \cdot c^2 = 0.29 \text{ MeV (allowed)}$$

$$^{230}\text{Pa} + \beta^- \rightarrow ^{230}\text{Th} :$$

$$Q = (230.034533 - 227.033127) \text{u} \cdot c^2 = 1.3 \text{ MeV (allowed)}$$

44. Looking at Figure 12.16 it appears the gamma ray energy can be 0.226 MeV or 0.230 MeV in a transition from E_x to the ground state. Clearly a 0.072 MeV transition is possible. There are also two possible transitions from E_x to the 0.072 MeV level. The energies of those gamma rays are $0.226 \text{ MeV} - 0.072 \text{ MeV} = 0.154 \text{ MeV}$ and $0.230 \text{ MeV} - 0.072 \text{ MeV} = 0.158 \text{ MeV}$.

45. After six days the number of radon atoms remaining is

$$N = N_0 \exp\left(-\frac{(\ln 2)(6 \text{ d})}{3.82 \text{ d}}\right) = 0.3367 N_0$$

Therefore the partial pressure of radon is 0.337 atm after 6 days. At first glance one might assume that the partial pressure of the helium is therefore 0.663 atm. But the problem leads to an interesting example.

Note that ^{222}Rn decays entirely by alpha decay. If we continue to follow the decay chain, however, the daughter ^{218}Po , can decay either by alpha or β^- with a half-life of 3.10 minutes; however, 99.98% of the time, ^{218}Po decays by emitting an alpha to ^{214}Pb . ^{214}Pb is also unstable and decays via β^- to ^{214}Bi with a half-life of 26.8 minutes. ^{214}Bi decays 99.9% of the time via β^- decay to ^{214}Po with a half-life of 19.9 minutes. Finally ^{214}Po decays by alpha decay to ^{210}Pb which has a half-life of 22.3 year which is quite long with respect to the 6 days of the problem. Notice that all of the other half-lives are short with respect to 6 days. Therefore the decay chain has produces 3alphas, not 1. Using standard chemistry, the ratio of the partial pressures is the same as the ratio of the molecules. The partial pressure of the helium is therefore $3 \times 0.663 \text{ atm} = 1.99 \text{ atm}$.

46. ^{60}Co decays by β^- to ^{60}Ni . The daughter nucleus is in an excited state and emits a gamma ray.

47. There are three explanations. Looking at the Q formulas, we see that two extra electron masses are required for β^+ decay, and this energy may not be available in “close calls.” Second, the product of β^- decay should be more stable with a greater N/A ratio. Finally, successive alpha decays place daughters farther and farther below the line of stability. Only β^- decays can move back in the direction of the stability line.

48. For the reaction $^{14}\text{O} \rightarrow ^{14}\text{N} + \beta^+$ we have

$$Q = (14.008595 - 14.003074 - 2(0.000549)) \text{u} \cdot c^2 = 4.1 \text{ MeV which is allowed.}$$

For the other reaction $p \rightarrow n + \beta^+$ we have

$$Q = (1.007825 - 1.008665 - 2(0.000549)) \text{u} \cdot c^2 = -1.8 \text{ MeV which is not allowed.}$$

49. From Equation (12.50) $R' = \frac{N(^{206}\text{Pb})}{N(^{238}\text{U})} = e^{\lambda t} - 1$. Rearranging we have $t = \frac{t_{1/2}}{\ln 2} \ln(R' + 1)$.

$$\text{For } R = 0.76: t = \frac{4.47 \times 10^9 \text{ y}}{\ln 2} \ln(1.76) = 3.65 \times 10^9 \text{ y}$$

$$\text{For } R = 3.1: t = \frac{4.47 \times 10^9 \text{ y}}{\ln 2} \ln(4.1) = 9.10 \times 10^9 \text{ y}$$

The large time difference implies different origins.

50. ^{40}K , ^{87}Rb , ^{138}La , and ^{147}Sm have half-lives that are close to the age of the earth. However, of these only ^{87}Rb and ^{147}Sm have reasonable abundances.

51. As in Problem 40 $t = \frac{4.47 \times 10^9 \text{ y}}{\ln 2} \ln(1.01) = 6.4 \times 10^7 \text{ y}$.

52. $R' = \frac{N(^{206}\text{Pb})}{N(^{238}\text{U})} = e^{\lambda t} - 1 = e^{\ln 2} - 1 = 1$ where the substitution for λ occurs since the time

given in the problem almost exactly matches the half-life of U-238. Thus

$\lambda t = (\ln(2) / t_{1/2})t = \ln(2)$. A more exact answer would be

$\lambda t = (\ln(2) / 4.47 \times 10^9) 4.6 \times 10^9 = 1.03[\ln(2)]$ and thus the ratio would be

$$R' = \frac{N(^{206}\text{Pb})}{N(^{238}\text{U})} = e^{\lambda t} - 1 = e^{(1.03)\ln 2} - 1 = 1.04 \text{ revealing a slightly higher amount of lead.}$$

53. From the A values it is clear that there are $28 / 4 = 7$ alpha decays. Seven alpha decays reduces Z from 92 to 78, so there must be four β^- decays in order to bring Z up to 82. There are other possible combinations of beta decays (including β^+ and electron capture), but the net result must be a change of four charge units. We would have to look at a table of nuclides to determine the exact chain(s).

54. $0.0072 = \frac{N(^{235}\text{U})}{N(^{238}\text{U})} = \frac{N_0(^{235}\text{U})\exp(-\lambda_{235}t)}{N_0(^{238}\text{U})\exp(-\lambda_{238}t)}$ so

$$\begin{aligned} \frac{N_0(^{235}\text{U})}{N_0(^{238}\text{U})} &= (0.0072) \frac{\exp(-\lambda_{238}t)}{\exp(-\lambda_{235}t)} \\ &= (0.0072) \exp\left(\left(\frac{\ln 2}{0.7038 \times 10^9 \text{ y}} - \frac{\ln 2}{4.468 \times 10^9 \text{ y}}\right)(4.5 \times 10^9 \text{ y})\right) = 0.301 \end{aligned}$$

or 30.1%.

55. (a) After one day $N = N_0 \exp\left(-\frac{(\ln 2)(24 \text{ h})}{25.39 \text{ h}}\right) = 0.519N_0$ so 51.9 g is still ^{252}Fm . The remainder (41.1 g) is almost all ^{248}Cf , because its half-life is 334 days.

(b) After one month the amount of fermium is

$$N = N_0 \exp\left(-\frac{(\ln 2)(30)(24 \text{ h})}{25.39 \text{ h}}\right) = N_0(2.9 \times 10^{-9}) \approx 0. \text{ Almost none of the fermium is}$$

left. It is more difficult to determine the remaining amount of ^{248}Cf . Of the amount created on the first day of the month the remaining amount is about

$$N_0 \exp\left(-\frac{(\ln 2)(29 \text{ d})}{333.5 \text{ d}}\right) = 0.942N_0$$

Of the amount created on the fourth day of the month the remaining amount is about

$$N_0 \exp\left(-\frac{(\ln 2)(25 \text{ d})}{333.5 \text{ d}}\right) = 0.949 N_0. \text{ Because most of the fermium decayed within the}$$

first four days, we can see that the sample should be between 94% and 95% remaining as ^{248}Cf by the end of the month.

(c) Five years is more than five half-lives of californium, so there is very little of it left. However, the half-life of curium is 18.1 years, so there is at least

$$N_0 \exp\left(-\frac{(\ln 2)(5 \text{ y})}{18.1 \text{ y}}\right) = 0.826 N_0 \text{ of } ^{244}\text{Cm after five years.}$$

(d) After 100 years more than five half-lives of ^{244}Cm have elapsed. The half life of its decay product ^{240}Pu is quite long, 6593 years, so most of the sample is ^{240}Pu after 100 years.

(e) After ^{240}Pu decays to ^{236}U with a half-life of 6563 years, the next decay is to ^{232}Th , with a half-life of $2.34 \times 10^7 \text{ y}$. It takes this amount of time for “most” of the sample to be thorium.

56. (a) $t_{1/2} = 7.7 \times 10^{24} \text{ y} = 2.4 \times 10^{32} \text{ s};$

$$R = \lambda N = \frac{\ln 2}{t_{1/2}} \frac{N_A}{0.128 \text{ kg}} = \frac{\ln 2}{2.4 \times 10^{32} \text{ s}} \frac{6.022 \times 10^{23}}{0.128 \text{ kg}} = 1.36 \times 10^{-8} \text{ s}^{-1} \cdot \text{kg}^{-1}$$

(b) $\frac{10 \text{ s}^{-1}}{1.36 \times 10^{-8} \text{ s}^{-1} \cdot \text{kg}^{-1}} = 7.36 \times 10^8 \text{ kg.}$ This is not a realistic sample size.

57. (a) The magnesium isotope has $Z > N$, which is generally not favorable to stability. On the other hand, the sodium isotope has $N - Z = 1$, which is favorable in this range of A . Therefore we expect ^{23}Mg to be less stable.

(b) Emitting an alpha particle would still leave $Z > N$, and emitting an electron would raise the Z value even further. Thus the only candidates are positron emission $^{23}\text{Mg} \rightarrow ^{23}\text{Na} + \beta^+$ and electron capture $^{23}\text{Mg} + \beta^- \rightarrow ^{23}\text{Na}$. We find the Q values for both reactions:

$$\text{Positron emission: } Q = [M(^{23}\text{Mg}) - M(^{23}\text{Na}) - 2m_e]c^2$$

$$Q = [22.994125 \text{ u} - 22.989770 \text{ u} - 2(0.000549 \text{ u})]c^2 = 0.003257 \text{ u} \cdot c^2 = 3.03 \text{ MeV}$$

$$\text{Electron capture: } Q = [M(^{23}\text{Mg}) - M(^{23}\text{Na})]c^2$$

$$Q = [22.994125 \text{ u} - 22.989770 \text{ u}]c^2 = 0.004355 \text{ u} \cdot c^2 = 4.06 \text{ MeV}$$

As expected, both processes are allowed.

58. For ^{36}Ar : $B = [18m_n + 18M(^1\text{H}) - M(^{36}\text{Ar})]c^2$; $B = 0.329274 \text{ u} \cdot c^2 = 306.7 \text{ MeV}$ so

$$B/A = 8.52 \text{ MeV/nucleon.}$$

For ^{76}Se : $B = [42m_n + 34M(^1\text{H}) - M(^{76}\text{Se})]c^2$; $B = 0.710766 \text{ u} \cdot c^2 = 662.1 \text{ MeV}$ so

$$B/A = 8.71 \text{ MeV/nucleon.}$$

This is the expected result, and matches the asymmetric shape of the curve in Figure 12.6.

59. The resulting nucleus is ^{16}O . If it experienced alpha decay, the resulting nucleus would be ^{12}C . The Q for this reaction is given by Equation (12.31):

$$Q = [M(^{16}\text{O}) - M(^{12}\text{C}) - M(^4\text{He})]c^2$$

$$Q = [15.994914 \text{ u} - 12.0 \text{ u} - 4.002603 \text{ u}]c^2 = -0.007689 \text{ u} \cdot c^2 = -7.16 \text{ MeV}$$

For β^- decay, $^{16}\text{O} \rightarrow ^{16}\text{F} + \beta^-$, and Q is given by

$$Q = [M(^{16}\text{O}) - M(^{16}\text{F})]c^2$$

$$Q = [15.994914 \text{ u} - 16.011466 \text{ u}]c^2 = -0.016552 \text{ u} \cdot c^2 = -15.4 \text{ MeV}$$

For β^+ decay, $^{16}\text{O} \rightarrow ^{16}\text{N} + \beta^+$, and Q is given by

$$Q = [M(^{16}\text{O}) - M(^{16}\text{N}) - 2m_e]c^2$$

$$Q = [15.994914 \text{ u} - 16.006101 \text{ u} - 2(0.000549 \text{ u})]c^2 = -0.012285 \text{ u} \cdot c^2 = -11.4 \text{ MeV}$$

For electron capture, $^{16}\text{O} + \beta^- \rightarrow ^{16}\text{N}$, and Q is given by

$$Q = [M(^{16}\text{O}) - M(^{16}\text{N})]c^2$$

$$Q = [15.994914 \text{ u} - 16.006101 \text{ u}]c^2 = -0.011187 \text{ u} \cdot c^2 = -10.4 \text{ MeV}$$

Q for each reaction is negative, proving that ^{16}O is stable.

60. The resulting nucleus is ^{13}N . If it experienced alpha decay, the resulting nucleus would be ^9B . The Q for this reaction is given by Equation (12.31):

$$Q = [M(^{13}\text{N}) - M(^9\text{B}) - M(^4\text{He})]c^2$$

$$Q = [13.005739 \text{ u} - 9.013329 \text{ u} - 4.002603 \text{ u}]c^2 = -0.010193 \text{ u} \cdot c^2 = -9.49 \text{ MeV}$$

For β^- decay, $^{13}\text{N} \rightarrow ^{13}\text{O} + \beta^-$, and Q is given by

$$Q = [M(^{13}\text{N}) - M(^{13}\text{O})]c^2$$

$$Q = [13.005739 \text{ u} - 13.024810 \text{ u}]c^2 = -0.019071 \text{ u} \cdot c^2 = -17.8 \text{ MeV}$$

For β^+ decay, $^{13}\text{N} \rightarrow ^{13}\text{C} + \beta^+$, and Q is given by

$$Q = [M(^{13}\text{N}) - M(^{13}\text{C}) - 2m_e]c^2$$

$$Q = [13.005739 \text{ u} - 13.003355 \text{ u} - 2(0.000549 \text{ u})]c^2 = 0.001286 \text{ u} \cdot c^2 = 1.20 \text{ MeV}$$

For electron capture, $^{13}\text{N} + \beta^- \rightarrow ^{13}\text{C}$, and Q is given by

$$Q = [M(^{13}\text{N}) - M(^{13}\text{C})]c^2$$

$$Q = [13.005739 \text{ u} - 13.003355 \text{ u}]c^2 = 0.002484 \text{ u} \cdot c^2 = 2.22 \text{ MeV}$$

Positron decay and electron capture are allowed, and this nucleus is unstable.

61. Adding charge dq to a solid sphere of radius r we have an energy change $dE = \frac{Q dq}{4\pi\epsilon_0 r}$

where $Q = \frac{4}{3}\rho\pi r^3$ is the charge already there and ρ is the charge density. Also we know

$$dq = \rho dV = 4\pi\rho r^2 dr. \text{ Making those substitutions gives } dE = \frac{16\pi^2\rho^2 r^4}{3(4\pi\epsilon_0)} dr.$$

Integrating from 0 to R we find $\Delta E = \int_0^R \frac{16\pi^2\rho^2 r^4}{3(4\pi\epsilon_0)} dr = \frac{16\pi^2\rho^2 R^5}{15(4\pi\epsilon_0)}$. The charge density is

$$\rho = Q/V = 3Q/4\pi R^3, \text{ so } \Delta E = \frac{16\pi^2 R^5}{15(4\pi\epsilon_0)} \left(\frac{3Q}{4\pi R^3} \right)^2 = \frac{3Q^2}{5(4\pi\epsilon_0)R} = \frac{3(Ze)^2}{5(4\pi\epsilon_0)R}.$$

62. For Δx use the diameter $2r = 2r_0 A^{1/3} = 2(1.2 \text{ fm})2^{1/3} = 3.02 \text{ fm}$. Then $p_{\min} = \hbar / 2\Delta x$ and

$$K_{\min} = \frac{p_{\min}^2}{2m} = \frac{\hbar^2}{8m(\Delta x)^2} = \frac{(197.3 \times 10^{-9} \text{ eV} \cdot \text{m})^2}{8(938 \times 10^6 \text{ eV})(3.02 \times 10^{-15} \text{ m})^2} = 569 \text{ keV}$$

$$\lambda = \frac{h}{p} = \frac{2h\Delta x}{\hbar} = 4\pi \Delta x = 4\pi(3.02 \text{ fm}) = 38 \text{ fm}.$$

These results are reasonable, because the energy is smaller than the deuteron's binding energy, and the wavelength computed is the *maximum* wavelength, so the actual wavelength may be less. Shorter wavelengths, corresponding to the size of the nucleus, are allowed by the uncertainty principle.

63. The parity term in the semi-empirical formula, along with the more negative coulomb and symmetry terms, keep larger odd-odd nuclei from being stable. The spins are significant in this case, because transitions with very large ΔJ are very forbidden.

64. With an even number of neutrons this isotope is apparently the most stable judging from the semi-empirical formula. The alpha decay would be $^{165}\text{Ho} \rightarrow ^{161}\text{Tb} + ^4\text{He}$ with $Q = [164.930319 - 160.927566 - 4.002603]u \cdot c^2 = 0.14 \text{ MeV}$. With a very low Q value, this process is allowed, but barely. With such a small Q value, it is difficult to overcome the nuclear attraction.

65. (a) Radon is created at a rate $\lambda_1 N_1$ and lost at a rate $\lambda_2 N_2$, so the net change is

$$dN_2 / dt = \lambda_1 N_1 - \lambda_2 N_2.$$

(b) Using separation of variables $\frac{dN_2}{\lambda_1 N_1 - \lambda_2 N_2} = dt$. Let $u = \lambda_1 N_1 - \lambda_2 N_2$ so $du = -\lambda_2 dN_2$

(because we assume N_1 is constant). Then $\frac{du}{u} = -\lambda_2 dt$ and $\ln u = -\lambda_2 t + \ln(\text{const.})$.

$u = (\text{const.})e^{-\lambda_2 t} = \lambda_1 N_1 - \lambda_2 N_2$. To determine the constant, set $N_2 = 0$ at $t = 0$, so the constant is $\lambda_1 N_1$. $\lambda_1 N_1 e^{-\lambda_2 t} = \lambda_1 N_1 - \lambda_2 N_2$ or rearranging $N_2 = \frac{\lambda_1 N_1}{\lambda_2} (1 - e^{-\lambda_2 t})$.

(c) For large t we have $e^{-\lambda_2 t} \approx 0$, and so $N_2 \approx \lambda_1 N_1 / \lambda_2$ (which is constant).

66. (a) From the uncertainty principle $\Delta E \Delta t = \hbar / 2$, so

$$\Delta E = \frac{\hbar}{2\Delta t} = \frac{6.582 \times 10^{-16} \text{ eV} \cdot \text{s}}{2(1.9 \times 10^{-10} \text{ s})} = 1.73 \times 10^{-6} \text{ eV}.$$

(b) We have $5\Delta E / E = v / c$, which we can solve for v :

$$v = \frac{5\Delta E}{E} c = \frac{5(1.73 \times 10^{-6} \text{ eV})}{129 \times 10^3 \text{ eV}} (2.998 \times 10^8 \text{ m/s}) = 2.01 \text{ cm/s}.$$

67. (a) Examining the mass numbers it must come from four alpha decays of ^{238}U .

(b) From the table of nuclides the decays are: $^{222}\text{Rn} \rightarrow ^{218}\text{Po} + ^4\text{He}$;
 $^{218}\text{Po} \rightarrow ^{214}\text{Pb} + ^4\text{He}$; $^{214}\text{Pb} \rightarrow ^{214}\text{Bi} + \beta^-$; $^{214}\text{Bi} \rightarrow ^{214}\text{Po} + \beta^-$;
 $^{214}\text{Po} \rightarrow ^{210}\text{Pb} + ^4\text{He}$

(c) The only half-life longer than an hour is the first decay, with $t_{1/2} = 3.8$ days. Therefore more than half the decays will occur in four days.

68. (a) Both ^4He and ^{16}O are “doubly magic”; that is both Z and N are magic numbers. These elements are more tightly bound than nuclei around them.

(b) Because ^{208}Pb has $Z = 82$ and $N = 126$ it is doubly magic. It is particularly stable and its binding energy is high. Other nuclei around it are unstable because of the large Coulomb energy.

(c) ^{40}Ca is doubly magic and is the most stable of the calcium isotopes. It is the heaviest nuclide with $Z = N$. ^{48}Ca is also stable and has $Z = 20$ and $N = 28$ which are both magic numbers. Six isotopes of calcium are stable with $Z = 20$ and various values of N .

69. ^4_2He ; $^{16}_8\text{O}$; $^{40}_{20}\text{Ca}$; $^{48}_{20}\text{Ca}$; $^{208}_{82}\text{Pb}$

70. (a) We use the atomic masses in Appendix 8 to find Q and then Equation (12.32) to find K_α for each isotope. For the decay of ^{218}Ra , the reaction is $^{218}\text{Ra} \rightarrow ^{214}\text{Rn} + ^4\text{He}$ and

$$Q = [M(^{218}\text{Ra}) - M(^{214}\text{Rn}) - M(^4\text{He})]c^2$$

$$Q = [218.007124 \text{ u} - 213.995346 \text{ u} - 4.002603 \text{ u}]c^2 = 0.009175 \text{ u} \cdot c^2 = 8.55 \text{ MeV}$$

$$K_\alpha = \left(\frac{A-4}{A} \right) Q = 8.39 \text{ MeV}$$

Similarly we find $K_\alpha = 7.46 \text{ MeV}$ for ^{220}Ra and $K_\alpha = 6.56 \text{ MeV}$ for ^{222}Ra .

(b) Notice that the graph in Figure 12.12 is logarithmic, suggesting that we try an exponential function of the form $t_{1/2} = Ae^{-BK_\alpha}$.

(The model of alpha decay in Section 6.7 also suggests an exponential function.) To fit the straight line between the 218 and 222 isotopes as suggested, take the natural logarithm:

$$\ln t_{1/2} = \ln A - BK_\alpha$$

Thus we can find the slope B :

$$B = -\frac{\Delta \ln t_{1/2}}{\Delta K_{\alpha}} = -\frac{-11.068 - 3.638}{8.39 - 6.56} = 8.04$$

where we used the half-lives given in Appendix 8. Now using the ^{222}Ra data point in the original equation we find $A = 3.06 \times 10^{24}$. The parameters A and B complete the model.

(c) Using $K_{\alpha} = 7.46 \text{ MeV}$ for ^{220}Ra , our function in part (b) estimates $t_{1/2} = 27 \text{ ms}$. This is within a factor of two of the actual value 17 ms. It is to be expected that the estimate will be high, given the concave nature of the graph.

Chapter 13

1. (a) ${}^4_2\text{He}$ (b) ${}^1_1\text{H}$ (c) ${}^{18}_9\text{F}$ (d) ${}^{76}_{34}\text{Se}$ (e) ${}^{108}_{48}\text{Cd}$ (f) ${}^3_2\text{He}$

2. (a) ${}^{16}\text{O}(d, n){}^{17}\text{F}$ (b) ${}^7\text{Li}(p, \alpha){}^4\text{He}$ (c) ${}^{15}\text{N}(\alpha, p){}^{18}\text{O}$ (d) ${}^{76}\text{Se}(d, n){}^{77}\text{Br}$
 (e) ${}^{107}\text{Ag}({}^3\text{He}, {}^4\text{He}){}^{106}\text{Ag}$ (f) ${}^{162}\text{Dy}({}^3\text{He}, p){}^{164}\text{Ho}$

3. Probability is equal to $nt\sigma$, or $nt\sigma = \frac{6.022 \times 10^{23}}{238 \text{ g}} \frac{19 \text{ g}}{\text{cm}^3} (3.2 \text{ cm}) (0.68 \times 10^{-24} \text{ cm}^2) = 0.10$

4. $p + {}^{21}\text{Ne}$, ${}^3\text{He} + {}^{19}\text{F}$, and ${}^6\text{Li} + {}^{16}\text{O}$

5. The probability is $nt\sigma$: $nt\sigma = \frac{6.022 \times 10^{23}}{108 \text{ g}} \frac{10.5 \text{ g}}{\text{cm}^3} (0.2 \text{ cm}) (17 \times 10^{-24} \text{ cm}^2) = 0.20$

6. We can find the probability of landing in a solid angle $\Delta\Omega$:

$$\frac{d\sigma}{d\Omega}(\Delta\Omega)nt = \left(0.25 \times 10^{-27} \frac{\text{cm}^2}{\text{sr}}\right) (3 \times 10^{-3} \text{ sr}) \left(\frac{6.022 \times 10^{23}}{12 \text{ g}}\right) \left(\frac{100 \times 10^{-6} \text{ g}}{\text{cm}^2}\right)$$

$$= 3.76 \times 10^{-12}$$

The rate of incident particles is $(0.20 \times 10^{-6} \text{ C/s}) \left(\frac{1}{2(1.602 \times 10^{-19} \text{ C})}\right) = 6.24 \times 10^{11} \text{ s}^{-1}$

so the rate of detected particles is $(6.24 \times 10^{11} \text{ s}^{-1})(3.76 \times 10^{-12}) = 2.35 \text{ s}^{-1}$ and the number detected in one hour is $2.35 \text{ s}^{-1}(3600 \text{ s}) = 8460$.

7. (a) ${}^{16}\text{O}(n, \alpha){}^{13}\text{C}$ (b) ${}^{16}\text{O}(d, n){}^{17}\text{F}$ (c) ${}^{16}\text{O}(\gamma, p){}^{15}\text{N}$
 (d) ${}^{16}\text{O}(\alpha, p){}^{19}\text{F}$ (e) ${}^{16}\text{O}(d, {}^3\text{He}){}^{15}\text{N}$ (f) ${}^{16}\text{O}({}^7\text{Li}, p){}^{22}\text{Ne}$

All the products listed above are stable except the one in part (b).

8. (a) $Q = [M({}^{16}\text{O}) + M({}^2\text{H}) - M({}^4\text{He}) - M({}^{14}\text{N})]u \cdot c^2 = 3.11 \text{ MeV}$ (exothermic)

(b) $Q = [M({}^{12}\text{C}) + M({}^{12}\text{C}) - M({}^2\text{H}) - M({}^{22}\text{Na})]u \cdot c^2 = -7.95 \text{ MeV}$ (endothermic)

(c) $Q = [M({}^{23}\text{Na}) + M({}^1\text{H}) - M({}^{12}\text{C}) - M({}^{12}\text{C})]u \cdot c^2 = -2.24 \text{ MeV}$ (endothermic)

9. (b) $K_{\text{th}} = (7.95 \text{ MeV}) \left(\frac{24}{12}\right) = 15.9 \text{ MeV}$

(c) $K_{\text{th}} = (2.24 \text{ MeV}) \left(\frac{24}{23}\right) = 2.34 \text{ MeV}$

10. In the center of mass system we require

$$9.63 \text{ MeV} + M(^{16}\text{O})c^2 = 9.63 \text{ MeV} + 14899.10 \text{ MeV} = 14908.73 \text{ MeV}$$

We also have

$$M(^4\text{He})c^2 + M(^{12}\text{C})c^2 = 3728.38 \text{ MeV} + 11177.93 \text{ MeV} = 14906.31 \text{ MeV}$$

Therefore we need $14908.73 \text{ MeV} - 14906.31 \text{ MeV} = 2.42 \text{ MeV}$ in bombarding energy.

In the lab frame $\frac{16}{12}(2.42 \text{ MeV}) = 3.23 \text{ MeV}$. The lifetime of the state can be

computed using the uncertainty principle: $\Gamma \tau = \hbar / 2$ so

$$\tau = \frac{\hbar}{2\Gamma} = \frac{6.582 \times 10^{-16} \text{ eV} \cdot \text{s}}{2(510 \times 10^3 \text{ eV})} = 6.45 \times 10^{-22} \text{ s}.$$

11. (a) $Q = K_y + K_Y - K_x = 1.1 \text{ MeV} + 8.4 \text{ MeV} - 5.5 \text{ MeV} = 4.0 \text{ MeV}$

(b) The Q value does not change for a particular reaction.

12. $Q = [M(^{20}\text{Ne}) + M(^4\text{He}) - M(^{12}\text{C}) - M(^{12}\text{C})]u \cdot c^2 = -4.62 \text{ MeV}$

$$K_{\text{th}} = \frac{24}{20}(4.62 \text{ MeV}) = 5.54 \text{ MeV}$$

The sum of the carbon kinetic energies is

$$K_y + K_Y = Q + K_x = -4.62 \text{ MeV} + 45 \text{ MeV} = 40.4 \text{ MeV}.$$

13. Letting $M = m_b + m_B$, $\gamma = (1 - v^2/c^2)^{-1/2}$, and $\gamma' = (1 - v_{\text{cm}}^2/c^2)^{-1/2}$, we have from conservation of energy and conservation of momentum:

$$m_A c^2 + \gamma m_a c^2 = \gamma' M c^2$$

$$\gamma m_a v = \gamma' M v_{\text{cm}}$$

Squaring the second expression and dividing by c^2 , we have: $\frac{\gamma^2 m_a^2 v^2}{c^2} = \frac{\gamma'^2 M^2 v_{\text{cm}}^2}{c^2}$

$$\gamma^2 m_a^2 (1 - \gamma^{-2}) = \gamma'^2 M^2 (1 - \gamma'^{-2}); \quad \gamma^2 m_a^2 - m_a^2 = \gamma'^2 M^2 - M^2$$

From the energy equation $\gamma' M = m_A + \gamma m_a$, so $\gamma^2 m_a^2 - m_a^2 = (m_A + \gamma m_a)^2 - M^2$.

After multiplying the binomial and rearranging we have $2\gamma m_a m_A = M^2 - m_a^2 - m_A^2$.

$$2(\gamma - 1)m_a m_A = M^2 - (m_a^2 + 2m_a m_A + m_A^2) = M^2 - (m_a + m_A)^2$$

$$\begin{aligned} K_{\text{th}} = (\gamma - 1)m_a c^2 &= \frac{(M^2 - (m_a + m_A)^2)c^2}{2m_A} = \frac{[(M - (m_a + m_A))(M + (m_a + m_A))]c^2}{2m_A} \\ &= -Q \frac{m_A + m_a + m_B + m_b}{2m_A} \end{aligned}$$

14. The energy available for the gamma ray is

$$Q = [M(^{10}\text{B}) + M(n) - M(^{11}\text{B})]u \cdot c^2 = 11.5 \text{ MeV}.$$

15. The kinetic energy is $Q + 5.0 \text{ MeV}$, or

$$K = [M(^9\text{Be}) + M(^4\text{He}) - M(n) - M(^{12}\text{C})]u \cdot c^2 + 4.61 \text{ MeV} = 5.70 \text{ MeV} + 4.61 \text{ MeV} \\ = 10.3 \text{ MeV}$$

16. (a) $Q = [M(^{16}\text{O}) + M(^4\text{He}) - M(^1\text{H}) - M(^{19}\text{F})]u \cdot c^2 = -8.11 \text{ MeV}$

$$K_{\text{th}} = \frac{20}{16}(8.11 \text{ MeV}) = 10.14 \text{ MeV}$$

(b) $Q = [M(^{12}\text{C}) + M(^2\text{H}) - M(^3\text{He}) - M(^{11}\text{B})]u \cdot c^2 = -10.46 \text{ MeV}$

$$K_{\text{th}} = \frac{14}{12}(10.46 \text{ MeV}) = 12.21 \text{ MeV}$$

17. (a) $N = \frac{6.022 \times 10^{23}}{59 \text{ g}}(40 \times 10^{-3} \text{ g}) = 4.083 \times 10^{20}$

$$\text{Activation rate} = n\sigma\phi = (4.083 \times 10^{20})(20 \times 10^{-28} \text{ m}^2)(1.0 \times 10^{18} \text{ m}^{-2} \cdot \text{s}^{-1}) = 8.166 \times 10^{11} \text{ s}^{-1}$$

Then in one week the number of ^{60}Co produced is

$$8.166 \times 10^{11} \text{ s}^{-1}(7(86400 \text{ s})) = 4.94 \times 10^{17}.$$

(b) Because the half-life of ^{60}Co is much longer than one week, we can assume almost all the nuclei produced are still present. The activity is

$$R = \lambda N = \frac{\ln 2}{t_{1/2}} N = \frac{\ln 2}{1.66 \times 10^8 \text{ s}}(4.94 \times 10^{17}) = 2.06 \times 10^9 \text{ Bq}$$

(c) $1.0 \times 10^{14} \text{ Bq} \frac{40 \times 10^{-3} \text{ g}}{2.06 \times 10^9 \text{ Bq}} = 1942 \text{ g}$

We would put 1.94 kg of ^{59}Co into the reactor for one week.

18. The equation for K_{cm} is correct because we know from classical mechanics that the system is equivalent to a mass $M_x + M_X$ moving with a speed v_{cm} , so the center

of mass kinetic energy is $K_{\text{cm}} = \frac{1}{2}(M_x + M_X)v_{\text{cm}}^2$. Letting $M = M_x + M_X$ we have by

conservation of momentum $Mv_{\text{cm}} = M_x v_x$, or $v_{\text{cm}} = M_x v_x / M$. Therefore

$$K_{\text{cm}} = \frac{1}{2} M v_{\text{cm}}^2 = \frac{1}{2} M \frac{M_x^2}{M^2} v_x^2 = \frac{1}{2} \frac{M_x^2}{M} v_x^2$$

$$K'_{\text{cm}} = K_{\text{lab}} - K_{\text{cm}} = \frac{M_x v_x^2}{2} - \frac{1}{2} \frac{M_x^2}{M} v_x^2 = \frac{M_x v_x^2}{2} \left(1 - \frac{M_x}{M}\right) = \frac{M_x v_x^2}{2} \left(\frac{M - M_x}{M}\right)$$

$$K'_{\text{cm}} = K_{\text{lab}} \left(\frac{M_x}{M_x + M_X}\right)$$

$$19. K'_{\text{cm}} = \frac{M_x}{M_x + M_X} K_{\text{lab}} = \frac{14}{18} (6.7 \text{ MeV}) = 5.21 \text{ MeV}$$

$$E^* = [M(^{14}\text{N}) + M(^4\text{He}) - M(^{18}\text{F})] \text{u} \cdot c^2 = 4.41 \text{ MeV}$$

$$E_x = E^* + K'_{\text{cm}} = 9.62 \text{ MeV}$$

20. A small amount of the neutron's energy must be used in recoil, so the amount of energy that can go into excitation is

$$14 \text{ MeV} \times \frac{208}{209} = 13.93 \text{ MeV}$$

The net excitation energy is this plus the available mass-energy, so

$$E^* = 13.93 \text{ MeV} + [M(^{208}\text{Pb}) + M(n) - M(^{209}\text{Pb})] \text{u} \cdot c^2 = 13.93 \text{ MeV} + 3.94 \text{ MeV}$$

or $E^* = 17.9 \text{ MeV}$. Because of the excited nucleus, we expect gamma decay.

21. From Chapter 12 the radius of the nucleus is $r(^{14}\text{N}) = (1.2 \text{ fm})14^{1/3} = 2.89 \text{ fm}$.

$$\text{At that radius the potential energy is } V = \frac{q_1 q_2}{4\pi\epsilon_0 r} = \frac{(2)(7)(1.44 \times 10^{-9} \text{ eV} \cdot \text{m})}{2.89 \times 10^{-15} \text{ m}} = 6.98 \text{ MeV}.$$

$$Q = [M(^{14}\text{N}) + M(^4\text{He}) - M(^1\text{H}) - M(^{17}\text{O})] \text{u} \cdot c^2 = -1.19 \text{ MeV}$$

$$K_{\text{exit}} = Q + K_{\alpha} = -1.19 \text{ MeV} + 7.7 \text{ MeV} = 6.51 \text{ MeV} = K(p) + K(^{17}\text{O}).$$

Most of the energy at the forward angle will be in the proton, so there could be sufficient energy to overcome the 6.98 MeV barrier in a tunneling process (though it appears to be forbidden classically).

22. From Chapter 12 Problem 29 we see that the mean lifetime is

$$\tau = \frac{t_{1/2}}{\ln 2} = \frac{109 \text{ ms}}{\ln 2} = 157 \text{ ms}$$

$$\text{Then from Equation (13.13)} \quad \Gamma = \frac{\hbar}{2\tau} = \frac{6.582 \times 10^{-16} \text{ eV} \cdot \text{s}}{2(0.157 \text{ s})} = 2.10 \times 10^{-15} \text{ eV}.$$

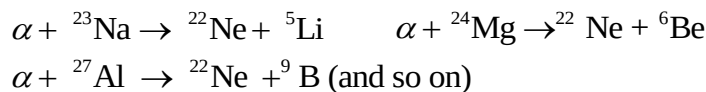
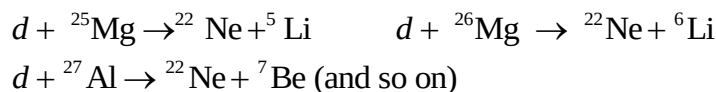
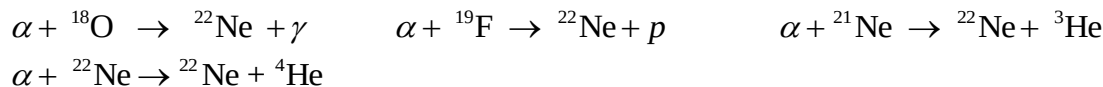
^{17}Ne can decay by positron decay or electron capture.

23. $E^* = [M(^1\text{H}) + M(^{16}\text{O}) - M(^{17}\text{F})] \text{u} \cdot c^2 = 0.60 \text{ MeV}$. No, this energy is greater than 0.495 MeV.

$$24. (a) E^* = [M(^{239}\text{Pu}) + M(n) - M(^{240}\text{Pu})] \text{u} \cdot c^2 = 6.53 \text{ MeV}$$

$$(b) K_{\text{cm}} = \frac{239}{240} (1.00 \text{ MeV}) = 1.00 \text{ MeV}; \quad E = 6.53 \text{ MeV} + 1.00 \text{ MeV} = 7.53 \text{ MeV}$$

$$25. d + ^{21}\text{Ne} \rightarrow ^{22}\text{Ne} + p \quad d + ^{22}\text{Ne} \rightarrow ^{22}\text{Ne} + d \quad d + ^{23}\text{Na} \rightarrow ^{22}\text{Ne} + ^3\text{He}$$



$$26. \Delta E = [M({}^{239}\text{Pu}) + M(n) - M({}^{95}\text{Zr}) - M({}^{142}\text{Xe}) - 3M(n)]u \cdot c^2 = 183.6 \text{ MeV}$$

$$27. (a) m = (5.5 \times 10^{-4})(10^6 \text{ kg}) = 550 \text{ kg}$$

$$(b) (550 \text{ kg}) \frac{6.022 \times 10^{23}}{0.238 \text{ kg}} = 1.4 \times 10^{27} \text{ atoms}$$

$$(c) R = (6.7 \text{ kg}^{-1} \cdot \text{s}^{-1})(550 \text{ kg}) = 3700 \text{ Bq}$$

$$(d) (3700 \text{ s}^{-1}) \frac{86400 \text{ s}}{d} = 3.2 \times 10^8 \text{ d}^{-1}$$

$$28. \frac{N({}^{235}\text{U})}{N({}^{238}\text{U})} = \frac{7.2}{993} = \frac{N_0({}^{235}\text{U})e^{-\lambda_1 t}}{N_0({}^{238}\text{U})e^{-\lambda_2 t}}$$

where the subscripts 1 and 2 refer to the 235 and 238 isotopes, respectively.

$$\frac{N_0({}^{235}\text{U})}{N_0({}^{238}\text{U})} = \frac{7.2}{993} \exp((\lambda_1 - \lambda_2)t)$$

$$(\lambda_1 - \lambda_2)t = \ln 2 \left(\frac{1}{t_{1/2}({}^{235}\text{U})} - \frac{1}{t_{1/2}({}^{238}\text{U})} \right) t$$

$$= \ln 2 \left(\frac{1}{7.04 \times 10^8 \text{ y}} - \frac{1}{4.47 \times 10^9 \text{ y}} \right) (2.0 \times 10^9 \text{ y}) = 1.659$$

$$\text{Then } \frac{N_0({}^{235}\text{U})}{N_0({}^{238}\text{U})} = \frac{7.2}{993} \exp((\lambda_1 - \lambda_2)t) = \frac{7.2}{993} \exp(1.659) = 0.0381$$

which is more than five times higher than today. Note that with the ratio of abundances equal to 0.0381, the percentage abundances needed to give that ratio are about 3.7% and 96.3%. Natural fission reactors cannot operate because of the relatively low abundance of ${}^{235}\text{U}$ today. The exception is a type of reactor known as the Canadian Deuterium Reactor (CANDU) which uses heavy water rather than regular or light water as the moderator.

29. Here are three possibilities, out of many:



Note that the fragments (two nuclei plus some free neutrons) have to add up to mass number $A = 236$ and atomic number $Z = 92$.

$$30. 1250 \text{ MWe} = (1.25 \times 10^9 \text{ J/s}) \frac{86400 \text{ s/d}}{1.602 \times 10^{-13} \text{ J/MeV}} = 6.74 \times 10^{26} \text{ MeV/d}$$

$$(6.74 \times 10^{26} \text{ MeV/d}) \left(\frac{1 \text{ (fission)}}{200 \text{ MeV}} \right) \frac{0.235 \text{ kg}}{6.022 \times 10^{23}} = 1.32 \text{ kg/d}$$

31. For uranium we assume as in the preceding problem that each fission produces 200 MeV of energy.

$$(1.0 \text{ kg}) \left(\frac{6.022 \times 10^{23} \text{ atoms}}{0.235 \text{ kg}} \right) \frac{200 \text{ MeV}}{\text{atom}} = 5.13 \times 10^{26} \text{ MeV}$$

Converting to kWh, we find $2.30 \times 10^7 \text{ kWh}$. For 1.0 kg of coal, Table 13.1 gives energy output $3 \times 10^7 \text{ J}$, which is equivalent to 8.33 kWh. Therefore we see that fission produces over one million times more energy per kilogram of fuel.

32. The USA consumes about $1 \times 10^{20} \text{ J}$ per year, according to most sources. With a population of 310 million, that's $3.3 \times 10^{11} \text{ J}$ per year per person, or roughly $9 \times 10^8 \text{ J}$ per day per person. The Rules of Golf specify the golf ball's maximum mass of 45.93 g (0.04593 kg). Multiplying this by c^2 gives $4.13 \times 10^{15} \text{ J}$ of energy. If each person consumes $9 \times 10^8 \text{ J}$, that's enough energy for 4.6 million people, a large city. The estimate is reasonable.

$$33. (a) 2.0 \times 10^{11} \text{ m}^3 \times \frac{6 \times 10^9 \text{ J}}{0.16 \text{ m}^3} = 7.5 \times 10^{21} \text{ J}$$

$$(b) 7.5 \times 10^{21} \text{ J} \times \frac{1 \text{ y}}{5 \times 10^{20} \text{ J}} = 15 \text{ y}$$

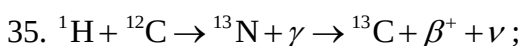
At this rate, it would not last long! Fortunately, other sources of energy are available, although oil is still the primary source of energy for transportation.

- (c) According to Table 13.1, 1 kg of uranium fuel produces 10^{14} J of energy, so the amount of uranium required is

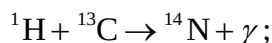
$$7.5 \times 10^{21} \text{ J} \times \frac{1 \text{ kg}}{10^{14} \text{ J}} = 7.5 \times 10^7 \text{ kg}$$

$$34. (a) \frac{3}{2} kT = \frac{3}{2} (8.617 \times 10^{-5} \text{ eV/K}) (300 \text{ K}) = 3.88 \times 10^{-2} \text{ eV}$$

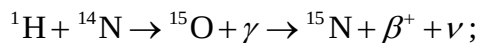
$$(b) \frac{3}{2} kT = \frac{3}{2} (8.617 \times 10^{-5} \text{ eV/K}) (15 \times 10^6 \text{ K}) = 1.94 \text{ keV}$$



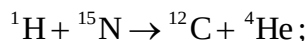
$$Q = [M({}^1\text{H}) + M({}^{12}\text{C}) - M({}^{13}\text{C})] u \cdot c^2 = 4.16 \text{ MeV}$$



$$Q = [M(^1\text{H}) + M(^{13}\text{C}) - M(^{14}\text{N})] \text{u} \cdot c^2 = 7.55 \text{ MeV}$$



$$Q = [M(^1\text{H}) + M(^{14}\text{N}) - M(^{15}\text{N})] \text{u} \cdot c^2 = 10.05 \text{ MeV}$$



$$Q = [M(^1\text{H}) + M(^{15}\text{N}) - M(^{12}\text{C}) - M(^4\text{He})] \text{u} \cdot c^2 = 4.97 \text{ MeV}$$

The total Q is $Q = 4.16 \text{ MeV} + 7.55 \text{ MeV} + 10.05 \text{ MeV} + 4.97 \text{ MeV} = 26.73 \text{ MeV}$.

36. (a) The temperature must be very high to overcome the Coulomb barrier so the two alpha particles can interact through the nuclear force. A high density is necessary so that the probability is high enough for a third alpha particle to interact with ^8Be before it decays.

(b) We use Equation (13.7) to find the Q value for the reaction:

$$Q = [3M(^4\text{He}) - M(^{12}\text{C})] \text{u} \cdot c^2 = [3(4.002603) - 12.0] \text{u} \cdot c^2 = 7.27 \text{ MeV}.$$

37. (a) The temperature must be very high to overcome the Coulomb barrier so that the two oxygen nuclei can interact through the nuclear force. As the value of Z increases, higher and higher temperatures are required.

(b) We use Equation (13.7) to find the Q value for the reaction:

$$Q = [2M(^{16}\text{O}) - M(^{32}\text{S})] \text{u} \cdot c^2 = [2(15.994915) - 31.972071] \text{u} \cdot c^2 = 16.5 \text{ MeV}.$$

This includes the energy of the γ ray in the released energy.

38. (a) This problem is similar to Example 13.9. First we will determine the Coulomb potential energy that must be overcome, using Equation (12.2) to determine the radius.

$$r = r_0 A^{1/3} = 1.2 \times 10^{-15} \text{ m} (28)^{1/3} = 3.6 \times 10^{-15} \text{ m}$$

Therefore the Coulomb barrier is:

$$V = \frac{q_1 q_2}{4\pi\epsilon_0 r} = \frac{(9 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(14)^2 (1.6 \times 10^{-19} \text{ C})^2}{3.6 \times 10^{-15} \text{ m}} = 1.3 \times 10^{-11} \text{ J}$$

We need at least this much kinetic energy to overcome the Coulomb barrier. We set this value equal to the thermal energy, namely $\frac{3}{2} kT$.

$$T = \frac{2V}{3k} = \frac{2(1.3 \times 10^{-11} \text{ J})}{3(1.38 \times 10^{-23} \text{ J/K})} = 6.3 \times 10^{11} \text{ K}$$

(b) We use Equation (13.7) to find the Q value for the reaction:

$$Q = [2M(^{28}\text{Si}) - M(^{56}\text{Ni})] \text{u} \cdot c^2 = [2(27.976927) - 55.942136] \text{u} \cdot c^2 = 10.9 \text{ MeV}$$

This includes the energy of the γ ray in the released energy.

39. (a) This problem is similar to Example 13.9. First we will determine the Coulomb potential energy that must be overcome, using Equation (12.2) to determine the radius.

$$r = r_0 A^{1/3} = 1.2 \times 10^{-15} \text{ m} (12)^{1/3} = 2.7 \times 10^{-15} \text{ m}$$

Therefore the Coulomb barrier is:

$$V = \frac{q_1 q_2}{4\pi\epsilon_0 r} = \frac{(9 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(6)^2 (1.6 \times 10^{-19} \text{ C})^2}{2.7 \times 10^{-15} \text{ m}} = 3.1 \times 10^{-12} \text{ J}$$

We need at least this much kinetic energy to overcome the Coulomb barrier. We set this value equal to the thermal energy, namely $\frac{3}{2} kT$.

$$T = \frac{2V}{3k} = \frac{2(3.1 \times 10^{-12} \text{ J})}{3(1.38 \times 10^{-23} \text{ J/K})} = 1.5 \times 10^{11} \text{ K}$$

- (b) We use Equation (13.7) to find the Q value for the reaction:

$$Q = [2M(^{12}\text{C}) - M(^{24}\text{Mg})]u \cdot c^2 = [2(12.0) - 23.985042]u \cdot c^2 = 13.9 \text{ MeV}$$

This includes the energy of the γ ray in the released energy.

40. If the depth is d and the radius of the earth is r and $2/3$ of the surface is covered with water, then the volume of the oceans is

$$V = \frac{2}{3}(4\pi r^2)d = \frac{8}{3}\pi(6.37 \times 10^6 \text{ m})^2(3000 \text{ m}) = 1.02 \times 10^{18} \text{ m}^3$$

With a density of 1000 kg/m^3 , the mass is $1.02 \times 10^{21} \text{ kg}$, so the number of water

$$\text{molecules is } N = (1.02 \times 10^{21} \text{ kg}) \frac{6.022 \times 10^{23}}{0.018 \text{ kg}} = 3.41 \times 10^{46}.$$

With a 0.015% abundance, the number of deuterium atoms (in D_2O molecules) is

$$(2)3.41 \times 10^{46} (0.00015) = 1.02 \times 10^{43}.$$

There will be at most half this number of fusions,

$$\text{or } 5.1 \times 10^{42}. \text{ The energy released is } (5.1 \times 10^{42})(4.0 \text{ MeV})(1.602 \times 10^{-13} \text{ J/MeV}) = 3.27 \times 10^{30} \text{ J}.$$

41. (a) We use temperatures because that is what we strive for experimentally. Kinetic energy is useful for comparison with the Coulomb barrier or Q values.

$$(b) K = \frac{3}{2} kT$$

$$(c) T = \frac{2K}{3k} = \frac{2(6000 \text{ eV})}{3(8.617 \times 10^{-5} \text{ eV/K})} = 4.64 \times 10^7 \text{ K}$$

42. ${}^4\text{He} + {}^3\text{He} \rightarrow {}^7\text{Be} + \gamma$; $Q = [M({}^4\text{He}) + M({}^3\text{He}) - M({}^7\text{Be})]u \cdot c^2 = 1.59 \text{ MeV}$
 ${}^2\text{H} + {}^2\text{H} \rightarrow {}^3\text{H} + {}^1\text{H}$; $Q = [M({}^2\text{H}) + M({}^2\text{H}) - M({}^3\text{H}) - M({}^1\text{H})]u \cdot c^2 = 4.03 \text{ MeV}$
 ${}^1\text{H} + {}^2\text{H} \rightarrow {}^3\text{He} + \gamma$; $Q = [M({}^1\text{H}) + M({}^2\text{H}) - M({}^3\text{He})]u \cdot c^2 = 5.49 \text{ MeV}$
 ${}^1\text{H} + {}^{12}\text{C} \rightarrow {}^{13}\text{N} + \gamma$; $Q = [M({}^1\text{H}) + M({}^{12}\text{C}) - M({}^{13}\text{N})]u \cdot c^2 = 1.94 \text{ MeV}$
 ${}^3\text{He} + {}^3\text{He} \rightarrow {}^4\text{He} + 2p$;
 $Q = [M({}^3\text{He}) + M({}^3\text{He}) - M({}^4\text{He}) - 2M({}^1\text{H})]u \cdot c^2 = 12.9 \text{ MeV}$
 ${}^7\text{Li} + {}^1\text{H} \rightarrow {}^4\text{He} + {}^4\text{He}$; $Q = [M({}^1\text{H}) + M({}^7\text{Li}) - 2M({}^4\text{He})]u \cdot c^2 = 17.3 \text{ MeV}$
 ${}^3\text{H} + {}^2\text{H} \rightarrow {}^4\text{He} + n$; $Q = [M({}^3\text{H}) + M({}^2\text{H}) - M({}^4\text{He}) - M(n)]u \cdot c^2 = 17.6 \text{ MeV}$
 ${}^3\text{He} + {}^2\text{H} \rightarrow {}^4\text{He} + {}^1\text{H}$; $Q = [M({}^3\text{He}) + M({}^2\text{H}) - M({}^4\text{He}) - M({}^1\text{H})]u \cdot c^2 = 18.4 \text{ MeV}$

43. The reaction is ${}^1\text{H} + {}^{12}\text{C} \rightarrow {}^{13}\text{N} + \gamma$.

$$Q = [M({}^{12}\text{C}) + M({}^1\text{H}) - M({}^{13}\text{N})]u \cdot c^2 = 1.94 \text{ MeV}$$

$$\text{Then } K_{\text{th}} = -Q \left(\frac{M({}^1\text{H}) + M({}^{12}\text{C})}{M({}^{12}\text{C})} \right)$$

which will be negative, so the threshold energy is not a concern.

However, the Coulomb barrier is

$$V = \frac{q_1 q_2}{4\pi\epsilon_0 r} = \frac{6e^2}{4\pi\epsilon_0 (r_p + r_0 A^{1/3})} = \frac{6(1.44 \times 10^{-9} \text{ eV} \cdot \text{m})}{(1.2 \times 10^{-15} \text{ m} + (1.2 \times 10^{-15} \text{ m})(12)^{1/3})} = 2.19 \text{ MeV}.$$

Setting this coulomb barrier energy equal to the thermal energy $\frac{3}{2}kT$, we have

$$T = \frac{2K}{3k} = \frac{2(2.19 \times 10^6 \text{ eV})}{3(8.617 \times 10^{-5} \text{ eV/K})} = 1.69 \times 10^{10} \text{ K}.$$

44. $Q = [3M({}^4\text{He}) - M({}^{12}\text{C})]u \cdot c^2 = 7.27 \text{ MeV}$

45. (a) $K = \frac{3}{2}kT = \frac{3}{2}(8.617 \times 10^{-5} \text{ eV/K})(300 \text{ K}) = 3.88 \times 10^{-2} \text{ eV}$

(b) $v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2(3.88 \times 10^{-2} \text{ eV})(1.602 \times 10^{-19} \text{ J/eV})}{1.675 \times 10^{-27} \text{ kg}}} = 2720 \text{ m/s}$

(c) $\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{(1.675 \times 10^{-27} \text{ kg})(2720 \text{ m/s})} = 1.45 \times 10^{-10} \text{ m}$

46. $R = \frac{512}{60} \text{ Bq} = 8.533 \text{ Bq}$. The initial rate was $R_0 = 4 \times 10^5 \text{ Bq}$. Over 60 days the activity

has decreased to $R'_0 = R_0 e^{-\lambda t} = (4 \times 10^5 \text{ Bq}) \exp\left(-\frac{\ln 2}{44.5 \text{ d}}(60 \text{ d})\right) = 1.571 \times 10^5 \text{ Bq}$.

Therefore the fraction f of the ring that has worn off into the oil is

$$f = \frac{R}{R'_0} = \frac{8.533 \text{ Bq}}{1.571 \times 10^5 \text{ Bq}} = 5.43 \times 10^{-5}$$

47. (a) With a half-life of 6.01 h, not much of the original material remains after one week.

(b) $R = R_0 e^{-\lambda t} = (10^{11} \text{ Bq}) \exp\left(-\frac{\ln 2}{6 \text{ h}}(216 \text{ h})\right) = 1.46 \text{ Bq}$

(c) $R = R_0 e^{-\lambda t} = (0.9 \times 10^{11} \text{ Bq}) \exp\left(-\frac{\ln 2}{6 \text{ h}}(96 \text{ h})\right) = 1.37 \times 10^6 \text{ Bq}$

48. (a) We need 1000 Bq at $t = 30$ minutes with $t_{1/2} = 83.1$ minutes.

$$R = R'_0 e^{-\lambda t} = (R'_0) \exp\left(-\frac{\ln 2}{83.1 \text{ min}}(30 \text{ min})\right) = 1000 \text{ Bq}$$

$$R'_0 = (1000 \text{ Bq}) \exp\left(\frac{\ln 2}{83.1 \text{ min}}(30 \text{ min})\right) = 1284 \text{ Bq}$$

Because only 27% go to the first excited state, we need activity $R_0 = \frac{R'_0}{0.27} = 4756 \text{ Bq}$.

Then $R_0 = \lambda N = N \frac{\ln 2}{t_{1/2}}$, so solving for N :

$$N = \frac{R_0 t_{1/2}}{\ln 2} = \frac{(4756 \text{ s}^{-1})(83.1 \text{ min})(60 \text{ s/min})}{\ln 2} = 3.42 \times 10^7$$

(b) $m = (3.42 \times 10^7) \frac{139 \text{ g}}{6.022 \times 10^{23}} = 7.89 \times 10^{-15} \text{ g}$. The fraction f activated is

$$f = \frac{7.89 \times 10^{-15} \text{ g}}{(0.717)(10^{-8} \text{ g})} = 1.10 \times 10^{-6}.$$

49. (a) $I = (0.15)(5.0 \times 10^5 \text{ s}^{-1})(2 \times 1.602 \times 10^{-19} \text{ C}) = 2.40 \times 10^{-14} \text{ A}$

(b) One-tenth of the above value is $2.40 \times 10^{-15} \text{ A}$.

50. (a) $R = (5000 \text{ J/s}) \left(\frac{1 \text{ MeV}}{1.602 \times 10^{-13} \text{ J}} \right) \left(\frac{1 \text{ particle}}{5.3 \text{ MeV}} \right) = 5.89 \times 10^{15} \text{ Bq}$

$$N = \frac{R}{\lambda} = \frac{(5.89 \times 10^{15} \text{ s}^{-1})(138 \text{ d})(86400 \text{ s/d})}{\ln 2} = 1.013 \times 10^{23}$$

$$m = (1.013 \times 10^{23}) \left(\frac{0.210 \text{ kg}}{6.022 \times 10^{23}} \right) = 3.53 \times 10^{-2} \text{ kg}$$

(b) For 7 kW we need a mass of $(7/5)(3.53 \times 10^{-2} \text{ kg}) = 0.0494 \text{ kg}$. Letting the initial mass be m_0 , we have the usual exponential decay and so $m = m_0 \exp(-\lambda t)$, or

$$m_0 = m \exp(\lambda t) = (0.0494 \text{ kg}) \exp\left(\frac{(\ln 2)(730 \text{ d})}{138 \text{ d}}\right) = 1.93 \text{ kg}.$$

51. The fraction absorbed is the patient's cross-sectional area divided by $4\pi(\text{distance})^2$.

$$\text{Note that there are two gammas per decay: } R = 2(3 \times 10^{14} \text{ Bq}) \frac{0.3 \text{ m}^2}{4\pi(4 \text{ m})^2} = 8.95 \times 10^{11} \text{ Bq}$$

52. From Chapter 12 the diameter of the uranium nucleus is

$$d = 2r(^{238}\text{U}) = 2(1.2 \text{ fm}) 238^{1/3} = 14.87 \text{ fm}. \text{ The momentum of the neutron is}$$

$$p = \frac{h}{\lambda} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{14.87 \times 10^{-15} \text{ m}} = 4.456 \times 10^{-20} \text{ kg} \cdot \text{m/s} \text{ and the kinetic energy is}$$

$$K = \frac{p^2}{2m} = \frac{(4.456 \times 10^{-20} \text{ kg} \cdot \text{m/s})^2}{2(1.675 \times 10^{-27} \text{ kg})} = 5.927 \times 10^{-13} \text{ J} = 3.70 \text{ MeV}.$$

This is still much less than the neutron's rest energy, which justifies the non-relativistic

$$\text{treatment. Then in thermal equilibrium } T = \frac{2K}{3k} = \frac{2(5.927 \times 10^{-13} \text{ J})}{3(1.381 \times 10^{-23} \text{ J/K})} = 2.86 \times 10^{10} \text{ K}.$$

Clearly this is not a realistic temperature in a reactor, so the proposed experiment is not possible.

53. The alpha decay reaction is $^{239}\text{Pu} \rightarrow ^{235}\text{U} + ^4\text{He}$, and the energy released in each decay is

$$Q = [M(^{239}\text{Pu}) - M(^{235}\text{U}) - M(^4\text{He})]u \cdot c^2 = 5.25 \text{ MeV}.$$

The activity of a 10 kg sample is

$$R = \lambda N = \frac{\ln 2}{(24110 \text{ y})(3.156 \times 10^7 \text{ s/y})} (10 \text{ kg}) \left(\frac{6.022 \times 10^{23}}{0.239 \text{ kg}} \right) = 2.30 \times 10^{13} \text{ Bq}. \text{ Then}$$

$$\text{the power is } P = (0.6)(5.25 \text{ MeV})(1.602 \times 10^{-13} \text{ J/MeV})(2.30 \times 10^{13} \text{ s}^{-1}) = 11.6 \text{ W}.$$

54. Perhaps the best source of information for this question is the Internet site for the International Union of Pure and Applied Chemistry (IUPAC) at www.iupac.org. As of early summer 2011, elements with $Z = 113, 114, 115, 116$, and 118 had been observed experimentally but not officially named by IUPAC. The current names for 114 and 116

are *ununquadium* and *ununhexium*, for example. Element 112 was recently named Copernicium.

55. An Internet search will yield current information. In addition the April 2000 issue of *Physics Today* magazine describes the relevance of radioisotopes for this procedure. There is still considerable debate about the efficacy of beta as compared to gamma emitters. Although the first study by Bottcher (1994) *et al.* used a gamma emitter, ^{192}Ir , recent studies have included beta emitters such as ^{32}P , ^{90}Sr , ^{90}Y , ^{188}Re , ^{99}Tc , and ^{137}Xe . Almost all of the trials have shown low-dose radiation to be effective in preventing keloids and benign vascular malformations.

56. (a) The weighted average is $580 \text{ barns} \times 0.0072 + 0 \text{ barns} \times 0.993 = 4.2 \text{ barns}$.
 (b) At the higher concentration, the weighted average becomes $580 \text{ barns} \times 0.020 + 0 \text{ barns} \times 0.98 = 11.6 \text{ barns}$.

57. The reaction is $n + {}^{235}_{92}\text{U} \rightarrow {}^{101}_{42}\text{Mo} + {}^{132}_{50}\text{Sn} + 3n$. Looking up the masses,

$$\text{mass}(n + {}^{235}_{92}\text{U}) = 1.008665 \text{ u} + 235.043924 \text{ u} = 236.052589 \text{ u}$$

$$\begin{aligned} \text{mass}({}^{101}_{42}\text{Mo} + {}^{132}_{50}\text{Sn} + 3n) &= 100.910347 \text{ u} + 131.917744 \text{ u} + 3(1.008665 \text{ u}) \\ &= 235.854086 \text{ u} \end{aligned}$$

The mass difference is $\Delta m = 0.198503 \text{ u}$.

$$\text{Energy released} = (\Delta m)c^2 = 0.198503 \text{ u} \left(\frac{931.49 \text{ MeV}/c^2}{\text{u}} \right) c^2 = 185 \text{ MeV}.$$

$$58. (a) (1 \text{ kg}) \frac{6.022 \times 10^{23}}{\text{mol}} \frac{1 \text{ mol}}{0.001 \text{ kg}} \frac{13.6 \text{ eV}}{1 \text{ atom}} = 8.19 \times 10^{27} \text{ eV}$$

$$(b) (1 \text{ kg}) \frac{6.022 \times 10^{23}}{\text{mol}} \frac{1 \text{ mol}}{0.002 \text{ kg}} \frac{2.224 \times 10^6 \text{ eV}}{1 \text{ atom}} = 6.70 \times 10^{32} \text{ eV}$$

$$(c) (1 \text{ kg}) \frac{1 \text{ u}}{1.66 \times 10^{-27} \text{ kg}} \frac{931.49 \times 10^6 \text{ MeV}}{\text{u}} = 5.61 \times 10^{35} \text{ eV}$$

(d) The results are as expected. The annihilation process in part (c) is the complete conversion of matter into energy, which maximizes energy yield. The energy in (b) is much greater than the energy in (a), because nuclear binding energies are so much larger than atomic binding energies.

59. (a) $Q = K_{\text{out}} - K_{\text{in}} = 86.63 \text{ MeV} - 100 \text{ MeV} = -13.37 \text{ MeV}$

This agrees very nearly with the Q computed using atomic masses (assuming all the masses are known).

$$Q = [M(^{18}\text{O}) + M(^{30}\text{Si}) - M(^{14}\text{O}) - M(^{34}\text{Si})] \text{u} \cdot c^2 = -13.27 \text{ MeV}$$

(b) $M = -\frac{Q}{c^2} + M(^{18}\text{O}) + M(^{30}\text{Si}) - M(^{14}\text{O}) = 33.979 \text{ u}$

which agrees very nearly with the value for the mass of ^{34}Si in Appendix 8.

60. (a) The number of atoms is $1000(100 \text{ kg}) \frac{1 \text{ mol}}{0.235 \text{ kg}} \frac{6.022 \times 10^{23}}{\text{mol}} (0.04) = 1.03 \times 10^{28}$

(b) $\frac{1.03 \times 10^{28}}{4\pi(6 \times 10^6 \text{ m})^2} = 2.28 \times 10^{13} \text{ m}^{-2}$

Then the activity for each square meter is

$$R = \lambda N = \frac{\ln 2}{28.8 \text{ y} (3.156 \times 10^7 \text{ s/y})} (2.28 \times 10^{13}) = 1.74 \times 10^4 \text{ Bq.}$$

61. (a) Starting with $F(E) = (\text{const.})e^{-E/kT} E^{1/2}$ and setting $dF/dE = 0$, we find that the most probable energy is $E^* = \frac{1}{2}kT$. $E^* = \frac{1}{2}kT = \frac{1}{2}(8.621 \times 10^{-5} \text{ eV/K})(2 \times 10^8 \text{ K}) = 8620 \text{ eV}$

(b) $\frac{n(2E)}{n(E)} = \exp(-\beta(2E - E)) = \exp(-\beta E) = \exp\left(-\frac{1}{2}\right) = 0.607$

$$\frac{n(5E)}{n(E)} = \exp(-\beta(5E - E)) = \exp(-4\beta E) = \exp(-2) = 0.135$$

$$\frac{n(10E)}{n(E)} = \exp(-\beta(10E - E)) = \exp(-9\beta E) = \exp\left(-\frac{9}{2}\right) = 0.0111.$$

62. (a) $P = \sigma n t = (90 \times 10^{-27} \text{ cm}^2)(1.85 \text{ g/cm}^3)(3.2 \text{ cm})(6.022 \times 10^{23} \text{ mol}^{-1})\left(\frac{1 \text{ mol}}{9 \text{ g}}\right) = 0.0357$

(b) With this probability the number of incident alpha particles is

$$\frac{1.6 \times 10^5 \text{ s}^{-1}}{0.0357} = 4.48 \times 10^6 \text{ s}^{-1}.$$

(c) $N = \frac{R}{\lambda} = (4.48 \times 10^6 \text{ s}^{-1})\left(\frac{2.41 \times 10^4 \text{ y}}{\ln 2}\right)\left(\frac{3.156 \times 10^7 \text{ s}}{\text{y}}\right) = 4.92 \times 10^{18}$ and

$$m = (4.92 \times 10^{18}) \frac{0.239 \text{ kg}}{6.022 \times 10^{23}} = 1.95 \times 10^{-6} \text{ kg, which is reasonable.}$$

$$63. m(^{40}\text{K}) = (65 \text{ kg})(0.0035)(0.00012) = 2.73 \times 10^{-5} \text{ kg}$$

$$N = (2.73 \times 10^{-5} \text{ kg}) \frac{6.022 \times 10^{23}}{0.040 \text{ kg}} = 4.11 \times 10^{20}$$

$$R = \lambda N = \frac{\ln 2}{1.28 \times 10^9 \text{ y}} \frac{1 \text{ y}}{3.156 \times 10^7 \text{ s}} (4.11 \times 10^{20}) = 7050 \text{ Bq}$$

The beta activity is $(7050 \text{ Bq})(0.893) = 6300 \text{ Bq}$.

64. We can determine the minimum mass as follows:

$$E = (15 \text{ kilotons}) (4.2 \times 10^{12} \text{ J/kiloton}) = 6.3 \times 10^{13} \text{ J} \left(\frac{1 \text{ MeV}}{1.6 \times 10^{-13} \text{ J}} \right) = 3.94 \times 10^{26} \text{ MeV}.$$

If each fission yields 200 MeV, then we have:

$$\text{Number of fissions} = \frac{3.94 \times 10^{26} \text{ MeV}}{200 \text{ MeV/fission}} = 1.97 \times 10^{24} \text{ fissions which is the minimum}$$

number of fissions. This requires a mass of:

$$M = (1.97 \times 10^{24}) \left(\frac{235 \text{ u}}{\text{atom}} \right) \left(\frac{1.66 \times 10^{-27} \text{ kg}}{\text{u}} \right) = 0.77 \text{ kg}.$$

The actual mass was larger since not every nucleus fissioned.

65. Using the mass of the 238 isotope, the rate of spontaneous fission per atom is

$$6.7 \frac{\text{decays}}{\text{kg} \cdot \text{s}} \times \frac{238 \text{ u}}{\text{atom}} \times \frac{1.66 \times 10^{-27} \text{ kg}}{1 \text{ u}} = 2.65 \times 10^{-24} \frac{\text{decays}}{\text{atom} \cdot \text{s}}$$

The overall decay rate for the 238 isotope is computed using the half-life of $4.47 \times 10^9 \text{ y}$:

$$\lambda = \frac{R}{N} = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{(4.47 \times 10^9 \text{ y})(3.15 \times 10^7 \text{ s/y})} = 4.92 \times 10^{-18} \frac{\text{decays}}{\text{atom} \cdot \text{s}}$$

Therefore the probability of spontaneous fission is

$$\frac{2.65 \times 10^{-24} \frac{\text{decays}}{\text{atom} \cdot \text{s}}}{4.92 \times 10^{-18} \frac{\text{decays}}{\text{atom} \cdot \text{s}}} = 5.4 \times 10^{-7}. \text{ This result agrees with published values.}$$

Chapter 14

1. By conservation of momentum the photons must have the same energy. For each one

$$E = hf = mc^2 \text{ and } f = \frac{mc^2}{h} = \frac{938.27 \times 10^6 \text{ eV}}{4.136 \times 10^{-15} \text{ eV} \cdot \text{s}} = 2.27 \times 10^{23} \text{ Hz.}$$

2. As in the text (see Example 14.1), let $mc^2 = \hbar c / R$, so

$$R = \frac{\hbar c}{mc^2} = \frac{197.3 \text{ eV} \cdot \text{nm}}{140 \times 10^6 \text{ eV}} = 1.41 \times 10^{-6} \text{ nm} = 1.41 \text{ fm.}$$

3. We will use a small nucleus (helium, diameter 3.8 fm) and a large nucleus (uranium, diameter 14.9 fm) to obtain a range of values.

$$\text{For helium: } \Delta t = \frac{3.8 \times 10^{-15} \text{ m}}{2.998 \times 10^8 \text{ m/s}} = 1.27 \times 10^{-23} \text{ s}$$

$$\text{For uranium: } \Delta t = \frac{14.9 \times 10^{-15} \text{ m}}{2.998 \times 10^8 \text{ m/s}} = 4.97 \times 10^{-23} \text{ s}$$

4. We know that $\lambda \leq D$ and the problem says specifically to choose $\lambda = 0.10D$ with the diameter $D = 0.15 \text{ fm}$. Therefore $\lambda = 0.1(1.5 \text{ fm}) = 0.15 \text{ fm}$. We know from de Broglie's relationship that $p = h / \lambda$ and we can determine the kinetic energy from this momentum

$$\text{as follows: } E^2 = (K + mc^2)^2 = (pc)^2 + (mc^2)^2$$

$$K = \sqrt{(pc)^2 + (mc^2)^2} - mc^2 = \sqrt{\left(\frac{\hbar c}{0.15 \text{ fm}}\right)^2 + (mc^2)^2} - mc^2$$

Electron:

$$K = \sqrt{\left(\frac{1239.8 \text{ eV} \cdot \text{nm}}{0.15 \text{ fm}(10^{-6} \text{ nm/1 fm})}\right)^2 + (0.511 \times 10^6 \text{ eV})^2} - 0.511 \times 10^6 \text{ eV} = 8.26 \text{ GeV}$$

Proton:

$$K = \sqrt{\left(\frac{1239.8 \text{ eV} \cdot \text{nm}}{0.15 \text{ fm}(10^{-6} \text{ nm/1 fm})}\right)^2 + (938.27 \times 10^6 \text{ eV})^2} - 938.27 \times 10^6 \text{ eV} = 7.38 \text{ GeV}$$

5. As in the preceding problem, we know that $p = h / \lambda$ and

$$K = \sqrt{(pc)^2 + (mc^2)^2} - mc^2 = \sqrt{\left(\frac{\hbar c}{1 \times 10^{-18} \text{ m}}\right)^2 + (mc^2)^2} - mc^2$$

$$\text{Electron: } K = \sqrt{\left(\frac{1239.8 \text{ eV} \cdot \text{nm}}{1 \times 10^{-9} \text{ nm}}\right)^2 + (0.511 \times 10^6 \text{ eV})^2} - 0.511 \times 10^6 \text{ eV} = 1.24 \text{ TeV}$$

$$\text{Proton: } K = \sqrt{\left(\frac{1239.8 \text{ eV} \cdot \text{nm}}{1 \times 10^{-9} \text{ nm}}\right)^2 + (938.27 \times 10^6 \text{ eV})^2} - 938.27 \times 10^6 \text{ eV} = 1.24 \text{ TeV}$$

The kinetic energies are so large that the rest energies do not affect our answer.

6. We use Equation (14.4) to estimate the range of the force.

$$R = \frac{\hbar c}{2mc^2} = \frac{197.33 \text{ eV} \cdot \text{nm}}{2(782 \times 10^6 \text{ eV})} = 1.26 \times 10^{-7} \text{ nm} = 1.26 \times 10^{-16} \text{ m}$$

7. We use Equation (14.4) to estimate the range of the force.

$$R = \frac{\hbar c}{2mc^2} = \frac{197.33 \text{ eV} \cdot \text{nm}}{2(150 \times 10^9 \text{ eV})} = 6.58 \times 10^{-10} \text{ nm} = 6.58 \times 10^{-19} \text{ m}$$

8. (a) $\bar{\nu}_\mu$ and ν_e (b) $\bar{\nu}_e$ (c) $\bar{\nu}_\mu$ (d) $\bar{\nu}_\mu$ (e) ν_μ

9. Assuming roughly an equal number of protons and neutrons, we can use the average of their masses per baryon, or $1.674 \times 10^{-27} \text{ kg}$. The mass of Earth is $5.98 \times 10^{24} \text{ kg}$, so the

$$\text{baryon number is } \frac{5.98 \times 10^{24} \text{ kg}}{1.674 \times 10^{-27} \text{ kg}} = 3.57 \times 10^{51}.$$

10. (a) We use Equation (14.6) to determine the width Γ from the mean lifetime.

$$\Gamma = \frac{\hbar}{\tau} = \frac{6.5821 \times 10^{-16} \text{ eV} \cdot \text{s}}{0.821 \times 10^{-10} \text{ s}} = 8.02 \times 10^{-6} \text{ eV}$$

(b) This value of Γ is much smaller than the experimental uncertainty of the measurement.

11. We use Equation (14.6) to determine the width Γ from the mean lifetime.

$$\text{J/Psi: } \Gamma = \frac{\hbar}{\tau} = \frac{6.5821 \times 10^{-16} \text{ eV} \cdot \text{s}}{7.1 \times 10^{-21} \text{ s}} = 9.3 \times 10^4 \text{ eV};$$

$$\text{Upsilon: } \Gamma = \frac{\hbar}{\tau} = \frac{6.5821 \times 10^{-16} \text{ eV} \cdot \text{s}}{1.2 \times 10^{-20} \text{ s}} = 5.5 \times 10^4 \text{ eV}$$

$$\text{The full-width of the charged pion is } \Gamma = \frac{\hbar}{\tau} = \frac{6.5821 \times 10^{-16} \text{ eV} \cdot \text{s}}{2.6 \times 10^{-8} \text{ s}} = 2.5 \times 10^{-8} \text{ eV}.$$

This value is more than 10^{12} times smaller than either of the previous two.

12. In the first reaction strangeness is violated ($+1 \rightarrow -1$). The second reaction is allowed.

The fact that the K^0 has strangeness $+1$ and $\bar{K}^0$ has strangeness -1 is consistent with the idea that it is not its own antiparticle. However, M. Gell-Mann and A. Pais found that

$K^0 \rightarrow \pi^+ + \pi^- \rightarrow \bar{K}^0$ (and the reverse operation $\bar{K}^0 \rightarrow \pi^+ + \pi^- \rightarrow K^0$) can occur by the laws of quantum mechanics (see *Feynman Lectures in Physics* vol. 3 p. 11-16). So in this case the K^0 does act as its own antiparticle.

13. In both (a) and (b) the baryon number is not conserved.

14. (a) μ and e lepton numbers are not conserved

(b) charge is not conserved

(c) momentum-energy is not conserved

15. Let subscript 1 refer to the Σ , subscript 2 to the Λ , and no subscript to the photon. From conservation of momentum $|p_1| = |p_2| = E/c$. From conservation of energy

$$m_1 c^2 = \sqrt{p_2^2 c^2 + (m_2 c^2)^2} + E = \sqrt{E^2 + (m_2 c^2)^2} + E, \text{ so}$$

$$E = \frac{(m_1 c^2)^2 - (m_2 c^2)^2}{2m_1 c^2} = \frac{(1193 \text{ MeV})^2 - (1116 \text{ MeV})^2}{2(1193 \text{ MeV})} = 74.5 \text{ MeV}.$$

16. The π^0 begins with energy $E = K + E_0 = 855 \text{ MeV}$. Because the photon energies are equal, we know by conservation of momentum that the two photons make equal angles θ above and below the original line of motion for the π^0 . Let each photon energy be E' .

The conservation of energy and momentum give: $E' = \frac{E}{2} = \frac{855}{2} \text{ MeV} = 427.5 \text{ MeV}$.

$$p = \frac{2E'}{c} \cos \theta = \frac{\sqrt{E^2 - E_0^2}}{c} = \frac{\sqrt{(855 \text{ MeV})^2 - (135 \text{ MeV})^2}}{c} = 844.3 \text{ MeV}/c.$$

$$\cos \theta = \frac{pc}{2E'} = \frac{844.3 \text{ MeV}}{2(427.5 \text{ MeV})} = 0.9875 \text{ from which we find } \theta = 9.1^\circ.$$

17.

(a) ν_μ is needed to conserve lepton number.

(b) A kaon is needed to conserve strangeness, and to conserve charge too the kaon must be a K^+ .

$$18. n(udd): \quad q = \frac{2e}{3} - \frac{e}{3} - \frac{e}{3} = 0; \quad B = 3\left(\frac{1}{3}\right) = 1; \quad S = 3(0) = 0$$

$$\Sigma^+(uus): \quad q = \frac{2e}{3} + \frac{2e}{3} - \frac{e}{3} = e; \quad B = 3\left(\frac{1}{3}\right) = 1; \quad S = 0 + 0 - 1 = -1$$

$$\Lambda_c^+(udc): \quad q = \frac{2e}{3} - \frac{e}{3} + \frac{2e}{3} = e; \quad B = 3\left(\frac{1}{3}\right) = 1; \quad S = 0 + 0 + 0 = 0$$

$$19. \pi^+(u\bar{d}): \quad q = \frac{2e}{3} + \frac{e}{3} = e; \quad B = \frac{1}{3} - \frac{1}{3} = 0; \quad S = 0 + 0 = 0$$

$$K^+(u\bar{s}): \quad q = \frac{2e}{3} + \frac{e}{3} = e; \quad B = \frac{1}{3} - \frac{1}{3} = 0; \quad S = 0 + 1 = 1$$

$$D^0(c\bar{u}): \quad q = \frac{2e}{3} - \frac{2e}{3} = 0; \quad B = \frac{1}{3} - \frac{1}{3} = 0; \quad S = 0 + 0 = 0$$

$$20. D^0: c\bar{u} \quad D^-: \bar{c}d \quad D^+: c\bar{d}$$

21. The B^+ and B^- have zero strangeness and zero charm, but they have masses greater than $5000 \text{ MeV}/c^2$ and charge $+1$ and -1 respectively. This leads one to conclude that the quark configuration should be $\bar{b}u$ for the B^+ and $b\bar{u}$ for the B^- . The B^0 has zero charge, so it should be either $b\bar{d}$ or $\bar{b}d$. The only way to distinguish between these two possibilities is by looking at the conservation laws in the appropriate decay reactions. It turns out that the B^0 is $\bar{b}d$ and the antiparticle $\bar{B}^0$ is $b\bar{d}$.
22. Charge, baryon number, strangeness, topness, and bottomness are zero. Charm $= -1$. From table 14.6 D^0 in the text has configuration $c\bar{u}$, so $\bar{c}u$ must be $\bar{D}^0$.
- 23.
- (a) The Ω^- , Λ^0 , and the K^- have mean lifetimes on the order of 10^{-10} s, which indicates the decays are due to the weak interaction. Remember that there can be quark transformations in weak interactions.
 - (b) $\Omega^- \rightarrow \Lambda^0 + K^-$, $sss \rightarrow uds + \bar{u}s$; an s quark transformed into a d quark and a $u\bar{u}$ quark-antiquark pair was created. $\Lambda^0 \rightarrow p + \pi^-$, $uds \rightarrow uud + \bar{u}d$; the s quark transformed into a d quark and a $u\bar{u}$ quark-antiquark pair was created. $K^- \rightarrow \mu^- + \bar{\nu}_\mu$, $\bar{u}s \rightarrow \text{no quarks}$; the s quark transformed into a u quark and the $\bar{u}u$ quark-antiquark pair annihilates. There are no quarks in the final decay products.
24. We use Table 14.5 for quark properties and Table 14.4 or Table 14.6 to identify the hadrons. The spin is $1/2$ for all quarks and antiquarks.
- (a) $c\bar{d}$; spin is 0 or 1; charge is 1; baryon number is 0; $C = 1$; $S, B, T = 0$. This is a D^+ meson.
 - (b) uds ; spin is $1/2$ or $3/2$; charge is 0; baryon number is 1; $S = -1$; $C, B, T = 0$. This could be a Σ^0 or a Λ baryon.
 - (c) $\bar{s}\bar{s}\bar{s}$; spin is $1/2$ or $3/2$; baryon number is -1 ; charge is 1; $S = 3$; $C, B, T = 0$. This is a Ω^+ baryon.
 - (d) $\bar{c}d$; spin is 0 or 1; charge is -1 ; baryon number is 0; $C = -1$; $S, B, T = 0$. This is a D^- meson.

25. Assuming an average nucleon mass of $1.674 \times 10^{-27} \text{ kg}$ and noting that 10 out of 18 nucleons in water are protons, we have

$$(3.5 \times 10^5 \text{ L}) \frac{10^{-3} \text{ m}^3}{\text{L}} \frac{1000 \text{ kg}}{\text{m}^3} \frac{1}{1.673 \times 10^{-27} \text{ kg}} \frac{10}{18} = 1.16 \times 10^{32} \text{ protons}$$

Then by the laws of radioactive decay

$$1.16 \times 10^{32} \text{ protons} \frac{\ln 2 \text{ decays}}{10^{33} \text{ y}} = 0.080 \text{ decays/y}$$

26. Baryon number is not conserved in any of the three. This is a common problem in proton decay schemes, because there are no lighter baryons. In addition, (a) violates electron lepton number, (b) violates muon lepton number, and (c) violates strangeness.

$$27. \frac{31 \text{ kg}}{1.673 \times 10^{-27} \text{ kg}} \frac{\ln 2 \text{ decays}}{10^{33} \text{ y}} \frac{1 \text{ y}}{365.25 \text{ d}} = 3.5 \times 10^{-8} \text{ decays/d}$$

28. The lifetime of the Σ^+ is $8.0 \times 10^{-11} \text{ s}$. Due to relativity it travels farther than one might expect.

$$\gamma = \frac{K + E_0}{E_0} = \frac{3.6 \text{ GeV} + 1.189 \text{ GeV}}{1.189 \text{ GeV}} = 4.028 = \frac{1}{\sqrt{1 - v^2/c^2}}$$

$$v = c\sqrt{1 - 1/\gamma^2} = c\sqrt{1 - 1/(4.028)^2} = 0.969c$$

$$d = vt' = \gamma vt = (4.028)(0.969)(2.998 \times 10^8 \text{ m/s})(8.0 \times 10^{-11} \text{ s}) = 9.4 \text{ cm}.$$

29. We begin with Equation (14.10). Since $K \gg mc^2$, this simplifies to $E_{\text{cm}} = \sqrt{2mc^2 K_{\text{lab}}}$.

From the problem statement, we know that $E_{\text{cm}} = 2K$ so $E_{\text{cm}} = 2K = \sqrt{2mc^2 K_{\text{lab}}}$. This can

be rearranged to find $4K^2 = 2mc^2 K_{\text{lab}}$ or $K_{\text{lab}} = \frac{2K^2}{mc^2}$.

30. From the preceding problem, we know that $K_{\text{lab}} = \frac{2K^2}{mc^2} = \frac{2(31 \text{ GeV})^2}{0.938 \text{ GeV}} = 2050 \text{ GeV}$.

$$31. \gamma = \frac{K + E_0}{E_0} = \frac{7000 \text{ GeV} + 0.938 \text{ GeV}}{0.938 \text{ GeV}} = 7464 = \frac{1}{\sqrt{1 - v^2/c^2}};$$

$$v = c\sqrt{1 - 1/\gamma^2} = c\sqrt{1 - 1/7464^2} = 0.999999991c$$

32. The maximum energies result from a head-on collision. Before the collision the proton has energy $E = K + E_0 = 1008.27 \text{ MeV}$. After the collision the proton has energy and momentum E_1, p_1 and the recoiling particle has E_2, p_2 . From conservation of momentum and energy

$$p = \frac{\sqrt{E^2 - E_0^2}}{c} = \frac{\sqrt{(1008.27 \text{ MeV})^2 - (938.27 \text{ MeV})^2}}{c} = 369.13 \text{ MeV}/c = p_1 + p_2; \text{ with}$$

$$E + E_0(d) = 1008.27 \text{ MeV} + 1875.61 \text{ MeV} = 2883.9 \text{ MeV} = E_1 + E_2.$$

Using the energy-momentum invariant $E_0^2 = E^2 - p^2 c^2$ for both the proton and deuteron, these four equations can be solved to yield $p_1 = -119 \text{ MeV}/c$, $p_2 = 488 \text{ MeV}/c$,

$E_1 = 946 \text{ MeV}$, $E_2 = 1938 \text{ MeV}$, so the deuteron's kinetic energy is

$$K = 1938 \text{ MeV} - 1876 \text{ MeV} = 62 \text{ MeV}.$$

Similarly for the triton (t) with $E_0 = 2809 \text{ MeV}$, we can follow the same procedure:

$$p = \frac{\sqrt{E^2 - E_0^2}}{c} = \frac{\sqrt{(1008.27 \text{ MeV})^2 - (938.27 \text{ MeV})^2}}{c} = 369.13 \text{ MeV}/c = p_1 + p_2; \text{ with}$$

$$E + E_0(t) = 1008.27 \text{ MeV} + 2809 \text{ MeV} = 3817 \text{ MeV} = E_1 + E_2.$$

Using the energy-momentum invariant $E_0^2 = E^2 - p^2 c^2$ for both the proton and triton, these two equations can be solved to yield $p_1 = -177 \text{ MeV}/c$, $p_2 = 546 \text{ MeV}/c$, $E_1 = 955 \text{ MeV}$, $E_2 = 2862 \text{ MeV}$, so the triton's kinetic energy is $K = 2862 \text{ MeV} - 2809 \text{ MeV} = 53 \text{ MeV}$.

Note: At this kinetic energy (70 MeV) you will get the same results to two significant figures using a non-relativistic calculation.

33. To conserve baryon number we must produce both a proton and antiproton, so in the cm system we need $E_{\text{cm}} = 4E_0$. We can use Equation (14.10) and solve for K to find

$$K = \frac{E_{\text{cm}}^2 - (m_1 c^2 + m_2 c^2)^2}{2m_2 c^2} = \frac{E_{\text{cm}}^2 - 4(mc^2)^2}{2mc^2} \text{ where we have used } m_1 = m_2 = m. \text{ With}$$

$$E_{\text{cm}} = 4mc^2 \text{ we have } K = \frac{(4mc^2)^2 - 4(mc^2)^2}{2mc^2} = 6mc^2 = 6(938.27 \text{ MeV}) = 5630 \text{ MeV}.$$

$$34. (a) \quad p = \frac{\sqrt{E^2 - E_0^2}}{c} = \frac{\sqrt{(948.27 \text{ MeV})^2 - (938.27 \text{ MeV})^2}}{c} = 137.35 \text{ MeV}/c$$

$$R = \frac{p}{qB} = \frac{137.35 \times 10^6 \text{ eV} \left(1.602 \times 10^{-19} \frac{\text{J}}{\text{eV}} \right)}{(2.998 \times 10^8 \text{ m/s})(e)(1.4 \text{ T})} = 0.327 \text{ m}.$$

$$(b) \text{ From Equation (14.9) we have } f = \frac{eB}{2\pi m} \sqrt{1 - v^2/c^2} = \frac{eB}{2\pi m\gamma} \text{ with}$$

$$\gamma = \frac{E}{E_0} = \frac{948.27 \text{ MeV}}{938.27 \text{ MeV}} = 1.0107 \text{ so}$$

$$f = \frac{(e)(1.4 \text{ T})}{2\pi(938.27 \times 10^6 \text{ eV})(1.0107)} (2.998 \times 10^8 \text{ m/s})^2 = 2.11 \times 10^7 \text{ Hz}.$$

35. With $p = \gamma mv$ we have $f = \frac{v}{2\pi R} = \frac{p}{2\pi \gamma m R}$. With $p = qBR$ we find

$$f = \frac{qBR}{2\pi \gamma m R} = \frac{qB}{2\pi m \gamma} = \frac{qB}{2\pi m} \sqrt{1 - v^2/c^2}.$$

$$36. E_{\text{lab}} = K + m_1 c^2 + m_2 c^2; \quad p_{\text{lab}} c = \sqrt{(K + m_1 c^2)^2 - (m_1 c^2)^2}$$

Using the fact that $E^2 - p^2 c^2$ is invariant, so

$$\begin{aligned} E_{\text{cm}}^2 &= E_{\text{lab}}^2 - (p_{\text{lab}} c)^2 = (K + m_1 c^2 + m_2 c^2)^2 - (K + m_1 c^2)^2 + (m_1 c^2)^2 \\ &= K^2 + (m_1 c^2 + m_2 c^2)^2 + 2K(m_1 c^2 + m_2 c^2) - K^2 - 2K m_1 c^2 - (m_1 c^2)^2 + (m_1 c^2)^2 \\ &= (m_1 c^2 + m_2 c^2)^2 + 2K m_2 c^2 \end{aligned}$$

For the nonrelativistic limit we have

$$E_{\text{cm}} = \sqrt{(m_1 c^2 + m_2 c^2)^2 + 2K m_2 c^2} = (m_1 c^2 + m_2 c^2) \sqrt{1 + \frac{2K m_2 c^2}{(m_1 c^2 + m_2 c^2)^2}}.$$

With $K \ll mc^2$ we can look at the binomial expansion of the square root:

$$\begin{aligned} E_{\text{cm}} &\approx (m_1 c^2 + m_2 c^2) \left(1 + \frac{K m_2 c^2}{(m_1 c^2 + m_2 c^2)^2} \right); \\ K_{\text{cm}} &= E_{\text{cm}} - (m_1 c^2 + m_2 c^2) \approx \frac{K m_2 c^2}{m_1 c^2 + m_2 c^2} = \frac{m_2}{m_1 + m_2} K. \end{aligned}$$

$$37. \text{ As in the preceding problem } E_{\text{cm}} = \sqrt{(m_1 c^2 + m_2 c^2)^2 + 2K m_2 c^2} = \sqrt{(2mc^2)^2 + 2K m_2 c^2}.$$

(a) If $K \ll mc^2$ we neglect K , so $E_{\text{cm}} \approx 2mc^2$.

(b) If $K \gg mc^2$ we neglect the first term and $E_{\text{cm}} \approx \sqrt{2K m_2 c^2} = \sqrt{2K m c^2}$.

In (a) we interpret the result to mean that at very low energies there is no extra energy available (beyond the masses of the two original particles). In (b) we see that the available center of mass energy increases only in proportion to $\sqrt{K}$, thus illustrating the great advantage of colliding beam experiments over fixed target experiments.

$$\begin{aligned} 38. \quad \gamma &= \frac{K + E_0}{E_0} = \frac{50000 \text{ MeV} + 0.511 \text{ MeV}}{0.511 \text{ MeV}} = 97848 = \frac{1}{\sqrt{1 - v^2 / c^2}}; \\ v &= c \sqrt{1 - 1 / \gamma^2} = c \sqrt{1 - 1 / 97848^2} = c (1 - 5.2 \times 10^{-11}) = 0.999999999948c \end{aligned}$$

39. In each case the desired ratio is simply equal to the relativistic factor γ .

$$(a) \quad \gamma = \frac{K + E_0}{E_0} = \frac{12 \text{ MeV} + 938.27 \text{ MeV}}{938.27 \text{ MeV}} = 1.01$$

$$(b) \quad \gamma = \frac{K + E_0}{E_0} = \frac{120 \text{ MeV} + 938.27 \text{ MeV}}{938.27 \text{ MeV}} = 1.13$$

$$(c) \quad \gamma = \frac{K + E_0}{E_0} = \frac{1200 \text{ MeV} + 938.27 \text{ MeV}}{938.27 \text{ MeV}} = 2.28$$

$$40. \quad \gamma = \frac{K + E_0}{E_0} = \frac{33000 \text{ MeV} + 938.27 \text{ MeV}}{938.27 \text{ MeV}} = 36.17$$

Therefore in the lab frame $v \approx c$ and the time for 160,000 revolutions is

$$t = N \frac{2\pi R}{v} \approx (160000) \frac{800 \text{ m}}{2.998 \times 10^8 \text{ m/s}} = 0.427 \text{ s. The statement is accurate.}$$

41. The spaceship of mass $m = 15,000 \text{ kg}$ has to escape the gravity of the sun, mass $M = 2.0 \times 10^{30} \text{ kg}$, starting from a distance of Earth's orbit, $r = 1.5 \times 10^{11} \text{ m}$. The gravitational potential energy goes from $U = -GMm/r$ to zero, a gain of GMm/r , so that is the amount of energy required. That amount of energy equals the annihilation energy $m_a c^2$, where m_a is the total amount of matter and antimatter involved. Thus $GMm/r = m_a c^2$, or

$$m_a = \frac{GMm}{rc^2} = \frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(2.0 \times 10^{30} \text{ kg})(1.5 \times 10^4 \text{ kg})}{(1.5 \times 10^{11} \text{ m})(3.0 \times 10^8 \text{ m/s})^2} = 1.48 \times 10^{-4} \text{ kg}$$

The amount (each) of matter and antimatter is half of this, or $7.4 \times 10^{-5} \text{ kg}$. That means only 74 mg of each! The matter-antimatter reaction is very efficient.

42.

- (a) At that kinetic energy the protons are traveling at nearly the speed of light, so to the stated precision the time required is the circumference divided by the speed of light:

$$t = \frac{2\pi r}{c} = \frac{6300 \text{ m}}{3.0 \times 10^8 \text{ m/s}} = 2.1 \times 10^{-5} \text{ s or } 21 \text{ } \mu\text{s}.$$

- (b) In a colliding-beam experiment, the full energy of both beams is available:
 $1.0 \text{ TeV} + 1.0 \text{ TeV} = 2.0 \text{ TeV}.$

- (c) To obtain the desired energy, 2.0 TeV, Equation 14.10 gives the energy required for a fixed-target experiment. We use $2.0 \text{ TeV} = 2000 \text{ GeV}$, so that we can use 0.938 GeV for the proton's mass. Solving Equation 14.10 for the required energy K ,

$$K = \frac{E_{\text{cm}}^2 - (m_1 c^2 + m_2 c^2)^2}{2m_2 c^2} = \frac{(2000 \text{ GeV})^2 - (2(0.938 \text{ GeV}))^2}{2(0.938 \text{ GeV})} = 2.1 \times 10^6 \text{ GeV}$$

or $2.1 \times 10^{15} \text{ eV}$. That is 300 times the beam energy and 150 times the cm energy of the LHC. It is unlikely that an accelerator this large could ever be built.

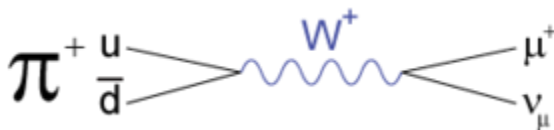
43. (a)



(b)



44. Here is the diagram:



The neutrino is a muon neutrino. The diagram for the first part has the pion as a single solid horizontal line, and the second part is as shown, with the quark structure included.

45.

- (a) With 100 GeV per nucleon, the total energy is $197 \times 100 \text{ GeV} = 19.7 \text{ TeV}$ per gold ion with $A = 197$. For a colliding-beam experiment, the center of mass energy is twice this amount, or 39.4 TeV.
- (b) We can consider each nucleon or the entire nucleus and get the same result. For each nucleon $K = 100 \text{ GeV}$ and $mc^2 = 938 \text{ MeV}$, so

$$\gamma = \frac{K + mc^2}{mc^2} = \frac{100.938 \text{ GeV}}{0.938 \text{ GeV}} = 107.6$$

Then from relativity $\beta = \sqrt{1 - \frac{1}{\gamma^2}} = 0.99996$ or $v = 0.99996c$.

46.

- (a) The quark configuration changes from uud to udd , which means that an up quark has changed into a down quark.
- (b) In this case uds has changed to $uud + \bar{u}d$. Note that strangeness is not conserved. The strange quark transforms via the weak interaction into an up quark and the anti-up down pair (the pion).
- (c) In this case $d\bar{s}$ has changed to $\bar{d}u + \bar{u}d$. Note that as in part (b) strangeness is not conserved. The strange quark transforms via the weak interaction into an anti-up quark and the up anti-down pair (the positive pion).

47. (a) Charge = 0; baryon number = 0; charm = 0; strangeness = 0; $L_e = +1$; spin = 1/2

(b) The unknown can be a single particle. The ν_e satisfies the requirements.

48. None of the decays are allowed.

(a) There is not enough mass-energy in Ξ^0 to produce π^- and Σ^+ and strangeness is not conserved.

(b) Strangeness is not conserved.

(c) There is not enough mass-energy in Ξ^0 to produce K^- and p .

49.

(a) For a stationary target the sum of the rest energies of the products equals the total center of mass energy, so

$$E_{\text{cm}} = \sqrt{2E_0(2E_0 + K)} = (m_p + m_\Lambda + m_K)c^2 \\ = 938 \text{ MeV} + 1116 \text{ MeV} + 494 \text{ MeV} = 2548 \text{ MeV}$$

Rearranging we have $\frac{E_{\text{cm}}^2}{2E_0} = 2E_0 + K$;

$$K = \frac{E_{\text{cm}}^2}{2E_0} - 2E_0 = \frac{(2548 \text{ MeV})^2}{2(938 \text{ MeV})} - 2(938 \text{ MeV}) = 1585 \text{ MeV}.$$

(b) In a colliding beam experiment the total momentum is zero, and we have by conservation of energy $2E_0 + 2K = E_0 + (m_\Lambda + m_K)c^2$;

$$K = \frac{-E_0 + (m_\Lambda + m_K)c^2}{2} = \frac{-938 \text{ MeV} + 1116 \text{ MeV} + 494 \text{ MeV}}{2} = 336 \text{ MeV}.$$

50.

(a) baryon number and μ lepton number are not conserved, as well as mass-energy

(b) allowed if neutrinos are added to conserve lepton number

(c) strangeness is not conserved

(d) baryon number is not conserved

51.

(a) allowed

(b) ν_e should be $\bar{\nu}_e$ to conserve electron lepton number

(c) strangeness is not conserved

(d) strangeness is not conserved in a strong interaction

(e) allowed

52.

- (a) baryon number and electron lepton number not conserved
- (b) not allowed; charge is not conserved
- (c) allowed
- (d) allowed

53.

- (a) strangeness is not conserved
- (b) charge is not conserved
- (c) baryon number is not conserved
- (d) strangeness is not conserved

54.

- (a) $\lambda = h / p$ and $p = \sqrt{2mK}$, so $\lambda = \frac{h}{\sqrt{2mK}}$
- (b) $p \approx E / c \approx K / c$, so $\lambda = \frac{hc}{K}$

55.

- (a) From Equation (14.10) we have $E_{\text{cm}} = \sqrt{(m_1 c^2 + m_2 c^2)^2 + 2K m_2 c^2}$ and with $m_1 = m_2 = m$ and $K \gg mc^2$, we have

$$E_{\text{cm}} \approx \sqrt{2Kmc^2} = \sqrt{2(0.938\text{GeV})(7000\text{ GeV})} = 114.5\text{ GeV}.$$
- (b) For colliding beams the available energy is the sum of the two beam energies, or 14.0 TeV. This is an improvement over the fixed-target result by a factor of

$$\frac{14000\text{ GeV}}{114.5\text{ GeV}} = 122.$$

Chapter 15

1. From Newton's second law we have for a pendulum of length L :

$$F = m_G g \sin \theta = m_I a = m_I L \frac{d^2 \theta}{dt^2}; \quad \frac{d^2 \theta}{dt^2} = \frac{m_G g}{m_I L} \sin \theta \approx \frac{m_G g}{m_I L} \theta, \text{ where we have made the}$$

small angle approximation $\sin \theta \approx \theta$. This is a simple harmonic oscillator equation with solution $\theta = \theta_0 \cos(\omega t)$ where θ_0 is the amplitude and the angular frequency is

$$\omega = \sqrt{\frac{m_G g}{m_I L}}. \text{ The period of oscillation is } T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m_I L}{m_G g}}. \text{ Therefore two masses}$$

with different ratios m_I / m_G will have different small-amplitude periods.

2. Beginning with Equation (15.3)

$$\Delta f = \frac{gHf}{c^2} = \frac{(9.80 \text{ m/s}^2)(4.80 \times 10^5 \text{ m})(10^8 \text{ s}^{-1})}{(2.998 \times 10^8 \text{ m/s})^2} = 5.2 \times 10^{-3} \text{ Hz}.$$

3. Beginning with Equation (15.4)

$$\frac{\Delta f}{f} = -\frac{GM}{c^2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = -\frac{GM}{r_1 r_2 c^2} (r_2 - r_1) = \frac{GM}{r_1 r_2 c^2} (r_1 - r_2). \text{ Use } r_1 - r_2 = H \text{ and let } r_1 \approx r_2 = r.$$

From classical mechanics $g = GM / r^2$, so $\frac{\Delta f}{f} = \frac{gH}{c^2}$.

4. $\frac{\Delta T}{T} = -\frac{GM}{c^2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$ We use $r_2 = 6378 \text{ km}$ and $r_1 = (6378 + 10) \text{ km}$.

$$\begin{aligned} \frac{\Delta T}{T} &= -\frac{(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(5.98 \times 10^{24} \text{ kg})}{(2.998 \times 10^8 \text{ m/s})^2} \left(\frac{1}{6388 \times 10^3 \text{ m}} - \frac{1}{6378 \times 10^3 \text{ m}} \right) \\ &= 1.09 \times 10^{-12} \end{aligned}$$

which is the same as in the example, to three significant digits.

5. The distance d is the sum of the radii of the earth's orbit and Venus's orbit (assuming circular orbits).

$$d = 149.6 \times 10^9 \text{ m} + 108.2 \times 10^9 \text{ m} = 257.8 \times 10^9 \text{ m}$$

$$\text{The round-trip time is } t = \frac{2d}{c} = \frac{2(257.8 \times 10^9 \text{ m})}{2.998 \times 10^8 \text{ m/s}} = 1719.8 \text{ s}.$$

Using Shapiro's measurement of $200 \mu\text{s}$, the percent change is therefore

$$\frac{200 \times 10^{-6} \text{ s}}{1719.8 \text{ s}} (100\%) = 1.16 \times 10^{-5} \%.$$

6. Using the mass and the radius of the neutron star and given that the detection is at a very large distance,

$$\frac{\Delta f}{f} = -\frac{GM}{c^2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = \frac{GM}{rc^2} = \frac{(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(5.8 \times 10^{30} \text{ kg})}{(1.0 \times 10^4 \text{ m})(2.998 \times 10^8 \text{ m/s})^2} = 4.31 \times 10^{-1}$$

The wavelength is affected by the same factor, so the redshift at the given wavelength is $\Delta\lambda = (4.31 \times 10^{-1})(550 \text{ nm}) = 240 \text{ nm}$, where we have rounded to the two significant figures given in the problem. Thus the new wavelength is $550 \text{ nm} + 240 \text{ nm} = 790 \text{ nm}$, which is above the visible range in the infrared.

7. Using the mass and radius of the sun

$$\frac{\Delta f}{f} = \frac{GM}{rc^2} = \frac{(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(1.99 \times 10^{30} \text{ kg})}{(6.96 \times 10^8 \text{ m})(2.998 \times 10^8 \text{ m/s})^2} = 2.123 \times 10^{-6}$$

The redshift of the wavelength is affected by the same factor, so the redshift at the two wavelengths equals: $\Delta\lambda = (2.123 \times 10^{-6})(400 \text{ nm}) = 8.49 \times 10^{-4} \text{ nm}$ and

$$\Delta\lambda = (2.123 \times 10^{-6})(700 \text{ nm}) = 1.49 \times 10^{-3} \text{ nm}.$$

8. We can assume that g is constant over this distance. We find the frequency of the gamma

$$\text{ray: } f = \frac{E}{h} = \frac{14.4 \times 10^3 \text{ eV}}{4.136 \times 10^{-15} \text{ eV} \cdot \text{s}} = 3.48 \times 10^{18} \text{ Hz}$$

We can use Equation (15.3) to find:

$$\frac{\Delta f}{f} = \frac{gH}{c^2} = \frac{(9.80 \text{ m/s}^2)(381 \text{ m})}{(2.998 \times 10^8 \text{ m/s})^2} = 4.15 \times 10^{-14} \text{ which equals a } 4.15 \times 10^{-12} \%$$

change. The change in frequency is $\Delta f = 4.15 \times 10^{-14} (3.48 \times 10^{18} \text{ Hz}) = 145 \text{ kHz}$.

9. We assume that g is constant over this distance. Using $E = hf$ we find:

$$\Delta f = \frac{gHf}{c^2} = \frac{gHE}{c^2 h} = \frac{(9.80 \text{ m/s}^2)(22.5 \text{ m})(14.4 \times 10^3 \text{ eV})}{(2.998 \times 10^8 \text{ m/s})^2 (4.136 \times 10^{-15} \text{ eV} \cdot \text{s})} = 8541 \text{ Hz}$$

From the previous problem we know the frequency of the gamma ray is $3.48 \times 10^{18} \text{ Hz}$ so the percentage change is $\frac{8541 \text{ Hz}}{3.48 \times 10^{18} \text{ Hz}} (100\%) = 2.45 \times 10^{-13} \%$.

10. We can ignore any blueshift due to each planet's gravity, which is small compared with the sun's gravity.

(a) Mercury's orbit is highly eccentric, but for this problem it will be good enough to use the mean orbital radius (semi-major axis) 5.79×10^{10} m. Then by Equation 15.4,

$$\begin{aligned}\frac{\Delta f}{f} &= -\frac{GM}{c^2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \\ &= -\frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(1.99 \times 10^{30} \text{ kg})}{(3.00 \times 10^8 \text{ m/s})^2} \left(\frac{1}{6.96 \times 10^8 \text{ m}} - \frac{1}{5.79 \times 10^{10} \text{ m}} \right)\end{aligned}$$

$$\frac{\Delta f}{f} = -2.09 \times 10^{-6}$$

(b) Now the distance at which the signal is received is Earth's mean orbital radius 1.50×10^{11} m.

$$\frac{\Delta f}{f} = -\frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(1.99 \times 10^{30} \text{ kg})}{(3.00 \times 10^8 \text{ m/s})^2} \left(\frac{1}{6.96 \times 10^8 \text{ m}} - \frac{1}{1.50 \times 10^{11} \text{ m}} \right)$$

$$\frac{\Delta f}{f} = -2.11 \times 10^{-6}$$

As we might expect, the added distance from Mercury to Earth contributes little to the redshift.

11. We need to rewrite the frequency shift in Equation 15.4 into a wavelength shift.

$f = c/\lambda$, so $df/d\lambda = -c/\lambda^2$. For small changes $df \approx \Delta f$ and $d\lambda \approx \Delta\lambda$, so

$$\Delta f = -\frac{c}{\lambda} \frac{\Delta\lambda}{\lambda} = -f \frac{\Delta\lambda}{\lambda} \text{ or } \frac{\Delta\lambda}{\lambda} = -\frac{\Delta f}{f}. \text{ Thus the laser's wavelength shift is}$$

$$\frac{\Delta\lambda}{\lambda} = -\frac{\Delta f}{f} = \frac{GM}{c^2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right). \text{ In this case } r_1 \text{ is large, so we may use } 1/r_1 = 0, \text{ and so}$$

$$\frac{\Delta\lambda}{\lambda} = -\frac{GM}{c^2 r_2} = -\frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(4.5 \times 10^{30} \text{ kg})}{(3.00 \times 10^8 \text{ m/s})^2 (1.2 \times 10^4 \text{ m})} = -0.278$$

Therefore $\Delta\lambda = -0.278(632.8 \text{ nm}) = -175.9 \text{ nm}$ and the new wavelength is

$\lambda + \Delta\lambda = 632.8 \text{ nm} - 175.9 \text{ nm} = 456.9 \text{ nm}$ or 460 nm to two significant figures.

$$12. r_s = \frac{2GM}{c^2} = \frac{2(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(7.35 \times 10^{22} \text{ kg})}{(2.998 \times 10^8 \text{ m/s})^2} = 1.09 \times 10^{-4} \text{ m}$$

$$13. r_s = \frac{2GM}{c^2} = \frac{2(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(1.90 \times 10^{27} \text{ kg})}{(2.998 \times 10^8 \text{ m/s})^2} = 2.82 \text{ m}$$

14.

(a) From Equation (15.7) we have

$$T = \frac{\hbar c^3}{8\pi kGM} = \frac{(1.0546 \times 10^{-34} \text{ J} \cdot \text{s})(2.998 \times 10^8 \text{ m/s})^3}{8\pi(1.381 \times 10^{-23} \text{ J/K})(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(1.99 \times 10^{30} \text{ kg})}$$

$$= 6.17 \times 10^{-8} \text{ K}$$

(b) Using Equation (15.7)

$$T = \frac{\hbar c^3}{8\pi kGM} = \frac{(1.05 \times 10^{-34} \text{ J} \cdot \text{s})(3.0 \times 10^8 \text{ m/s})^3}{8\pi(1.38 \times 10^{-23} \text{ J/K})(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(6 \times 10^9)(2.0 \times 10^{30} \text{ kg})}$$

$$= 1.0 \times 10^{-17} \text{ K}$$

Such low temperatures, especially for part (b) would make the object extremely difficult to observe.

15. Rearranging Equation (15.7), we have

$$M = \frac{\hbar c^3}{8\pi kGT} = \frac{(1.0546 \times 10^{-34} \text{ J} \cdot \text{s})(2.998 \times 10^8 \text{ m/s})^3}{8\pi(1.381 \times 10^{-23} \text{ J/K})(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(293 \text{ K})} = 4.19 \times 10^{20} \text{ kg}$$

which is about $\frac{4.19 \times 10^{20} \text{ kg}}{1.99 \times 10^{30} \text{ kg}} = 2.11 \times 10^{-10}$ solar masses.

$$r_s = \frac{2GM}{c^2} = \frac{2(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(4.19 \times 10^{20} \text{ kg})}{(2.998 \times 10^8 \text{ m/s})^2} = 6.22 \times 10^{-7} \text{ m}$$

16.

(a) The mass is $1 \times 10^9 (M_{\text{Sun}}) = 1 \times 10^9 (1.99 \times 10^{30} \text{ kg}) = 1.99 \times 10^{39} \text{ kg}$. We use Equation (15.5) to determine the Schwarzschild radius.

$$r_s = \frac{2GM}{c^2} = \frac{2(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(1.99 \times 10^{30} \text{ kg})}{(2.998 \times 10^8 \text{ m/s})^2} = 2.95 \times 10^{12} \text{ m}$$

Neptune's orbit has a semi-major axis of $4.50 \times 10^{12} \text{ m}$, so this black hole would be smaller than our solar system.

- (b) We use the result of Example 15.3, with the value of α from the example (or from problem 16 below).

$$t = \frac{M_0^3}{3\alpha} = \frac{(1.99 \times 10^{39})^3}{3(3.965 \times 10^{15} \text{ kg}^3/\text{s})} = 6.63 \times 10^{101} \text{ s}$$

This is much, much greater than the age of our universe!

17. (a) We use the results of Example 15.3 that give the time in terms of the mass. Rearranging the equation, using the age of the universe as 13.7 billion years, and using the value of α from Problem 16, we have

$$M_0 = (3\alpha t)^{1/3} = \left[3(3.965 \times 10^{15} \text{ kg}^3/\text{s}) \left(13.7 \times 10^9 \text{ y} \frac{3.156 \times 10^7 \text{ s}}{1 \text{ y}} \right) \right]^{1/3} = 1.73 \times 10^{11} \text{ kg}.$$

- (b) Current evidence for the smallest black holes require a mass of about 5 to 20 times the mass of the Sun. The mass from part (a) is only $M_0 \approx 10^{-19} M_{\text{Sun}}$ which is too small to form a black hole.

18. From Equation (15.10) we see

$$\begin{aligned} \alpha &= \frac{2\sigma\hbar^4 c^6}{(8\pi)^3 k^4 G^2} = \frac{2(5.6705 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4})(1.0546 \times 10^{-34} \text{ J} \cdot \text{s})^4 (2.9979 \times 10^8 \text{ m/s})^6}{(8\pi)^3 (1.3807 \times 10^{-23} \text{ J/K})^4 (6.6726 \times 10^{-11} \text{ N} \cdot \text{m}^2 \cdot \text{kg}^{-2})^2} \\ &= 3.965 \times 10^{15} \text{ kg}^3/\text{s} \end{aligned}$$

19. We could use Equation 15.9, but it is simpler to start at Equation 15.10 and use the fact the energy E released by the loss of mass M is $E = Mc^2$. Then by Equation 15.10

$$\frac{dE}{dt} = c^2 \frac{dM}{dt} = -\frac{c^2 \alpha}{M^2}$$

where $\alpha = 3.96 \times 10^{15} \text{ kg}^3/\text{s}$ was given in Example 15.3. With energy being lost at a rate (negative, for a loss) $-4.2 \times 10^{15} \text{ J/s}$, we solve for M to find

$$M = \left(-\frac{c^2 \alpha}{dE/dt} \right)^{1/2} = \left[-\frac{(3.0 \times 10^8 \text{ m/s})^2 (3.96 \times 10^{15} \text{ kg}^3/\text{s})}{-4.2 \times 10^{15} \text{ J/s}} \right]^{1/2} = 2.9 \times 10^8 \text{ kg}.$$

This is a fairly modest mass—much smaller than a planet.

20.

$$(a) \quad r_s = \frac{2GM}{c^2} = \frac{2(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(7.3 \times 10^{22} \text{ kg})}{(3.0 \times 10^8 \text{ m/s})^2} = 1.1 \times 10^{-4} \text{ m} = 0.11 \text{ mm}$$

$$(b) \quad T = \frac{\hbar c^3}{8\pi kGM} = \frac{(1.05 \times 10^{-34} \text{ J} \cdot \text{s})(3.0 \times 10^8 \text{ m/s})^3}{8\pi(1.38 \times 10^{-23} \text{ J/K})(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(7.3 \times 10^{22} \text{ kg})} = 1.7 \text{ K}$$

$$(c) \quad U = -\frac{GMm}{R} = -\frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(7.3 \times 10^{22} \text{ kg})(6.0 \times 10^{24} \text{ kg})}{6.4 \times 10^6 \text{ m}} = -4.6 \times 10^{30} \text{ J}$$

This is a huge amount of gravitational energy.

21. We use Equation (15.4) and find

$$\begin{aligned} \frac{\Delta f}{f} &= -\frac{GM}{c^2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \\ &= -\frac{(6.6726 \times 10^{-11} \text{ N} \cdot \text{m}^2 \cdot \text{kg}^{-2})(5.976 \times 10^{24} \text{ kg})}{(2.998 \times 10^8 \text{ m/s})^2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \\ &= -(4.439 \times 10^{-3} \text{ m}) \left[\frac{1}{(2.02 \times 10^7 + 6.37 \times 10^6) \text{ m}} - \frac{1}{6.37 \times 10^6 \text{ m}} \right] \\ &= 5.30 \times 10^{-10}. \end{aligned}$$

The frequency change is then $\Delta f = (5.30 \times 10^{-10})(1575.42 \text{ MHz}) = 0.834 \text{ Hz}$. This is a small, but significant change with the precision required of the GPS system.

22. Set the change in the photon's energy equal to the change in gravitational potential

$$\text{energy: } \Delta E = h\Delta f = -\frac{GMm}{r_1} - \left(-\frac{GMm}{r_2} \right) = -GMm \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \text{ where } M \text{ is the mass of the}$$

earth and m is the equivalent mass of the photon. Now $m = E/c^2 = hf/c^2$, so

$$h\Delta f = -\frac{GMhf}{c^2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \text{ and } \frac{\Delta f}{f} = -\frac{GM}{c^2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right).$$

23. The problem statement advises us to assume that g is constant. Therefore we can use

Equation (15.3) to determine the change in frequency. We have $\frac{\Delta f}{f} = \frac{gH}{c^2}$ so

$$\Delta f = \left(\frac{gH}{c^2} \right) f = \left(\frac{(9.80 \text{ m/s}^2)(3.52 \times 10^5 \text{ m})}{(2.998 \times 10^8 \text{ m/s})^2} \right) (259.7 \times 10^6 \text{ Hz}) = 9.97 \times 10^{-3} \text{ Hz}.$$

We can obtain a more precise answer using the formula from the previous problem and recalling that the earth's radius is 6378 km,

$$\begin{aligned}\frac{\Delta f}{f} &= -\frac{GM}{c^2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \\ &= -\frac{(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(5.976 \times 10^{24} \text{ kg})}{(2.998 \times 10^8 \text{ m/s})^2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \\ &= -(4.437 \times 10^{-3} \text{ m}) \left(\frac{1}{6.730 \times 10^6 \text{ m}} - \frac{1}{6.378 \times 10^6 \text{ m}} \right) \\ &= 3.639 \times 10^{-11}.\end{aligned}$$

Therefore $\Delta f = (3.639 \times 10^{-11})(259.7 \times 10^6 \text{ Hz}) = 9.45 \times 10^{-3} \text{ Hz}$, which is about a 5% difference.

24. $g = GM / r^2$ which can be differentiated to give $dg = -\frac{2GM}{r^3} dr$. For a small change let

$dg \approx |dg|$ and $dr \approx \Delta r = 3 \text{ m}$. Also notice that the distance from the center of the earth is $6.378 \times 10^6 \text{ m} + 3.50 \times 10^5 \text{ m} = 6.728 \times 10^6 \text{ m}$. Then

$$\Delta g \approx \frac{2(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(5.98 \times 10^{24} \text{ kg})}{(6.728 \times 10^6 \text{ m})^3} (3 \text{ m}) = 8 \times 10^{-6} \text{ m/s}^2.$$

This is about $10^{-6} g$, so it is a very small effect.

25. If we use $\lambda = h / mc$ for a relativistic particle of mass m , we have $\lambda = \frac{h}{mc} = \pi r_s = \frac{2\pi Gm}{c^2}$.

Solving for m we have

$$m = \sqrt{\frac{hc}{2\pi G}} = \sqrt{\frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(2.998 \times 10^8 \text{ m/s})}{2\pi(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})}} = 2.18 \times 10^{-8} \text{ kg}.$$

The Planck energy is

$$E_{\text{Pl}} = mc^2 = (2.18 \times 10^{-8} \text{ kg})(2.998 \times 10^8 \text{ m/s})^2 = 1.96 \times 10^9 \text{ J} = 1.22 \times 10^{28} \text{ eV}.$$

- 26.

(a) The combination of G , h , and c that has the right units is

$$\lambda_{\text{Pl}} = \sqrt{\frac{Gh}{c^3}} = \sqrt{\frac{(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(6.626 \times 10^{-34} \text{ J} \cdot \text{s})}{(2.998 \times 10^8 \text{ m/s})^3}} = 4.05 \times 10^{-35} \text{ m}.$$

- (b) $\lambda = \frac{h}{mc} = \frac{hc}{mc^2} = \frac{1240 \times 10^{-9} \text{ eV} \cdot \text{m}}{1.22 \times 10^{28} \text{ eV}} = 1.02 \times 10^{-34} \text{ m}$ which is the same order of magnitude as (a) and close to the known value $1.62 \times 10^{-35} \text{ m}$.

27. The combination of constants that gives units of time is

$$t_{\text{pl}} = \sqrt{\frac{Gh}{c^5}} = \sqrt{\frac{(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(6.626 \times 10^{-34} \text{ J} \cdot \text{s})}{(2.998 \times 10^8 \text{ m/s})^5}} = 1.35 \times 10^{-43} \text{ s}.$$

The time for light to travel the Planck length is $t = \frac{\lambda_{\text{pl}}}{c} = \sqrt{\frac{Gh}{c^5}} = 1.35 \times 10^{-43} \text{ s}$, as we found above.

28. As in Problem 23,

$$\begin{aligned} \frac{\Delta f}{f} &= -\frac{GM}{c^2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \\ &= -\frac{(6.673 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(5.976 \times 10^{24} \text{ kg})}{(2.998 \times 10^8 \text{ m/s})^2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \\ &= -(4.439 \times 10^{-3} \text{ m}) \left(\frac{1}{3.587 \times 10^7 \text{ m} + 6.378 \times 10^6 \text{ m}} - \frac{1}{6.378 \times 10^6 \text{ m}} \right) \\ &= 5.91 \times 10^{-10}. \end{aligned}$$

Therefore $\Delta f = (5.91 \times 10^{-10})(2.0 \times 10^9 \text{ Hz}) = 1.18 \text{ Hz}$.

29. Rewrite the entropy in terms of U using $M = U/c^2$: $S = \frac{8\pi^2 GU^2 k}{hc^5}$

Then $\frac{1}{T} = \frac{\partial S}{\partial U} = \frac{16\pi^2 GUk}{hc^5}$; solving for T and using $\hbar = h/2\pi$, we find

$T = \frac{hc^5}{16\pi^2 GUk} = \frac{\hbar c^5}{8\pi GUk} = \frac{\hbar c^3}{8\pi GMk}$, where in the last step we again used $U = Mc^2$. This is the desired result shown in Equation (15.7).

30. We assume a constant downward acceleration g . From classical mechanics, the magnitude of the fall assuming only an initial horizontal velocity is $\Delta y = \frac{1}{2}gt^2$, where the time of flight is $t = L/c$. Making this substitution:

$$\Delta y = \frac{1}{2}gt^2 = \frac{1}{2}g\left(\frac{L}{c}\right)^2 = \frac{gL^2}{2c^2}.$$

The distance L between New York and Los Angeles is roughly $4000 \text{ km} = 4 \times 10^6 \text{ m}$.

Inserting numerical values:
$$\Delta y = \frac{(9.8 \text{ m/s}^2)(4.0 \times 10^6 \text{ m})^2}{2(3 \times 10^8 \text{ m/s})^2} = 8.7 \times 10^{-4} \text{ m} = 0.87 \text{ mm}.$$

Chapter 16

1. $1 \text{ pc} = \frac{1 \text{ au}}{\tan 1''} (1.496 \times 10^{11} \text{ m/au}) = 3.086 \times 10^{16} \text{ m}$
 $1 \text{ ly} = (2.9979 \times 10^8 \text{ m/s})(365.25 \text{ d/y})(86400 \text{ s/d}) = 9.461 \times 10^{15} \text{ m}$
 $1 \text{ pc} = \frac{3.086 \times 10^{16} \text{ m}}{9.461 \times 10^{15} \text{ m/ly}} = 3.26 \text{ ly}$
2. As in Example 16.3 we have $7.0 = \exp(\Delta mc^2 / kT)$ so $\ln 7.0 = \Delta mc^2 / kT$. Then

$$T = \frac{\Delta mc^2}{k \ln 7.0} = \frac{939.56563 \text{ MeV} - 938.27231 \text{ MeV}}{(8.617 \times 10^{-11} \text{ MeV/K})(\ln 7.0)} = 7.71 \times 10^9 \text{ K}.$$

3. Using $mc^2 = 135 \text{ MeV} = kT$: $T = \frac{135 \text{ MeV}}{8.617 \times 10^{-11} \text{ MeV/K}} = 1.57 \times 10^{12} \text{ K}.$

From Figure 16.6 the time associated with this temperature is about 10^{-4} s .

$$4. \frac{\text{number of d}}{\text{number of u}} = \exp\left(\frac{10 \text{ MeV} - 6 \text{ MeV}}{(8.617 \times 10^{-11} \text{ MeV/K})(10^{14} \text{ K})}\right) = 1.00;$$

$$\frac{\text{number of u}}{\text{number of c}} = \exp\left(\frac{1300 \text{ MeV} - 10 \text{ MeV}}{(8.617 \times 10^{-11} \text{ MeV/K})(10^{14} \text{ K})}\right) = 1.16;$$

$$\frac{\text{number of u}}{\text{number of s}} = \exp\left(\frac{115 \text{ MeV} - 10 \text{ MeV}}{(8.617 \times 10^{-11} \text{ MeV/K})(10^{14} \text{ K})}\right) = 1.01;$$

$$\text{Similarly, } \frac{\text{number of d}}{\text{number of s}} = \exp\left(\frac{115 \text{ MeV} - 6 \text{ MeV}}{(8.617 \times 10^{-11} \text{ MeV/K})(10^{14} \text{ K})}\right) = 1.01;$$

$$\frac{\text{number of u or d}}{\text{number of c}} = \exp\left(\frac{1300 \text{ MeV} - 6 \text{ MeV}}{(8.617 \times 10^{-11} \text{ MeV/K})(10^{14} \text{ K})}\right) = 1.16;$$

$$\frac{\text{number of u or d}}{\text{number of t}} = \exp\left(\frac{174000 \text{ MeV} - 6 \text{ MeV}}{(8.617 \times 10^{-11} \text{ MeV/K})(10^{14} \text{ K})}\right) = 5.9 \times 10^8;$$

$$\frac{\text{number of u or d}}{\text{number of b}} = \exp\left(\frac{4250 \text{ MeV} - 6 \text{ MeV}}{(8.617 \times 10^{-11} \text{ MeV/K})(10^{14} \text{ K})}\right) = 1.64;$$

$$\frac{\text{number of s}}{\text{number of c}} = \exp\left(\frac{1300 \text{ MeV} - 115 \text{ MeV}}{(8.617 \times 10^{-11} \text{ MeV/K})(10^{14} \text{ K})}\right) = 1.15 ;$$

$$\frac{\text{number of s}}{\text{number of t}} = \exp\left(\frac{174000 \text{ MeV} - 115 \text{ MeV}}{(8.617 \times 10^{-11} \text{ MeV/K})(10^{14} \text{ K})}\right) = 5.8 \times 10^8;$$

$$\frac{\text{number of s}}{\text{number of b}} = \exp\left(\frac{4250 \text{ MeV} - 115 \text{ MeV}}{(8.617 \times 10^{-11} \text{ MeV/K})(10^{14} \text{ K})}\right) = 1.62;$$

$$\frac{\text{number of c}}{\text{number of t}} = \exp\left(\frac{174000 \text{ MeV} - 1300 \text{ MeV}}{(8.617 \times 10^{-11} \text{ MeV/K})(10^{14} \text{ K})}\right) = 5.1 \times 10^8;$$

$$\frac{\text{number of c}}{\text{number of b}} = \exp\left(\frac{4250 \text{ MeV} - 1300 \text{ MeV}}{(8.617 \times 10^{-11} \text{ MeV/K})(10^{14} \text{ K})}\right) = 1.41;$$

$$\frac{\text{number of b}}{\text{number of t}} = \exp\left(\frac{174000 \text{ MeV} - 4250 \text{ MeV}}{(8.617 \times 10^{-11} \text{ MeV/K})(10^{14} \text{ K})}\right) = 3.6 \times 10^8.$$

5. $mc^2 = kT$ so we have:

$$\text{electron: } T = \frac{mc^2}{k} = \frac{0.5110 \text{ MeV}}{8.617 \times 10^{-11} \text{ MeV/K}} = 5.93 \times 10^9 \text{ K};$$

$$\text{muon: } T = \frac{mc^2}{k} = \frac{105.66 \text{ MeV}}{8.617 \times 10^{-11} \text{ MeV/K}} = 1.23 \times 10^{12} \text{ K}$$

6. As in the preceding problem we have for the $2.2 \text{ eV} / c^2$ mass,

$$T = \frac{mc^2}{k} = \frac{2.2 \text{ eV}}{8.617 \times 10^{-5} \text{ eV/K}} = 2.55 \times 10^4 \text{ K}.$$

Using Figure 16.6, this temperature corresponds to a time of approximately 10^8 seconds.

If the mass of the neutrino is $1.0 \times 10^{-4} \text{ eV} / c^2$ then

$$T = \frac{mc^2}{k} = \frac{10^{-4} \text{ eV}}{8.617 \times 10^{-5} \text{ eV/K}} = 1.16 \text{ K}.$$

Note this answer is lower than the present cosmic background temperature!

7. The π^+ ($E_0 = 140 \text{ MeV}$) is more massive than the π^0 ($E_0 = 135 \text{ MeV}$), so the π^+ would have been formed for a longer time. With $\Delta mc^2 = k\Delta T$ we have

$$\Delta T = \frac{\Delta mc^2}{k} = \frac{5 \text{ MeV}}{8.617 \times 10^{-11} \text{ MeV/K}} = 5.80 \times 10^{10} \text{ K}.$$

8. Set the deuteron binding energy 2.22 MeV equal to kT :

$$T = \frac{2.22 \text{ MeV}}{k} = \frac{2.22 \text{ MeV}}{8.617 \times 10^{-11} \text{ MeV/K}} = 2.58 \times 10^{10} \text{ K}.$$

9. Set the hydrogen binding energy 13.6 eV equal to kT :

$$T = \frac{13.6 \text{ eV}}{k} = \frac{13.6 \text{ eV}}{8.617 \times 10^{-5} \text{ eV/K}} = 1.58 \times 10^5 \text{ K}.$$

10. Beginning with Equation (16.15), $\frac{0.3GN^2m^2}{V^{4/3}} = \frac{3.9\hbar^2N^{5/3}}{2mV^{5/3}}$, one can solve for V and find:

$$V^{1/3} = \frac{3.9\hbar^2}{0.6m^3N^{1/3}G} = \frac{6.5\hbar^2}{N^{1/3}m^3G}.$$

11. Notice that Example 16.5 indicates a neutron star with a mass of two solar masses.

$$\rho = \frac{\text{mass}}{\text{volume}} = \frac{2M}{\frac{4}{3}\pi R^3} = \frac{3M}{2\pi R^3} \text{ where } M = 1.99 \times 10^{30} \text{ kg (mass of sun) and } R = 11 \text{ km}.$$

$$\rho = \frac{3M}{2\pi R^3} = \frac{3(1.99 \times 10^{30} \text{ kg})}{2\pi(11 \times 10^3 \text{ m})^3} = 7.14 \times 10^{17} \text{ kg/m}^3$$

For the nucleon we have $\rho = \frac{m_p}{\frac{4}{3}\pi r_0^3} = \frac{3(1.67 \times 10^{-27} \text{ kg})}{4\pi(1.2 \times 10^{-15} \text{ m})^3} = 2.31 \times 10^{17} \text{ kg/m}^3$ and the

neutron star is about three times as dense.

12. We see from Equation (16.16) that $V^{1/3}$ is proportional to $N^{-1/3}$. However, $V^{1/3}$ is proportional to R , so R is proportional to $N^{-1/3}$. The reason this makes sense is that the gravitational pressure increases in proportion to N^2 [see Equation (16.12)] while the neutron pressure increase in proportion to $N^{5/3}$ [see Equation (16.14)]. The gravitational pressure increases more rapidly than the neutron pressure so that the radius of the neutron star decreases as the number of neutrons increases.

13. (a) Beginning with Equation (16.12):

$$\begin{aligned} P &= 0.3G \frac{(Nm)^2}{V^{4/3}} = 0.3G \frac{M^2}{\left(\frac{4}{3}\pi R^3\right)^{4/3}} \\ &= 0.3(6.67 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2}) \frac{(1.99 \times 10^{30} \text{ kg})^2}{\left(\frac{4}{3}\pi(6.96 \times 10^8 \text{ m})^3\right)^{4/3}} = 5.00 \times 10^{13} \text{ N/m}^2. \end{aligned}$$

(b)

$$P = 0.3G \frac{M^2}{\left(\frac{4}{3}\pi R^3\right)^{4/3}} = 0.3(6.67 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2}) \frac{4(1.99 \times 10^{30} \text{ kg})^2}{\left(\frac{4}{3}\pi (1.1 \times 10^4 \text{ m})^3\right)^{4/3}}$$

$$= 3.21 \times 10^{33} \text{ N/m}^2.$$

14. We begin with Equation (16.1).

$$v = HR = \frac{71 \text{ km/s}}{\text{Mpc}} (4.0 \times 10^9 \text{ ly}) \left(\frac{1 \text{ Mpc}}{3.26 \times 10^6 \text{ ly}} \right) = 8.71 \times 10^4 \text{ km/s} = 0.29c.$$

$$15. \quad R = \frac{v}{H} = \frac{15000 \text{ km/s}}{71 \text{ km/s/Mpc}} = 211 \text{ Mpc or about } 689 \text{ Mly}.$$

16.

(a) From Equation (16.18) we see that for a redshift of 3.8 we have $\Delta\lambda / \lambda_0 = 3.8$, so

$$3.8 = \sqrt{\frac{1+\beta}{1-\beta}} - 1 \text{ which can be solved to find } \beta = 0.917. \text{ Then}$$

$$R = \frac{v}{H} = \frac{0.917(299790 \text{ km/s})}{71 \text{ km/s/Mpc}} = 3872 \text{ Mpc}.$$

$$(b) \quad 3872 \text{ Mpc} \left(\frac{3.26 \text{ Mly}}{1 \text{ Mpc}} \right) = 12.6 \text{ Gly}.$$

17. The redshift is $z = \Delta\lambda / \lambda_0 = 3.8$, so $\Delta\lambda = z\lambda_0 = 3.8(21 \text{ cm}) = 80 \text{ cm}$ and the new wavelength is $21 \text{ cm} + 80 \text{ cm} = 101 \text{ cm}$. There could be another spectral line at this wavelength, but it is unlikely to lead to confusion. As Figure 16.1 shows, the pattern of spectral lines shifts as a group, with each line experiencing the same redshift factor z . This makes it possible to identify the 21-cm line within the pattern of the entire spectrum.

18. Beginning with Equation (16.18), the speed is given by $z = 6.42 = \sqrt{\frac{1+\beta}{1-\beta}} - 1$ which can be solved to find $\beta = 0.964$ or $v = 0.964c$. The distance is then given by Hubble's law

$$R = \frac{v}{H} = \frac{0.964(299790 \text{ km/s})}{71 \text{ km/s/Mpc}} = 4070 \text{ Mpc}.$$

$$R = 4070 \text{ Mpc} \times \frac{3.26 \text{ ly}}{\text{Mpc}} = 13.3 \text{ Gly}.$$

19.

(a) From Equation (16.18), for a redshift of 8.6 we have $\Delta\lambda / \lambda_0 = 8.6$, so

$$8.6 = \sqrt{\frac{1+\beta}{1-\beta}} - 1 \text{ which can be solved to find } \beta = 0.979 \text{ or } v = 0.979c.$$

$$(b) \quad R = \frac{v}{H} = \frac{0.979(299790 \text{ km/s})}{71 \text{ km/s/Mpc}} = 4130 \text{ Mpc};$$

$$4130 \text{ Mpc} \left(\frac{3.26 \text{ Mly}}{1 \text{ Mpc}} \right) = 13.5 \text{ Gly}.$$

20.

(a) Type I supernova have no hydrogen spectral lines. Type Ia has strong silicon spectral lines but no helium lines. Type Ib has very weak or no silicon lines and has neutral helium lines. Type Ic has weak or no helium lines and weak or no silicon lines.

(b) All supernovas other than Type Ia are caused by gravitational contraction of stars much more massive than our Sun.

21.

$$(a) \quad (3 \times 10^{-28} \text{ kg/m}^3) \left(\frac{1 \text{ nucleon}}{1.67 \times 10^{-27} \text{ kg}} \right) = 0.18 \text{ nucleons/m}^3$$

$$(b) \quad R = (16.6 \text{ ly}) (9.46 \times 10^{15} \text{ m/ly}) = 1.57 \times 10^{17} \text{ m}$$

$$60 (1.99 \times 10^{30} \text{ kg}) \frac{1 \text{ nucleon}}{1.67 \times 10^{-27} \text{ kg}} = 7.15 \times 10^{58} \text{ nucleons}$$

The nucleon density is the number of nucleons divided by the volume, or

$$\frac{7.15 \times 10^{58}}{\frac{4}{3} \pi (1.57 \times 10^{17} \text{ m})^3} = 4.41 \times 10^6 \text{ nucleons/m}^3. \text{ The nucleon density in our neighborhood}$$

is much larger than it is for the universe as a whole.

22.

(a) The solid blue curve represents the current, best idea of the expansion of the universe. The expansion of the universe is accelerating. Current models suggest that about 5% of the mass-energy is ordinary matter, about 23% is dark matter, and about 72% is dark energy. Most of the mass is represented by dark energy. The solid black curve represents the known mass of the universe. It is an open, low density universe, and the expansion of the universe is slowing down. The blue dashed curve represents a flat, critical density universe. The expansion rate slows down until the curve becomes more horizontal. The black dashed curve represents a closed, high density universe which will expand for a few more billion years, but eventually will turn around and collapse. There is not enough mass in the universe for this to happen.

(b) Yes, the black dashed curve represents a closed universe. See the explanation in part (a).

$$23. H = (22 \text{ km/s/Mly}) \frac{1 \text{ Mly}}{10^6 (9.461 \times 10^{12} \text{ km})} = 2.32 \times 10^{-18} \text{ s}^{-1}$$

$$\rho_c = \frac{3H^2}{8\pi G} = \frac{3(2.32 \times 10^{-18} \text{ s}^{-1})^2}{8\pi (6.67 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})} = 9.63 \times 10^{-27} \text{ kg/m}^3 = 9.63 \times 10^{-30} \text{ g/cm}^3.$$

$$24. \text{ From Wien's law } \lambda_{\text{max}} T = 2.898 \times 10^{-3} \text{ m} \cdot \text{K}. \quad \lambda_{\text{max}} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{2.725 \text{ K}} = 1.06 \text{ mm}.$$

25. For $H = 64 \text{ km/s/Mpc}$ we have

$$H = (64000 \text{ m/s/Mpc}) \left(\frac{1 \text{ Mpc}}{3.086 \times 10^{22} \text{ m}} \right) = 2.07 \times 10^{-18} \text{ s}^{-1};$$

$$\rho_c = \frac{3H^2}{8\pi G} = \frac{3(2.07 \times 10^{-18} \text{ s}^{-1})^2}{8\pi (6.67 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})} = 7.7 \times 10^{-27} \text{ kg/m}^3.$$

For $H = 78 \text{ km/s/Mpc}$ we have

$$H = (78000 \text{ m/s/Mpc}) \left(\frac{1 \text{ Mpc}}{3.086 \times 10^{22} \text{ m}} \right) = 2.53 \times 10^{-18} \text{ s}^{-1};$$

$$\rho_c = \frac{3H^2}{8\pi G} = \frac{3(2.53 \times 10^{-18} \text{ s}^{-1})^2}{8\pi (6.67 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})} = 1.2 \times 10^{-26} \text{ kg/m}^3.$$

26.

(a) The only combination of the constants that gives time is $m = \frac{1}{2}$, $n = \frac{1}{2}$, and $l = -\frac{5}{2}$.

$$(b) \quad t_p = \sqrt{\frac{Gh}{c^5}} = \sqrt{\frac{(6.67 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})(6.626 \times 10^{-34} \text{ J} \cdot \text{s})}{(2.998 \times 10^8 \text{ m/s})^5}} = 1.35 \times 10^{-43} \text{ s}$$

27. A complete derivation can be found in Section 2.10.

28. Let dm be the mass of a spherical shell of thickness dr , so $dm = \rho d(\text{vol}) = 4\pi r^2 dr$.

$$dV = -\frac{GM'dm}{r} \text{ where } M' = \frac{4}{3}\rho\pi r^3 \text{ is the mass inside radius } r. \text{ Thus}$$

$$dV = -G\rho^2 \frac{16\pi^2}{3} r^4 dr \text{ and therefore } V = \int_0^R dV = -\frac{16\pi^2 \rho^2 R^5 G}{15}.$$

$$\text{Now } \rho = 3M / 4\pi R^3, \text{ so } V = -\frac{16\pi^2 \rho^2 R^5 G}{15} \left(\frac{9M^2}{16\pi^2 R^6} \right) = -\frac{3GM^2}{5R}.$$

29. From the slopes it appears that $t_0 \approx \tau / 2$.

30. In general, as in Example 16.7, we have $\Delta t = \left(\frac{d}{2c} \right) \left(\frac{mc^2}{E} \right)^2$. For a distance of 50 kpc,

$$\frac{d}{2c} = \frac{(50 \text{ kpc})}{2(3.00 \times 10^8 \text{ m/s})} \frac{3.086 \times 10^{19} \text{ m}}{\text{kpc}} = 2.57 \times 10^{12} \text{ s. Note that}$$

$$2.57 \times 10^{12} \text{ s} = (2.57 \text{ s})(10 \text{ MeV}/10 \text{ eV})^2, \text{ so for any distance (in kpc)}$$

$$\Delta t = (2.57 \text{ s}) \left(\frac{\text{distance}}{50 \text{ kpc}} \right) \left(\frac{m_e c^2}{10 \text{ eV}} \right)^2 \left(\frac{10 \text{ MeV}}{E} \right)^2.$$

31. We know from Section 9.7 [see the comment after Equation (9.56)] that the energy flux rate $\Phi = \sigma T^4$ is related to the energy density ρ_e by a factor of $c/4$, such that

$$\Phi = (c/4)\rho_e. \text{ But using } E = mc^2 \text{ we have } \rho_e = \rho_{\text{rad}} c^2, \text{ so } \sigma T^4 = \frac{c}{4} \rho_e = \frac{c}{4} \rho_{\text{rad}} c^2 \text{ or,}$$

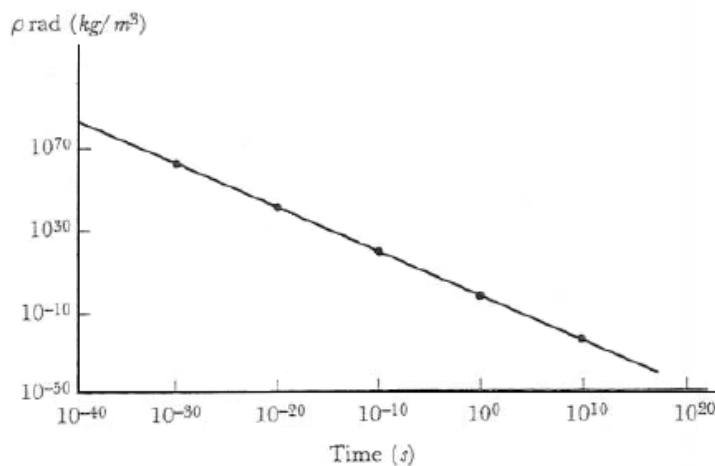
$$\text{rearranging } \rho_{\text{rad}} = \frac{4\sigma T^4}{c^3}.$$

$$32. \rho_{\text{rad}} = \frac{4\sigma T^4}{c^3} = \frac{4(5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4})(2.725 \text{ K})^4}{(3.00 \times 10^8 \text{ m/s})^3} = 4.63 \times 10^{-31} \text{ kg/m}^3$$

This is substantially less than the matter density of about $3 \times 10^{-28} \text{ kg/m}^3$, so the universe is matter dominated.

33. At 380,000 years the temperature is about 10^2 K . Then

$$\rho_{\text{rad}} = \frac{4\sigma T^4}{c^3} = \frac{4(5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4})(10^2 \text{ K})^4}{(3.00 \times 10^8 \text{ m/s})^3} = 8.4 \times 10^{-25} \text{ kg/m}^3.$$



34. The 300 day mark is almost exactly 200 days after the peak, so use $t = 200$ d in the exponential decay formula $\exp(-(\ln 2)t / t_{1/2})$.

$$^{56}\text{Ni}: \exp(-(\ln 2)t / t_{1/2}) = \exp(-(\ln 2)(200 \text{ d}) / (6.1 \text{ d})) = 1.35 \times 10^{-10}$$

$$^{56}\text{Co}: \exp(-(\ln 2)t / t_{1/2}) = \exp(-(\ln 2)(200 \text{ d}) / (77.1 \text{ d})) = 0.166$$

The cobalt must be primarily responsible, with almost no contribution from the nickel.

$$35. \text{Redshift} = \frac{\Delta\lambda}{\lambda_0} = \frac{580.0 \text{ nm} - 121.6 \text{ nm}}{121.6 \text{ nm}} = 3.77; \quad 3.77 = \sqrt{\frac{1+\beta}{1-\beta}} - 1$$

Therefore $\beta = 0.916$ and $v = 0.916c$.

$$36. \text{By definition } \frac{\Delta\lambda}{\lambda_0} = z. \text{ From Equation (16.18) we know } \frac{\Delta\lambda}{\lambda_0} = \sqrt{\frac{1+\beta}{1-\beta}} - 1; \text{ this is}$$

$$\text{equivalent to } \frac{\Delta\lambda}{\lambda_0} + 1 = z + 1 = \sqrt{\frac{1+\beta}{1-\beta}}.$$

37. Using the binomial expansion with the result of the preceding problem

$$1 + z \approx \left(1 + \frac{\beta}{2} + \dots\right) \left(1 + \frac{\beta}{2} + \dots\right) \approx 1 + \beta \text{ so we see that to first order } z = \beta.$$

$$38. 5.34 = \sqrt{\frac{1+\beta}{1-\beta}} - 1 \text{ so } \beta = 0.951 \text{ and } v = 0.951c;$$

$$R = \frac{v}{H} = \frac{0.951(299790 \text{ km/s})}{71 \text{ km/s/Mpc}} = 4016 \text{ Mpc}.$$

$$39. Q = (2m_p - m_d - m_e)u \cdot c^2 = 0.43 \text{ MeV}.$$

There are three particles in the final state, but it is possible for the deuteron and positron to have negligible energy, in which case the neutrino energy is 0.43 MeV.

We begin with Equation (16.21):

$$\rho_c = \frac{3H^2}{8\pi G} = \left(\frac{H \text{ in (km/s)/Mpc}}{100}\right)^2 \frac{3 \times 10^4}{8\pi(6.6726 \times 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2})}$$

$$40. = \left(\frac{H}{100}\right)^2 (1.789 \times 10^{13} \text{ km}^2 \cdot \text{s}^{-2} \cdot \text{Mpc}^{-2} \cdot \text{m}^{-3} \cdot \text{kg}^{-1} \cdot \text{s}^2) \left(\frac{\text{Mpc}}{3.086 \times 10^{22} \text{ m}}\right)^2 \left(10^3 \frac{\text{m}}{\text{km}}\right)^2$$

$$= \left(\frac{H}{100}\right)^2 (1.88 \times 10^{-26} \text{ kg/m}^3) = \left(\frac{H}{100}\right)^2 (1.88 \times 10^{-29} \text{ g/m}^3)$$

41.

$$(a) \quad a = Ct^n; \quad da/dt = nCt^{n-1}; \quad d^2a/dt^2 = n(n-1)Ct^{n-2};$$

$$q = -a \frac{d^2a/dt^2}{\left(\frac{da}{dt}\right)^2} = -\frac{an(n-1)Ct^{n-2}}{n^2C^2t^{2n-2}} = -\frac{an(n-1)}{n^2Ct^n} = -\frac{n(n-1)}{n^2}$$

where in the last step we used the definition $a = Ct^n$. From this we see that there is deceleration ($q > 0$) only if $0 < n < 1$.

$$(b) \quad H = \frac{1}{a} \frac{da}{dt} = \frac{nCt^{n-1}}{Ct^n} = \frac{n}{t} \text{ with } 0 < n < 1. \text{ There is an inverse dependence on time.}$$

42.

$$(a) \quad \frac{X_n}{X_p} = \frac{\exp(-E_n/kT)}{\exp(-E_p/kT)} = \exp\left[-(E_n - E_p)/kT\right] = \exp\left[-(m_n - m_p)c^2/kT\right] \text{ where is}$$

last step uses the fact that since $E = K + mc^2$, the kinetic energy term will cancel.

The difference in rest energy equals 1.3 MeV so we have $\frac{X_n}{X_p} = \exp(-1.3 \text{ MeV}/kT)$.

$$(b) \quad \frac{X_n}{X_p} = \frac{1}{6.7} = \exp(-1.3 \text{ MeV}/kT)$$

Take the natural logarithm of both sides and solve for kT . We find $kT = 0.68 \text{ MeV}$ and thus $T = 7.9 \times 10^9 \text{ K}$.

(c) Using the equipartition theorem, this temperature corresponds to an energy of $K = \frac{3}{2}kT = 1.02 \text{ MeV}$. According to Figure 16.6, this temperature corresponds to a time of perhaps 0.1 s or 1 s, which is before the time at which nuclei could form.

43. For a uniform distribution assume a cubical lattice of stars, each separated by distance d . Then each star occupies a cube of volume d^3 , and the density of that cube and the universe as a whole is $\rho = M/d^3$. Solving for d ,

$$d = \left(\frac{M}{\rho}\right)^{1/3} = \left(\frac{2.0 \times 10^{30} \text{ kg}}{9 \times 10^{-27} \text{ kg/m}^3}\right)^{1/3} = 6.0 \times 10^{18} \text{ m. Converting to lightyears,}$$

$$d = 6.0 \times 10^{18} \text{ m} \left(\frac{1 \text{ ly}}{9.45 \times 10^{15} \text{ m}}\right) = 640 \text{ ly.}$$

Our local region is much denser than this, because we know there are many stars closer than 640 ly to the sun. (Other than the sun, the closest star system, Alpha Centauri, is just over 4 ly away.) This density is to be expected, because galaxies are relatively dense collections of stars.